

Fundamental Kinetics Database Utilizing Shock Tube Measurements

Section A: Ignition Delay Time Measurements

Section B: Species Time-History Measurements

Section C: Reaction Rate Constant Measurements

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Preface

The Fundamental Kinetic Database Utilizing Shock Tube Measurements summarizes the published experimental shock tube research performed under the supervision of Prof. Ronald K. Hanson and Dr. David Davidson of the Mechanical Engineering Department at Stanford University. The database covers the years from 1974 to 2019 inclusively. The database is divided into three types of data: ignition delay times, species time-history measurements, and reaction rate measurements.

Section A (previously Volumes 1, 4 and 7) include summaries of the ignition delay time data measured in shock tubes for various fuel/oxidizer mixtures.

Section B (previously Volumes 2, 5 and 8) include summaries of measured species concentration time-histories for various fuel and components.

Section C (previously Volumes 3, 6 and 9) include summaries of reaction rate constants measured using shock tubes and laser absorption.

The database is also available in the Stanford Digital Repository. It can be accessed at <https://purl.stanford.edu/kb621cw6967>.

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Introduction

There is a critical need for standardized experimental data that can be used as targets in the validation and refinement of reaction mechanisms for hydrocarbon fuels. In our laboratory at Stanford University, we are able to provide some of this data in the form of shock tube experiments.

The data from shock tube experiments generally take three forms: ignition delay times, species concentration time-histories and reaction rate measurements.

Ignition delay times are a measure of the time from initial shock wave heating to a defined ignition point, often a rapid change in pressure or radical species population. These targets place a constraint on the overall predictive behavior of the reaction mechanism. Does the mechanism predict the time of ignition properly for a particular initial temperature, pressure and mixture composition? These ignition delay times can also be provided in the form of correlation equations which provide similar information in a compact form.

Species concentration time-histories are a measure of the concentration of a particular species as a function of time during the entire experiment. These targets place strong constraints on the internal workings of the reaction mechanism. Concentration time-histories for OH, for example, are strongly related to the concentrations of other small radical species including: H-atoms, O-atoms, and HO₂. The production and removal rates of these species have an important role in the reaction progress to ignition.

Reaction rate measurements provide the basic rate data that reaction mechanisms are comprised of. Accurate measurements are needed of the rates of critical reactions that important system parameters are sensitive to, such as ignition delay times, heat release rates, and product species. These are necessary as it is not yet possible to accurately predict these rates (nor is it likely that they will ever be reliably predicted) without experimental verification.

Shock tube data are well suited for comparison with computation models. Shock wave experiments can provide near constant-volume test conditions, generally over the entire time period before ignition, and in many cases for longer times. Shock tube experiments can provide test conditions over a wide range of temperatures, pressure and gas mixtures, typically over temperatures of 500 to 4000 K, pressures from sub-atmospheric to 1000 atm, and fuel concentrations from ppm to percent levels with test times in the 1-20 ms range. Methods have been developed to extend these ranges if need be. The nature of planar shock wave flows as they are formed in conventional shock tubes means that the test gas mixtures are effectively instantaneously compressed and heated, providing very simple initial conditions for modeling. The spatial uniformity of the stationary heated test gas mixture behind reflected shock waves means that only chemistry need be modeled, and fluid mechanical effects such as diffusion, mixing, and fluid

movement are not significant in most cases. And finally, the time scales and physical dimensions of shock tube experiments means that the test gas volume can be considered to be adiabatically isolated from its surroundings.

Database Format

The database is comprised of three sections: Section A, ignition delay time measurements; Section B, species concentration time histories; and Section C, reaction rate measurements. The formal final cut-off point for inclusion into the database for all sections is June 2019.

Section A. The data in Section A is limited to ignition delay times and ignition delay times derived from species time histories. Each data set includes the literature source of the data, a table describing the range of the data, a short description of the data type, and the data table. All data in this database have been previously published in refereed journals, conference proceedings or Ph.D. theses. The short description of the data type includes information on the diagnostic used in the measurement, as well as the type of carrier gas (generally argon or nitrogen). The data table includes: the initial reflected shock temperature and pressure, mixture composition, and equivalence ratio, the ignition delay time and/or discrete species concentration points (such as time and concentration at the profile peak or plateau). Further information on each dataset can be derived from the literature sources of the data.

Measurements can be separated into two groups: low and high pressure. The low pressure measurements (generally up to 10 atm) were performed in the Stanford 15.2 and 14.3 cm diameter shock tubes; the high pressure measurements (generally above 10 atm) were performed in the Stanford 5 cm diameter shock tube.

It should be noted that the ignition times described in these table have different definitions depending on the diagnostic used. In general, for the conditions of the experiments described here, the differences in ignition time from different definitions are not significant. However, these differences should be reviewed before comparison with other measurements. The discrete species concentration measurements included for some fuels in this volume can be related to ignition delay times by comparison with the modeled concentrations for these fuels. A discussion of ignition delay time data and other technicalities of shock tube work is given in Davidson and Hanson (2004).

D. F. Davidson and R. K. Hanson, "Interpreting Shock Tube Ignition Data," International Journal of Chemical Kinetics Vol. 36, pp. 510-523 (2004).

Section B. The data in Section B is limited to species concentration time-histories in pyrolysis and oxidative systems. Each data set includes the literature source of the data. All data in this database have been previously published in refereed journals, conference proceedings or Ph.D. theses. The short description of the species measured and the wavelength used is included. An overview of measurement techniques of species concentration time-histories using laser absorption diagnostics is available in Davidson and Hanson (2001).

Laser absorption measurements of species concentration are based on the Beer-Lambert Law:

$$I/I_0 = \exp(-k_\lambda P X L)$$

where I/I_0 is the ratio of the transmitted laser power to the incident laser power, k_λ is the absorption coefficient (generally in units of $[\text{atm}^{-1}\text{cm}^{-1}]$), P is the total pressure in $[\text{atm}]$, X is the absorbing species mole fraction $[\text{n.d.}]$, and L is the optical path length in $[\text{cm}]$ (the shock tube diameter).

The majority of the measurements in the current database were performed in low pressure shock tubes (Stanford 15.2 and 14.3 cm diameter shock tubes). In later years, substantially more measurements were performed in the high pressure Stanford 5 cm diameter shock tube.

D. F. Davidson, R. K. Hanson, "Spectroscopic Diagnostics," Chapter 5.2 in Handbook of Shock Waves, Volume 1, G. Ben-Dor, O. Igara, T. Elperin eds. Academic Press, San Diego (2001).

Section C. The data in Section C is limited to reaction rate constant measurements acquired using shocks tubes and laser absorption. Generally, temperatures are given in Kelvin and activation energies are given in calories/mole and using $R = 1.987$ $[\text{mole/calories/K}]$ when a conversion from a published activation energy in Kelvins is needed and 4.18 Joules = 1 calorie when the published activation energy is in Joules. Reaction rate constants are given in units of $[1/\text{s}]$, $[\text{cm}^3/\text{mol/s}]$, or $[\text{cm}^6/\text{mol}^2/\text{s}]$. Pressures are quoted in atm. The publications are listed approximately in chronological order.

IDT Measurements up to January 2005

Small Fuels

Hydrogen

Literature Source of Data:

E. L. Petersen, D. F. Davidson, M. Rohrig, R. K. Hanson, “*Shock-Induced Ignition of High-Pressure H_2 - O_2 -Ar and CH_4 - O_2 -Ar Mixtures*,” AIAA 95-3113, 31st AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit, San Diego (1995).

E. L. Petersen, D. F. Davidson, M. Rohrig, R. K. Hanson, *High-Pressure Shock-Tube Measurements of Ignition Times in Stoichiometric H_2 - O_2 -Ar Mixtures*,” Proceedings of the 20th International Symposium on Shock Waves, pp. 941-946, Pasadena (1995).

Range of Data:

Temperature [K]	1189	1930
Pressure [atm]	33	87
Fuel Mole Fraction [%]	0.1	2.0
Oxygen Mole Fraction [%]	0.05	1.0
Equivalence Ratio	1	1

Type of Data:

Hydrogen Table 1: Ignition delay time measurement in argon based on $\{d[OH]/dt\}_{\max}$ from laser absorption of OH at 306 nm.

Hydrogen Table 1:

T_5 [K]	P_5 [atm]	Hydrogen [%]	Oxygen [%]	ϕ (E.R.)	Ignition Time [μs]
1684	33	0.5	0.25	1	12
1709	33	0.5	0.25	1	9
1709	33	0.5	0.25	1	12
1715	33	0.5	0.25	1	11
1724	33	0.5	0.25	1	11
1730	33	0.5	0.25	1	11
1736	33	0.5	0.25	1	11
1739	33	0.5	0.25	1	11
1754	33	0.5	0.25	1	10
1764	33	0.5	0.25	1	10
1773	33	0.5	0.25	1	10
1792	33	0.5	0.25	1	9

Hydrogen Table 1 (continued):

T_5 [K]	P_5 [atm]	Hydrogen [%]	Oxygen [%]	ϕ (E.R.)	Ignition Time [μ s]
1802	33	0.5	0.25	1	9
1855	33	0.5	0.25	1	9
1189	33	2	1	1	393
1206	33	2	1	1	89
1206	33	2	1	1	203
1221	33	2	1	1	133
1252	33	2	1	1	57
1264	33	2	1	1	29
1289	33	2	1	1	13
1300	33	2	1	1	13
1655	57	0.5	0.25	1	12
1669	57	0.5	0.25	1	11
1672	57	0.5	0.25	1	7
1678	57	0.5	0.25	1	7
1681	57	0.5	0.25	1	7
1683	57	0.5	0.25	1	13
1684	57	0.5	0.25	1	6
1685	57	0.5	0.25	1	6
1700	57	0.5	0.25	1	6
1710	57	0.5	0.25	1	10
1713	57	0.5	0.25	1	6
1714	57	0.5	0.25	1	6
1733	57	0.5	0.25	1	6
1748	57	0.5	0.25	1	7
1757	57	0.5	0.25	1	5
1779	57	0.5	0.25	1	8
1930	57	0.5	0.25	1	4
1684	64	0.33	0.167	1	12
1701	64	0.33	0.167	1	9
1709	64	0.33	0.167	1	10
1712	64	0.33	0.167	1	9
1715	64	0.33	0.167	1	10
1733	64	0.33	0.167	1	9
1779	64	0.33	0.167	1	8
1361	64	0.1	0.05	1	145
1366	64	0.1	0.05	1	202
1462	64	0.1	0.05	1	94
1481	64	0.1	0.05	1	53
1524	64	0.1	0.05	1	49

Hydrogen Table 1 (continued):

T_5 [K]	P_5 [atm]	Hydrogen [%]	Oxygen [%]	ϕ (E.R.)	Ignition Time [μ s]
1553	64	0.1	0.05	1	39
1555	64	0.1	0.05	1	49
1560	64	0.1	0.05	1	47
1575	64	0.1	0.05	1	49
1577	64	0.1	0.05	1	35
1577	64	0.1	0.05	1	31
1577	64	0.1	0.05	1	42
1585	64	0.1	0.05	1	38
1585	64	0.1	0.05	1	40
1616	64	0.1	0.05	1	29
1876	64	0.1	0.05	1	21
1279	64	0.5	0.25	1	265
1314	64	0.5	0.25	1	152
1344	64	0.5	0.25	1	69
1715	87	0.5	0.25	1	6
1704	87	0.5	0.25	1	6
1701	87	0.5	0.25	1	5
1706	87	0.5	0.25	1	6
1712	87	0.5	0.25	1	6
1715	87	0.5	0.25	1	6

Methane

Literature Source of Data:

E. J. Chang, “*Shock Tube Experiments for the Development and Validation of Models of Hydrocarbon Combustion*,” M. Eng. Thesis, (also published as HTGL Report No. T-320), Mechanical Engineering Department, Stanford University, Stanford CA (1995).

M. Frenklach, H. Wang, M. Goldenberg, G. P. Smith, D. M. Golden, C. T. Bowman, R. K. Hanson, W. C. Gardiner, V. Lissianski, “*GRI-Mech – An Optimized Detailed Chemical Reaction Mechanism for Methane Combustion*,” Topical Report GRI-95/0058, Gas Research Institute (1995).

Range of Data:

Temperature [K]	1933	2470
Pressure [atm]	0.37	1.22
Fuel Mole Fraction [%]	0.1	0.3
Oxygen Mole Fraction [%]	0.1	1.21
Equivalence Ratio	0.5	4

Type of Data:

Methane Table 1: CO₂ species concentration time history IR laser absorption measurements at 4.200 μm (2380.72 cm^{-1}) in argon.

Methane Table 2: CH₃ species concentration time history UV laser absorption measurements at 216 nm in argon.

Methane Table 3: OH species concentration time history UV laser absorption measurements at 306.7 nm (32606.56 cm^{-1}) in argon.

Methane Table 1:

T ₂ [K]	P ₂ [atm]	Methane [ppm]	Oxygen [ppm]	ϕ (E.R.)	CO ₂ @1ms [ppm]	Plateau CO ₂ [ppm]	¼ Plat [μs]
2164	0.395	3007	6055	1	941	1980	600
2238	0.541	3007	6055	1	1057	1828	340
2317	0.541	3007	6055	1	1123	1588	235
2470	0.512	3007	6055	1	998	1144	141
2140	0.616	3005	12079	0.5	1764	2661	304
2298	0.559	3005	12079	0.5	1599	2270	193
2374	0.370	3005	12079	0.5	1532	1937	238
2408	0.496	3005	12079	0.5	1477	1913	155

Methane Table 2:

T₅ [K]	P₅ [atm]	Methane [ppm]	Oxygen [ppm]	φ (E.R.)	Peak CH₃ [ppm]	Time [μs]
1951	1.066	2012	1000	4	236	530
2075	1.011	2012	1000	4	337	256
2264	1.024	2012	1000	4	451	83
2401	0.939	2012	1000	4	550	42
1942	1.092	1000	2011	0.99	145	625
1989	1.090	994	2021	0.98	171	443
1997	1.032	994	2021	0.98	187	281
2053	1.016	994	2021	0.98	203	230
2169	0.990	1000	2011	0.99	216	129
2224	1.039	994	2021	0.98	270	109
2338	1.219	994	2021	0.98	342	34
2503	0.924	1000	2011	0.99	325	35
1933	1.099	1000	4034	0.5	134	508
2210	1.006	1000	4034	0.5	240	89
2460	0.910	1000	4034	0.5	333	27.5

Methane Table 3:

T₅ or T₂ [K]	P₅ or P₂ [atm]	Methane [ppm]	Oxygen [ppm]	φ (E.R.)	Max OH [ppm]	½ Max [μs]
2056	1.036	997	1996	1	254	632
2145	1.063	1002	2000	1	295	287
2166	1.098	1002	2000	1	290	305
2190	1.130	997	1996	1	303	230
2216	1.081	1002	2000	1	321	195
2169	0.497	1007	2014	1	286	590
2223	0.348	1007	2014	1	304	650
2279	0.490	1007	2014	1	323	403
2299	0.472	1007	2014	1	336	355
2623	0.413	1007	2014	1	429	140

Methane (continued)

Literature Source of Data:

E. L. Petersen, "A Shock Tube and Diagnostics For Chemistry Measurements at Elevated Pressures with Application to Methane Ignition," Ph.D. Thesis, (also published as Topical Report No. TSD-111,) Mechanical Engineering Department, Stanford University, Stanford CA (1998),

E. L. Petersen, D. F. Davidson, R. K. Hanson, "Ignition Delay Times of Ram Accelerator CH_4/O_2 /Diluent Mixtures," Journal of Propulsion and Power 15: 82-91 (1999).

E. L. Petersen, D. F. Davidson, R. K. Hanson, "Kinetics Modeling of Shock-Induced Ignition in Low-Dilution CH_4/O_2 Mixtures at High Pressures and Intermediate Temperatures," Paper 97S-066, Western States Section/The Combustion Institute Spring Meeting, Sandia (1997),

E. L. Petersen, D. F. Davidson, R. K. Hanson, "Ram Accelerator Mixture Chemistry: Kinetics Modeling and Ignition Measurements," CPIA Publication #653, JANNAF 33rd Combustion Subcommittee Meeting, Monterey (1996).

E. L. Petersen, D. F. Davidson, R. K. Hanson, "Ignition Delay Times of Ram Accelerator Mixtures," AIAA 96-2681, 32nd AIAA/ASME/SAE/ASEE Joint Propulsion Conference, Lake Buena Vista (1996).

Range of Data:

Temperature [K]	1041	1607
Pressure [atm]	12.0	263.6
Fuel Mole Fraction [%]	3.8	50.0
Oxygen Mole Fraction [%]	11.8	19.2
Equivalence Ratio	0.4	6

Type of Data:

Methane Table 4: High pressure ignition delay time data with argon as the bulk carrier gas using PZT pressure measurements of the time between the arrival of the reflected shock (the center of the reflected shock bifurcation feature) and the distinct ignition pressure rise (the time of the intersection of the linear extrapolation of the pressure rise with the pre-ignition pressure floor). These ignition delay time values have been corrected to account for blast wave arrival effects at the shock tube sidewall measurement location.

Methane Table 4: Ignition delay times based on emission measurements are also reported for visible (Si photo diode) and selected other wavelengths.

Methane Table 5: Same as Methane Table 4 but with nitrogen as the bulk carrier gas.

Methane Table 6: Same as Methane Table 4 but with helium or nitrogen and helium as the bulk carrier gas.

Methane Table 4:

T_5 [K]	P_5 [atm]	Methane [%]	Oxygen [%]	Argon [%]	$\phi_{\square\square}$ (E.R.)	Ignition (Pressure) [μ s]	Ignition (Emission) [μ s]
1203	50.5	3.8	19.2	77	0.4	694	689
1232	50.0	3.8	19.2	77	0.4	555	565
1268	49.0	3.8	19.2	77	0.4	406	
1285	44.4	3.8	19.2	77	0.4	415	407
1319	48.9	3.8	19.2	77	0.4	261	273
1359	47.3	3.8	19.2	77	0.4	206	202
1137	100.9	3.8	19.2	77	0.4	661	
1169	98.8	3.8	19.2	77	0.4	541	532
1198	96.2	3.8	19.2	77	0.4	440	
1207	94.7	3.8	19.2	77	0.4	392	
1268	98.2	3.8	19.2	77	0.4	241	239
1292	100.0	3.8	19.2	77	0.4	191	188
1312	94.5	3.8	19.2	77	0.4	169	161
1361	94.7	3.8	19.2	77	0.4	114	108
1198	161.2	3.8	19.2	77	0.4	253	244
1250	150.8	3.8	19.2	77	0.4	177	175
1325	39.8	20	13.3	66.7	3.0	654	
1346	47.5	20	13.3	66.7	3.0	361	
1383	38.5	20	13.3	66.7	3.0	391	
1431	34.5	20	13.3	66.7	3.0	250	
1536	35.6	20	13.3	66.7	3.0	102	
1167	91.0	20	13.3	66.7	3.0	572	563
1196	80.9	20	13.3	66.7	3.0	681	
1203	77.7	20	13.3	66.7	3.0	622	607
1206	83.0	20	13.3	66.7	3.0	541	541
1363	91.8	20	13.3	66.7	3.0	152	
1418	92.9	20	13.3	66.7	3.0	95	
1179	106.0	20	13.3	66.7	3.0	414	415
1182	132.2	20	13.3	66.7	3.0	280	274
1182	102.8	20	13.3	66.7	3.0	419	419
1156	114.0	20	13.3	66.7	3.0	421	430
1244	114.4	20	13.3	66.7	3.0	254	260
1380	120.0	20	13.3	66.7	3.0	77	71

Methane Table 4 (continued):

T ₅ [K]	P ₅ [atm]	Methane [%]	Oxygen [%]	Argon [%]	$\phi_{\square\square}$ (E.R.)	Ignition (Pressure) [μs]	Ignition (Emission) [μs]
1209	178.4	20	13.3	66.7	3.0	170	
1254	163.0	20	13.3	66.7	3.0	157	
1372	164.0	20	13.3	66.7	3.0	71	
1176	262.5	20	13.3	66.7	3.0	98	
1239	263.6	20	13.3	66.7	3.0	70	
1301	256.2	20	13.3	66.7	3.0	55	
1191	51.5	27.3	18.2	54.5	3.0	847	854
1295	46.6	27.3	18.2	54.5	3.0	530	530
1362	44.9	27.3	18.2	54.5	3.0	221	230
1215	71.3	27.3	18.2	54.5	3.0	411	
1283	64.3	27.3	18.2	54.5	3.0	288	
1346	60.4	27.3	18.2	54.5	3.0	190	
1089	125.5	27.3	18.2	54.5	3.0	331	338
1156	128	27.3	18.2	54.5	3.0	231	231

Methane Table 5:

T ₅ [K]	P ₅ [atm]	Methane [%]	Oxygen [%]	N ₂ [%]	$\phi_{\square\square}$ (E.R.)	Ignition (Pressure) [μs]	Ignition (Emission) [μs]
1358	47.3	20	13.3	66.7	3	327	
1362	41.4	20	13.3	66.7	3	360	
1381	36.7	20	13.3	66.7	3	425	
1402	43.3	20	13.3	66.7	3	265	
1409	48.2	20	13.3	66.7	3	214	
1414	42.2	20	13.3	66.7	3	238	
1418	40.4	20	13.3	66.7	3	234	
1454	39.5	20	13.3	66.7	3	182	
1486	44.3	20	13.3	66.7	3	115	
1511	39.3	20	13.3	66.7	3	106	
1290	72.1	20	13.3	66.7	3	431	416
1344	76.1	20	13.3	66.7	3	259	
1366	74.0	20	13.3	66.7	3	194	
1408	76.5	20	13.3	66.7	3	124	112
1466	75.7	20	13.3	66.7	3	74	74
1227	92.0	20	13.3	66.7	3	504	508
1273	86.9	20	13.3	66.7	3	445	468
1316	84.1	20	13.3	66.7	3	330	344
1390	85.0	20	13.3	66.7	3	152	163

Methane Table 5 (continued):

T ₅ [K]	P ₅ [atm]	Methane [%]	Oxygen [%]	N ₂ [%]	φ _{□□} (E.R.)	Ignition (Pressure) [μs]	Ignition (Emission) [μs]
1437	84.4	20	13.3	66.7	3	80	86
1149	113.0	20	13.3	66.7	3	533	537
1238	116.8	20	13.3	66.7	3	282	
1286	103.1	20	13.3	66.7	3	233	
1323	117.4	20	13.3	66.7	3	155	
1186	145.0	20	13.3	66.7	3	261	
1216	139.6	20	13.3	66.7	3	227	
1293	148.2	20	13.3	66.7	3	130	
1317	134.9	20	13.3	66.7	3	135	
1196	55.3	27.3	18.2	54.5	3	738	742
1258	54.4	27.3	18.2	54.5	3	453	453
1371	59.6	27.3	18.2	54.5	3	151	
1242	86.6	27.3	18.2	54.5	3	228	232
1272	82.8	27.3	18.2	54.5	3	200	195
1041	128.4	27.3	18.2	54.5	3	433	
1138	139.9	27.3	18.2	54.5	3	194	
1186	138.5	27.3	18.2	54.5	3	182	186
1249	122.8	27.3	18.2	54.5	3	142	
1250	133.0	27.3	18.2	54.5	3	118	
1304	136.0	27.3	18.2	54.5	3	89	
1327	129.7	27.3	18.2	54.5	3	81	
1150	176.6	27.3	18.2	54.5	3	120	111
1230	188.7	27.3	18.2	54.5	3	71	
1235	194.0	27.3	18.2	54.5	3	67	
1268	169.7	27.3	18.2	54.5	3	72	

Methane Table 6:

T ₅ [K]	P ₅ [atm]	Methane [%]	Oxygen [%]	He [%]	N ₂ [%]	φ _{□□} (E.R.)	Ignition (Pressure) [μs]	Ignition (Emission) [μs]
1453	20.3	50	16.7	33.3	0	6	466	
1520	13.8	50	16.7	33.3	0	6	420	
1523	18.6	50	16.7	33.3	0	6	333	
1529	13.8	50	16.7	33.3	0	6	391	
1548	15.2	50	16.7	33.3	0	6	345	
1607	12.0	50	16.7	33.3	0	6	265	
1456	57.2	50	16.7	33.3	0	6	86	82
1547	51.2	50	16.7	33.3	0	6	58	58

Methane Table 6 (continued):

T₅ [K]	P₅ [atm]	Methane [%]	Oxygen [%]	He [%]	N₂ [%]	φ_{□□} (E.R.)	Ignition (Pressure) [μs]	Ignition (Emission) [μs]
1324	79.0	50	16.7	33.3	0	6	185	187
1367	72.8	50	16.7	33.3	0	6	135	138
1395	68.4	50	16.7	33.3	0	6	111	105
1128	90.7	50	16.7	33.3	0	6	332	334
1203	91.3	50	16.7	33.3	0	6	315	297
1290	81.6	50	16.7	33.3	0	6	249	241
1272	35.5	17.6	11.8	33.3	23.6	3	875	
1338	29.5	17.6	11.8	33.3	23.6	3	591	651
1398	28.9	17.6	11.8	33.3	23.6	3	418	425
1458	27.0	17.6	11.8	33.3	23.6	3	260	269
1342	49.5	17.6	11.8	33.3	23.6	3	379	399
1469	47.1	17.6	11.8	33.3	23.6	3	149	151
1269	57.2	17.6	11.8	33.3	23.6	3	529	572
1366	61.5	17.6	11.8	33.3	23.6	3	181	180
1391	60.5	17.6	11.8	33.3	23.6	3	152	151
1347	76.4	17.6	11.8	33.3	23.6	3	233	233
1340	92.5	17.6	11.8	33.3	23.6	3	128	128
1358	106.6	17.6	11.8	33.3	23.6	3	111	111

Methane (continued)

Literature Source of Data:

D. Woiki, M. Votsmeier, D. F. Davidson, R. K. Hanson, C. T. Bowman, “CH-Radical Concentration Measurements in Fuel-Rich CH₄/O₂/Ar and CH₄/O₂/NO/Ar Mixtures Behind Shock Waves,” Combustion and Flame 113: 624-626 (1998).

Range of Data:

Temperature [K]	2644	3254
Pressure [atm]	1.7	1.8
Fuel Mole Fraction [ppm]	80	80
Oxygen Mole Fraction [ppm]	100	100
Nitric Oxide Mole Fraction [ppm]	0	400
Equivalence Ratio	0.53	1.6

Type of Data:

Methane Table 7: CH species concentration time history measurements in argon using laser absorption at 431 nm. Data was derived from a digitization of Fig. 2 of Woiki et al. (1998).

Methane Table 7:

T ₅ [K]	P ₅ [atm]	Methane [ppm]	Oxygen [ppm]	NO [ppm]	φ _{CH} (E.R.)	CH Peak [ppm]
3038	1.7	80	100	0	1.6	6.1
3018	1.7	80	100	0	1.6	6.4
2875	1.8	80	100	0	1.6	3.9
2773	1.7	80	100	0	1.6	3.0
2773	1.7	80	100	0	1.6	2.8
2656	1.7	80	100	0	1.6	1.8
2644	1.7	80	100	0	1.6	1.9
3254	1.7	80	100	400	0.53	4.9
3212	1.7	80	100	400	0.53	4.0
2970	1.7	80	100	400	0.53	2.9
2896	1.8	80	100	400	0.53	2.0
2806	1.7	80	100	400	0.53	1.6
2687	1.7	80	100	400	0.53	1.2

Methane (continued)

Literature Source of Data:

E. L. Petersen, M. Rohrig, D. F. Davidson, R. K. Hanson, C. T. Bowman, "*High Pressure Methane Oxidation Behind Reflected Shock Waves*," Proceedings of the Combustion Institute 26: 799-806 (1996).

E. L. Petersen, D. F. Davidson, M. Rohrig, R. K. Hanson, C. T. Bowman, "*A Shock Tube Study of High Pressure Methane Oxidation*," Paper 95F-153, Western States Section/The Combustion Institute Fall Meeting, Stanford (1995).

E. L. Petersen, D. F. Davidson, M. Rohrig, R. K. Hanson, "*Shock-Induced Ignition of High-Pressure H_2 - O_2 -Ar and CH_4 - O_2 -Ar Mixtures*," AIAA 95-3113, 31st AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit, San Diego (1995).

Range of Data:

Temperature [K]	1408	2043
Pressure [atm]	9.2	86.8
Fuel Mole Fraction [%]	0.25	3.4
Oxygen Mole Fraction [%]	0.14	6.7
Equivalence Ratio	0.5	4.0

Type of Data:

Methane Table 8: OH species concentration time history measurements in argon using laser absorption at 306.7 nm.

Methane Table 8: High pressure ignition delay time data in argon using same definition as found in Methane Table 4.

Methane Table 9: CH_4 species concentration time history induction time and end time are determined from the interception points (times) of a linear fit through the decaying CH_4 signal and the initial and final signal levels respectively.

Methane Table 9: High pressure ignition delay time data using same definition as found in Methane Table 4.

Methane Table 8:

T ₅ [K]	P ₅ [atm]	Methane [%]	O ₂ [%]	ϕ (E.R.)	OH First Rise [μs]	OH Peak [μs]	Ignition (Pressure) [μs]
1680	85.3	0.25	1	0.5	92	118	
1770	31.2	0.25	1	0.5	62	89	
1730	30.3	0.5	2	0.5	64	73	60
1659	80.9	0.5	2	0.5	75	82	74
1665	81.2	0.5	2	0.5			62
1830	9.8	0.5	2	0.5	68	98	66
1853	10.5	0.5	2	0.5	51	75	47
1706	83.9	0.28	0.56	1		200	160
1811	31.3	0.28	0.56	1		128	
1687	30.5	0.28	0.56	1		401	
1761	36.8	0.28	0.56	1		177	
1821	79.4	0.28	0.56	1		74	
1778	79.1	0.56	1.14	1	47	53	46
1957	9.7	0.56	1.14	1	40	58	
2043	9.4	0.56	1.14	1	26	41	
1534	30.1	3.4	6.7	1			206
1535	61.9	3.4	6.7	1			113
1633	31.7	3.4	6.7	1			51
1584	34.1	3.4	6.7	1			96
1496	34.4	3.4	6.7	1			229
1408	34.6	3.4	6.7	1			554
1408	34.6	3.4	6.7	1			554

Methane Table 9:

T ₅ [K]	P ₅ [atm]	Methane [%]	O ₂ [%]	ϕ (E.R.)	CH ₄ Induction [μs]	CH ₄ End [μs]	Ignition (Pressure) [μs]
1673	84.4	0.25	1	0.5	95	127	
1590	86.8	0.25	1	0.5	209	278	
1671	32.8	0.25	1	0.5	142	184	140
1627	31.7	1	4	0.5	88	95	92
1532	32.3	1	4	0.5	227	274	237
1512	34.5	1	4	0.5	314	346	317
1426	34.1	1	4	0.5	725	810	747
1739	10.1	1	4	0.5	82	94	74
1988	30.2	0.25	0.25	2	14	63	
2031	9.3	0.25	0.25	2	20	97	
1710	39.5	0.5	0.5	2	169	491	

Methane Table 9 (continued):

T ₅ [K]	P ₅ [atm]	Methane [%]	O ₂ [%]	$\phi_{\square\square}$ (E.R.)	CH ₄ Induction [μs]	CH ₄ End [μs]	Ignition (Pressure) [μs]
1896	77.1	0.5	0.5	2	15	56	
1926	9.4	0.5	0.5	2	63	144	
1887	9.4	0.5	0.5	2	81	178	
1807	35.5	0.5	0.5	2	63	165	
1425	36.6	5	5	2	783		785
1579	34.6	5	5	2	173		165
1536	33.7	5	5	2	376		352
1455	33.9	5	5	2	671		645
1555	30.4	5	5	2	268		271
1633	29.1	5	5	2	118		110
1635	10.2	5	5	2	333		336
1744	9.8	5	5	2	110		114
1478	60.3	5	5	2	321		307
1706	80.9	0.25	0.14	3.6	174		
1742	29.9	0.25	0.14	3.6	163		
1977	32.3	0.25	0.14	3.6	16		
1692	79.9	0.25	0.14	3.6	163		
1949	9.5	0.25	0.14	3.6	53	266	
1802	9.2	0.25	0.14	3.6	238	910	
1690	85.7	0.5	0.25	4	177	702	
1818	31.4	0.5	0.25	4	60	283	

Ethane

Literature Source of Data:

E. J. Chang, "Shock Tube Experiments for the Development and Validation of Models of Hydrocarbon Combustion," M. Eng. Thesis, (also published as HTGL Report No. T-320), Mechanical Engineering Department, Stanford University, Stanford CA (1995).

M. Frenklach, H. Wang, M. Goldenberg, G. P. Smith, D. M. Golden, C. T. Bowman, R. K. Hanson, W. C. Gardiner, V. Lissianski, "GRI-Mech – An Optimized Detailed Chemical Reaction Mechanism for Methane Combustion," Topical Report GRI-95/0058, Gas Research Institute (1995).

Range of Data:

Temperature [K]	1717	2497
Pressure [atm]	0.54	1.23
Fuel Mole Fraction [ppm]	247	2012
Oxygen Mole Fraction [ppm]	1055	7062
Methane Mole Fraction [ppm]	0	503
Equivalence Ratio	0.98	1.01

Type of Data:

Ethane Table 1: CO₂ species concentration time history IR laser absorption measurements at 4.200 μm (2380.72 cm^{-1}) in argon. Low pressure experiments were performed behind incident shock waves.

Ethane Table 2: CH₃ species concentration time history UV laser absorption measurements at 216 nm in argon. Includes mixtures with added methane.

Ethane Table 3: OH species concentration time history UV laser absorption measurements at 306.7 nm (32606.56 cm^{-1}) in argon.

Ethane Table 1:

T2 [K]	P2 [atm]	Ethane [ppm]	Oxygen [ppm]	ϕ (E.R.)	CO ₂ @1ms [ppm]	Plateau CO ₂ [ppm]	¼ Plat [μs]
2216	0.62	2012	7062	1.0	1455	2568	133
2497	0.537	2012	7062	1.0	1123	1475	68

Ethane Table 2:

T₅ [K]	P₅ [atm]	Ethane [ppm]	Methane [ppm]	Oxygen [ppm]	φ (E.R.)	Peak CH₃ [ppm]	Time [μs]
1794	1.166	295	0	1055	0.98	342	59
1717	1.201	247	503	1855	1.01	80	256
1759	1.163	247	503	1855	1.01	55	302
1785	1.181	247	503	1855	1.01	67	297
1844	1.233	247	503	1855	1.01	48	321

Ethane Table 3:

T₅ [K]	P₅ [atm]	Ethane [ppm]	Oxygen [ppm]	φ (E.R.)	Max OH [ppm]	½ Max [μs]
1757	1.179	300	1052	1.0	94	250
1817	1.209	300	1052	1.0	100	195
2012	1.183	300	1052	1.0	135	115

Ethane (continued)

Literature Source of Data:

M. Rohrig, E. L. Petersen, D. F. Davidson, R. K. Hanson, C. T. Bowman, "Measurement of the Rate Coefficient of the Reaction $CH+O_2 = \text{Products}$ in the Temperature Range 2200 to 2600 K," International Journal Chemical Kinetics 29: 781-789 (1997).

Range of Data:

Temperature [K]	2121	2450
Pressure [atm]	1.1	1.2
Fuel Mole Fraction [%]	100	150
Oxygen Mole Fraction [%]	972	1002
Equivalence Ratio	0.35	0.54

Type of Data:

Ethane Table 4: CH species concentration time history measurements based on laser absorption measurements at 431 nm in argon. CH data was derived from a digitization of Figs. 1 and 5 of Rohrig et al. (1997).

Ethane Table 4:

T ₅ [K]	P ₅ [atm]	Ethane [ppm]	Oxygen [ppm]	φ (E.R.)	1 st Plateau [ppm]	Peak CH [ppm]	Peak [μs]
2450	1.2	100	1002	0.35	0.5	7.5	70
2121	1.1	150	972	0.54		6.5	100

Ethylene

Literature Source of Data:

D. C. Horning, "A Study of the High Temperature Autoignition and Thermal Decomposition of Hydrocarbons," Ph.D. Thesis, (also published as Report No. TSD-135), Mechanical Engineering Department, Stanford University, Stanford CA (2001).

Range of Data:

Temperature [K]	1253	1572
Pressure [atm]	1.10	3.98
Fuel Mole Fraction [%]	1	4
Oxygen Mole Fraction [%]	3	12
Equivalence Ratio	1	1

Type of Data:

Ethylene Table 1: Ignition delay time defined as the peak of the CH* emission at 431 nm. Carrier gas is argon.

Ethylene Table 1:

T ₅ [K]	P ₅ [atm]	Ethylene [%]	O ₂ [%]	φ (E.R.)	Ignition Time [μs]
1541	1.10	2.0	6.0	1.0	66
1465	1.14	2.0	6.0	1.0	93
1409	1.19	2.0	6.0	1.0	115
1316	1.23	2.0	6.0	1.0	200
1286	2.17	2.0	6.0	1.0	190
1329	2.13	2.0	6.0	1.0	132
1360	2.10	2.0	6.0	1.0	105
1395	2.08	2.0	6.0	1.0	84
1294	3.86	2.0	6.0	1.0	134
1253	3.98	2.0	6.0	1.0	221
1354	3.80	2.0	6.0	1.0	76
1314	1.29	4.0	12	1.0	99
1336	1.19	4.0	12	1.0	89
1311	1.35	4.0	12	1.0	101
1276	1.32	4.0	12	1.0	125
1280	1.33	4.0	12	1.0	121
1257	2.15	4.0	12	1.0	129
1283	2.11	4.0	12	1.0	101

Ethylene Table 1 (continued):

T₅ [K]	P₅ [atm]	Ethylene [%]	O₂ [%]	φ_{□□} (E.R.)	Ignition Time [μs]
1296	2.06	4.0	12	1.0	88
1477	1.29	1.0	3.0	1.0	148
1402	1.23	1.0	3.0	1.0	220
1517	1.26	1.0	3.0	1.0	120
1572	1.25	1.0	3.0	1.0	96
1347	2.14	1.0	3.0	1.0	248
1396	2.12	1.0	3.0	1.0	182
1450	2.11	1.0	3.0	1.0	123
1498	2.04	1.0	3.0	1.0	95

Normal Alkanes

Propane

Literature Source of Data:

D. C. Horning, "A Study of the High Temperature Autoignition and Thermal Decomposition of Hydrocarbons," Ph.D. Thesis, (also published as Report No. TSD-135,) Mechanical Engineering Department, Stanford University, Stanford CA (2001).

D. C. Horning, D. F. Davidson, R. K. Hanson, "Study of the High-Temperature Autoignition of *n*-Alkane/O₂/Ar Mixtures," Journal of Propulsion and Power 18: 363-371 (2002).

D. C. Horning, D. F. Davidson, R. K. Hanson, "Ignition Time Correlations for *n*-Alkane/O₂/Ar Mixtures," Paper 5732, pp. 208-214, Proceedings of the 23rd International Symposium on Shock Waves, Fort Worth TX (2001).

Range of Data:

Temperature [K]	1376	1504
Pressure [atm]	1.12	1.26
Fuel Mole Fraction [%]	4	4
Oxygen Mole Fraction [%]	20	20
Equivalence Ratio	1	1

Type of Data:

Propane Table 1: Ignition delay time measurements derived from the peak of CH* emission measurements at 431 nm. Carrier gas is argon.

Propane Table 1:

T5 [K]	P5 [atm]	Propane [%]	O2 [%]	ϕ (E.R.)	Ignition Time [μ s]
1376	1.19	4.0	20	1	357
1461	1.15	4.0	20	1	140
1413	1.16	4.0	20	1	246
1377	1.12	4.0	20	1	115
1444	1.26	4.0	20	1	152
1504	1.14	4.0	20	1	93

Propane (continued)

Literature Source of Data:

D. F. Davidson, J. T. Herbon, D. C. Horning, R. K. Hanson, "*OH Concentration Time Histories in n-Alkane Oxidation*," International Journal of Chemical Kinetics 33: 775-783 (2001).

D. F. Davidson, J. T. Herbon, D. C. Horning, R. K. Hanson, "*OH Concentration Time Histories in n-Alkane Oxidation*," Paper 01F-49, Western States Section/The Combustion Institute Fall Meeting, Salt Lake City (2001).

Range of Data:

Temperature [K]	1497	1687
Pressure [atm]	2.13	2.24
Fuel Mole Fraction [ppm]	500	1500
Oxygen Mole Fraction [ppm]	2500	7500
Equivalence Ratio	1	1

Type of Data:

Propane Table 2: OH concentration time history measurements using laser absorption at 306.7 nm. $T_1(90\%)$ and $T_2(50\%)$ are the time for the OH mole fraction to reach 90% of the initial plateau level and 50% of the final plateau level respectively. For those conditions where no 1st plateau is indicated, no distinct plateau existed because of the rapid rise to the final plateau level. Carrier gas is argon.

Propane Table 2:

T_5 [K]	P_5 [atm]	Propane [ppm]	O ₂ [ppm]	ϕ (E.R.)	$T_1(90\%)$ [μ s]	1 st Plateau [ppm]	$T_2(50\%)$ [μ s]	2 nd Plateau [ppm]
1505	2.21	500	2500	1	28	9	534	105
1547	2.17	500	2500	1	28	14	344	129
1608	2.19	500	2500	1	27	30	174	146
1628	2.24	500	2500	1	25	30	156	157
1687	2.13	500	2500	1			89	190
1497	2.15	1500	7500	1	20	11	420	337

***n*-Butane**

Literature Source of Data:

D. C. Horning, "A Study of the High Temperature Autoignition and Thermal Decomposition of Hydrocarbons," Ph.D. Thesis, (also published as Report No. TSD-135,) Mechanical Engineering Department, Stanford University, Stanford CA (2001).

D. C. Horning, D. F. Davidson, R. K. Hanson, "Study of the High-Temperature Autoignition of *n*-Alkane/O₂/Ar Mixtures," Journal of Propulsion and Power 18: 363-371 (2002).

D. C. Horning, D. F. Davidson, R. K. Hanson, "Ignition Time Correlations for *n*-Alkane/O₂/Ar Mixtures," Paper 5732, pp. 208-214, Proceedings of the 23rd International Symposium on Shock Waves, Fort Worth TX (2001).

Range of Data:

Temperature [K]	1400	1483
Pressure [atm]	1.14	3.80
Fuel Mole Fraction [%]	1	1
Oxygen Mole Fraction [%]	6.5	6.5
Equivalence Ratio	1	1

Type of Data:

n-Butane Table 1: Ignition delay time measurements derived from the peak of CH* emission measurements at 431 nm. Carrier gas is argon.

n-Butane Table 1:

T ₅ [K]	P ₅ [atm]	<i>n</i> -Butane [%]	O ₂ [%]	φ (E.R.)	Ignition Time [μs]
1405	1.19	1.0	6.5	1.0	415
1405	1.18	1.0	6.5	1.0	420
1409	1.17	1.0	6.5	1.0	390
1447	1.14	1.0	6.5	1.0	244
1478	1.20	1.0	6.5	1.0	171
1483	1.32	1.0	6.5	1.0	161
1407	1.99	1.0	6.5	1.0	297
1400	2.03	1.0	6.5	1.0	335
1445	2.02	1.0	6.5	1.0	186
1454	2.00	1.0	6.5	1.0	170
1415	3.80	1.0	6.5	1.0	189

n-Butane Table 1 (continued):

T5 [K]	P5 [atm]	n-Butane [%]	O2 [%]	ϕ (E.R.)	Ignition Time [μ s]
1399	3.81	1.0	6.5	1.0	243
1433	3.75	1.0	6.5	1.0	151
1468	1.97	1.0	6.5	1.0	150
1499	1.14	1.0	6.5	1.0	151
1505	1.12	1.0	6.5	1.0	140
1557	1.13	1.0	6.5	1.0	87
1561	1.17	0.5	3.25	1.0	131
1513	1.18	0.5	3.25	1.0	186
1478	1.20	0.5	3.25	1.0	275
1532	1.33	0.5	3.25	1.0	156
1546	1.37	0.5	3.25	1.0	136
1433	1.27	0.5	3.25	1.0	475
1432	1.17	2.0	13	1.0	179
1490	1.19	2.0	13	1.0	100
1385	1.18	2.0	13	1.0	353
1434	1.15	2.0	13	1.0	180
1455	1.15	2.0	13	1.0	149
1517	1.16	2.0	13	1.0	83
1389	1.12	1.0	13	0.5	204
1422	1.08	1.0	13	0.5	131
1385	1.11	1.0	13	0.5	206
1412	1.11	1.0	13	0.5	149
1453	1.07	1.0	13	0.5	96
1352	1.14	1.0	13	0.5	307
1666	1.03	1.0	3.25	2.0	177
1627	1.06	1.0	3.25	2.0	220
1541	1.07	1.0	3.25	2.0	441
1734	1.08	1.0	3.25	2.0	105
1618	1.13	1.0	3.25	2.0	218
1577	1.13	1.0	3.25	2.0	304

***n*-Butane (continued)**

Literature Source of Data:

D. F. Davidson, J. T. Herbon, D. C. Horning, R. K. Hanson, "*OH Concentration Time Histories in n-Alkane Oxidation*," International Journal of Chemical Kinetics 33: 775-783 (2001).

D. F. Davidson, J. T. Herbon, D. C. Horning, R. K. Hanson, "*OH Concentration Time Histories in n-Alkane Oxidation*," Paper 01F-49, Western States Section/The Combustion Institute Fall Meeting, Salt Lake City (2001).

Range of Data:

Temperature [K]	1531	1761
Pressure [atm]	2.02	2.06
Fuel Mole Fraction [ppm]	500	500
Oxygen Mole Fraction [ppm]	3250	3250
Equivalence Ratio	1	1

Type of Data:

Butane Table 2: OH concentration time history measurements using laser absorption at 306.7 nm. $T_1(90\%)$ and $T_2(50\%)$ are the time for the OH mole fraction to reach 90% of the initial plateau level and 50% of the final plateau level respectively. For those conditions where no 1st plateau is indicated, no distinct plateau existed because of the rapid rise to the final plateau level. Carrier gas is argon.

n-Butane Table 2:

T_5 [K]	P_5 [atm]	n-Butane [ppm]	O ₂ [ppm]	ϕ (E.R.)	$T_1(90\%)$ [μ s]	1 st Plateau [ppm]	$T_2(50\%)$ [μ s]	2 nd Plateau [ppm]
1531	2.16	500	3250	1	24	23	260	175
1598	2.06	500	3250	1	21	40	144	192
1761	2.02	500	3250	1			40	266

***n*-Heptane**

Literature Source of Data:

D. C. Horning, "A Study of the High Temperature Autoignition and Thermal Decomposition of Hydrocarbons," Ph.D. Thesis, (also published as Report No. TSD-135,) Mechanical Engineering Department, Stanford University, Stanford CA (2001).

D. C. Horning, D. F. Davidson, R. K. Hanson, "Study of the High-Temperature Autoignition of *n*-Alkane/O₂/Ar Mixtures," Journal of Propulsion and Power 18: 363-371 (2002).

D. C. Horning, D. F. Davidson, R. K. Hanson, "Ignition Time Correlations for *n*-Alkane/O₂/Ar Mixtures," Paper 5732, pp. 208-214, Proceedings of the 23rd International Symposium on Shock Waves, Fort Worth TX (2001).

Range of Data:

Temperature [K]	1329	1547
Pressure [atm]	1.15	5.71
Fuel Mole Fraction [%]	0.36	1.75
Oxygen Mole Fraction [%]	4.4	19.7
Equivalence Ratio	0.5	2.0

Type of Data:

n-Heptane Table 1: Ignition delay time measurements derived from the peak of CH* emission measurements at 431 nm. Carrier gas is argon.

n-Heptane Table 1:

T5 [K]	P5 [atm]	n-Heptane [%]	O2 [%]	ϕ (E.R.)	Ignition Time [μ s]
1446	1.15	0.4	4.8	0.9	234
1381	1.18	0.4	4.8	0.9	488
1491	1.23	0.4	4.8	0.9	137
1438	1.22	0.4	4.8	0.9	238
1492	1.22	0.36	4.4	0.9	141
1501	1.25	0.36	4.4	0.9	125
1476	1.25	0.36	4.4	0.9	163
1486	1.23	0.4	4.4	1.0	159
1458	1.27	0.4	4.4	1.0	208
1428	1.27	0.4	4.4	1.0	296
1455	1.20	0.4	8.8	0.5	86
1413	1.32	0.4	8.8	0.5	121
1347	1.30	0.4	8.8	0.5	286
1676	1.22	0.4	2.2	2.0	147
1529	1.26	0.4	2.2	2.0	442
1619	1.22	0.4	2.2	2.0	210
1396	1.23	0.4	8.8	0.5	161
1347	1.29	0.4	8.8	0.5	308
1345	1.34	0.4	8.8	0.5	305
1432	1.21	0.4	8.8	0.5	110
1347	1.22	0.4	8.8	0.5	323
1455	2.13	0.4	4.4	1.0	165
1384	2.12	0.4	4.4	1.0	403
1434	2.15	0.4	4.4	1.0	205
1478	2.08	0.4	4.4	1.0	131
1503	2.06	0.4	4.4	1.0	100
1412	2.14	0.4	4.4	1.0	265
1476	4.03	0.4	4.4	1.0	90
1420	4.07	0.4	4.4	1.0	163
1394	4.22	0.4	4.4	1.0	227
1463	4.11	0.4	4.4	1.0	103
1413	4.16	0.4	4.4	1.0	183

n-Heptane Table 1 (continued):

T5 [K]	P5 [atm]	n-Heptane [%]	O2 [%]	ϕ (E.R.)	Ignition Time [μ s]
1378	1.15	0.8	8.8	1.0	385
1426	1.17	0.8	8.8	1.0	208
1335	1.30	1.2	13.2	1.0	466
1455	1.22	1.2	13.2	1.0	115
1421	1.19	1.2	13.2	1.0	163
1349	1.20	1.2	13.2	1.0	411
1409	1.27	1.2	13.2	1.0	179
1465	1.14	1.2	8.8	1.5	189
1461	1.23	1.2	8.8	1.5	189
1409	1.27	1.2	8.8	1.5	321
1411	1.29	1.2	8.8	1.5	311
1386	1.24	1.75	19.7	1.0	177
1329	1.19	1.75	19.7	1.0	382
1330	1.16	1.75	19.7	1.0	382
1501	2.08	0.2	2.2	1.0	150
1468	2.13	0.2	2.2	1.0	216
1432	2.15	0.2	2.2	1.0	345
1547	2.07	0.2	2.2	1.0	100
1375	5.68	0.4	4.4	1.0	271
1431	5.71	0.4	4.4	1.0	120
1400	5.63	0.4	4.4	1.0	190
1421	5.69	0.4	4.4	1.0	138
1372	5.57	0.4	4.4	1.0	279
1442	5.65	0.4	4.4	1.0	117
1419	5.41	0.4	4.4	1.0	158
1423	1.20	1.2	6.6	2.0	486
1516	1.21	1.2	6.6	2.0	205
1456	1.20	1.2	6.6	2.0	361

***n*-Heptane (continued)**

Literature Source of Data:

D. F. Davidson, J. T. Herbon, D. C. Horning, R. K. Hanson, "*OH Concentration Time Histories in *n*-Alkane Oxidation*," International Journal of Chemical Kinetics 33: 775-783 (2001).

D. F. Davidson, J. T. Herbon, D. C. Horning, R. K. Hanson, "*OH Concentration Time Histories in *n*-Alkane Oxidation*," Paper 01F-49, Western States Section/The Combustion Institute Fall Meeting, Salt Lake City (2001).

Range of Data:

Temperature [K]	1540	1784
Pressure [atm]	2.04	3.80
Fuel Mole Fraction [ppm]	300	300
Oxygen Mole Fraction [ppm]	3300	3300
Equivalence Ratio	1	1

Type of Data:

n-Heptane Table 2: OH concentration time history measurements using laser absorption at 306.7 nm. $T_1(90\%)$ and $T_2(50\%)$ are the time for the OH mole fraction to reach 90% of the initial plateau level and 50% of the final plateau level respectively. For those conditions where no 1st plateau is indicated, no distinct plateau existed because of the rapid rise to the final plateau level. Carrier gas is argon.

n-Heptane Table 2:

T_5 [K]	P_5 [atm]	<i>n</i> -Heptane [ppm]	O ₂ [ppm]	ϕ (E.R.)	$T_1(90\%)$ [μ s]	1 st Plateau [ppm]	$T_2(50\%)$ [μ s]	2 nd Plateau [ppm]
1551	2.14	300	3300	1	6	20	207	160
1640	2.04	300	3300	1			99	191
1784	2.12	300	3300	1			47	255
1540	3.67	300	3300	1			181	149
1649	3.80	300	3300	1			73	187
1710	3.75	300	3300	1			49	212

***n*-Heptane (continued)**

Literature Source of Data:

B.M. Gauthier, D.F. Davidson, R.K. Hanson, "*Shock Tube Determination of Ignition Delay Times in Full-Blend and Surrogate Fuel Mixtures*," Combustion and Flame, in press (2004).

Range of Data:

Temperature [K]	806	1378
Pressure [atm]	1.85	60.6
Fuel Mole Fraction [ppm]	1.874	1.874
Oxygen Mole Fraction [ppm]	20.6	20.6
Equivalence Ratio	1	1

Type of Data:

n-Heptane Table 3: Ignition delay time data with Nitrogen as the bulk carrier gas using PZT pressure measurements of the time between the arrival of the reflected shock (the center of the reflected shock bifurcation feature) and the distinct ignition pressure rise (the time of the intersection of the linear extrapolation of the pressure rise with the pre-ignition pressure floor). Similar ignition delay times were recovered from CH and OH emission measurements.

n-Heptane Table 4: Ignition delay time data with Nitrogen as the bulk carrier gas using the same ignition time definition as n-Heptane Table 3.

n-Heptane Table 3:

T5 [K]	P5 [atm]	n-Heptane [%]	Oxygen [%]	ϕ (E.R.)	Ignition Time [μ s]
1249	1.97	1.874	20.6	1	529
1299	1.89	1.874	20.6	1	311
1358	1.85	1.874	20.6	1	152
1378	1.99	1.874	20.6	1	117
1305	10.62	1.874	20.6	1	117
1236	11.24	1.874	20.6	1	207
1299	10.25	1.874	20.6	1	122
1344	10.27	1.874	20.6	1	85
1290	11.8	1.874	20.6	1	118

n-Heptane Table 4:

T5 [K]	P5 [atm]	n-Heptane [%]	Oxygen [%]	ϕ (E.R.)	Ignition Time [μ s]
806	20.0	1.874	20.6	1	1377
850	19.92	1.874	20.6	1	1653
906	19.87	1.874	20.6	1	1836
1012	18.13	1.874	20.6	1	944
1048	16.72	1.874	20.6	1	854
909	53.9	1.874	20.6	1	254
923	60.0	1.874	20.6	1	244
926	60.6	1.874	20.6	1	233
932	55.4	1.874	20.6	1	323
985	48.6	1.874	20.6	1	364
1007	57.7	1.874	20.6	1	232
1013	53.6	1.874	20.6	1	292
1023	54.2	1.874	20.6	1	261
1027	59.1	1.874	20.6	1	237
1057	50.0	1.874	20.6	1	194
1063	53.1	1.874	20.6	1	179
1115	52.3	1.874	20.6	1	102

n-Decane

Literature Source of Data:

D. C. Horning, "A Study of the High Temperature Autoignition and Thermal Decomposition of Hydrocarbons," Ph.D. Thesis, (also published as Report No. TSD-135,) Mechanical Engineering Department, Stanford University, Stanford CA (2001).

D. C. Horning, D. F. Davidson, R. K. Hanson, "Study of the High-Temperature Autoignition of *n*-Alkane/O₂/Ar Mixtures," Journal of Propulsion and Power 18: 363-371 (2002).

D. C. Horning, D. F. Davidson, R. K. Hanson, "Ignition Time Correlations for *n*-Alkane/O₂/Ar Mixtures," Paper 5732, pp. 208-214, Proceedings of the 23rd International Symposium on Shock Waves, Fort Worth TX (2001).

Range of Data:

Temperature [K]	1397	1516
Pressure [atm]	1.22	1.26
Fuel Mole Fraction [%]	0.2	0.2
Oxygen Mole Fraction [%]	3.1	3.1
Equivalence Ratio	1	1

Type of Data:

n-Decane Table 1: Ignition delay time measurements derived from the peak of CH* emission measurements at 431 nm. Carrier gas is argon.

n-Decane Table 1:

T ₅ [K]	P ₅ [atm]	<i>n</i> -Decane [%]	O ₂ [%]	φ (E.R.)	Ignition Time [μs]
1470	1.23	0.2	3.1	1	180
1445	1.24	0.2	3.1	1	290
1397	1.26	0.2	3.1	1	480
1456	1.24	0.2	3.1	1	235
1434	1.23	0.2	3.1	1	320
1516	1.22	0.2	3.1	1	124

n-Decane (continued)

Literature Source of Data:

D. F. Davidson, J. T. Herbon, D. C. Horning, R. K. Hanson, "*OH Concentration Time Histories in n-Alkane Oxidation*," International Journal of Chemical Kinetics 33: 775-783 (2001).

D. F. Davidson, J. T. Herbon, D. C. Horning, R. K. Hanson, "*OH Concentration Time Histories in n-Alkane Oxidation*," Paper 01F-49, Western States Section/The Combustion Institute Fall Meeting, Salt Lake City (2001).

Range of Data:

Temperature [K]	1357	1706
Pressure [atm]	2.03	2.28
Fuel Mole Fraction [ppm]	300	2000
Oxygen Mole Fraction [ppm]	3875	3100
Equivalence Ratio	0.8	1.2

Type of Data:

n-Decane Table 2: OH concentration time history measurements using laser absorption at 306.7 nm. $T_1(90\%)$ and $T_2(50\%)$ are the time for the OH mole fraction to reach 90% of the initial plateau level and 50% of the final plateau level respectively. For those conditions where no 1st plateau is indicated, no distinct plateau existed because of the rapid rise to the final plateau level. Carrier gas is argon.

n-Decane Table 2:

T_5 [K]	P_5 [atm]	n-Decane [ppm]	O_2 [ppm]	ϕ (E.R.)	$T_1(90\%)$ [μs]	1 st Plateau [ppm]	$T_2(50\%)$ [μs]	2 nd Plateau [ppm]
1479	2.19	300	4650	1	11	14	375	166
1525	2.21	300	4650	1		34	154	206
1574	2.12	300	4650	1			119	222
1661	2.08	300	4650	1			87	273
1706	2.15	300	4650	1			37	228
1462	2.11	300	5817	0.8	19	24	206	197
1526	2.03	300	5817	0.8			83	209
1602	2.25	300	3875	1.2			124	196
1706	2.17	300	3875	1.2			43	222
1357	2.28	2000	31000	1		9	341	872
1404	2.20	2000	31000	1	6	13	238	1051

Branched Alkanes

Iso-Butane

Literature Source of Data:

M. A. Oehlschlaeger, D. F. Davidson, J. T. Herbon, R. K. Hanson, "*Shock Tube Measurements of Branched Alkane Ignition Times and OH Concentration Time Histories*," International Journal of Chemical Kinetics 36: 67-78 (2004).

M. A. Oehlschlaeger, D. F. Davidson, J. T. Herbon, R. K. Hanson, "*Shock Tube Measurements of Branched Alkane Ignition Times and OH Concentration Time Histories*," AIAA paper 2003-0830, 41st AIAA Aerospace Sciences Meeting and Exhibit, Reno NV (2003).

Range of Data:

Temperature [K]	1291	1864
Pressure [atm]	1.33	12.58
Fuel Mole Fraction [%]	0.01	1
Oxygen Mole Fraction [%]	0.065	6.5
Equivalence Ratio	0.25	2.0

Type of Data:

Iso-Butane Table 1: Ignition delay time measurement in argon based on the maximum slope of the CH* emission at 431 nm extrapolated to the zero baseline.
Iso-Butane Table 2: OH concentration time history measurements in argon using narrow-linewidth ring-dye laser absorption of the R₁(5) line of the OH A-X (0,0) band at 306.7 nm. "Sat." indicates laser transmission of ~0.

Iso-Butane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1526	1.53	0.01	0.065	1	1114
1571	1.60	0.01	0.065	1	754
1616	1.52	0.01	0.065	1	474
1679	1.43	0.01	0.065	1	265
1737	1.43	0.01	0.065	1	168
1864	1.33	0.01	0.065	1	72
1444	1.59	0.05	0.325	1	1731
1546	1.58	0.05	0.325	1	507
1630	1.49	0.05	0.325	1	195
1703	1.40	0.05	0.325	1	115
1442	2.71	0.05	0.325	1	1338
1540	2.78	0.05	0.325	1	512
1603	2.70	0.05	0.325	1	224
1625	2.54	0.05	0.325	1	184
1723	2.47	0.05	0.325	1	75
1765	2.00	0.05	0.325	1	62
1406	5.53	0.05	0.325	1	1515
1545	5.61	0.05	0.325	1	338
1577	5.03	0.05	0.325	1	231
1750	4.78	0.05	0.325	1	45
1404	11.12	0.05	0.325	1	1281
1473	10.69	0.05	0.325	1	554
1542	10.32	0.05	0.325	1	305
1607	12.58	0.05	0.325	1	144
1684	9.94	0.05	0.325	1	73
1411	1.55	0.1	0.65	1	1807
1488	1.57	0.1	0.65	1	741
1491	1.50	0.1	0.65	1	639
1522	1.56	0.1	0.65	1	458
1593	1.46	0.1	0.65	1	246
1720	1.41	0.1	0.65	1	72
1291	1.67	0.125	3.25	0.25	1404
1341	1.63	0.125	3.25	0.25	672
1390	1.56	0.125	3.25	0.25	329
1474	1.51	0.125	3.25	0.25	113
1339	1.69	0.25	3.25	0.5	1135
1402	1.62	0.25	3.25	0.5	493
1488	1.58	0.25	3.25	0.5	166
1541	1.47	0.25	3.25	0.5	96
1366	1.61	0.5	3.25	1	1475
1426	1.55	0.5	3.25	1	694
1520	1.49	0.5	3.25	1	258
1646	1.46	0.5	3.25	1	77
1461	1.52	1.0	3.25	2	1159
1566	1.45	1.0	3.25	2	450
1569	1.47	1.0	3.25	2	453
1681	1.35	1.0	3.25	2	155
1346	1.58	1.0	6.5	1	1162
1351	1.56	1.0	6.5	1	1096
1379	1.55	1.0	6.5	1	675
1383	1.56	1.0	6.5	1	764
1403	1.53	1.0	6.5	1	634
1434	1.56	1.0	6.5	1	381
1445	1.49	1.0	6.5	1	379
1455	1.54	1.0	6.5	1	324
1545	1.49	1.0	6.5	1	118
1555	1.44	1.0	6.5	1	120
1565	1.48	1.0	6.5	1	118

Iso-Butane Table 2:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	ϕ E.R.	First Peak [μ s]	First Peak [ppm]	Minimum [μ s]	Minimum [ppm]	50% Peak [μ s]	Peak [ppm]
1366	1.61	0.5	3.25	1	16	5	346	3	1429	Sat.
1426	1.55	0.5	3.25	1	9	13	198	6	680	Sat.
1520	1.49	0.5	3.25	1	7	34	76	17	233	Sat.
1646	1.46	0.5	3.25	1	4	96	20	63	65	Sat.
1339	1.69	0.25	3.25	0.5	27	4	353	2	1128	641
1402	1.62	0.25	3.25	0.5	19	10	156	6	488	646
1488	1.58	0.25	3.25	0.5	11	28	55	19	165	777
1541	1.47	0.25	3.25	0.5	9	45	30	36	93	831
1291	1.67	0.125	3.25	0.25	65	3	381	2	1476	342
1341	1.63	0.125	3.25	0.25	45	9	212	6	735	386
1390	1.56	0.125	3.25	0.25	34	19	130	15	372	466
1474	1.51	0.125	3.25	0.25	26	56	40	55	120	608
1411	1.55	0.1	0.65	1	30	6	550	2	1810	268
1488	1.57	0.1	0.65	1	18	15	212	4	758	303
1491	1.50	0.1	0.65	1	19	17	178	6	662	315
1593	1.46	0.1	0.65	1	13	37	61	22	237	375
1720	1.41	0.1	0.65	1	12	71	20	69	87	470
1444	1.59	0.05	0.325	1	33	7	390	2	1770	143
1546	1.58	0.05	0.325	1	23	17	155	7	572	177
1630	1.49	0.05	0.325	1	18	29	61	23	248	214
1703	1.40	0.05	0.325	1	18	42	31	41	146	242
1603	2.70	0.05	0.325	1	12	21	69	11	255	187
1625	2.50	0.05	0.325	1	12	23	55	14	224	187
1723	2.50	0.05	0.325	1	11	37	24	35	100	255
1545	5.61	0.05	0.325	1	7	9	85	3	368	137
1577	5.03	0.05	0.325	1	7	16	64	5	255	180
1750	4.78	0.05	0.325	1	5	42	12	40	49	254

Iso-Pentane

Literature Source of Data:

M. A. Oehlschlaeger, D. F. Davidson, J. T. Herbon, R. K. Hanson, "*Shock Tube Measurements of Branched Alkane Ignition Times and OH Concentration Time Histories*," International Journal of Chemical Kinetics 36: 67-78 (2004).

M. A. Oehlschlaeger, D. F. Davidson, J. T. Herbon, R. K. Hanson, "*Shock Tube Measurements of Branched Alkane Ignition Times and OH Concentration Time Histories*," AIAA paper 2003-0830, 41st AIAA Aerospace Sciences Meeting and Exhibit, Reno NV (2003).

Range of Data:

Temperature [K]	1300	1726
Pressure [atm]	1.56	5.67
Fuel Mole Fraction [%]	0.025	1.25
Oxygen Mole Fraction [%]	0.8	20
Equivalence Ratio	0.25	2

Type of Data:

Iso-Pentane Table 1: Ignition delay time measurement in argon based on the maximum slope of the CH* emission at 431 nm extrapolated to the zero baseline.
Iso-Pentane Table 2: OH concentration time history measurements in argon using narrow-linewidth ring-dye laser absorption of the R₁(5) line of the OH A-X (0,0) band at 306.7 nm. "Sat." indicates laser transmission of ~0.

Iso-Pentane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1300	1.64	0.025	0.8	0.25	1679
1399	1.74	0.025	0.8	0.25	342
1431	1.74	0.025	0.8	0.25	215
1507	1.72	0.025	0.8	0.25	92
1362	1.82	0.05	0.8	0.5	1175
1426	1.81	0.05	0.8	0.5	468
1460	1.75	0.05	0.8	0.5	314
1487	1.71	0.05	0.8	0.5	254
1556	1.68	0.05	0.8	0.5	121
1413	1.74	0.1	0.8	1	1726
1496	1.75	0.1	0.8	1	619
1517	1.74	0.1	0.8	1	499
1606	1.68	0.1	0.8	1	191
1708	1.62	0.1	0.8	1	83
1448	4.68	0.1	0.8	1	556
1576	4.54	0.1	0.8	1	182
1609	5.41	0.1	0.8	1	96
1635	5.67	0.1	0.8	1	77
1668	4.09	0.1	0.8	1	57
1726	5.22	0.1	0.8	1	35
1517	1.70	0.2	0.8	2	1203
1574	1.69	0.2	0.8	2	731
1672	1.67	0.2	0.8	2	239
1717	1.59	0.2	0.8	2	181
1426	1.71	0.5	4.0	1	622
1466	1.69	0.5	4.0	1	400
1534	1.68	0.5	4.0	1	179
1552	1.62	0.5	4.0	1	162
1342	1.72	1.0	8.0	1	924
1366	1.75	1.0	8.0	1	747
1393	1.67	1.0	8.0	1	486
1473	1.56	1.0	8.0	1	198
1310	1.85	1.25	20	0.5	318
1392	1.81	1.25	20	0.5	109

Iso-Pentane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	First Peak [μs]	First Peak [ppm]	Minimum [μs]	Minimum [ppm]	50% Peak [μs]	Peak [ppm]
1534	1.68	0.5	4.0	1	7	21	117	12	235	Sat.
1552	1.62	0.5	4.0	1	5	28	113	15	212	Sat.
1413	1.74	0.1	0.8	1	19	9	458	2	1788	344
1517	1.74	0.1	0.8	1	9	26	112	8	551	396
1606	1.68	0.1	0.8	1	7	44	44	19	219	459
1708	1.62	0.1	0.8	1	6	71	22	56	95	571
1609	5.40	0.1	0.8	1	3	18	17	4	183	281
1635	5.67	0.1	0.8	1	3	35	13	7	146	308
1726	5.22	0.1	0.8	1	2	39	10	19	103	355

Iso-Octane

Literature Source of Data:

M. A. Oehlschlaeger, D. F. Davidson, J. T. Herbon, R. K. Hanson, "*Shock Tube Measurements of Branched Alkane Ignition Times and OH Concentration Time Histories*," International Journal of Chemical Kinetics 36: 67-78 (2004)

D. F. Davidson, M. A. Oehlschlaeger, J. T. Herbon, R. K. Hanson, "*Shock Tube Measurements of Iso-Octane Ignition Times and OH Concentration Time Histories*," Proceedings of the Combustion Institute 29: 1295-1301 (2002).

M. A. Oehlschlaeger, D. F. Davidson, J. T. Herbon, R. K. Hanson, "*Shock Tube Measurements of Branched Alkane Ignition Times and OH Concentration Time Histories*," AIAA paper 2003-0830, 41st AIAA Aerospace Sciences Meeting and Exhibit, Reno NV (2003).

Range of Data:

Temperature [K]	1177	2009
Pressure [atm]	1.10	8.17
Fuel Mole Fraction [%]	0.01	1.0
Oxygen Mole Fraction [%]	0.125	12.5
Equivalence Ratio	0.25	2.0

Type of Data:

Iso-Octane Table 1: Ignition delay time measurement in argon based on the maximum slope of the CH* emission at 431 nm extrapolated to the zero baseline. Iso-Octane Table 2: OH concentration time history measurements in argon using narrow-linewidth ring-dye laser absorption of the R₁(5) line of the OH A-X (0,0) band at 306.7 nm. "Sat." indicates laser transmission of ~0.

Iso-Octane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1545	1.44	0.01	0.125	1	1966
1592	1.42	0.01	0.125	1	1260
1652	1.39	0.01	0.125	1	741
1714	1.39	0.01	0.125	1	413
1810	1.34	0.01	0.125	1	197
1922	1.30	0.01	0.125	1	100
2009	1.29	0.01	0.125	1	63
1491	1.31	0.025	0.3125	1	1954
1495	1.50	0.025	0.3125	1	1994
1516	1.45	0.025	0.3125	1	1648
1564	1.47	0.025	0.3125	1	940
1601	1.44	0.025	0.3125	1	689
1647	1.46	0.025	0.3125	1	406
1703	1.37	0.025	0.3125	1	259
1738	1.37	0.025	0.3125	1	214
1809	1.40	0.025	0.3125	1	120
1899	1.35	0.025	0.3125	1	76
1902	1.10	0.025	0.3125	1	71
1455	1.45	0.05	0.625	1	1954
1523	1.44	0.05	0.625	1	976
1534	1.37	0.05	0.625	1	907
1603	1.34	0.05	0.625	1	410
1613	1.38	0.05	0.625	1	346
1646	1.34	0.05	0.625	1	231
1667	1.36	0.05	0.625	1	238
1733	1.30	0.05	0.625	1	144
1767	1.34	0.05	0.625	1	102
1773	1.26	0.05	0.625	1	95
1823	1.30	0.05	0.625	1	74
1455	5.12	0.05	0.625	1	1290
1484	7.50	0.05	0.625	1	767
1521	4.89	0.05	0.625	1	576
1618	6.88	0.05	0.625	1	199
1672	4.74	0.05	0.625	1	141
1699	6.51	0.05	0.625	1	96
1800	4.48	0.05	0.625	1	52
1467	1.44	0.1	1.25	1	1454
1480	1.47	0.1	1.25	1	1302
1484	1.49	0.1	1.25	1	1334
1543	1.49	0.1	1.25	1	651
1551	1.42	0.1	1.25	1	573
1642	1.30	0.1	1.25	1	269
1642	1.38	0.1	1.25	1	227
1823	1.33	0.1	1.25	1	68
1236	1.61	0.125	6.25	0.25	2700
1337	1.55	0.125	6.25	0.25	756
1394	1.52	0.125	6.25	0.25	345
1476	1.43	0.125	6.25	0.25	130
1598	1.40	0.125	6.25	0.25	47
1421	1.56	0.25	3.125	0.5	1576
1487	1.46	0.25	3.125	0.5	758
1598	1.42	0.25	3.125	0.5	223
1646	1.35	0.25	3.125	0.5	147
1763	1.37	0.25	3.125	0.5	64
1297	1.57	0.25	6.25	0.5	2370
1361	1.50	0.25	6.25	0.5	1115
1460	1.48	0.25	6.25	0.5	321
1556	1.41	0.25	6.25	0.5	117
1710	1.35	0.25	6.25	0.5	38

Iso-Octane Table 1 (continued):

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1318	1.53	0.5	6.25	1	2768
1386	1.27	0.5	6.25	1	1365
1393	1.48	0.5	6.25	1	1293
1442	1.29	0.5	6.25	1	646
1493	1.43	0.5	6.25	1	375
1554	1.28	0.5	6.25	1	236
1595	1.33	0.5	6.25	1	161
1695	1.28	0.5	6.25	1	74
1700	1.18	0.5	6.25	1	66
1766	1.26	0.5	6.25	1	51
1206	5.53	0.5	6.25	1	2911
1215	8.03	0.5	6.25	1	1621
1251	7.69	0.5	6.25	1	1272
1265	5.44	0.5	6.25	1	1624
1305	5.33	0.5	6.25	1	1167
1363	7.34	0.5	6.25	1	498
1385	5.08	0.5	6.25	1	477
1437	6.90	0.5	6.25	1	226
1474	4.83	0.5	6.25	1	236
1606	7.07	0.5	6.25	1	57
1360	1.44	1.0	6.25	2	2099
1412	1.43	1.0	6.25	2	1170
1512	1.46	1.0	6.25	2	486
1589	1.35	1.0	6.25	2	199
1177	8.17	1.0	6.25	2	2142
1269	8.07	1.0	6.25	2	952
1352	7.91	1.0	6.25	2	356
1238	1.46	1.0	12.5	1	1852
1289	1.43	1.0	12.5	1	1248
1302	1.32	1.0	12.5	1	1253
1317	1.44	1.0	12.5	1	836
1402	1.44	1.0	12.5	1	303
1439	1.40	1.0	12.5	1	245
1498	1.32	1.0	12.5	1	139
1535	1.38	1.0	12.5	1	108
1610	1.36	1.0	12.5	1	44

Iso-Octane Table 2:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	ϕ E.R.	First Peak [μ s]	First Peak [ppm]	Minimum [μ s]	Minimum [ppm]	50% Peak [μ s]	Peak [ppm]
1322	1.58	1.0	6.25	2	3	12		1		Sat.
1398	1.43	1.0	6.25	2		31		1	1422	Sat.
1506	1.42	1.0	6.25	2		26		3	589	Sat.
1656	1.34	1.0	6.25	2		103	17	12	127	Sat.
1330	1.55	0.5	6.25	1	4	17		4	1870	Sat.
1408	1.50	0.5	6.25	1	3	42		2	880	Sat.
1507	1.43	0.5	6.25	1		100		7	280	Sat.
1641	1.37	0.5	6.25	1		149	12	34	62	Sat.
1299	1.63	0.25	6.25	.5	6	16		1	1825	Sat.
1394	1.53	0.25	6.25	.5	4	49	49	2	621	Sat.
1467	1.48	0.25	6.25	.5	3	93	14	12	226	Sat.
1570	1.41	0.25	6.25	.5		159		42	62	Sat.
1449	1.59	0.05	0.625	1	8	16		1	2182	245
1511	1.51	0.05	0.625	1	6	22		2	1152	288
1614	1.43	0.05	0.625	1	5	30	86	9	388	352
1736	1.40	0.05	0.625	1	5	44	27	36	137	456

Iso-Octane (continued)

Literature Source of Data:

D. F. Davidson, B. M. Gauthier, R. K. Hanson, “*Shock Tube Ignition Measurements of Iso-Octane/Air and Toluene/Air at High Pressures,*” Proceedings of the Combustion Institute, in press (2004).

B. M. Gauthier, D. F. Davidson, R. K. Hanson, “*A Shock Tube Study of Iso-Octane and Toluene Ignition at High Pressures,*” Western States Section/ Combustion Institute Fall 2003 Meeting, Paper 03F-26 (2003).

Range of Data:

Temperature [K]	855	1159
Pressure [atm]	14.9	59.3
Fuel Mole Fraction [%]	0.829	1.652
Oxygen Mole Fraction [%]	20.8	20.7
Equivalence Ratio	0.5	1.0

Type of Data:

Iso-Octane Table 3: Ignition delay time measurement in air based on sidewall PZT pressure measurements and confirmed with CH* (at 431 nm) and OH* (at 306nm) emission measurements.

Iso-Octane Table 3:

T₅ [K]	P₅ [atm]	IsoOctane [%]	φ (E.R.)	Ignition Time [μs]
984	18.1	1.652	1	1511
995	16.3	1.652	1	1535
1043	17.1	1.652	1	927
1077	18.4	1.652	1	604
1109	15.9	1.652	1	516
1159	14.9	1.652	1	214
855	56.4	1.652	1	1719
867	59.3	1.652	1	1755
894	58.5	1.652	1	1193
927	55.7	1.652	1	1067
975	51.2	1.652	1	871
1006	51.1	1.652	1	625
1015	47.8	1.652	1	505
1098	47.5	1.652	1	222
1043	15.0	0.826	0.5	1747
1071	14.9	0.826	0.5	1222
1095	14.4	0.826	0.5	916
1171	13.5	0.826	0.5	329
1009	51.1	0.826	0.5	1075
1057	51.0	0.826	0.5	598
1099	48.6	0.826	0.5	348
1147	47.8	0.826	0.5	178

Cyclo-Alkanes

JP-10

Literature Source of Data:

D. F. Davidson, D. C. Horning, J. T. Herbon, R. K. Hanson, "Shock Tube Measurements of JP-10 Ignition," Proceedings of the Combustion Institute 28: 1687-1692 (2000).

D. F. Davidson, D. C. Horning, R. K. Hanson, "Shock Tube Ignition Time Measurements for N-Heptane/O₂/Ar and JP-10/O₂/Ar Mixtures," AIAA 99-2216 (1999).

Range of Data:

Temperature [K]	1352	1671
Pressure [atm]	1.15	9.20
Fuel Mole Fraction [%]	0.2	0.4
Oxygen Mole Fraction [%]	1.39	5.89
Equivalence Ratio	0.48	2.01

Type of Data:

JP-10 Table 1: OH concentration time history measurements in argon using narrow-linewidth ring-dye laser absorption of the R₁(5) line of the OH A-X (0,0) band at 306.7 nm.

JP-10 Table 2: Ignition delay time measurement in argon based on the peak of the sidewall CH* emission at 431 nm and confirmed where possible using PZT pressure traces.

JP-10 Table 1:

T5 [K]	P5 [atm]	JP-10 [%]	O2 [%]	ϕ (E.R.)	Time-to-peak [μ s]	Peak OH [ppm]
1383	3.03	.40	5.66	0.99	444	2797
1391	2.95	.395	5.57	0.99	355	2748
1447	3.18	0.198	2.76	1.00	462	1056
1460	3.02	0.20	2.83	0.99	429	1159
1502	3.11	0.198	2.76	1.00	276	1266
1475	3.05	.20	1.39	2.01	887	808

JP-10 Table 2:

T5 [K]	P5 [atm]	JP-10 [%]	O2 [%]	ϕ (E.R.)	Ignition Time [μ s]
1469	1.25	0.20	2.82	0.99	580
1492	1.16	0.20	2.79	1.00	425
1527	1.15	0.20	2.81	1.00	280
1584	1.19	0.20	2.81	1.00	139
1588	1.23	0.20	2.74	1.02	140
1629	1.20	0.20	2.87	0.97	90
1671	1.05	0.20	2.99	0.94	63
1378	6.17	0.20	2.89	0.97	675
1406	6.03	0.20	2.81	1.00	465
1407	6.26	0.20	2.79	1.00	455
1414	6.34	0.20	2.78	1.01	405
1428	6.13	0.20	2.63	1.06	365
1453	5.99	0.20	2.86	0.98	240
1456	6.05	0.20	2.86	0.98	271
1490	6.40	0.20	2.76	1.01	187
1365	6.25	0.20	5.89	0.48	410
1352	8.69	0.20	2.78	1.01	875
1358	8.52	0.20	2.85	0.98	735
1390	9.20	0.20	2.83	0.99	485
1406	3.30	0.20	2.82	0.99	755
1451	3.08	0.20	2.81	1.00	435
1457	3.18	0.20	2.76	1.01	395
1469	3.02	0.20	2.83	0.99	365
1473	3.16	0.20	2.76	1.01	305
1493	3.11	0.20	2.81	1.00	265
1507	3.03	0.20	2.82	0.99	212
1507	3.11	0.20	2.76	1.01	215
1512	3.32	0.20	2.82	0.99	225
1527	3.23	0.20	2.76	1.01	170
1535	3.23	0.20	2.81	1.00	175
1580	3.17	0.20	2.76	1.01	90
1364	3.18	0.20	5.60	0.50	520
1367	3.02	0.20	5.54	0.51	455
1405	3.08	0.20	5.54	0.51	300
1434	3.03	0.20	5.54	0.51	235
1463	3.01	0.20	5.54	0.51	161
1474	3.05	0.20	5.54	0.51	111
1482	3.05	0.20	1.39	2.01	805
1559	3.06	0.20	1.39	2.01	310
1620	3.00	0.20	1.39	2.01	150
1632	2.92	0.20	1.39	2.01	121
1400	3.03	0.40	5.66	0.99	380
1407	2.95	0.40	5.57	1.01	349
1433	3.05	0.40	5.48	1.02	285
1480	2.87	0.40	5.57	1.01	162

Olefins

1,3-Butadiene

Literature Source of Data:

C. S. Libby, D. F. Davidson, R. K. Hanson, "A Shock Tube Study of the Oxidation of 1,3-Butadiene," AIAA 2004-1322 (2004).

Range of Data:

Temperature [K]	1389	1813
Pressure [atm]	1.392	3.909
Fuel Mole Fraction [ppm]	250	5000
Oxygen Mole Fraction [%]	0.55	5.5
Equivalence Ratio	0.25	1.0

Type of Data:

1,3-Butadiene Table 1: OH concentration time history measurements in argon using narrow-linewidth ring-dye laser absorption of the R₁(5) line of the OH A-X (0,0) band at 306.7 nm.

1,3-Butadiene Table 2: Ignition delay time measurements in argon based on the time to reach 50% maximum OH absorption.

1,3-Butadiene Table 1:

T ₅ [K]	P ₅ [atm]	C ₄ H ₆ [ppm]	φ (E.R.)	1/8 Peak [μs]	1/8 Peak [ppm]	1/4 Peak [μs]	1/4 Peak [ppm]	Plateau [ppm]	50% peak [μs]	50% peak [ppm]	Peak [μs]	Peak [ppm]
1408	1.712	250	0.25	86	1	173	4	111	529	56	690	111
1508	1.688	250	0.25	52	1	103	4	132	295	66	412	132
1582	1.682	250	0.25	34	3	68	8	117	178	81	270	161
1405	1.513	1000	1	287	3.2	574	3.5	175	1624	88	2294	175
1481	1.496	1000	1	207	7.8	414	9.6	218	906	109	1655	218
1585	1.449	1000	1	72	1.7	144	14.5	266	416	129	576	257
1642	1.416	1000	1	48	14.2	97	24.8	273	257	137	386	273
1710	1.392	1000	1	45	0.5	90	18	324	223	162	359	324
1808	1.401	1000	1	41	0.5	82	33.5	391	152	196	329	391

1,3-Butadiene Table 2:

T₅ [K]	P₅ [atm]	Fuel [ppm]	φ (E.R.)	Ignition Time [μs]
1408	1.712	250	0.25	529
1508	1.688	250	0.25	295
1582	1.682	250	0.25	178
1672	1.586	250	0.25	120
1389	1.783	500	0.5	920
1422	1.781	500	0.5	772
1501	1.69	500	0.5	374
1571	1.639	500	0.5	227
1813	1.537	500	0.5	62
1405	1.513	1000	1	1624
1481	1.496	1000	1	906
1509	1.683	1000	1	580
1564	1.604	1000	1	376
1571	1.634	1000	1	408
1585	1.449	1000	1	416
1625	1.558	1000	1	252
1642	1.416	1000	1	257
1710	1.392	1000	1	223
1748	1.544	1000	1	119
1808	1.401	1000	1	152
1459	3.823	1000	1	422
1534	3.909	1000	1	258
1594	3.731	1000	1	176
1684	3.604	1000	1	96
1430	1.816	5000	1	372
1523	1.762	5000	1	163
1532	1.681	5000	1	150
1643	1.67	5000	1	65

Aromatics

Toluene

Literature Source of Data:

V. Vasudevan, D. F. Davidson, R. K. Hanson, "*Shock Tube Measurements of Toluene Ignition Times and OH Concentration Time Histories*," Proceedings of the Combustion Institute, in press (2004).

Range of Data:

Temperature [K]	1458	1847
Pressure [atm]	1.75	4.54
Fuel Mole Fraction [ppm]	250	5000
Oxygen Mole Fraction [%]	0.225	4.5
Equivalence Ratio	0.5	1.5

Type of Data:

Toluene Table 1: OH concentration time history measurements in argon using narrow-linewidth ring-dye laser absorption of the R₁(5) line of the OH A-X (0,0) band at 306.7 nm. "n.d." indicates that the plateau was not well defined and "sat." indicated a saturated signal with laser transmission ~0.

Toluene Table 1:

T ₅ [K]	P ₅ [atm]	C ₇ H ₈ [ppm]	O ₂ [%]	φ (E.R.)	First Plateau [μs]	First Plateau [ppm]	Ignition Time [μs]	Peak [ppm]
1607	2.03	250	0.225	1	121	12	949	87
1648	2.03	250	0.225	1	104	15	608	96
1700	1.89	250	0.225	1	64	23	369	109
1783	1.84	250	0.225	1	24	36	136	118
1564	1.95	1000	0.9	1	127	31	1068	324
1586	1.90	1000	0.9	1	97	32	702	377
1614	1.80	1000	0.9	1	64	40	389	388
1689	1.79	1000	0.9	1	43	56	209	460
1527	4.54	1000	0.9	1	n.d.	32	798	306
1541	4.43	1000	0.9	1	n.d.	36	651	318
1697	4.26	1000	0.9	1	37	53	150	421
1458	1.99	1000	1.8	0.5	n.d.	23	1123	424
1504	1.98	1000	1.8	0.5	230	26	725	448
1540	1.96	1000	1.8	0.5	110	31	501	455
1550	1.94	1000	1.8	0.5	54	31	386	501
1666	1.92	1000	1.8	0.5	28	65	153	591
1616	1.82	1000	0.6	1.5	100	35	1090	162
1627	1.92	1000	0.6	1.5	47	36	922	201
1714	1.77	1000	0.6	1.5	26	55	384	259
1847	1.75	1000	0.6	1.5	12	83	143	298
1434	2.03	5000	4.5	1	n.d.	75	1070	sat
1454	1.66	5000	4.5	1	n.d.	87	750	sat.
1618	1.88	5000	4.5	1	n.d.	177	130	sat.
1635	1.83	5000	4.5	1	n.d.	182	107	sat.

Toluene (continued)

Literature Source of Data:

D. F. Davidson, B. M. Gauthier, R. K. Hanson, “*Shock Tube Ignition Measurements of Iso-Octane/Air and Toluene/Air at High Pressures,*” Proceedings of the Combustion Institute, in press (2004).

B. M. Gauthier, D. F. Davidson, R. K. Hanson, “*A Shock Tube Study of Iso-Octane and Toluene Ignition at High Pressures,*” Western States Section/ Combustion Institute Fall 2003 Meeting, Paper 03F-26 (2003).

Range of Data:

Temperature [K]	971	1269
Pressure [atm]	13.1	54.3
Fuel Mole Fraction [%]	1.15	2.28
Oxygen Mole Fraction [%]	20.8	20.5
Equivalence Ratio	0.5	1.0

Type of Data:

Toluene Table 2: Ignition delay time data in air using PZT pressure measurements of the time between the arrival of the reflected shock (the center of the reflected shock bifurcation feature) and the distinct ignition pressure rise (the time of the intersection of the linear extrapolation of the pressure rise with the pre-ignition pressure floor). Similar ignition delay times were recovered from CH and OH emission measurements.

Toluene Table 2:

T₅ [K]	P₅ [atm]	Toluene [%]	φ (E.R.)	Ignition Time [μs]
971	17.2	2.28	1	1314
995	16.5	2.28	1	1100
1053	17.1	2.28	1	1288
1067	16.1	2.28	1	1646
1076	16.5	2.28	1	1174
1097	16.6	2.28	1	1223
1138	22.2	2.28	1	716
1173	14.9	2.28	1	767
977	54.3	2.28	1	1173
990	50.5	2.28	1	952
1005	50.4	2.28	1	870
1068	48.0	2.28	1	605
1111	46.7	2.28	1	397
1166	44.6	2.28	1	276
1179	41.5	2.28	1	273
1183	46.2	2.28	1	191
1242	42.4	2.28	1	121
1112	15.3	1.15	0.5	1250
1125	13.1	1.15	0.5	1454
1206	14.1	1.15	0.5	598
1269	14.4	1.15	0.5	279
1091	50.5	1.15	0.5	1186
1135	46.5	1.15	0.5	669
1149	44.4	1.15	0.5	579
1211	44.4	1.15	0.5	250

Other Fuels

Gasoline

Literature Source of Data:

B.M. Gauthier, D.F. Davidson, R.K. Hanson, "*Shock Tube Determination of Ignition Delay Times in Full-Blend and Surrogate Fuel Mixtures*," Combustion and Flame, accepted for publication (2004).

Range of Data:

Temperature [K]	902	1280
Pressure [atm]	14.9	60.9
Fuel Mole Fraction [%]		
Oxygen Mole Fraction [%]		
Equivalence Ratio	0.5	2.1

Type of Data:

Gasoline Table 1: Ignition delay time data in air using PZT pressure measurements of the time between the arrival of the reflected shock (the center of the reflected shock bifurcation feature) and the distinct ignition pressure rise (the time of the intersection of the linear extrapolation of the pressure rise with the pre-ignition pressure floor). Similar ignition delay times were recovered from CH and OH emission measurements. Gasoline identified as RD387 by General Motors Research and Development Center, Warren Michigan. This is an 87 Octane Number, $(RON+MON)/2$, gasoline with an H/C ratio of 1.85 blended to represent a "customer average" regular-grade reformulated gasoline without added oxygenates. EGR ratio is defined such that a mixture of X% EGR corresponds to a mixture of $(100-X)$ mol.% of the fuel/air mixture and X mol.% of the products that result from the complete conversion of the fuel/air mixture to CO_2 , H_2O , O_2 and N_2 .

Gasoline Table 1:

T_5 [K]	P_5 [atm]	EGR [%]	ϕ (E.R.)	Ignition Time [μ s]
986	23.6	0	1.0	1317
1011	18.7	0	1.0	1395
1048	21.6	0	1.0	908
1053	16.5	0	1.0	1068
1153	23.6	0	1.0	208
1158	17.3	0	1.0	271
1280	15.4	0	1.0	51
1014	17.9	20	1.0	1651
1048	18.0	20	1.0	1122
1081	17.9	20	1.0	759
1191	15.6	20	1.0	241
1042	18.1	30	1.0	1321
1081	18.0	30	1.0	841
1143	18.3	30	1.0	421
1160	16.3	30	1.0	390
1031	16.1	0	0.5	1805
1125	15.3	0	0.5	661
1067	15.8	20	0.5	1445
1227	15.5	20	0.5	186
1023	16.5	0	1.9	1367
1144	14.9	0	2.1	411
902	53.4	0	1.0	1558
903	48.1	0	1.0	1299
929	54.4	0	1.0	1265
947	60.9	0	1.0	890
977	54.0	0	1.0	734
1015	54.6	0	1.0	465
1045	47.8	0	1.0	355
1083	48.6	0	1.0	229
988	49.5	0	0.5	1243
1111	45.0	0	0.5	281
1091	46.9	20	0.5	463
1115	51.1	20	0.5	341
916	55.2	0	1.6	928
1048	53.0	0	1.8	233

Gasoline Surrogate

Literature Source of Data:

B.M. Gauthier, D.F. Davidson, R.K. Hanson, "*Shock Tube Determination of Ignition Delay Times in Full-Blend and Surrogate Fuel Mixtures*," Combustion and Flame, accepted for publication (2004).

Range of Data:

Temperature [K]	859	1214
Pressure [atm]	14.2	58.3
Fuel Mole Fraction [%]		
Oxygen Mole Fraction [%]		
Equivalence Ratio	0.5	2.0

Type of Data:

Gasoline Surrogate Table 1: Ignition delay time data in air using PZT pressure measurements of the time between the arrival of the reflected shock (the center of the reflected shock bifurcation feature) and the distinct ignition pressure rise (the time of the intersection of the linear extrapolation of the pressure rise with the pre-ignition pressure floor). Similar ignition delay times were recovered from CH and OH emission measurements. Gasoline Surrogate A is composed of: 56% iso-octane, 28% toluene and 17% n-heptane by mole fraction. EGR ratio definition is the same as in Gasoline Table 1.

Gasoline Surrogate Table 2: Ignition time definition is the same as Gasoline Surrogate Table 1. Gasoline Surrogate B is composed of: 63% iso-octane, 20% toluene and 17% n-heptane by mole fraction.

Gasoline Surrogate Table 1:

T₅ [K]	P₅ [atm]	EGR [%]	φ (E.R.)	Ignition Time [μs]
1023	16.6	0	1.0	1288
1035	17.4	0	1.0	1041
1065	19.6	0	1.0	746
1116	15.9	0	1.0	550
1118	18.4	0	1.0	443
1137	19.4	0	1.0	268
1051	16.8	20	1.0	1233
1080	16.4	20	1.0	914
1131	14.6	20	1.0	568
1157	14.6	20	1.0	381
1036	17.5	30	1.0	1754
1085	17.4	30	1.0	949
1104	14.7	30	1.0	869
1059	18.3	0	0.5	1326
1183	16.9	0	0.5	252
1068	16.0	20	0.5	1348
1195	14.3	20	0.5	263
1049	16.2	0	2.0	1003
1214	14.2	0	2.0	126
859	48.7	0	1.0	1428
930	52.1	0	1.0	1198
1049	51.8	0	1.0	376
992	54.3	0	0.5	1122
1021	49.7	0	0.5	893
1138	53.7	0	0.5	185
1053	51.4	20	0.5	696
840	53.7	0	2.0	911
974	54.3	0	2.0	576
1054	48.3	0	2.0	236

Gasoline Surrogate Table 2:

T₅ [K]	P₅ [atm]	EGR [%]	φ (E.R.)	Ignition Time [μs]
972	24.6	0	1.0	1528
1101	24.8	0	1.0	368
1136	25.0	0	1.0	237
907	53.4	0	1.0	1152
914	58.3	0	1.0	918
947	56.3	0	1.0	866
980	49.2	0	1.0	767
985	58.9	0	1.0	637
1005	53.7	0	1.0	537
1011	50.0	0	1.0	569
1065	49.7	0	1.0	297
1092	55.1	0	1.0	182
1103	49.3	0	1.0	180

IDT Measurements up to January 2014

Small Fuels

Hydrogen

Literature Source of Data:

G. A. Pang, D. F. Davidson, R. K. Hanson, "*Shock tube ignition delay times for hydrogen-oxygen-argon mixtures at low temperatures and elevated pressures,*" Paper 07F-12, Western States Section/Combustion Institute Fall Meeting, Sandia National Laboratory, October 16-17, 2007.

G. A. Pang, D. F. Davidson, R. K. Hanson, "*Experimental study and modeling of shock tube ignition delay times for hydrogen-oxygen-argon mixtures at low temperatures,*" Western States Section/Combustion Institute Spring Meeting, University Southern California, Los Angeles, March 17-18, 2008.

G. A. Pang, D. F. Davidson, R. K. Hanson, "*Experimental study and modeling of shock tube ignition delay times for hydrogen-oxygen-argon mixtures at low temperatures,*" Proceedings of the Combustion Institute 32 (2009) 181-188.

Range of Data:

Temperature [K]	908	1118
Pressure [atm]	3.0	3.7
Fuel Mole Fraction [%]	4	15
Oxygen Mole Fraction [%]	2	18
Equivalence Ratio	0.42	1.0

Type of Data:

Hydrogen Table 1: Ignition delay time measurement using sidewall pressure, and OH* emission near 306 nm in argon. Details of the pressure profiles and shock attenuations for each experiment are included in the paper.

Hydrogen Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
924	3.42	4	2	1	10050
930	3.586	4	2	1	7912
930	3.534	4	2	1	13090
930	3.635	4	2	1	10420
931	3.622	4	2	1	7851
938	3.651	4	2	1	7723
938	3.535	4	2	1	6973
938	3.52	4	2	1	6133
939	3.622	4	2	1	6803
939	3.598	4	2	1	7274
944	3.651	4	2	1	5627
944	3.525	4	2	1	6819
949	3.512	4	2	1	4740
957	3.541	4	2	1	3652
958	3.706	4	2	1	3737
964	3.566	4	2	1	3882
967	3.515	4	2	1	3408
967	3.711	4	2	1	3352
971	3.627	4	2	1	2377
973	3.624	4	2	1	2260
975	3.687	4	2	1	3229
977	3.589	4	2	1	2893
981	3.646	4	2	1	2920
984	3.679	4	2	1	1625
987	3.397	4	2	1	2075
992	3.448	4	2	1	1641
1000	3.651	4	2	1	1103
1003	3.572	4	2	1	1060
1010	3.454	4	2	1	628
1014	3.437	4	2	1	637
1021	3.374	4	2	1	618.4
1102	3.393	4	2	1	183.7
1118	3.392	4	2	1	149.7
906	3.5	15	18	0.417	9130
908	3.4	15	18	0.417	6210
909	3.4	15	18	0.417	6185
919	3.4	15	18	0.417	6645
928	3.4	15	18	0.417	3595
943	3.4	15	18	0.417	3280
951	3.3	15	18	0.417	2810
963	3.3	15	18	0.417	1614
969	3.2	15	18	0.417	1268
992	3.2	15	18	0.417	582
999	3.1	15	18	0.417	352
1027	3.1	15	18	0.417	82.0
1049	3.0	15	18	0.417	46.6

Hydrogen

Literature Source of Data:

S. S. Vasu, D. F. Davidson, R. K. Hanson, “*Shock Tube Study of Syngas Ignition in Rich CO₂ Mixtures and Determination of the Rate of $H+O_2+CO_2=HO_2+CO_2$,*” Energy & Fuels 25 (2011) 990-997.

Range of Data:

Temperature [K]	974	1091
Pressure [atm]	1.1	2.6
Fuel Mole Fraction [%]	See	Below
Oxygen Mole Fraction [%]	See	Below
Equivalence Ratio	See	Below

Type of Data:

Hydrogen Table 2: Ignition delay time measurement using sidewall pressure, and OH* emission near 306 nm. The mixture is syngas/air with the following composition: 8.91% H₂, 10.25% O₂, 11.58% CO, 24.44% CO₂, and 44.83% N₂.

Hydrogen Table 2:

T ₅ [K]	P ₅ [atm]	Ign. Time [μs]
974	1.38	1630
974	1.35	1907
981	1.24	1360
988	1.28	1225
1000	1.27	820
1015	1.36	420
1016	1.29	490
1029	1.25	358
1032	1.28	239
1065	1.3	182
1069	1.22	144
1076	1.16	143
1081	1.13	146
1081	1.19	139
1095	1.19	131
1120	1.2	93
1135	1.2	80
1135	1.19	83
1151	1.1	66
1160	1.22	73
975	1.72	1720
979	1.74	1256
983	1.76	1630
999	1.8	951
1010	1.59	640
1020	1.84	482
1048	1.7	199
1076	1.69	150
1145	1.51	56
1017	1.92	823
1024	2.59	641
1054	2.38	362
1091	2.31	97

Hydrogen

Literature Source of Data:

R. K. Hanson, S. Chakarborty, G. A. Pang, W. Ren, S. Wang, D. F. Davidson, "Constrained Reaction Volume: A New Approach to Studying Reactive Systems in Shock Tubes," Paper 27, 29th International Symposium on Shock Waves, 2013.

R. K. Hanson, S. Chakraborty, G. A. Pang, W. Ren, S. Wang, D. F. Davidson, "Constrained Reaction Volume: A Strategy for Reflected Shock Wave Kinetics Experiments," Paper 29, 24th ICDERS Meeting, July 28-August 2, 2013, Taipei, Taiwan.

R. K. Hanson, G. A. Pang, S. Chakraborty, W. Ren, D. F. Davidson, "Constrained Reaction Volume Approach for Studying Chemical Kinetics behind Reflected Shock Waves," Combustion and Flame 160 (2013) 1550-1558.

Range of Data:

Temperature [K]	960	967
Pressure [atm]	3.4	3.5
Fuel Mole Fraction [%]	3.1	3.3
Oxygen Mole Fraction [%]	1.55	1.65
Equivalence Ratio	1.0	1.0

Type of Data:

Hydrogen Table 3: **Constrained Reaction Volume (CRV) experiments** of ignition delay time using OH* emission near 306 nm in argon. Details of the pressure profiles are included in the paper.

Hydrogen Table 3:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
960	3.40	3.1	1.55	1	3300
967	3.50	3.3	1.65	1	2500

Methane

Literature Source of Data:

D. F. Davidson, W. Ren, R. K. Hanson, "*Experimental Database for Development of a HiFiRE JP-7 Surrogate Fuel Mechanism*," AIAA-2012-0620, 50th AIAA Aerospace Sciences Meeting, Tennessee, January 2012.

Range of Data:

Temperature [K]	1369	1760
Pressure [atm]	13.1	37.5
Fuel Mole Fraction [%]	1	4
Oxygen Mole Fraction [%]	4	4
Equivalence Ratio	0.5	2

Type of Data:

Methane Table 1: Ignition delay time measurements in argon using sidewall PZT pressure and CH* emission near 431 nm.

Methane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1760	13.19	2	4	1	67
1616	13.70	2	4	1	284
1691	13.44	2	4	1	132
1457	13.88	2	4	1	1289
1575	14.06	2	4	1	422
1521	14.16	2	4	1	749
1469	14.37	2	4	1	1205
1621	37.35	2	4	1	142
1499	35.26	2	4	1	535
1584	36.79	2	4	1	208
1565	37.49	2	4	1	242
1522	36.20	2	4	1	403
1369	15.03	1	4	0.5	1714
1557	14.29	1	4	0.5	360
1663	13.80	1	4	0.5	130
1587	13.76	1	4	0.5	288
1505	14.65	1	4	0.5	640
1411	14.28	1	4	0.5	1589
1447	14.26	1	4	0.5	1201
1438	14.06	4	4	2	1770
1452	13.74	4	4	2	1593
1522	13.43	4	4	2	1018
1621	13.58	4	4	2	380
1665	13.18	4	4	2	243
1716	13.10	4	4	2	153
1564	13.18	4	4	2	676

Ethylene

Literature Source of Data:

W. Ren, D. F. Davidson, R. K. Hanson, “*IR Laser Absorption Diagnostic for C₂H₄ in Shock Tube Kinetics Studies*,” International Journal of Chemical Kinetics 44 (2012) 423-432.

Range of Data:

Temperature [K]	1199	1550
Pressure [atm]	2.91	3.44
Fuel Mole Fraction [%]	0.5	0.5
Oxygen Mole Fraction [%]	0.75	3.0
Equivalence Ratio	0.5	2.0

Type of Data:

Ethylene Table 1: Ignition delay time measurement in argon using removal of ethylene concentration (extrapolated to zero baseline) from measurements of ethylene time-histories using 10.5 micron laser absorption. The following table includes unpublished data from the same study.

Ethylene Table 1:

T _s [K]	P _s [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1267	3.27	0.5	1.5	1.0	804
1353	3.04	0.5	1.5	1.0	344
1440	2.95	0.5	1.5	1.0	160
1199	3.44	0.5	3.0	0.5	662
1221	3.42	0.5	3.0	0.5	480
1466	3.09	0.5	3.0	0.5	50
1534	2.91	0.5	3.0	0.5	38
1550	2.87	0.5	0.75	2.0	305

Ethylene

Literature Source of Data:

R. K. Hanson, S. Chakarborty, G. A. Pang, W. Ren, S. Wang, D. F. Davidson, "Constrained Reaction Volume: A New Approach to Studying Reactive Systems in Shock Tubes," Paper 27, 29th International Symposium on Shock Waves, 2013.

R. K. Hanson, S. Chakraborty, G. A. Pang, W. Ren, S. Wang, D. F. Davidson, "Constrained Reaction Volume: A Strategy for Reflected Shock Wave Kinetics Experiments," Paper 29, 24th ICDERS Meeting, July 28-August 2, 2013, Taipei, Taiwan.

R. K. Hanson, G. A. Pang, S. Chakarborty, W. Ren, S. Wang, D. F. Davidson, "Constrained Reaction Volume Approach for Studying Chemical Kinetics Behind Reflected Shock Waves," Combustion and Flame 160 (2013) 1550-1558.

Range of Data:

Temperature [K]	1135	11390
Pressure [atm]	2.3	2.3
Fuel Mole Fraction [%]	0.38	0.4
Oxygen Mole Fraction [%]	3.8	4.0
Equivalence Ratio	0.33	0.33

Type of Data:

Ethylene Table 2: **CONSTRAINED REACTION VOLUME STRATEGY** experiments for ignition delay time measurement in argon base on temperature measurement (time of 50% temperature rise) using CO₂ laser absorption near 2.7 microns and using OH laser absorption (time of 50% peak OH value) near 306.7 nm. Mixtures include 1.5-1.6% CO₂ for temperature monitoring.

Ethylene Table 2:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1135	2.3	0.38	3.8	0.33	1624
1139	2.3	0.40	4.0	0.33	1661

Ethylene

Literature Source of Data:

D. F. Davidson, W. Ren, R. K. Hanson, "*Experimental Database for Development of a HiFiRE JP-7 Surrogate Fuel Mechanism*," AIAA-2012-0620, 50th AIAA Aerospace Sciences Meeting, Tennessee, January 2012.

Range of Data:

Temperature [K]	1099	1268
Pressure [atm]	15.5	40.8
Fuel Mole Fraction [%]	0.375	1.5
Oxygen Mole Fraction [%]	4	4
Equivalence Ratio	0.5	2

Type of Data:

Ethylene Table 3: Ignition delay time measurements in argon using sidewall PZT pressure and CH* emission near 431 nm.

Ethylene Table 3:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1268	15.54	1.33	4	1	115
1167	16.34	1.33	4	1	579
1234	15.59	1.33	4	1	197
1197	16.17	1.33	4	1	391
1133	16.12	1.33	4	1	998
1130	16.25	1.33	4	1	1060
1246	40.44	1.33	4	1	152
1208	40.05	1.33	4	1	263
1199	40.76	1.33	4	1	291
1165	40.37	1.33	4	1	453
1130	40.16	1.33	4	1	708
1171	39.09	1.33	4	1	429
1170	15.78	0.667	4	0.5	645
1245	15.84	0.667	4	0.5	175
1224	15.98	0.667	4	0.5	262
1202	15.94	0.667	4	0.5	385
1152	16.22	0.667	4	0.5	899
1114	16.43	0.667	4	0.5	1601
1139	16.41	0.667	4	0.5	1062
1126	16.23	2.667	4	2	925
1136	15.97	2.667	4	2	819
1099	16.23	2.667	4	2	1258
1177	15.65	2.667	4	2	501
1168	15.92	2.667	4	2	556
1217	15.59	2.667	4	2	301

Methane/Ethylene 36/64 Blend

Literature Source of Data:

D. F. Davidson, W. Ren, R. K. Hanson, "*Experimental Database for Development of a HiFiRE JP-7 Surrogate Fuel Mechanism*," AIAA-2012-0620, 50th AIAA Aerospace Sciences Meeting, Tennessee, January 2012.

Range of Data:

Temperature [K]	1106	1340
Pressure [atm]	15.2	42.4
Fuel Mole Fraction [%]	0.75	3.02
Oxygen Mole Fraction [%]	4	4
Equivalence Ratio	0.5	2

Type of Data:

Methane/Ethylene 36/64 Table 1: Ignition delay time measurements in argon using sidewall PZT pressure and CH* emission near 431 nm. This fuel mixture is a surrogate blend for JP-7 (G. L. Pellett et al. AIAA Paper 2007-5664, July 2007) comprised of 36% mole fraction methane and 64% mole fraction ethylene.

Methane/Ethylene 36/64 Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1340	15.26	1.51	4	1	108
1276	15.20	1.51	4	1	258
1212	15.72	1.51	4	1	600
1165	16.05	1.51	4	1	1073
1133	16.32	1.51	4	1	1669
1238	15.38	1.51	4	1	425
1283	40.30	1.51	4	1	167
1239	40.99	1.51	4	1	327
1163	42.42	1.51	4	1	738
1196	41.83	1.51	4	1	554
1106	42.36	1.51	4	1	1541
1301	15.65	0.75	4	0.5	145
1192	16.07	0.75	4	0.5	822
1161	16.15	0.75	4	0.5	1216
1244	15.84	0.75	4	0.5	362
1210	15.77	0.75	4	0.5	607
1267	15.62	0.75	4	0.5	257
1196	16.06	3.02	4	2	702
1237	15.79	3.02	4	2	446
1281	15.59	3.02	4	2	277
1331	15.35	3.02	4	2	151
1167	15.99	3.02	4	2	956
1147	16.22	3.02	4	2	1202

Propane

Literature Source of Data:

K.-Y. Lam, D. Vinh, S. Wang, Z. Hong, D. F. Davidson, R. K. Hanson, “*Shock Tube Ignition Delay Time Measurements in Propane/O₂/Argon Mixtures at Near-Constant-Volume Conditions*,” Paper 09F-66, Western States Section/Combustion Institute Fall Meeting, September 2009.

K.-Y. Lam, Z. Hong, D. F. Davidson, R. K. Hanson, “*Shock Tube Ignition Measurements in Propane/O₂/Argon Mixtures at Near-Constant-Volume Conditions*,” Proceedings of the Combustion Institute 33 (2011) 251-258.

K.-Y. Lam, “*Shock Tube Measurements of Oxygenated Fuel Combustion using Laser Absorption Spectroscopy*,” Ph.D. Thesis, Mechanical Engineering Department, Stanford University, June 2013.

Range of Data:

Temperature [K]	980	1400
Pressure [atm]	6	60
Fuel Mole Fraction [%]	0.8	0.8
Oxygen Mole Fraction [%]	8.0	8.0
Equivalence Ratio	0.5	0.5

Type of Data:

Propane Table 1: Ignition delay time measurement in argon using sidewall PZT pressure and OH* emission near 306 nm. All shock wave experiments use a mixture of 0.8% C₃H₈, and 8% O₂ in argon. Details of the pressure profiles are included in the paper and outlined here.

Propane Table 1:

T_5 [K]	P_5 [atm]	dP_5/dt [%/ms]	Ign. Time [ms]
1405	5.46	0	0.083
1366	5.60	0	0.17
1339	5.68	0	0.23
1303	5.70	0	0.40
1281	5.88	0	0.59
1250	5.98	0	0.89
1234	6.99	0	0.94
1223	6.08	0	1.27
1194	6.13	0	1.79
1189	7.17	0	1.81
1184	7.17	0	1.82
1147	7.44	3.5	2.98
1125	7.29	3.1	4.26
1081	6.64	2.6	6.18
1034	7.06	2.0	10.51
1006	7.16	1.4	15.66
991	7.16	1.2	17.79
1381	5.29	0	0.15
1341	5.41	0	0.27
1273	6.24	0	0.68
1260	6.37	0	0.83
1208	6.69	0	1.76
1202	5.26	0	1.75
1170	6.65	0	2.95
1157	5.64	0	4.04
1141	5.65	0	5.45
1126	6.04	0	5.46
1095	6.39	0	7.10
1069	6.77	0	10.19
1052	6.79	0	12.68
1051	6.82	0	12.47
1044	6.67	0	16.38
1033	6.73	0	17.90
1027	6.80	0	21.23
1021	6.59	0	20.94
1289	27.1	0	0.20
1260	21.9	0	0.33
1221	25.2	0	0.51
1189	26.0	0	0.68
1173	22.4	0	0.96
1160	23.3	0	1.01
1155	26.1	0	0.99
1123	28.8	0	1.34
1114	29.5	7.4	1.43
1113	23.1	8.0	1.72
1062	22.7	6.8	2.67

1053	21.1	6.7	2.99
1052	25.5	6.5	2.65
1009	22.5	5.9	4.47
1005	21.9	6.5	4.29
1004	22.0	7.2	4.51
975	21.4	6.9	5.54
968	29.4	6.6	4.77
950	21.4	6.8	6.79
1273	23.6	0	0.26
1254	24.4	0	0.34
1247	26.8	0	0.35
1206	27.0	0	0.60
1180	24.2	0	0.90
1154	24.4	0	1.24
1137	26.4	0	1.42
1114	24.3	0	2.10
1101	30.7	0	1.84
1088	29.1	0	2.10
1086	25.8	0	2.95
1077	22.8	0	3.30
1077	27.4	0	3.17
1060	28.1	0	3.70
1049	30.7	0	3.36
1042	31.1	0	3.53
1009	30.9	0	8.49
988	29.2	0	9.28
1244	66.8	0	0.17
1219	59.8	0	0.26
1145	64.0	0	0.53
1093	55.4	0	1.00
1071	62.2	8.5	1.19
1026	57.7	7.4	2.07
1022	61.2	6.8	1.95
996	54.7	7.1	2.71
983	56.0	6.7	3.14
980	54.6	6.2	3.26
1247	59.5	0	0.20
1202	55.9	0	0.34
1178	63.4	0	0.39
1129	67.4	0	0.65
1114	60.4	0	0.82
1100	63.9	0	0.91
1071	65.0	0	1.37
1061	63.8	0	1.60
1043	67.8	0	2.01
1035	54.1	0	2.61
1022	59.4	0	2.94
1008	53.7	0	3.89
994	54.0	0	4.75

Propene

Literature Source of Data:

S. Burke, U. Burke, et al., “An Experimental and Modeling Study of Propene Oxidation. Part 2: Ignition Delay Time and Flame Speed Measurements,” Paper submitted to Combustion and Flame, April 2014.

Range of Data:

Temperature [K]	980	1400
Pressure [atm]	2	4
Fuel Mole Fraction [%]	0.44	1.6
Oxygen Mole Fraction [%]	4.0	7.2
Equivalence Ratio	0.5	2.0

Type of Data:

Propene Table 1: Ignition delay time measurement in argon using corrected endwall OH* emission near 306 nm. High pressure shocks (40+ atm) used sidewall OH* emission. This paper includes a comparison of shock tube data from several laboratories.

Propene Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	E.R.	Ign. Time [μs]
1195	49.8	0.888	4.0	1	1454
1226	47.4	0.888	4.0	1	1044
1371	46.4	0.888	4.0	1	251
1242	43.9	0.888	4.0	1	930
1175	51.2	0.888	4.0	1	1877
1327	46.4	0.888	4.0	1	372
1428	45.6	0.888	4.0	1	134
1318	47.1	0.888	4.0	1	394
1302	47.5	0.888	4.0	1	456
1223	46.2	1.78	4.0	2	1044
1281	46.4	1.78	4.0	2	555
1210	46.8	1.78	4.0	2	1300
1316	44.8	1.78	4.0	2	457
1371	44.0	1.78	4.0	2	280
1200	45.7	1.78	4.0	2	1543
1432	44.2	1.78	4.0	2	158
1249	45.2	1.78	4.0	2	910

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	E.R.	Ign. Time [μs]
1554	2.08	0.888	4.0	1	268
1432	2.19	0.888	4.0	1	1072
1597	1.94	0.888	4.0	1	159
1587	1.92	0.888	4.0	1	183
1720	1.97	0.888	4.0	1	52
1656	1.95	0.888	4.0	1	95
1438	2.28	0.888	4.0	1	1003
1408	2.26	0.888	4.0	1	1527
1644	2.18	0.444	4.0	0.5	63
1557	2.26	0.444	4.0	0.5	150
1493	2.32	0.444	4.0	0.5	323
1446	2.40	0.444	4.0	0.5	583
1370	2.45	0.444	4.0	0.5	1635
1360	2.49	0.444	4.0	0.5	2002
1698	2.14	0.444	4.0	0.5	37
1485	2.29	1.78	4.0	2	792
1554	2.22	1.78	4.0	2	417
1660	2.14	1.78	4.0	2	164
1756	2.04	1.78	4.0	2	76
1409	2.34	1.78	4.0	2	1809
1388	2.41	1.78	4.0	2	2324
1566	4.35	0.888	4.0	1	163
1413	4.08	0.888	4.0	1	904
1408	4.25	0.888	4.0	1	920
1465	4.74	0.888	4.0	1	438
1405	4.80	0.888	4.0	1	884
1333	4.76	0.888	4.0	1	2472
1486	4.43	1.6	7.2	1	227
1496	4.24	1.6	7.2	1	205
1567	4.33	1.6	7.2	1	86
1609	4.20	1.6	7.2	1	62
1431	4.47	1.6	7.2	1	404

Normal Alkanes

n-Pentane

Literature Source of Data:

D. F. Davidson, S. C. Ranganath, K.-Y. Lam, M. Liaw, Z. Hong, R. K. Hanson, "Ignition Delay Time Measurements of Normal Alkanes and Simple Oxygenates," Journal of Propulsion and Power 26 (2010) 280-287.

K.-Y. Lam, "Shock Tube Measurements of Oxygenated Fuel Combustion using Laser Absorption Spectroscopy," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, June 2013.

Range of Data:

Temperature [K]	1261	1489
Pressure [atm]	1.62	3.61
Fuel Mole Fraction [%]	0.25	0.5
Oxygen Mole Fraction [%]	4.0	4.0
Equivalence Ratio	0.5	1.0

Type of Data:

n-Pentane Table 1: Ignition delay time measurement in argon using sidewall PZT pressure and sidewall and endwall OH* emission near 306 nm.

n-Pentane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1489	1.74	0.5	4	1	145
1426	1.83	0.5	4	1	281
1360	1.91	0.5	4	1	603
1304	1.95	0.5	4	1	1227
1267	2.00	0.5	4	1	1845
1303	3.61	0.5	4	1	832
1261	3.75	0.5	4	1	1455
1395	3.47	0.5	4	1	316
1449	3.41	0.5	4	1	162
1268	3.61	0.5	4	1	1296
1396	1.65	0.25	4	0.5	255
1263	1.73	0.25	4	0.5	1391
1533	1.71	0.25	4	0.5	49
1428	1.62	0.25	4	0.5	153
1397	1.75	0.25	4	0.5	232

***n*-Hexane**

Literature Source of Data:

D. F. Davidson, S. C. Ranganath, K.-Y. Lam, M. Liaw, Z. Hong, R. K. Hanson, *“Ignition Delay Time Measurements of Normal Alkanes and Simple Oxygenates,”* Journal of Propulsion and Power 26 (2010) 280-287.

K.-Y. Lam, *“Shock Tube Measurements of Oxygenated Fuel Combustion using Laser Absorption Spectroscopy,”* Ph.D. Thesis, Mechanical Engineering Department, Stanford University, June 2013.

Range of Data:

Temperature [K]	1237	1475
Pressure [atm]	1.77	3.60
Fuel Mole Fraction [%]	0.21	0.42
Oxygen Mole Fraction [%]	4.0	4.0
Equivalence Ratio	0.5	1.0

Type of Data:

n-Hexane Table 1: Ignition delay time measurement in argon using sidewall PZT pressure and sidewall and endwall OH* emission near 306 nm.

n-Hexane Table 1:

T_5 [K]	P_5 [atm]	Fuel [%]	O ₂ [%]	ϕ E.R.	Ign. Time [μ s]
1284	1.92	0.42	4	1	1317
1353	1.90	0.42	4	1	623
1358	1.76	0.42	4	1	688
1470	1.77	0.42	4	1	176
1375	1.86	0.42	4	1	483
1237	1.93	0.42	4	1	2706
1273	3.32	0.42	4	1	1046
1396	3.60	0.42	4	1	301
1250	3.10	0.42	4	1	1805
1386	3.48	0.42	4	1	291
1407	3.43	0.42	4	1	263
1475	1.77	0.21	4	0.5	78
1274	1.76	0.21	4	0.5	1225
1361	1.89	0.21	4	0.5	303
1367	1.67	0.21	4	0.5	308
1403	1.69	0.21	4	0.5	196

***n*-Heptane**

Literature Source of Data:

D. F. Davidson, M. A. Oehlschlaeger, R. K. Hanson, "*Methyl Concentration Time Histories during iso-Octane and n-Heptane Oxidation*," Paper 05F-61, WSS/CI Fall Meeting, Stanford CA, October 17-18, 2005.

D. F. Davidson, M. A. Oehlschlaeger, R. K. Hanson, "*Methyl Concentration Time Histories during iso-Octane and n-Heptane Oxidation and Pyrolysis*," Proceedings of the Combustion Institute 31 (2007) 321-328.

Range of Data:

Temperature [K]	1395	1546
Pressure [atm]	1.61	1.72
Fuel Mole Fraction [%]	0.05	0.05
Oxygen Mole Fraction [%]	0.550	0.550
Equivalence Ratio	1.0	1.0

Type of Data:

n-Heptane Table 1: Ignition delay time measurement in argon based on the time of full consumption of CH₃ radicals measured by laser absorption at 216 nm.

n-Heptane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1395	1.72	0.05	0.550	1	878
1461	1.67	0.05	0.550	1	366
1546	1.61	0.05	0.550	1	162

***n*-Heptane**

Literature Source of Data:

S. S. Vasu, D. F. Davidson, R. K. Hanson, "*OH time-history during oxidation of n-heptane and methylcyclohexane at high pressures and temperatures,*" Combustion and Flame 156 (2009) 736-749.

S. S. Vasu, "*Measurements of Ignition Times, OH Time-Histories, and Reaction Rates in Jet Fuel and Surrogate Oxidation System,*" Ph.D. Thesis, Mechanical Engineering Department, Stanford University, August 2010.

http://hanson.stanford.edu/dissertations/Vasu_2010.pdf

Range of Data:

Temperature [K]	1229	1271
Pressure [atm]	15.00	15.81
Fuel Mole Fraction [%]	0.10	0.10
Oxygen Mole Fraction [%]	2.2	2.2
Equivalence Ratio	0.5	0.5

Type of Data:

n-Heptane Table 2: OH concentration time history measurements using laser absorption at 306.7 nm. The ignition delay time is the time for the OH mole fraction to reach 50% of the final plateau level. Carrier gas is argon.

n-Heptane Table 2:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1229	15.13	0.10	2.2	0.5	708
1230	15.81	0.10	2.2	0.5	698
1236	15.28	0.10	2.2	0.5	652
1271	15.00	0.10	2.2	0.5	377

***n*-Heptane**

Literature Source of Data:

D. F. Davidson, Z. Hong, G. L. Pilla, A. Farooq, R. D. Cook, R. K. Hanson, "*Species Time-History Measurements During n-Heptane Oxidation Behind Reflected Shock Waves*," Paper 31F1, U.S. Combustion Meeting, Ann Arbor MI, May, 2009.

D. F. Davidson, Z. Hong, G. L. Pilla, A. Farooq, R. D. Cook, R. K. Hanson, "*Multi-Species Time-History Measurements During n-Heptane Oxidation Behind Reflected Shock Waves*," Combustion and Flame 157 (2010) 1899-1905.

D. F. Davidson, Z. Hong, G. L. Pilla, A. Farooq, R. D. Cook, R. K. Hanson, "*Multi-Species Measurements Behind Reflected Shock Waves in Hydrocarbons using Laser Absorption*," Paper AIAA-2010-198, 48th AIAA Aerospace Sciences Meeting, January 2010, Orlando FL.

Range of Data:

Temperature [K]	1358	1506
Pressure [atm]	2.16	2.44
Fuel Mole Fraction [%]	0.03	0.03
Oxygen Mole Fraction [%]	0.33	0.33
Equivalence Ratio	1.0	1.0

Type of Data:

n-Heptane Table 3: Ignition delay time measurement in argon based on the time to reach 50% removal (C₂H₄) or 50% formation (H₂O, CO₂, OH) of the initial/final plateau value of the species mole fractions measured by laser absorption.

n-Heptane Table 3:

T ₅ [K]	P ₅ [atm]	Fuel [ppm]	O ₂ [ppm]	φ (E.R.)	Species	T(50%) [μs]
1358	2.436	300	3300	1	C ₂ H ₄	1280
1368	2.86	300	3300	1	H ₂ O	1810
1375	2.282	300	3300	1	CO ₂	1661
1378	2.326	300	3300	1	OH	1330
1494	2.155	300	3300	1	CO ₂	1330
1494	2.155	300	3300	1	OH	308
1502	2.262	300	3300	1	H ₂ O	280
1506	2.360	300	3300	1	C ₂ H ₄	189

***n*-Octane**

Literature Source of Data:

D. F. Davidson, S. C. Ranganath, K.-Y. Lam, M. Liaw, Z. Hong, R. K. Hanson, *"Ignition Delay Time Measurements of Normal Alkanes and Simple Oxygenates,"* Journal of Propulsion and Power 26 (2010) 280-287.

K.-Y. Lam, *"Shock Tube Measurements of Oxygenated Fuel Combustion using Laser Absorption Spectroscopy,"* Ph.D. Thesis, Mechanical Engineering Department, Stanford University, June 2013.

Range of Data:

Temperature [K]	1252	1455
Pressure [atm]	1.87	3.81
Fuel Mole Fraction [%]	0.16	0.32
Oxygen Mole Fraction [%]	4.0	4.0
Equivalence Ratio	0.5	1.0

Type of Data:

n-Octane Table 1: Ignition delay time measurement in argon using sidewall PZT pressure and sidewall and endwall OH* emission near 306 nm.

n-Octane Table 1:

T_5 [K]	P_5 [atm]	Fuel [%]	O_2 [%]	ϕ E.R.	Ign. Time [μ s]
1325	1.95	0.32	4	1	859
1265	2.05	0.32	4	1	1569
1289	2.01	0.32	4	1	1198
1385	1.96	0.32	4	1	390
1455	1.87	0.32	4	1	166
1282	3.37	0.32	4	1	1020
1421	3.59	0.32	4	1	210
1384	3.63	0.32	4	1	285
1319	3.75	0.32	4	1	614
1273	3.81	0.32	4	1	1064
1434	1.91	0.16	4	0.5	104
1307	1.98	0.16	4	0.5	663
1273	2.07	0.16	4	0.5	988
1252	2.19	0.16	4	0.5	1269
1380	2.01	0.16	4	0.5	220

***n*-Nonane**

Literature Source of Data:

D. F. Davidson, S. C. Ranganath, K.-Y. Lam, M. Liaw, Z. Hong, R. K. Hanson, *“Ignition Delay Time Measurements of Normal Alkanes and Simple Oxygenates,”* Journal of Propulsion and Power 26 (2010) 280-287.

K.-Y. Lam, *“Shock Tube Measurements of Oxygenated Fuel Combustion using Laser Absorption Spectroscopy,”* Ph.D. Thesis, Mechanical Engineering Department, Stanford University, June 2013.

Range of Data:

Temperature [K]	1263	1470
Pressure [atm]	1.68	3.63
Fuel Mole Fraction [%]	0.143	0.286
Oxygen Mole Fraction [%]	4.0	4.0
Equivalence Ratio	0.5	1.0

Type of Data:

n-Nonane Table 1: Ignition delay time measurement in argon using sidewall PZT pressure and sidewall and endwall OH* emission near 306 nm.

n-Nonane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1307	1.85	0.286	4	1	1002
1288	1.68	0.286	4	1	1351
1384	1.80	0.286	4	1	382
1367	1.86	0.286	4	1	499
1330	1.91	0.286	4	1	780
1348	3.63	0.286	4	1	405
1263	3.46	0.286	4	1	1237
1372	3.58	0.286	4	1	318
1359	3.45	0.286	4	1	349
1403	1.67	0.143	4	0.5	190
1399	1.78	0.143	4	0.5	180
1331	1.75	0.143	4	0.5	421
1350	1.87	0.143	4	0.5	317
1299	1.81	0.143	4	0.5	730
1305	1.92	0.143	4	0.5	536
1284	1.96	0.143	4	0.5	781
1385	1.84	0.143	4	0.5	231
1470	1.82	0.143	4	0.5	93

***n*-Decane**

Literature Source of Data:

D. F. Davidson, S. Li, K.-Y. Lam, J. T. Harmon, R. K. Hanson, "*Shock Tube/Laser Absorption Measurements of JP-8 Ignition Delay Times and Multi-Species Time-Histories*," Paper 2128, JANNAF Meeting, December 2011, Huntsville AL.

P. Gokulakrishnan, C. Fuller, M. Klassen, D. F. Davidson, R. K. Hanson, B. Kiel, "*Experimental and Modeling of JP-8, F-T and HRJ Fuel Ignition under Vitiated Conditions*," 49th AIAA Joint Propulsion Conference, July 15-17, 2013, San Jose, CA.

K.-Y. Lam, "*Shock Tube Measurements of Oxygenated Fuel Combustion using Laser Absorption Spectroscopy*," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, June 2013.

Range of Data:

Temperature [K]	1327	1524
Pressure [atm]	1.6	1.6
Fuel Mole Fraction [ppm]	334	365
Oxygen Mole Fraction [%]	0.813	0.813
Equivalence Ratio	0.64	0.70

Type of Data:

n-Decane Table 4: Ignition delay time measurement in argon using an ignition delay time definition based on the time to achieve 50% of the peak OH or CO concentration or the removal of 50% of the plateau/peak C₂H₄ concentration.

n-Decane Table 4:

T ₅ [K]	P ₅ [atm]	Fuel [ppm]	O ₂ [%]	φ E.R.	OH T(50) [μs]	C ₂ H ₄ (50) [μs]	CO T(50) [μs]
1327	1.69	334	0.813	0.64	1099	1140	912
1382	1.68	358	0.813	0.68	621	715	475
1524	1.61	365	0.813	0.70	144	162	102

n-Decane

Literature Source of Data:

D. R. Haylett, “*The Development and Application of Aerosol Shock Tube Methods for the Study of Low-Vapor-Pressure Fuels*,” Ph.D. Thesis, Mechanical Engineering Department, Stanford University, March 2011.
http://hanson.stanford.edu/dissertations/Haylett_2011.pdf

Range of Data:

Temperature [K]	1081	1173
Pressure [atm]	4.56	5.15
Fuel Mole Fraction [%]	1.4	2.5
Oxygen Mole Fraction [%]	21	21
Equivalence Ratio	1.60	1.89

Type of Data:

n-Decane Table 1: Ignition delay time measurement using sidewall pressure, and confirmed with CH* emission near 430 nm, in nitrogen from **Aerosol Shock Tube experiments**. Data is summarized in D. R. Haylett: Ph.D. thesis (2011).

n-Decane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
Diluent = N2					
1081	5.14	1.440	21	1.06	1696
1131	4.87	1.494	21	1.10	1252
1154	5.15	1.895	21	1.40	921
1109	4.56	1.472	21	1.09	1358
1173	4.93	2.567	21	1.89	950

***n*-Dodecane**

Literature Source of Data:

D. R. Haylett, D. F. Davidson, R. K. Hanson, "*Development of an Aerosol Shock Tube for Kinetic Studies of Low-Vapor-Pressure Fuels*," Paper AIAA-2007-5678, 43rd Joint Propulsion Conference, July 8-11 2007.

D. R. Haylett, D. F. Davidson, R. K. Hanson, "*Development of an Aerosol Shock Tube for Kinetic Studies of Low-Vapor-Pressure Fuels*," Combustion and Flame 155 (2008) 108-117.

D. R. Haylett, D. F. Davidson, R. K. Hanson, "*A Second Generation Aerosol Shock Tube and its Application to Combustion Research*," Paper AIAA-2010-196, 48th Aerospace Meeting, January 4-7, 2010, Orland FL.

D. R. Haylett, D. F. Davidson, R. K. Hanson, "*Second Generation Aerosol Shock Tube: An Improved Design*," Shock Waves 22 (2012) 483-493.

D. R. Haylett, D. F. Davidson, R. K. Hanson, "*Ignition Delay Times of Low-Vapor-Pressure Fuels Measured Using an Aerosol Shock Tube*," Combustion and Flame 159 (2012) 552-561.

D. R. Haylett, "*The Development and Application of Aerosol Shock Tube Methods for the Study of Low-Vapor-Pressure Fuels*," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, March 2011.
http://hanson.stanford.edu/dissertations/Haylett_2011.pdf

Range of Data:

Temperature [K]	1395	1546
Pressure [atm]	1.61	1.72
Fuel Mole Fraction [%]	0.05	0.05
Oxygen Mole Fraction [%]	0.550	0.550
Equivalence Ratio	1.0	1.0

Type of Data:

n-Dodecane Table 1: Ignition delay time measurements using sidewall pressure, and confirmed with CH* emission near 430 nm, from **Aerosol Shock Tube experiments**. Data is summarized in D. R. Haylett: Ph.D. thesis (2011).

n-Dodecane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
Diluent = N2					
1127	6.12	0.6844	21	0.60	797
1050	7.88	0.9365	21	0.82	1034
1084	8.02	0.7871	21	0.69	738
1114	8.50	0.6885	21	0.61	640
1179	8.53	0.9497	21	0.84	220
1167	7.88	1.0246	21	0.90	215

n-Dodecane Table 1 (continued):

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
Diluent = Argon					
1168	6.37	0.7339	21	0.65	484
1209	4.64	0.1860	21	0.16	926
1163	4.93	0.2534	21	0.22	1800
1192	5.06	0.3437	21	0.30	1110
1078	4.00	0.2074	21	0.18	5605
976	4.87	0.4056	21	0.36	6220
903	5.00	0.9297	21	0.82	7825
1241	4.79	0.3488	21	0.31	337
1351	4.51	0.2064	21	0.18	233
979	5.04	2.138	21	1.88	306
1025	6.06	1.659	21	1.46	543
988	6.46	1.990	21	1.75	260
1112	5.65	0.1905	21	0.17	2281
838	6.26	1.504	21	1.32	5352
939	6.31	1.802	21	1.59	544
964	5.82	1.289	21	1.14	1175
932	6.01	1.494	21	1.32	1559
1156	5.34	0.0558	21	0.05	1526
1134	5.20	0.0662	21	0.06	2006
1172	6.19	0.5336	21	0.47	524
1191	6.40	0.589	21	0.52	550
1249	6.23	0.7572	21	0.67	114
1284	6.42	0.8276	21	0.73	80
1053	6.56	0.5649	21	0.5	2105
1045	6.71	0.6098	21	0.54	2753
1068	6.4	0.516	21	0.45	2126
1078	6.32	0.5023	21	0.44	1624
1086	6.32	0.5655	21	0.5	1422
1107	6.25	0.5533	21	0.49	1200

1118	6.18	0.5678	21	0.5	1081
1140	6.07	0.576	21	0.51	820
1345	5.92	0.6297	21	0.55	123
1319	5.78	0.4595	21	0.40	181
1117	8.09	0.7901	21	0.70	412
1124	8.24	0.6972	21	0.61	702
993	4.23	0.6795	21	0.60	5732
921	4.86	1.231	21	1.08	6085
943	5.21	1.280	21	1.13	3804
932	4.98	1.240	21	1.09	4627
954	5.05	1.158	21	1.02	4169
1061	4.10	1.116	21	0.98	959
1019	4.36	0.9624	21	0.85	1828
976	4.49	1.000	21	0.88	2555
1192	4.57	0.875	21	0.77	246
1118	4.88	0.8929	21	0.79	587
1146	4.01	0.5656	21	0.50	1027
1035	4.43	0.83	21	0.73	1803
1003	4.68	0.742	21	0.65	4081
1147	4.97	1.276	21	1.12	357
1220	8.63	0.4879	21	0.43	292
1172	5.33	0.7078	21	0.62	663
1011	6.57	1.179	21	1.04	1100
940	7.00	1.166	21	1.03	2599
958	6.57	1.169	21	1.03	2041
921	6.25	0.9985	21	0.88	5556
1037	5.69	0.5463	21	0.48	2607
1049	5.76	0.5734	21	0.51	2581

***n*-Dodecane**

Literature Source of Data:

S. S. Vasu, D. F. Davidson, R. K. Hanson, "*High-Pressure Shock Tube Experiments and Modeling of n-Dodecane/Air Ignition*," Paper P2730, 26th International Symposium on Shock Waves-ISSW26, Gottingen Germany, July 15-20, 2007.

S. S. Vasu, D. F. Davidson, Z. Hong, V. Vasudevan, R. K. Hanson, "*n-Dodecane Oxidation at High-Pressures: Measurements of Ignition Delay Times and OH Concentration Time-Histories*," Proceedings of the Combustion Institute 32 (2009) 173-180.

S. S. Vasu, "*Measurements of Ignition Times, OH Time-Histories, and Reaction Rates in Jet Fuel and Surrogate Oxidation System*," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, August 2010.

http://hanson.stanford.edu/dissertations/Vasu_2010.pdf

Range of Data:

Temperature [K]	727	1422
Pressure [atm]	15.0	30.9
Fuel Mole Fraction [%]	0.051	1.123
Oxygen Mole Fraction [%]	1.9	20.9
Equivalence Ratio	0.5	1.0

Type of Data:

n-Dodecane Table 2: Ignition delay time measurements in air in a heated shock tube (100 C) using sidewall pressure, and confirmed with OH* emission near 306 nm. Data summarized in S. S. Vasu: Ph.D. thesis (2010).

n-Dodecane Table 2 continued: Ignition delay time measurements in air in a heated shock tube (100 C). OH concentration time history measurements using laser absorption at 306.7 nm. T(50%) is the time for the OH mole fraction to reach 50% of the final plateau level. . Data is summarized in S. S. Vasu: Ph.D. thesis (2010).

n-Dodecane Table 2:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
Diluent = N2					
747	26.0	0.565	20.89	0.5	2243
806	25.0	0.565	20.89	0.5	1522
873	23.1	0.565	20.89	0.5	1918
910	22.7	0.565	20.89	0.5	2134
943	18.1	0.565	20.89	0.5	2276
996	20.3	0.565	20.89	0.5	1245
1039	21.0	0.565	20.89	0.5	826
1049	18.9	0.565	20.89	0.5	839
1054	20.1	0.565	20.89	0.5	788
1082	18.4	0.565	20.89	0.5	566
1087	19.6	0.565	20.89	0.5	527
1117	30.9	0.565	20.89	0.5	266
1118	20.3	0.565	20.89	0.5	346
1149	21.0	0.565	20.89	0.5	261
1163	21.2	0.565	20.89	0.5	213
1177	20.8	0.565	20.89	0.5	165
727	27.0	1.123	20.77	1.0	809
773	28.7	1.123	20.77	1.0	556
818	26.9	1.123	20.77	1.0	881
822	23.3	1.123	20.77	1.0	805
855	25.9	1.123	20.77	1.0	875
869	22.5	1.123	20.77	1.0	1040
907	23.4	1.123	20.77	1.0	1081
942	21.0	1.123	20.77	1.0	1116
953	20.0	1.123	20.77	1.0	1141
957	19.8	1.123	20.77	1.0	1064
976	20.5	1.123	20.77	1.0	800
978	22.9	1.123	20.77	1.0	912
987	22.0	1.123	20.77	1.0	699
991	21.8	1.123	20.77	1.0	645
992	20.4	1.123	20.77	1.0	739
1008	22.4	1.123	20.77	1.0	570
1015	22.3	1.123	20.77	1.0	508
1036	22.1	1.123	20.77	1.0	432
1087	19.4	1.123	20.77	1.0	403
1102	20.8	1.123	20.77	1.0	268
1107	23.3	1.123	20.77	1.0	298
1109	23.0	1.123	20.77	1.0	278
1118	33.7	1.123	20.77	1.0	183
1123	24.0	1.123	20.77	1.0	262
1125	18.7	1.123	20.77	1.0	188
1135	20.6	1.123	20.77	1.0	178

n-Dodecane Table 2 continued:

T_s [K]	P_s [atm]	Fuel [ppm]	O_2 [%]	ϕ E.R.	T(50%) [μ s]
Diluent = N2					
1230	16.73	1000	3.596	0.5	429
1217	16.07	1000	3.596	0.5	508
1196	15.77	1000	3.596	0.5	701
1186	15.38	1000	3.596	0.5	822
1158	15.19	1000	3.596	0.5	1181
1322	15.52	514.6	1.911	0.5	293
1258	15.31	514.6	1.911	0.5	682
1252	15.08	514.6	1.911	0.5	713
1219	16.10	1000	3.725	0.5	485
1211	16.13	1000	3.725	0.5	534
1179	15.93	1000	3.725	0.5	847
1222	15.84	750	2.786	0.5	625
1221	15.45	750	2.786	0.5	685
1192	14.98	750	2.786	0.5	1056
1422	15.50	1000	3.700	0.5	30
1280	16.57	1000	3.700	0.5	195

***n*-Dodecane**

Literature Source of Data:

D. E. Jackson, D. F. Davidson, R. K. Hanson, “*Application of an Aerosol Shock Tube for the Kinetic Studies of n-Dodecane/Nano-Aluminum Slurries*,” Paper AIAA-2008-4767, 44th AIAA Joint Propulsion Conference, Hartford, CT, July 2008.

Range of Data:

Temperature [K]	1108	1249
Pressure [atm]	4.7	9.4
Fuel Mole Fraction [%]	0.57	0.57
Oxygen Mole Fraction [%]	21	21
Equivalence Ratio	0.5	0.5

Type of Data:

n-Dodecane Table 3: Ignition delay time measurements in argon using CH* emission near 430 nm, from **Aerosol Shock Tube experiments**. Mixtures include nano-particles of aluminum. Tabulated data has been digitized from Figure 10 of the paper. Two data points are from the example figures.

n-Dodecane Table 3:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	%wt Nano Al	Ign. Time [μs]
Diluent = N2						
1126	6.7	0.57	21	0.5	0%	1069
1145	6.7	0.57	21	0.5	0%	605
1173	6.7	0.57	21	0.5	0%	475
1175	6.7	0.57	21	0.5	0%	675
1187	7.1	0.21	21	0.18	0%	793
1141	6.7	0.57	21	0.5	5%	680
1187	6.7	0.57	21	0.5	5%	395
1224	6.7	0.57	21	0.5	5%	253
1243	7.53	0.39	21	0.34	5%	130
1249	6.7	0.57	21	0.5	5%	84
1108	6.7	0.57	21	0.5	20%	1339
1196	6.7	0.57	21	0.5	20%	229
1238	6.7	0.57	21	0.5	20%	109

***n*-Dodecane**

Literature Source of Data:

D. F. Davidson, Z. Hong, G. L. Pilla, A. Farooq, R. D. Cook, R. K. Hanson, "*Multi-Species Time-History Measurements During n-Dodecane Oxidation Behind Reflected Shock Waves*," Proceedings of the Combustion Institute 33 (2011) 151-157.

Range of Data:

Temperature [K]	1330	1652
Pressure [atm]	2.07	2.47
Fuel Mole Fraction [ppm]	371	510
Oxygen Mole Fraction [%]	0.73	0.77
Equivalence Ratio	1.0	1.0

Type of Data:

n-Dodecane Table 4: Ignition delay time measurements in argon based on the time to half peak concentrations measured by laser absorption.

n-Dodecane Table 4:

T_5 [K]	P_5 [atm]	Fuel [ppm]	O ₂ [ppm]	ϕ (E.R.)	Species	T(50%) [μ s]
1330	2.25	445	7500	1	C ₂ H ₄	1592
1403	2.41	445	7500	1	C ₂ H ₄	586
1406	2.37	470	7500	1	C ₂ H ₄	571
1430	2.33	445	7500	1	C ₂ H ₄	349
1477	2.25	500	7500	1	C ₂ H ₄	216
1528	2.24	470	7500	1	C ₂ H ₄	116
1585	2.23	400	7500	1	C ₂ H ₄	58
1637	2.26	430	7500	1	C ₂ H ₄	36
1652	2.23	371	7500	1	C ₂ H ₄	27
1536	2.17	440	7500	1	H ₂ O	140
1657	2.12	485	7500	1	H ₂ O	56
1418	2.35	470	7500	1	H ₂ O	614
1607	2.17	460	7500	1	H ₂ O	84
1583	2.14	450	7500	1	H ₂ O	90
1612	2.09	470	7500	1	H ₂ O	80
1450	2.26	475	7500	1	H ₂ O	461
1496	2.22	480	7500	1	H ₂ O	242
1381	2.32	485	7500	1	H ₂ O	1020
1553	2.25	500	7776	1	OH	276
1407	2.28	460	7617	1	OH	791
1387	2.34	500	7489	1	OH	1197
1466	2.26	476	7525	1	OH	438
1427	2.07	510	7572	1	OH	655
1545	2.07	505	7427	1	OH	262
1645	2.13	428	7452	1	OH	91
1618	2.14	480	7635	1	OH	96
1394	2.30	430	7310	1	OH	1463
1415	2.47	456	7657	1	OH	790

***n*-Hexadecane**

Literature Source of Data:

D. R. Haylett, R. D. Cook, D. F. Davidson, R. K. Hanson, "*OH and C₂H₄ species time-histories during hexadecane and diesel ignition behind reflected shock waves*," Proceedings of the Combustion Institute 33 (2011) 167-173. (Note that the supplementary material for this paper has the wrong citation listed at the beginning.)

D. R. Haylett, D. F. Davidson, R. K. Hanson, "*Ignition Delay Times of Low-Vapor-Pressure Fuels Measured Using an Aerosol Shock Tube*," Combustion and Flame 159 (2012) 552-561.

D. R. Haylett, D. F. Davidson, R. D. Cook, Z. Hong, W. Ren, S. H. Pyun, R. K. Hanson, "*Multi-species time-histories measurements during hexadecane oxidation behind reflected shock waves*," Proceedings of the Combustion Institute 34 (2013) 369-376.

D. R. Haylett, "*The Development and Application of Aerosol Shock Tube Methods for the Study of Low-Vapor-Pressure Fuels*," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, March 2011.
http://hanson.stanford.edu/dissertations/Haylett_2011.pdf

Range of Data:

Temperature [K]	1159	1355
Pressure [atm]	1.7	6.8
Fuel Mole Fraction [%]	0.09	0.18
Oxygen Mole Fraction [%]	1.0	4.0
Equivalence Ratio	0.56	1.22

Type of Data:

n-Hexadecane Table 1: Ignition delay time measurement in argon from **Aerosol Shock Tube experiments** using sidewall PZT pressure and CH* emission near 430 nm. Selected ignition delay times are derived from OH species time-history measurements using laser absorption at 306.7 nm.

n-Hexadecane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [ms]
1267	6.54	0.0497	1	1.22	1.902
1172	4.44	0.0312	1	0.76	4.577
1170	4.6	0.0326	1	0.80	5.022
1333	6.77	0.0493	1	1.21	0.917
1327	6.46	0.1832	4	1.12	0.36
1355	4.19	0.1023	4	0.63	0.212
1183	2.05	0.1207	4	0.74	4.264
1159	2.06	0.092	4	0.56	3.87
1181	2.13	0.1776	4	1.09	5.536
1266	1.94	0.1168	4	0.72	1.122
1212	1.91	0.0909	4	0.56	2.27
1247	1.83	0.1485	4	0.91	2.06
1285	1.89	0.1582	4	0.97	1.386
1353	1.71	0.1565	4	0.96	0.708
1165	2.08	0.129	4	0.79	4.548
1177	2.03	0.1571	4	0.96	4.778
1218	1.96	0.1165	4	0.71	2.657
1231	1.86	0.1254	4	0.77	2.137

Branched Alkanes

2,4-Dimethyl Pentane

Literature Source of Data:

S. Li, A. Campos, D. F. Davidson, R. K. Hanson, “*Shock Tube Measurements of Branched Alkane Ignition Delay Times*,” Fuel 118 (2014) 398-405.

Range of Data:

Temperature [K]	1367	1506
Pressure [atm]	1.5	3.2
Fuel Mole Fraction [%]	0.18	0.36
Oxygen Mole Fraction [%]	4.0	4.0
Equivalence Ratio	0.5	1.0

Type of Data:

2,4-Dimethyl Pentane Table 1: Ignition delay time measurement in argon using the onset of sidewall OH* emission near 306 nm.

2,4-Dimethyl Pentane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1391	1.52	0.364	4	1	1251
1407	1.55	0.364	4	1	1189
1439	1.57	0.364	4	1	812
1445	1.54	0.364	4	1	764
1446	1.5	0.364	4	1	763
1470	1.5	0.364	4	1	536
1479	1.59	0.364	4	1	509
1506	1.51	0.364	4	1	371
1514	1.49	0.364	4	1	320
1324	1.61	0.182	4	0.5	1357
1349	1.41	0.182	4	0.5	1294
1352	1.59	0.182	4	0.5	999
1380	1.58	0.182	4	0.5	779
1419	1.57	0.182	4	0.5	507
1420	1.58	0.182	4	0.5	506
1439	1.57	0.182	4	0.5	400
1449	1.55	0.182	4	0.5	336
1470	1.54	0.182	4	0.5	256
1488	1.53	0.182	4	0.5	205
1367	3.22	0.364	4	1	1158
1374	3.13	0.364	4	1	1098
1404	3.08	0.364	4	1	770
1413	2.88	0.364	4	1	655
1415	3.01	0.364	4	1	683
1433	3	0.364	4	1	556
1455	2.9	0.364	4	1	466
1466	2.95	0.364	4	1	388

2,5-Dimethyl Hexane

Literature Source of Data:

S. Li, A. Campos, D. F. Davidson, R. K. Hanson, “*Shock Tube Measurements of Branched Alkane Ignition Delay Times*,” Fuel 118 (2014) 398-405.

Range of Data:

Temperature [K]	1313	1517
Pressure [atm]	1.4	3.1
Fuel Mole Fraction [%]	0.16	0.32
Oxygen Mole Fraction [%]	4.0	4.0
Equivalence Ratio	0.5	1.0

Type of Data:

2,4-Dimethyl Hexane Table 1: Ignition delay time measurement in argon using the onset of sidewall OH* emission near 306 nm.

2,4-Dimethyl Hexane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1360	1.47	0.32	4	1	1501
1391	1.57	0.32	4	1	1046
1405	1.47	0.32	4	1	942
1428	1.56	0.32	4	1	747
1437	1.51	0.32	4	1	733
1442	1.45	0.32	4	1	632
1450	1.43	0.32	4	1	580
1455	1.41	0.32	4	1	568
1497	1.48	0.32	4	1	341
1517	1.49	0.32	4	1	280
1313	1.43	0.18	4	0.5	1589
1328	1.41	0.18	4	0.5	1211
1341	1.45	0.18	4	0.5	1092
1354	1.52	0.18	4	0.5	824
1379	1.54	0.18	4	0.5	630
1384	1.52	0.18	4	0.5	592
1387	1.53	0.18	4	0.5	540
1395	1.54	0.18	4	0.5	509
1403	1.53	0.18	4	0.5	456
1409	1.52	0.18	4	0.5	452
1419	1.5	0.18	4	0.5	382
1420	1.52	0.18	4	0.5	354
1333	3.03	0.32	4	1	1234
1350	3.09	0.32	4	1	1057
1370	3.14	0.32	4	1	933
1371	2.98	0.32	4	1	851
1402	2.93	0.32	4	1	613
1409	2.92	0.32	4	1	608
1410	3.11	0.32	4	1	629
1433	2.94	0.32	4	1	485
1451	3.02	0.32	4	1	416

Iso-Octane

Literature Source of Data:

D. F. Davidson, M. A. Oehlschlaeger, R. K. Hanson, "*Methyl Concentration Time Histories during iso-Octane and n-Heptane Oxidation*," Paper 05F-61, WSS/CI Fall Meeting, Stanford CA, October 17-18, 2005.

D. F. Davidson, M. A. Oehlschlaeger, R. K. Hanson, "*Methyl Concentration Time Histories during iso-Octane and n-Heptane Oxidation and Pyrolysis*," Proceedings of the Combustion Institute 31 (2007) 321-328.

Range of Data:

Temperature [K]	1300	1559
Pressure [atm]	1.55	1.74
Fuel Mole Fraction [%]	0.05	0.05
Oxygen Mole Fraction [%]	0.625	0.625
Equivalence Ratio	1.0	1.0

Type of Data:

Iso-Octane Table 1: Ignition delay time measurement in argon based on the time of full consumption of CH₃ radicals measured by laser absorption at 216 nm.

Iso-Octane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1300	1.74	0.05	0.625	1	1910
1470	1.61	0.05	0.625	1	1504
1559	1.55	0.05	0.625	1	667

Iso-Octane

Literature Source of Data:

S. S. Vasu, D. F. Davidson, R. K. Hanson, “*Shock Tube Measurements and Modeling of Ignition Delay Times in Lean Iso-Octane/Air*,” 25th International Symposium on Shock Waves-ISSW25, July 18, 2005, Bangalore, India, published Orient Blackswan (January 25, 2006).

Range of Data:

Temperature [K]	1021	1302
Pressure [atm]	21	51
Fuel Mole Fraction [%]	0.418	0.418
Oxygen Mole Fraction [%]	20.9	20.9
Equivalence Ratio	0.25	0.25

Type of Data:

Iso-Octane Table 2: Ignition delay time measurement in air based on sidewall PZT pressure measurements and confirmed with OH* (at 306nm) emission measurements. Table includes data values revised after submission of paper.

Iso-Octane Table 2:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1021	45.1	0.418	20.9	0.25	1545
1064	43.5	0.418	20.9	0.25	937
1117	51.1	0.418	20.9	0.25	437
1117	43.8	0.418	20.9	0.25	510
1126	47.2	0.418	20.9	0.25	436
1144	45.5	0.418	20.9	0.25	294
1144	45.1	0.418	20.9	0.25	233
1152	45.8	0.418	20.9	0.25	267
1156	44.7	0.418	20.9	0.25	280
1165	47.4	0.418	20.9	0.25	199
1175	49.0	0.418	20.9	0.25	169
1176	46.7	0.418	20.9	0.25	186
1194	48.3	0.418	20.9	0.25	132
1206	47.1	0.418	20.9	0.25	116
1207	43.8	0.418	20.9	0.25	117
1232	43.7	0.418	20.9	0.25	77
1105	26.5	0.418	20.9	0.25	705
1185	21.8	0.418	20.9	0.25	306
1187	24.5	0.418	20.9	0.25	293
1211	23.7	0.418	20.9	0.25	198
1302	23.5	0.418	20.9	0.25	64

Iso-Octane

Literature Source of Data:

S. Li, A. Campos, D. F. Davidson, R. K. Hanson, “*Shock Tube Measurements of Branched Alkane Ignition Delay Times*,” Fuel 118 (2014) 398-405.

Range of Data:

Temperature [K]	1356	1554
Pressure [atm]	1.4	3.1
Fuel Mole Fraction [%]	0.16	0.32
Oxygen Mole Fraction [%]	4.0	4.0
Equivalence Ratio	0.5	1.0

Type of Data:

Iso-Octane Table 3: Ignition delay time measurement in argon using the onset of sidewall OH* emission near 306 nm.

Iso-Octane Table 3:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1356	1.5	0.32	4	1	2031
1370	1.49	0.32	4	1	1695
1397	1.48	0.32	4	1	1323
1427	1.48	0.32	4	1	978
1461	1.45	0.32	4	1	706
1479	1.59	0.32	4	1	509
1494	1.42	0.32	4	1	490
1501	1.34	0.32	4	1	510
1537	1.35	0.32	4	1	305
1554	1.37	0.32	4	1	263
1336	1.39	0.16	4	0.5	1690
1369	1.49	0.16	4	0.5	1041
1384	1.55	0.16	4	0.5	935
1389	1.56	0.16	4	0.5	852
1410	1.49	0.16	4	0.5	684
1414	1.49	0.16	4	0.5	623
1442	1.5	0.16	4	0.5	436
1456	1.49	0.16	4	0.5	383
1474	1.48	0.16	4	0.5	298
1499	1.48	0.16	4	0.5	222
1502	1.44	0.16	4	0.5	208
1367	3.05	0.32	4	1	1223
1387	3.04	0.32	4	1	1096
1428	3	0.32	4	1	709
1453	2.96	0.32	4	1	546
1461	2.9	0.32	4	1	487
1472	2.86	0.32	4	1	431
1486	2.81	0.32	4	1	358
1500	2.86	0.32	4	1	331

iso-Octane

Literature Source of Data:

D. F. Davidson, S. Li, K.-Y. Lam, J. T. Harmon, R. K. Hanson, "*Shock Tube/Laser Absorption Measurements of JP-8 Ignition Delay Times and Multi-Species Time-Histories*," Paper 2128, JANNAF Meeting, December 2011, Huntsville AL.

P. Gokulakrishnan, C. Fuller, M. Klassen, D. F. Davidson, R. K. Hanson, B. Kiel, "*Experimental and Modeling of JP-8, F-T and HRJ Fuel Ignition under Vitiated Conditions*," 49th AIAA Joint Propulsion Conference, July 15-17, 2013, San Jose, CA.

K.-Y. Lam, "*Shock Tube Measurements of Oxygenated Fuel Combustion using Laser Absorption Spectroscopy*," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, June 2013.

Range of Data:

Temperature [K]	1474	1610
Pressure [atm]	1.58	1.59
Fuel Mole Fraction [ppm]	511	511
Oxygen Mole Fraction [%]	0.813	0.813
Equivalence Ratio	0.79	0.79

Type of Data:

Iso-Octane Table 4: Ignition delay time measurement in argon using an ignition delay time definition based on the time to achieve 50% of the peak OH or CO concentration or the removal of 50% of the plateau/peak C₂H₄ concentration.

Iso-Octane Table 4:

T ₅ [K]	P ₅ [atm]	Fuel [ppm]	O ₂ [%]	φ E.R.	OH T(50) [μs]	CO T(50) [μs]
1474	1.58	511	0.813	0.79	968	844
1610	1.59	511	0.813	0.79	244	182

Cyclic Fuels

Cyclohexane

Literature Source of Data:

S. S. Vasu, K. -Y. Lam, D. F. Davidson, R. K. Hanson, "A Comparative Study of the Oxidation Characteristics of Cyclohexane, Methylcyclohexane, and *n*-Butylcyclohexane at high temperatures," Combustion and Flame 158 (2011) 1456-1468.

K.-Y. Lam, "Shock Tube Measurements of Oxygenated Fuel Combustion using Laser Absorption Spectroscopy," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, June 2013.

S. S. Vasu, "Measurements of Ignition Times, OH Time-Histories, and Reaction Rates in Jet Fuel and Surrogate Oxidation System," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, August 2010.

http://hanson.stanford.edu/dissertations/Vasu_2010.pdf

Range of Data:

Temperature [K]	1359	1543
Pressure [atm]	2.2	2.2
Fuel Mole Fraction [ppm]	420	467
Oxygen Mole Fraction [ppm]	4200	4200
Equivalence Ratio	0.9	1.0

Type of Data:

Cyclohexane Table 1: Ignition delay time measurement in argon using H₂O and OH laser absorption at 2.55 microns and 306.7 nm respectively. Ignition delay time defined as time to 50% peak H₂O or OH mole fraction.

Cyclohexane Table 1:

T _s [K]	P _s [atm]	Fuel [ppm]	O ₂ [ppm]	φ E.R.	Species	T(50%) [μs]
1359	2.2	467	4200	1.0	H ₂ O	2400
1359	2.2	467	4200	1.0	OH	2631
1441	2.2	467	4200	1.0	H ₂ O	945
1441	2.2	467	4200	1.0	OH	1027
1543	2.2	420	4200	0.9	H ₂ O	338
1543	2.2	420	4200	0.9	OH	370

Butylcyclohexane

Literature Source of Data:

S. S. Vasu, K. -Y. Lam, D. F. Davidson, R. K. Hanson, "A Comparative Study of the Oxidation Characteristics of Cyclohexane, Methylcyclohexane, and *n*-Butylcyclohexane at high temperatures," Combustion and Flame 158 (2011) 1456-1468.

K.-Y. Lam, "Shock Tube Measurements of Oxygenated Fuel Combustion using Laser Absorption Spectroscopy," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, June 2013.

S. S. Vasu, "Measurements of Ignition Times, OH Time-Histories, and Reaction Rates in Jet Fuel and Surrogate Oxidation System," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, August 2010.

http://hanson.stanford.edu/dissertations/Vasu_2010.pdf

Range of Data:

Temperature [K]	1346	1534
Pressure [atm]	2.1	2.2
Fuel Mole Fraction [ppm]	280	310
Oxygen Mole Fraction [ppm]	4200	4200
Equivalence Ratio	0.9	1.0

Type of Data:

Butylcyclohexane Table 1: Ignition delay time measurement in argon using H₂O and OH laser absorption at 2.55 microns and 306.7 nm respectively. Ignition delay time defined as time to 50% peak H₂O or OH mole fraction.

Butylcyclohexane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [ppm]	O ₂ [ppm]	φ E.R.	Species	T(50%) [μs]
1346	2.2	280	4200	1.0	H ₂ O	2973
1346	2.2	280	4200	1.0	OH	3195
1450	2.2	310	4200	1.1	H ₂ O	1074
1450	2.2	310	4200	1.1	OH	1250
1534	2.1	280	4200	1.0	H ₂ O	381
1534	2.1	280	4200	1.0	OH	486

Methylcyclohexane

Literature Source of Data:

S. S. Vasu, N. N. Parikh, D. F. Davidson, R. K. Hanson, “*Methylcyclohexane Oxidation: Shock Tube Experiments and Modeling over a Wide Range of Pressures and Temperatures*,” Paper D17 5th U.S. Combustion Meeting, University of California at San Diego, March 25-27, 2007.

S. S. Vasu, D. F. Davidson, R. K. Hanson, “*Shock Tube Study of Methylcyclohexane Ignition over a Wide Range of Pressures and Temperatures*,” *Energy and Fuels* 23 (2009) 175-185.

S. S. Vasu, “*Measurements of Ignition Times, OH Time-Histories, and Reaction Rates in Jet Fuel and Surrogate Oxidation System*,” Ph.D. Thesis, Mechanical Engineering Department, Stanford University, August 2010.

http://hanson.stanford.edu/dissertations/Vasu_2010.pdf

Range of Data:

Temperature [K]	795	1098
Pressure [atm]	17.3	49.2
Fuel Mole Fraction [%]	1.96	1.96
Oxygen Mole Fraction [%]	20.6	20.6
Equivalence Ratio	1	1

Type of Data:

Methylcyclohexane Table 1: Ignition delay time measurement in argon and air using sidewall PZT pressure (using the time corresponding to the intersection of the maximum slope extrapolated to the baseline pressure) and OH* emission.

Methylcyclohexane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
795	48.1	1.96	20.60	1	1620
827	49.1	1.96	20.60	1	1384
837	48.8	1.96	20.60	1	1337
854	49.2	1.96	20.60	1	1393
857	48.6	1.96	20.60	1	1509
876	47.8	1.96	20.60	1	1484
893	49.2	1.96	20.60	1	1238
918	46.4	1.96	20.60	1	1185
922	45.4	1.96	20.60	1	1084
962	44.8	1.96	20.60	1	664
968	41.6	1.96	20.60	1	722
1019	44.6	1.96	20.60	1	345
1029	42.7	1.96	20.60	1	318
1046	42.2	1.96	20.60	1	223
1057	38.2	1.96	20.60	1	226
1081	40.8	1.96	20.60	1	133
1085	38.2	1.96	20.60	1	136
912	20.5	1.96	20.60	1	1995
926	19.8	1.96	20.60	1	1940
937	18.8	1.96	20.60	1	1485
958	19.5	1.96	20.60	1	1226
994	19.2	1.96	20.60	1	842
1002	17.5	1.96	20.60	1	637
1049	17.2	1.96	20.60	1	411
1060	22.0	1.96	20.60	1	358
1098	17.3	1.96	20.60	1	273
948	44.5	1.96	20.60	1	670
953	39.0	1.96	20.60	1	696
988	43.7	1.96	20.60	1	440
992	43.5	1.96	20.60	1	418
1007	44.8	1.96	20.60	1	337

Methylcyclohexane

Literature Source of Data:

S. S. Vasu, D. F. Davidson, R. K. Hanson, "*OH Time-History Absorption Measurements at High Pressures and Temperatures behind Reflected Shock Waves during Methylcyclohexane Oxidation*," Paper 08S-13 Western States Section/Combustion Institute Spring Meeting, University of Southern California, Los Angeles, March 17-18, 2008.

S. S. Vasu, D. F. Davidson, R. K. Hanson, "*OH Time-Histories during Oxidation of n-Heptane and Methylcyclohexane at High Pressures and Temperatures*," Combustion and Flame 156 (2009) 736-749.

S. S. Vasu, "*Measurements of Ignition Times, OH Time-Histories, and Reaction Rates in Jet Fuel and Surrogate Oxidation System*," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, August 2010.

http://hanson.stanford.edu/dissertations/Vasu_2010.pdf

Range of Data:

Temperature [K]	1205	1332
Pressure [atm]	14.44	15.83
Fuel Mole Fraction [%]	0.075	0.1
Oxygen Mole Fraction [%]	1.575	2.1
Equivalence Ratio	0.5	0.5

Type of Data:

Methylcyclohexane Table 2: Ignition delay time measurement in argon using OH laser absorption at 306.7 nm. Ignition delay time defined as time to 50% peak OH mole fraction.

Methylcyclohexane Table 2:

T_5 [K]	P_5 [atm]	Fuel [%]	O_2 [%]	ϕ E.R.	Ign. Time [μ s]
1332	15.64	0.1	2.1	0.5	301
1285	15.16	0.1	2.1	0.5	529
1271	15.32	0.1	2.1	0.5	643
1269	15.80	0.1	2.1	0.5	643
1265	15.22	0.1	2.1	0.5	718
1262	15.45	0.1	2.1	0.5	722
1213	14.44	0.1	2.1	0.5	1284
1205	14.47	0.1	2.1	0.5	1372
1304	15.92	0.075	1.575	0.5	500
1303	15.63	0.075	1.575	0.5	499
1276	15.77	0.075	1.575	0.5	690
1266	15.83	0.075	1.575	0.5	764
1283	15.43	0.075	1.575	0.5	617

Methylcyclohexane

Literature Source of Data:

S. S. Vasu, K. -Y. Lam, D. F. Davidson, R. K. Hanson, "A Comparative Study of the Oxidation Characteristics of Cyclohexane, Methylcyclohexane, and *n*-Butylcyclohexane at high temperatures," Combustion and Flame 158 (2011) 1456-1468.

K.-Y. Lam, "Shock Tube Measurements of Oxygenated Fuel Combustion using Laser Absorption Spectroscopy," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, June 2013.

S. S. Vasu, "Measurements of Ignition Times, OH Time-Histories, and Reaction Rates in Jet Fuel and Surrogate Oxidation System," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, August 2010.

http://hanson.stanford.edu/dissertations/Vasu_2010.pdf

Range of Data:

Temperature [K]	1359	1541
Pressure [atm]	2.1	2.2
Fuel Mole Fraction [ppm]	320	380
Oxygen Mole Fraction [ppm]	4200	4200
Equivalence Ratio	0.80	0.95

Type of Data:

Methylcyclohexane Table 3: Ignition delay time measurement in argon using H₂O and OH laser absorption at 2.55 microns and 306.7 nm respectively. Ignition delay time defined as time to 50% peak H₂O or OH mole fraction.

Methylcyclohexane Table 3:

T ₅ [K]	P ₅ [atm]	Fuel [ppm]	O ₂ [ppm]	φ E.R.	Species	T(50%) [μs]
1359	2.2	380	4200	0.95	H ₂ O	3889
1359	2.2	380	4200	0.95	OH	4162
1435	2.2	340	4200	0.85	H ₂ O	1238
1435	2.2	340	4200	0.85	OH	1330
1541	2.1	320	4200	0.80	H ₂ O	344
1541	2.1	320	4200	0.80	OH	391

Toluene

Literature Source of Data:

S. S. Vasu, D. F. Davidson, R. K. Hanson, “*Some Aspects of Toluene Ignition: Experiments and Modeling*,” Paper 12F1 Proceedings of the 6th U.S. National Combustion Meeting, 2009.

S. S. Vasu, D. F. Davidson, R. K. Hanson, “*Shock-Tube Experiments and Kinetic Modeling of Toluene Ignition*,” Journal of Propulsion and Power 26 (2010) 776-783.

Range of Data:

Temperature [K]	965	1211
Pressure [atm]	41.5	54.2
Fuel Mole Fraction [%]	2.28	2.28
Oxygen Mole Fraction [%]	20.5	20.5
Equivalence Ratio	1	1

Type of Data:

Toluene Table 1: Ignition delay time measurement in air using sidewall PZT pressure (using the time corresponding to the intersection of the maximum slope extrapolated to the baseline pressure). Various cleaning methods are employed to understand the influence of shock tube preparation on ignition delay time.

* All mixtures are Toluene in air, except for the last two points in the table (T= 1048 and 1018K) that have a mixture of 2.26% toluene with 0.163% n-heptane in air.

Toluene Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1052	49.4	2.26	20.5	1.0	530
1010	44.4	2.26	20.5	1.0	906
1103	51.5	2.26	20.5	1.0	498
1010	43.5	2.26	20.5	1.0	921
1044	47.1	2.26	20.5	1.0	790
1055	47.6	2.26	20.5	1.0	750
1096	51.4	2.26	20.5	1.0	514
1101	51.3	2.26	20.5	1.0	513
1059	48.3	2.26	20.5	1.0	686
1043	48.8	2.26	20.5	1.0	921
1106	51.9	2.26	20.5	1.0	481
1084	50.4	2.26	20.5	1.0	530
1091	50.9	2.26	20.5	1.0	636
1062	48.1	2.26	20.5	1.0	780
1059	47.2	2.26	20.5	1.0	729
1057	49.1	2.26	20.5	1.0	742
1051	49.7	2.26	20.5	1.0	774
975	45.5	2.26	20.5	1.0	1262
1006	48.8	2.26	20.5	1.0	940
966	46.2	2.26	20.5	1.0	1311
1066	48.1	2.26	20.5	1.0	807
1050	47.1	2.26	20.5	1.0	853
1082	48.8	2.26	20.5	1.0	600
1049	45.8	2.26	20.5	1.0	754
965	47.4	2.26	20.5	1.0	1192
1065	47.7	2.26	20.5	1.0	711
1060	47.3	2.26	20.5	1.0	613
1112	52.2	2.26	20.5	1.0	435
1075	48.6	2.26	20.5	1.0	588
999	41.5	2.26	20.5	1.0	1366
1055	47.6	2.26	20.5	1.0	884
1114	49.7	2.26	20.5	1.0	519
1041	45.9	2.26	20.5	1.0	889
1048	46.6	2.26	20.5	1.0	917
1073	48.0	2.26	20.5	1.0	832
1056	46.1	2.26	20.5	1.0	886
1096	49.9	2.26	20.5	1.0	597
1087	49.3	2.26	20.5	1.0	585
1211	52.0	2.26	20.5	1.0	111
1116	54.2	2.26	20.5	1.0	349
1054	53.5	2.26	20.5	1.0	744
1048	53.4	2.26*	20.5	1.0	705
1018	53.4	2.26*	20.5	1.0	840

Toluene

Literature Source of Data:

D. F. Davidson, S. Li, K.-Y. Lam, J. T. Harmon, R. K. Hanson, "Shock Tube/Laser Absorption Measurements of JP-8 Ignition Delay Times and Multi-Species Time-Histories," Paper 2128, JANNAF Meeting, December 2011, Huntsville AL.

K.-Y. Lam, "Shock Tube Measurements of Oxygenated Fuel Combustion using Laser Absorption Spectroscopy," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, June 2013.

P. Gokulakrishnan, C. Fuller, M. Klassen, D. F. Davidson, R. K. Hanson, B. Kiel, "Experimental and Modeling of JP-8, F-T and HRJ Fuel Ignition under Vitiated Conditions," 49th AIAA Joint Propulsion Conference, July 15-17, 2013, San Jose, CA.

Range of Data:

Temperature [K]	1524	1602
Pressure [atm]	1.5	1.5
Fuel Mole Fraction [ppm]	640	640
Oxygen Mole Fraction [%]	0.813	0.813
Equivalence Ratio	0.7	0.7

Type of Data:

Toluene Table 2: Ignition delay time measurement in argon using an ignition delay time definition based on the time to achieve 50% of the peak OH or CO concentration or the removal of 50% of the plateau/peak C₂H₄ concentration.

Toluene Table 2:

T ₅ [K]	P ₅ [atm]	Fuel [ppm]	O ₂ [%]	φ E.R.	OH T(50) [μs]	CO T(50) [μs]
1524	1.52	640	0.813	0.71	1310	1057
1602	1.52	640	0.813	0.71	503	379

Decalin

Literature Source of Data:

Y. Zhu, D. F. Davidson, R. K. Hanson, “*Pyrolysis and Oxidation of Decalin at Elevated Pressures*,” Combustion and Flame 161 (2014) 371-383.

Range of Data:

Temperature [K]	769	1202
Pressure [atm]	11.7	56.6
Fuel Mole Fraction [%]	0.72	2.26
Oxygen Mole Fraction [%]	20.5	20.9
Equivalence Ratio	0.41	1.60

Type of Data:

Decalin Table 1: Ignition delay time measurement in air using sidewall OH* emission near 306 nm and sidewall PZT pressure.

Decalin Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1202	16.6	1.33	20.73	0.93	109
1193	18.8	1.33	20.73	0.93	106
1145	20.5	1.51	20.69	1.06	157
1100	18.9	1.30	20.74	0.91	284
1090	19.9	1.57	20.68	1.10	261
1068	19.2	1.36	20.72	0.95	381
1059	20.5	1.44	20.71	1.01	396
1051	19.3	1.29	20.74	0.90	440
1014	26.8	1.46	20.70	1.02	553
1012	23.3	1.29	20.74	0.90	788
990	22.1	1.62	20.67	1.14	917
951	24.1	1.43	20.71	1.00	1271
914	19.5	1.37	20.72	0.96	2644
876	20.2	1.43	20.71	1.00	3291
830	24.0	1.40	20.71	0.98	3049
802	19.3	1.20	20.76	0.84	5341
769	15.5	1.24	20.75	0.87	8746
1153	18.5	0.75	20.85	0.52	228
1147	19.3	0.73	20.85	0.51	235
1121	20.5	0.60	20.88	0.42	318
1091	20.6	0.68	20.87	0.47	406
1074	21.2	0.60	20.88	0.42	533
1072	19.4	0.59	20.88	0.41	603

1062	20.3	0.72	20.86	0.50	604
1046	21.9	0.70	20.86	0.49	701
995	20.8	0.66	20.87	0.46	1531
992	20.0	0.66	20.87	0.46	1553
905	25.5	0.65	20.87	0.45	4324
864	19.9	0.66	20.87	0.46	5964
849	22	0.62	20.88	0.43	6114
818	16.7	0.70	20.86	0.49	10380
1071	19.5	2.09	20.57	1.47	295
1014	27.5	2.00	20.59	1.41	435
1013	26.5	2.27	20.53	1.60	412
959	24.5	2.02	20.58	1.42	849
922	24.7	1.89	20.61	1.33	1657
1043	46.0	1.23	20.75	0.86	264
1011	50.4	1.16	20.76	0.81	413
1006	46.4	1.20	20.76	0.84	443
962	47.9	1.19	20.76	0.83	891
961	50.1	1.23	20.75	0.86	804
941	50.6	1.24	20.75	0.87	1024
912	48.9	1.32	20.73	0.92	903
883	56.6	1.43	20.71	1.00	638
852	37.8	1.40	20.71	0.98	1984
831	48.3	0.99	20.80	0.69	3411
1007	49.4	0.68	20.87	0.47	685
978	48.4	0.73	20.85	0.51	885
974	51.2	0.69	20.86	0.48	895
964	48.8	0.76	20.85	0.53	1081
958	50.8	0.73	20.85	0.51	1143
930	49.8	0.72	20.86	0.5	1662
1141	11.7	1.34	20.73	0.94	268
1061	12.8	1.43	20.71	1.00	591
1025	12.8	1.37	20.72	0.96	1011
988	13.1	1.29	20.74	0.9	1509

Distillate Fuels

Jet Fuel (JP-7)

Literature Source of Data:

D. F. Davidson, D. R. Haylett, R. K. Hanson, "Development of an Aerosol Shock Tube for Kinetic Studies of Low-Vapor-Pressure Fuels," Combustion and Flame 155 (2008) 108-117.

Range of Data:

Temperature [K]	1066	1351
Pressure [atm]	6.0	7.8
Fuel Mole Fraction [%]	0.35	0.98
Oxygen Mole Fraction [%]	21	21
Equivalence Ratio	0.30	0.82

Type of Data:

Jet Fuel JP-7 Table 1: Ignition delay time measurement from **Aerosol Shock Tube experiments** using sidewall PZT pressure and CH* emission near 431 nm. The JP-7 (POSF 3327) was provided by T. Edwards at Wright-Patterson AFRL.

The molecular formula for JP-7 is given by J. L. Ross as C_{12.3}H_{25.5} with a molecular weight of 173.4. The molecular weight of JP-7 is given by J. L. Convery, as 181 gm/mole. In the current paper a molecular formula of C₁₂H₂₄ was used for the equivalence ratio calculations.

J. L. Ross, A Fuel Data Standardization Study for JP-4, JP-5, JP-7 and RJ-5 Combustion in Air, AD-783 308 March 1974 AF Aero Propulsion Laboratory WPAFB, Ohio
<http://oai.dtic.mil/oai/oai?verb=getRecord&metadataPrefix=html&identifier=AD0783308>

J. L. Convery, M. Sc. Thesis, Virginia State University (2005)
http://scholar.lib.vt.edu/theses/available/etd-11212005-135852/unrestricted/JLC_thesis_FINAL_2.pdf).

Jet Fuel (JP-7) Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1248	6.05	0.469	21	0.40	278
1237	5.98	0.551	21	0.47	306
1236	6.12	0.469	21	0.40	413
1066	6.03	0.484	21	0.41	2791
1068	6.31	0.531	21	0.45	2043
1095	6.14	0.513	21	0.43	1689
1070	6.54	0.835	21	0.71	1162
1154	6.08	0.688	21	0.58	585
1167	6.55	0.833	21	0.70	739
1239	6.62	0.713	21	0.60	258
1080	6.94	0.713	21	0.60	1557
1037	7.78	0.975	21	0.82	2068
1058	7.26	0.800	21	0.68	2154
1108	7.46	0.746	21	0.63	1687
1092	7.28	0.805	21	0.68	1448
1121	7.02	0.690	21	0.58	1368
1137	7.12	0.729	21	0.62	1174
1159	6.96	0.668	21	0.56	931
1197	7.21	0.689	21	0.58	519
1204	6.94	0.657	21	0.56	532
1327	6.68	0.332	21	0.28	115
1327	6.58	0.353	21	0.30	139
1351	6.51	0.350	21	0.30	100
1344	6.46	0.426	21	0.36	54

Jet Fuel (JP-8)

Literature Source of Data:

D. F. Davidson, S. Li, K.-Y. Lam, J. T. Harmon, R. K. Hanson, "Shock Tube/Laser Absorption Measurements of JP-8 Ignition Delay Times and Multi-Species Time-Histories," Paper 2128, JANNAF Meeting, December 2011, Huntsville AL.

P. Gokulakrishnan, C. Fuller, M. Klassen, D. F. Davidson, R. K. Hanson, B. Kiel, "Experimental and Modeling of JP-8, F-T and HRJ Fuel Ignition under Vitiated Conditions," 49th AIAA Joint Propulsion Conference, July 15-17, 2013, San Jose, CA.

Range of Data:

Temperature [K]	919	1594
Pressure [atm]	1.2	35
Fuel Mole Fraction [%]	0.05	1.2
Oxygen Mole Fraction [%]	0.8	21
Equivalence Ratio	0.5	2.0

Type of Data:

Jet Fuel (JP-8) Table 2: Ignition delay time measurement in synthetic air from heated high pressure shock tube experiments using sidewall PZT pressure and OH* emission near 306 nm. The JP-8 used in this study had AF identifier POSF-6169, was composed of 84% saturates, had a molecular formula of $C_{11}H_{25}$, and a molecular weight of 181. Small amounts of nitric oxide were added to selected experiments. For certain experiments only normalized values are available.

Jet Fuel (JP-8) Table 3: These experiments are at lower pressures and use an ignition delay time definition based on the time to achieve 50% of the peak OH concentration or the removal of 50% of the plateau/peak C_2H_4 concentration. Small amounts of nitric oxide were added to selected experiments. The experiments in this table all use argon as the buffer gas except for the 21% O_2 experiment (1182 and 1144 K) that are in air.

Jet Fuel (JP-8) Table 2:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	NO ppm	Ign. Time [μs]
1133	15.72	1.203	21	1.0	500	187
1057	16.73	1.203	21	1.0	500	459
1097	16.29	1.203	21	1.0	500	270
1157	15.11	1.203	21	1.0	500	141
985	15.45	1.203	21	1.0	500	973
1009	15.10	1.203	21	1.0	500	767
973	15.11	1.203	21	1.0	500	1132
1174	18.14	1.203	21	1.0	0	97
1159	19.11	1.203	21	1.0	0	107
1145	19.15	1.203	21	1.0	0	146
1115	18.92	1.203	21	1.0	0	253
1084	19.30	1.203	21	1.0	0	413
1077	20.52	1.203	21	1.0	0	481
1053	20.74	1.203	21	1.0	0	766
1032	20.60	1.203	21	1.0	0	910
995	20.33	1.203	21	1.0	0	1427
990	21.17	1.203	21	1.0	0	1433
975	21.68	1.203	21	1.0	0	1935
919	18.02	1.203	21	1.0	0	5345
925	17.21	1.203	21	1.0	0	5148
929	16.38	1.203	21	1.0	0	4520
1205	17.82	1.203	21	0.5	0	138
1185	17.76	1.203	21	0.5	0	179
1170	18.05	1.203	21	0.5	0	222
1159	19.11	1.203	21	0.5	0	287
1131	18.15	1.203	21	0.5	0	437
1108	18.94	1.203	21	0.5	0	592
1099	19.07	1.203	21	0.5	0	653
1065	19.06	1.203	21	0.5	0	920
1046	19.89	1.203	21	0.5	0	1214
1008	22.66	1.203	21	2.0	0	646
1070	21.93	1.203	21	2.0	0	191
959	23.66	1.203	21	2.0	0	1202
1105	20.55	1.203	21	2.0	0	104
1096	33.73	1.203	21	1.0	0	233
1090	36.69	1.203	21	1.0	0	275
1059	38.00	1.203	21	1.0	0	414
1018	37.84	1.203	21	1.0	0	748
1141	34.21	1.203	21	1.0	0	107
1147	36.06	1.203	21	1.0	0	106
1039	38.10	1.203	21	1.0	0	560

Jet Fuel (JP-8) Table 3:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	NO ppm	OH T(50) [μs]	C ₂ H ₄ T(50) [μs]
1594	1.48	500	0.8125	1.0	0	200	66
1544	1.49	500	0.8125	1.0	0	534	315
1477	1.6	500	0.8125	1.0	0	1075	836
1421	1.52	500	0.8125	1.0	0	2071	1768
1241	1.75	500	0.8125	1.0	0		9237
1396	1.26	500	0.8125	1.0	40	1493	
1461	1.33	500	0.8125	1.0	40	856	441
1527	1.24	500	0.8125	1.0	40	252	
1418	1.50	500	0.8125	0.5	0	421	
1181	1.50	500	0.8125	0.5	0	8444	
1182	5.60	1.08	21	0.9	0		366
1144	6.30	1.08	21	0.9	0		478
1568	2.12	424	0.8125	0.9	0	232	179
1454	2.22	426	0.8125	0.9	0	825	723
1545	2.13	474	0.8125	0.9	40	356	310
1426	2.19	439	0.8125	0.9	40	1262	1174

Jet Fuel (JP-8 and Jet-A)

Literature Source of Data:

S. S. Vasu, D. F. Davidson, R. K. Hanson, "*Shock Tube Ignition Delay Times and Modeling of Jet Fuel Mixtures*," Paper AIAA-2006-4402, 42nd AIAA Joint Propulsion Conference, July 9-12, 2006, Sacramento, CA.

S. S. Vasu, D. F. Davidson, R. K. Hanson, "*Shock Tube Ignition Delay Times and Modeling of Jet Fuel Mixtures*," Paper AIAA-2007-5671, 43rd AIAA Joint Propulsion Conference, July 8-11, 2007, Cincinnati, OH.

S. S. Vasu, D. F. Davidson, R. K. Hanson, "*Jet Fuel Ignition Delay Times: Shock Tube Investigations at High Pressures*," 21st ICDERS Meeting, July 23-27, 2007, Poitiers, France.

S. S. Vasu, D. F. Davidson, R. K. Hanson, "*Jet Fuel Ignition Delay Times: Shock Tube Experiments over Wide Conditions and Surrogate Model Predictions*," Combustion and Flame 152 (2008) 125-143.

S. S. Vasu, "*Measurements of Ignition Times, OH Time-Histories, and Reaction Rates in Jet Fuel and Surrogate Oxidation System*," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, August 2010.
http://hanson.stanford.edu/dissertations/Vasu_2010.pdf

Range of Data:

Temperature [K]	715	1220
Pressure [atm]	17	50
Fuel Mole Fraction [%]	0.6	1.3
Oxygen Mole Fraction [%]	10	21
Equivalence Ratio	0.5	1.0

Type of Data:

Jet Fuel (JP-8 and Jet-A) Table 1: Ignition delay time measurement in synthetic air from heated high pressure shock tube experiments using sidewall PZT pressure and OH* emission near 306 nm. These studies assumed a molecular formula of C₁₁H₂₁ and a density of 0.81 gm/cm³ for all three jet fuels tested.

The three fuels tested are:

- 1) Zhukov et al. Jet-A is the same fuel as used by V. P. Zhukov, V. A. Sechenov, A. Y. Starikovskiy, “*Autoignition of Kerosene (Jet-A)/air Mixtures Behind Reflected Shock*,” Fuel 126 (2014) 169-176; and Dean, A. J.; Penyazkov, O. G.; Sevruck, K. L.; Varatharajan, B. “*Ignition of Aviation Kerosene at High Temperatures*”; 20th Int. Colloquium on the Dynamics of Explosions and Reactive Systems (ICDERS), 2005, Montreal, Canada.
- 2) Jet-A Composite Blend AF identifier was #04POSF4658 supplied by T. Edwards, WPAFB.
- 3) JP-8 fuel was supplied by P. Schihl, ARL and had a cetane number of 43.3, density of 0.8 kg/l, hydrogen content of 13.9 (wt. %) and aromatics composition of 13.9 (vol %).

Jet Fuel (JP-8 and Jet-A) Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [ms]
Jet A: 04POSF4658					
874	23.3	1.276	20.74	1	3109
880	25.1	1.276	20.74	1	2355
934	31.4	1.276	20.74	1	1230
952	24.0	1.276	20.74	1	1358
958	22.5	1.276	20.74	1	1287
963	21.7	1.276	20.74	1	1412
974	32.5	1.276	20.74	1	777
987	21.3	1.276	20.74	1	984
997	32.1	1.276	20.74	1	641
1022	36.1	1.276	20.74	1	400
1035	21.9	1.276	20.74	1	539
1042	22.0	1.276	20.74	1	486
1073	28.3	1.276	20.74	1	278
1130	24.8	1.276	20.74	1	138
1145	29.4	1.276	20.74	1	101
1192	27.0	1.276	20.74	1	52
1220	29.2	1.276	20.74	1	34
961	49.3	1.276	20.74	1	528
1009	50.9	1.276	20.74	1	286
1090	50.5	1.276	20.74	1	115
1124	47.8	1.276	20.74	1	80
1025	18.9	0.615	10	1	1820
1057	17.5	0.615	10	1	1340
1067	19.7	0.615	10	1	1109
1072	17.7	0.615	10	1	1137
1081	18.3	0.615	10	1	1005
1092	20.2	0.615	10	1	818
1111	17.7	0.615	10	1	732
1124	17.3	0.615	10	1	646
1127	17.7	0.615	10	1	569

715	21.7	1.276	20.74	1	3286
753	21.8	1.276	20.74	1	2584
763	21.5	1.276	20.74	1	2423
774	19.4	1.276	20.74	1	2566
785	21.1	1.276	20.74	1	2484
786	19.7	1.276	20.74	1	2742
808	20.8	1.276	20.74	1	2628

Jet A: 04POSF4658					
970	19.6	0.642	20.87	0.5	1961
1032	21.0	0.642	20.87	0.5	917
1039	19.9	0.642	20.87	0.5	873
1081	19.9	0.642	20.87	0.5	530
1081	22.2	0.642	20.87	0.5	463
1187	22.1	0.642	20.87	0.5	110
Jet-A: Zhukov et al.					
1000	26.4	1.276	20.74	1	973
1100	26.2	1.276	20.74	1	208
1137	25.5	1.276	20.74	1	123
1145	21.8	1.276	20.74	1	131
1172	26.1	1.276	20.74	1	72
1187	21.9	1.276	20.74	1	70
1194	22.2	1.276	20.74	1	64
1229	23.7	1.276	20.74	1	28
JP-8: Schihl ARL					
910	30.9	1.276	20.74	1	1532
949	20.4	1.276	20.74	1	1504
960	21.0	1.276	20.74	1	1305
1011	22.3	1.276	20.74	1	685
1017	30.7	1.276	20.74	1	475
1019	19.4	1.276	20.74	1	674
1019	22.4	1.276	20.74	1	616
1073	19.1	1.276	20.74	1	373
1104	19.3	1.276	20.74	1	259
1124	18.4	1.276	20.74	1	191
1128	18.9	1.276	20.74	1	190
1146	19.0	1.276	20.74	1	145

Jet Fuel (JP-8 and Alternative Fuels)

Literature Source of Data:

Y. Zhu, S. Li, D. F. Davidson, R. K. Hanson, *"Ignition Delay Times of Conventional and Alternative Fuels behind Reflected Shock Waves," accepted for publication, Proceedings of the Combustion Institute 35 (2015).*

Range of Data:

Temperature [K]		
Pressure [atm]		
Fuel Mole Fraction [%]		
Oxygen Mole Fraction [%]		
Equivalence Ratio		

Type of Data:

Jet Fuel (JP-8 and Alternative Fuels) Table 1: Ignition delay time measurement in synthetic air from heated high pressure shock tube experiments using sidewall PZT pressure and OH* emission near 306 nm.

Detailed properties of the individual fuels are given in the Table A.

Properties of Neat Fuels, Table A:

Name	Label	Molecular Weight ^{a,b} (MW, g/mol)	Nominal Formula	Derived Cetane Number ^{a,b} (DCN)	H/C Ratio ^{a,b}	Threshold Sooting Index ^a (TSI)	General Specifics ^{a,b} n-alkanes, iso-alkanes, cyclo-alkanes, aromatics
JP-8	POSF 6169	154	C _{10.9} H _{22.0}	47.3	2.017	19.3	n- > iso- > cyclo- > aro
SASOL IPK	POSF 7629	149	C _{10.5} H _{23.0}	31.1	2.195	17.3	iso- > cyclo- > n-
SHELL SPK	POSF 5729	137	C _{9.6} H _{21.4}	58.4	2.237	8.4	iso- > n- > cyclo-
HRJ-Tallow	POSF 6308	161	C _{11.3} H _{24.6}	58.1	2.176	≤11.6	iso- > n- > cyclo-
HRJ-Camelina	POSF 7720	165	C _{11.6} H _{25.5}	58.9	2.202	≤12.0	iso- > n- > cyclo-
GEVO ATJ	POSF10151	173	C _{12.2} H _{26.4}	15.5	2.168	22.48	N/A
Swedish BioJet	POSF10244	154	C _{11.0} H _{22.0}	N/A	2.005	N/A	N/A
F-76	N/A	205	C _{14.8} H _{26.9}	51	1.82	N/A	N/A
F-76 Blend	N/A	205	C _{14.6} H _{29.5}	59	2.021	N/A	50% F-76/50% Hydro-refined Algal Oil
DIM	N/A	182	C _{13.4} H _{21.4}	N/A	1.597	N/A	50% Terpene dimers (TD) /50% JP-10
103-2	N/A	185	C _{13.6} H _{21.4}	N/A	1.574	N/A	50% TD/50% Hydrogenated alpha pinene

Jet Fuel (JP-8 and Alternative Fuels) Table 1:

T_5 [K]	P_5 [atm]	Fuel [%]	O_2 [%]	ϕ E.R.	Ign. Time [μ s]
JP-8 POSF6169 in Air					
1246	2.46		21	0.95	496
1179	2.82		21	0.94	935
1341	2.66		21	0.87	177
1246	2.75		21	0.97	376
1153	2.45		21	1.00	1271
1182	2.16		21	0.86	1075
1163	2.07		21	0.97	1426
1248	2.80		21	0.48	297
1225	3.20		21	0.49	371
1169	3.16		21	0.62	786
1121	2.98		21	0.65	1381
1164	2.54		21	1.42	1304
1212	3.13		21	1.33	705
1196	5.66		21	0.91	338
1192	5.70		21	0.95	410
1151	6.00		21	1.03	661
1233	5.69		21	0.90	228
1156	6.27		21	0.87	496
1159	6.37		21	0.79	532
1103	6.08		21	0.95	1115

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
GEVO POSF10151 in Air					
1290	2.69		21	1.01	567
1232	3.08		21	1.00	778
1147	2.99		21	0.99	1417
1210	3.09		21	0.99	878
1331	2.52		21	0.96	368
1221	3.23		21	1.07	659
1363	6.64		21	0.98	109.5
1244	6.08		21	1.02	268
1364	6.81		21	1.05	106.2
1168	6.26		21	1.01	492
1168	7.39		21	1.07	393
1127	7.69		21	1.11	570
1070	8.27		21	1.15	1118
1217	7.34		21	1.22	265
1069	8.15		21	0.96	1227
1070	8.07		21	1.00	1164
1278	6.59		21	0.45	196
1173	7.37		21	0.50	451
1126	7.74		21	0.54	714
1053	8.19		21	0.57	1515
1070	8.20		21	2.19	1231

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
GEVO/JP-8 50/50 Blend in Air					
1142	2.98		21	0.93	1246
1134	2.46		21	0.91	1564
1244	2.57		21	0.95	575
1215	2.75		21	1.01	718
1314	2.54		21	0.88	323
1254	6.10		21	0.88	221
1232	7.04		21	1.01	214
1211	7.47		21	0.98	241
1135	7.11		21	0.89	605
1072	7.75		21	1.20	1087
1060	7.61		21	0.92	1474
1181	7.55		21	1.17	290
1287	6.54		21	0.50	138
1096	7.35		21	0.50	1085
1191	7.00		21	0.49	369
1086	7.81		21	1.82	946
1157	7.22		21	1.90	463

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
Swedish Biojet POSF10244 in Air					
1170	2.95		21	0.91	1263
1186	2.75		21	0.90	1128
1278	2.51		21	0.96	486
1269	2.91		21	0.96	435
1226	2.82		21	0.96	743
1296	2.30		21	1.09	451
1327	2.19		21	1.20	358
1212	6.10		21	1.03	363
1047	6.59		21	0.98	2347
1271	6.51		21	1.08	195
1101	7.16		21	1.14	1093
1140	6.98		21	1.12	788
1188	6.77		21	1.12	456
1144	6.66		21	0.42	689
1322	5.53		21	0.44	106
1520	4.90		21	1.93	30
1264	5.93		21	1.59	360
1221	6.46		21	1.77	440

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
Swedish Biojet/JP-8 50/50 Blend in Air					
1181	2.51		21	1.07	1124
1249	2.37		21	0.97	515
1298	2.28		21	0.94	376
1256	2.49		21	0.82	428
1160	2.60		21	1.19	1528
1141	2.57		21	0.89	1678
1121	7.34		21	0.91	830
1273	6.98		21	0.91	176
1258	7.31		21	0.94	188
1128	7.05		21	0.98	858
1189	7.08		21	0.96	398
1066	7.38		21	0.99	1663
1254	6.29		21	0.86	186
1268	5.71		21	0.74	217
1238	7.63		21	0.89	206
1117	7.40		21	0.78	921
1235	6.66		21	0.54	219
1155	7.02		21	0.52	560
1140	7.58		21	1.50	769
1194	7.40		21	1.81	463

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
SASOL IPK POSF7629 in Air					
1201	5.98		21	1.07	348
1167	6.12		21	1.13	517
1128	6.01		21	1.14	831

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
SASOL IPK/JP-8 50/50 Blend in Air					
1169	5.70		21	0.90	514
1126	5.70		21	0.87	838
1122	5.94		21	0.89	859
1083	5.87		21	0.83	1372
1125	5.30		21	0.96	864

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
SHELL SPK POSF5729 in Air					
1232	5.92		21	0.93	257
1175	6.04		21	1.18	533
1193	5.98		21	1.28	471
1065	5.57		21	1.36	2488
1136	6.17		21	1.12	859

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
SHELL SPK/JP-8 50/50 Blend in Air					
1148	5.31		21	1.10	893
1185	4.62		21	1.11	704
1230	4.64		21	1.08	348
1255	5.09		21	1.23	242
1225	5.14		21	1.15	383

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
HRJ-Tallow POSF6308 in Air					
1210	6.14		21	1.00	272
1148	6.27		21	0.91	511
1055	5.73		21	1.35	1681
1123	6.36		21	1.17	703

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
HRJ-Tallow/JP-8 50/50 Blend in Air					
1225	5.71		21	1.10	178
1095	5.84		21	1.17	1199
1159	5.69		21	1.16	549
1087	6.14		21	1.00	1245
1191	5.57		21	1.10	330

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
HRJ-Camelina POSF7720 in Air					
1095	5.26		21	0.81	1422
1162	5.23		21	1.12	633
1225	5.22		21	1.12	279
1165	5.76		21	0.94	518
1133	5.77		21	0.89	833

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
HRJ-Camelina/JP-8 50/50 Blend in Air					
1178	5.62		21	0.83	439
1114	5.37		21	0.92	1059
1220	5.59		21	0.92	233
1109	5.71		21	0.92	1079

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
F-76 in 4% O2/Argon					
1289	16.9		4	0.25	371
1354	16.9		4	0.26	185
1228	17.5		4	0.27	1221
1384	16.9		4	0.29	103
1250	17.2		4	0.46	551
1215	17.5		4	0.47	893
1288	17.2		4	0.48	409
1237	17.1		4	0.50	605
1356	16.6		4	0.52	182
1154	17.3		4	0.54	2149
1220	17.6		4	0.57	811
1351	17.0		4	0.60	210
1309	17.2		4	0.62	317
1226	16.5		4	0.69	703
1263	17.1		4	0.78	570
1136	17.3		4	0.83	1685
1277	16.7		4	0.86	496
1174	17.0		4	0.86	1152
1170	17.1		4	0.86	1189
1313	16.9		4	0.95	370
1251	17.4		4	1.02	595
1232	16.7		4	1.05	673
1355	15.9		4	0.98	222
1347	17.1		4	1.00	257
1345	17.2		4	1.08	273
1165	17.4		4	1.14	1301
1208	17.1		4	1.22	893
1311	16.4		4	1.37	368
1410	43.9		4	1.00	51
1239	43.9		4	0.93	310
1080	44.0		4	1.09	1548

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
F-76 Blend in 4% O₂/Argon					
1243	17.0		4	0.82	524
1331	16.5		4	0.82	212
1143	17.4		4	0.78	1468
1257	16.8		4	1.12	479
1155	17.1		4	0.95	1309
1210	16.7		4	0.77	738
1298	16.8		4	0.97	297
1369	16.2		4	0.96	142

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
DIM in 4% O₂/Argon					
1328	16.4		4	0.81	283
1326	16.7		4	0.82	280
1268	17.0		4	0.92	509
1335	16.5		4	0.94	262
1171	17.4		4	1.05	1168
1222	17.1		4	1.03	774
1120	17.3		4	0.98	1749
1352	16.3		4	1.04	226

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
103-2 in 4% O₂/Argon					
1345	16.5		4	1.10	187
1261	17.1		4	1.15	436
1213	17.0		4	1.18	654
1182	17.7		4	1.19	860
1128	17.8		4	1.17	1395

Diesel Fuel (DF-2)

Literature Source of Data:

D. R. Haylett, P. P. Lappas, D. F. Davidson, R. K. Hanson, “*Application of an Aerosol Shock Tube to the Measurement of Diesel Ignition Delay Times,*” Proceedings of the Combustion Institute 32 (2009) 477-484.

Range of Data:

Temperature [K]	902	1305
Pressure [atm]	2.67	7.99
Fuel Mole Fraction [%]	.34	1.6
Oxygen Mole Fraction [%]	21	21
Equivalence Ratio	0.29	1.35

Type of Data:

Diesel (DF-2) Table 1: Ignition delay time measurement in argon from **Aerosol Shock Tube experiments** using sidewall PZT pressure and CH* emission near 430 nm.

DF-2 fuel was supplied by P. Schiul of the U.S. Army Propulsion Laboratory. This fuel has a density of 0.8618 kg/liter at 15C, and a calculated cetane index of 43.4, an assumed H/C ratio of 1.92, and an equivalent molecular formula of C₁₂H₂₃.

Diesel (DF-2) Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
902	5.70	0.5561	21	0.47	8440
922	5.84	0.4969	21	0.42	10540
970	5.56	0.3786	21	0.32	5360
990	4.51	1.0885	21	0.92	2080
1001	5.19	0.4377	21	0.37	4010
1044	4.27	0.6980	21	0.59	3500
1045	4.99	0.4377	21	0.37	2730
1058	7.64	0.7808	21	0.66	955
1069	7.47	0.9583	21	0.81	500
1078	7.99	0.7808	21	0.66	1080
1081	2.67	1.5972	21	1.35	835
1084	4.68	0.7217	21	0.61	1230
1087	6.11	0.4969	21	0.42	1690
1095	7.54	0.7690	21	0.65	740
1098	4.33	0.8282	21	0.70	1230
1113	6.33	0.4141	21	0.35	1630
1124	7.21	0.3431	21	0.29	754
1129	6.58	0.4023	21	0.34	1340
1130	7.59	0.7572	21	0.64	514
1141	6.26	0.4259	21	0.36	1060
1150	2.33	1.5735	21	1.33	619
1153	6.11	0.4969	21	0.42	763
1154	7.08	0.5206	21	0.44	508
1155	7.58	0.4377	21	0.37	574
1170	7.32	0.4259	21	0.36	517
1175	4.06	0.6270	21	0.53	541
1185	7.40	0.4614	21	0.39	331
1190	6.37	0.3549	21	0.30	600
1191	4.35	0.7335	21	0.62	430
1197	7.21	0.5679	21	0.48	348
1198	7.24	0.3668	21	0.31	328
1200	6.49	0.4377	21	0.37	508
1215	7.38	0.6152	21	0.52	162
1228	6.35	0.4259	21	0.36	349
1240	3.86	0.7335	21	0.62	300
1245	6.88	0.5206	21	0.44	141
1248	6.57	0.5206	21	0.44	275
1251	5.98	0.5206	21	0.44	202
1251	5.92	0.4259	21	0.36	252
1305	7.00	0.3431	21	0.29	104

Diesel Fuel (DF-2)

Literature Source of Data:

D. R. Haylett, R. D. Cook, D. F. Davidson, R. K. Hanson, "*OH and C₂H₄ species time-histories during hexadecane and diesel ignition behind reflected shock waves*," Proceedings of the Combustion Institute 33 (2011) 167-173. (Note that the supplementary material for this paper has the wrong citation wording listed at the beginning.)

D. R. Haylett, D. F. Davidson, R. K. Hanson, "*Ignition Delay Times of Low-Vapor-Pressure Fuels Measured Using an Aerosol Shock Tube*," Combustion and Flame 159 (2012) 552-561.

D. R. Haylett, "*The Development and Application of Aerosol Shock Tube Methods for the Study of Low-Vapor-Pressure Fuels*," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, March 2011.

http://hanson.stanford.edu/dissertations/Haylett_2011.pdf

Range of Data:

Temperature [K]	947	1381
Pressure [atm]	3.5	8.0
Fuel Mole Fraction [%]	0.12	1.58
Oxygen Mole Fraction [%]	4	21
Equivalence Ratio	0.18	2.35

Type of Data:

Diesel (DF-2) Table 2: Ignition delay time measurement in argon from **Aerosol Shock Tube experiments** using sidewall PZT pressure and CH* emission near 430 nm. Selected ignition delay times are derived from OH species time-history measurements using laser absorption at 306.7 nm.

Details of the fuel composition can be found in Haylett (2011) and Haylett et al (2012). The US DF-2 has a cetane index of 43; the Europe DF2 has an estimated cetane index of 55; the low aromatic DF-2 is 16.2% aromatic v/v with a cetane index of 46; the high aromatic DF-2 is 38.8% aromatic v/v with a cetane index of 42.

Diesel (DF-2) Table 2:

T ₅ [K]	P ₅ [atm]	Type of Fuel	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [ms]
US DF2						
1128	6.42		0.521	21	0.46	1.065
1212	6.52		0.531	21	0.47	0.350
1185	6.69		0.538	21	0.47	0.509
1178	6.51		0.445	21	0.39	0.600
1236	6.07		0.518	21	0.46	0.252
1232	6.17		0.643	21	0.57	0.202
1073	6.29		0.605	21	0.53	1.690
1138	6.29		0.609	21	0.54	0.763
1170	7.32		0.430	21	0.38	0.517
1185	7.4		0.460	21	0.41	0.331
1198	7.24		0.366	21	0.32	0.328
1095	7.54		0.770	21	0.68	0.741
1078	7.99		0.775	21	0.68	1.082
1155	7.58		0.441	21	0.39	0.574
1154	7.08		0.522	21	0.46	0.508
1130	7.59		0.756	21	0.67	0.515
1197	7.21		0.567	21	0.5	0.288
1191	4.35		0.734	21	0.65	0.430
1240	3.86		0.734	21	0.65	0.301
1098	4.33		0.830	21	0.73	1.232
1175	4.06		0.624	21	0.55	0.542
1044	4.27		0.703	21	0.62	3.510
1150	2.33		1.577	21	1.39	0.620
1084	4.68		0.722	21	0.64	1.233
Europe DF2						
1261	7.28		0.198	21	0.18	0.160
1193	7.39		0.180	21	0.17	0.493
1174	3.5		0.195	21	0.18	0.601
1158	3.78		0.204	21	0.19	0.881
1129	4.02		0.246	21	0.23	1.232
1047	3.73		0.264	21	0.24	3.240
998	4.55		0.263	21	0.24	6.228
990	4.32		0.344	21	0.32	7.125
962	4.83		0.262	21	0.24	11.40
1036	5.24		0.251	21	0.23	4.067
1025	4.96		0.238	21	0.22	4.116
947	4.82		0.414	21	0.38	12.54

Low Aromatic DF2					
1373	6.33	0.138	4	0.61	0.243
1327	6.1	0.128	4	0.57	0.402
1263	6.77	0.146	4	0.65	0.931
1206	6.88	0.116	4	0.51	1.317
1198	6.69	0.123	4	0.54	1.251
1193	6.6	0.120	4	0.53	1.400
1182	6.76	0.171	4	0.76	1.406
1177	6.61	0.101	4	0.45	2.116
1168	6.54	0.170	4	0.76	2.007
1119	4.63	0.176	4	0.78	4.251
1124	4.06	0.195	4	0.86	5.040
1120	4.11	0.177	4	0.79	5.866
High Aromatic DF2					
1381	5.88	0.097	4	0.43	0.641
1344	6.15	0.167	4	0.74	0.731
1291	5.5	0.237	4	1.05	1.013
1203	2.05	0.317	4	1.41	4.621
1166	2.09	0.530	4	2.35	4.644

Small Oxygenates

Dimethyl Ether

Literature Source of Data:

R. D. Cook, D. F. Davidson, R. K. Hanson, “*Measurements of Ignition Delay Times and OH Species Concentrations in DME/O₂/Ar*,” Paper 2970P, 26th Shock Wave Symposium, July 15-20, 2007, Gottingen Germany.

R. D. Cook, D. F. Davidson, R. K. Hanson, “*Shock Tube Measurements of Ignition Delay Times and OH Species Concentrations in Dimethyl Ether Oxidation*,” Proceedings of the Combustion Institute 32 (2009) 189-196.

Range of Data:

Temperature [K]	1175	1900
Pressure [atm]	1.8	6.6
Fuel Mole Fraction [%]	1	1
Oxygen Mole Fraction [%]	6	1.5
Equivalence Ratio	0.5	3

Type of Data:

Dimethyl Ether Table 1: Ignition delay time measurement in argon using OH emission near 306 nm.

$$\tau(ign) = 4.21 \times 10^{-5} P^{-0.66} \phi^{0.72} \exp\left(\frac{22690}{T}\right) [s]$$

Dimethyl Ether Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign Time [μs]
1298	1.8	1	6	0.5	787
1241	1.8	1	6	0.5	1816
1376	1.8	1	6	0.5	237
1387	1.8	1	6	0.5	203
1324	1.8	1	6	0.5	489
1266	1.8	1	6	0.5	1200
1241	1.8	1	3	1	2517
1442	1.8	1	3	1	153
1468	1.8	1	3	1	112
1372	1.8	1	3	1	368
1308	1.8	1	3	1	913
1381	1.8	2	1.5	2	775
1253	1.8	2	1.5	2	2832
1302	1.8	2	1.5	2	1651
1376	1.8	2	1.5	2	711
1451	1.8	2	1.5	2	314
1551	1.8	2	1.5	2	140
1449	1.6	1	3	1	187
1395	1.6	1	3	1	347
1357	1.6	1	3	1	592
1314	1.6	1	3	1	979
1441	3.3	1	3	1	111
1385	3.3	1	3	1	234
1348	3.3	1	3	1	360
1331	3.3	1	3	1	462
1279	3.3	1	3	1	940
1248	3.3	1	3	1	1506
1247	3.3	1	3	1	1485
1365	6.6	1	3	1	210
1281	6.6	1	3	1	614
1224	6.6	1	3	1	1283
1392	5.4	1	3	1	159
1280	5.4	1	3	1	757

Revised September 29, 2016 by dfd.

Notes:

I do not have a record of the exact pressures of these experiments.

What I do have are the ignition delay times normalized by $P^{(-0.66)}$.

All experiments had test pressures within about 15% of the listed pressures.

Acetone

Literature Source of Data:

D. F. Davidson, S. C. Ranganath, K.-Y. Lam, M. Liaw, Z. Hong, R. K. Hanson, *"Ignition Delay Time Measurements of Normal Alkanes and Simple Oxygenates,"* Journal of Propulsion and Power 26 (2010) 280-287.

Range of Data:

Temperature [K]	1169	1585
Pressure [atm]	1.3	20.3
Fuel Mole Fraction [%]	1	2
Oxygen Mole Fraction [%]	4	4
Equivalence Ratio	1	2

Type of Data:

Acetone Table 1: Ignition delay time measurement in argon using sidewall and endwall OH emission near 306 nm (using a time based on extrapolating the maximum signal slope to the intersection with the baseline), and sidewall PZT pressure.

Acetone Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign Time [μs]
1417	1.38	1	4	1	456
1471	1.30	1	4	1	242
1540	1.30	1	4	1	115
1322	1.45	1	4	1	1548
1373	1.46	1	4	1	885
1230	1.53	1	4	1	5114
1249	1.50	1	4	1	4220
1292	1.44	1	4	1	2557
1329	1.53	1	4	1	1394
1363	1.50	1	4	1	966
1411	1.57	1	4	1	531
1447	1.53	1	4	1	321
1496	1.55	1	4	1	178
1506	1.50	1	4	1	150
1169	21.1	1	4	1	2580
1181	20.3	1	4	1	2166
1240	19.7	1	4	1	1101
1276	19.4	1	4	1	707
1285	18.5	1	4	1	632
1443	1.55	2	4	2	553
1475	1.50	2	4	2	403
1359	1.55	2	4	2	1317
1327	1.62	2	4	2	1856
1548	1.63	2	4	2	171
1269	1.48	2	4	2	4105
1310	1.53	2	4	2	2791
1500	1.66	2	4	2	292
1256	1.46	2	4	2	4548
1374	1.51	2	4	2	1138
1466	1.49	2	4	2	453
1585	1.70	2	4	2	122

Acetone

Literature Source of Data:

K.-Y. Lam, W. Ren, Z. Hong, D. F. Davidson, R. K. Hanson, “*Shock Tube Measurements of 3-Pentanone Pyrolysis and Oxidation*,” Combustion and Flame 159 (2012) 3251-3263.

Range of Data:

Temperature [K]	1355	1551
Pressure [atm]	1.3	2.7
Fuel Mole Fraction [%]	0.131	0.344
Oxygen Mole Fraction [%]	0.525	1.38
Equivalence Ratio	1	1

Type of Data:

Acetone Table 2: Ignition delay time measurement in argon time to 50% peak of H₂O or OH mole fraction time-history based on laser absorption measurements.

Acetone Table 2:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Species	T[50] [μs]
1551	2.43	0.131	0.525	1	H ₂ O	174
1509	2.55	0.131	0.525	1	H ₂ O	245
1449	2.59	0.131	0.525	1	H ₂ O	509
1392	2.67	0.131	0.525	1	H ₂ O	1033
1372	2.68	0.131	0.525	1	H ₂ O	1234
1567	1.32	0.344	1.38	1	OH	177
1468	1.34	0.344	1.38	1	OH	442
1439	1.40	0.344	1.38	1	OH	598
1372	1.41	0.344	1.38	1	OH	1494
1355	1.48	0.344	1.38	1	OH	1909

2-Pentanone

Literature Source of Data:

K.-Y. Lam, W. Ren, Z. Hong, D. F. Davidson, R. K. Hanson, “*Shock Tube Measurements of 3-Pentanone Pyrolysis and Oxidation*,” Combustion and Flame 159 (2012) 3251-3263.

Range of Data:

Temperature [K]	1357	1511
Pressure [atm]	2.52	2.65
Fuel Mole Fraction [ppm]	750	750
Oxygen Mole Fraction [%]	0.525	0.525
Equivalence Ratio	1	1

Type of Data:

2-Pentanone Table 1: Ignition delay time measurement in argon based on the time to 50% peak of OH mole fraction time-history using laser absorption measurements.

2-Pentanone Table 1:

T _s [K]	P _s [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign Time [μs]
1511	2.52	750	0.525	1	233
1449	2.57	750	0.525	1	410
1409	2.63	750	0.525	1	659
1374	2.62	750	0.525	1	919
1357	2.65	750	0.525	1	1160

3-Pentanone

Literature Source of Data:

K.-Y. Lam, Z. Hong, S. H. Durrstein, D. F. Davidson, R. K. Hanson, C. Schulz, *"Shock Tube Measurements of Ignition Delay Times and OH and H₂O Species Time-Histories in 3-Pentanone/O₂/Ar Mixtures,"* 7th International Conference on Chemical Kinetics, July 10-14, 2011, MIT Boston MA.

K.-Y. Lam, W. Ren, Z. Hong, D. F. Davidson, R. K. Hanson, *"Shock Tube Measurements of 3-Pentanone Pyrolysis and Oxidation,"* Combustion and Flame 159 (2012) 3251-3263.

K.-Y. Lam, *"Shock Tube Measurements of Oxygenated Fuel Combustion using Laser Absorption Spectroscopy,"* Ph.D. Thesis, Mechanical Engineering Department, Stanford University, June 2013.

E. E. Dames, K.-Y. Lam, D. F. Davidson, R. K. Hanson, *"An improved kinetic mechanism for 3-pentanone pyrolysis and oxidation developed using multispecies time histories in shock-tubes,"* Combustion and Flame 161 (2014) 1135-1145.

Range of Data:

Temperature [K]	1127	1678
Pressure [atm]	1.0	2.79
Fuel Mole Fraction [ppm]	400	8750
Oxygen Mole Fraction [%]	0.28	12.25
Equivalence Ratio	0.5	1

Type of Data:

3-Pentanone Table 2: Ignition delay time measurement in argon using the time to 50% of peak value of CO, OH and H₂O or the complete removal of CH₃ mole fraction time-histories.

3-Pentanone Table 2:

T ₅ [K]	P ₅ [atm]	Fuel [ppm]	O ₂ [%]	φ E.R.	Species	T _{ign} or T[50] [μs]
1470	1.55	750	0.525	1	CH3	255
1427	1.64	750	0.525	1	CH3	301
1321	1.72	750	0.525	1	CH3	1188
1387	1.61	400	0.56	0.5	CO	125
1336	1.64	400	0.56	0.5	CO	201
1277	1.74	400	0.56	0.5	CO	393
1678	1.35	400	0.187	1.5	H2O	167
1563	1.41	400	0.187	1.5	H2O	301
1475	1.51	400	0.187	1.5	H2O	637
1425	1.58	400	0.56	0.5	H2O	164
1305	1.66	400	0.56	0.5	H2O	527
1263	1.72	400	0.56	0.5	H2O	996
1217	1.74	400	0.56	0.5	H2O	1808
1425	1.58	400	0.56	0.5	OH	147
1305	1.66	400	0.56	0.5	OH	613
1263	1.72	400	0.56	0.5	OH	1237
1217	1.74	400	0.56	0.5	OH	2188
1283	1.70	400	0.56	0.5	OH	864
1542	1.42	400	0.28	1	H2O	199
1486	1.52	400	0.28	1	H2O	290
1377	1.54	400	0.28	1	H2O	734
1343	1.61	400	0.28	1	H2O	1187
1542	1.42	400	0.28	1	OH	247
1486	1.52	400	0.28	1	OH	405
1377	1.54	400	0.28	1	OH	1032
1343	1.61	400	0.28	1	OH	1627
1548	1.43	750	0.525	1	H2O	118
1473	1.54	750	0.525	1	H2O	269
1395	1.58	750	0.525	1	H2O	449
1349	1.61	750	0.525	1	H2O	670
1548	1.43	750	0.525	1	OH	158
1473	1.54	750	0.525	1	OH	276
1395	1.58	750	0.525	1	OH	650
1349	1.61	750	0.525	1	OH	913
1293	1.66	750	0.525	1	OH	1668
1232	1.048	2857	4	0.5	OH	746
1180	1.092	2857	4	0.5	OH	1698
1208	1.076	2857	4	0.5	OH	1137
1152	1.158	2857	4	0.5	OH	2770
1145	1.140	2857	4	0.5	OH	2697
1313	1.008	2857	4	0.5	OH	260

1280	1.035	2857	4	0.5	OH	390
1127	1.083	8750	12.25	0.5	OH	2187
1172	1.068	8750	12.25	0.5	OH	992
1235	1.021	8750	12.25	0.5	OH	359
1289	1.025	8750	12.25	0.5	OH	178
1211	1.056	8750	12.25	0.5	OH	518
1225	1.037	8750	12.25	0.5	OH	414
1278	0.965	8750	12.25	0.5	OH	224
1306	0.999	5710	4	1	OH	446
1469	0.946	5710	4	1	OH	116
1428	0.970	5710	4	1	OH	150
1246	1.038	5710	4	1	OH	920
1351	1.000	5710	4	1	OH	290
1214	1.048	5710	4	1	OH	1404
1787	1.017	5710	4	1	OH	549
1371	0.986	5710	4	1	OH	266
1474	2.51	750	0.525	1	OH	183
1419	2.64	750	0.525	1	OH	319
1354	2.64	750	0.525	1	OH	554
1338	2.73	750	0.525	1	OH	694
1302	2.79	750	0.525	1	OH	1076

3-Pentanone

Literature Source of Data:

D. F. Davidson, S. C. Ranganath, K.-Y. Lam, M. Liaw, Z. Hong, R. K. Hanson, "Ignition Delay Time Measurements of Normal Alkanes and Simple Oxygenates," Journal of Propulsion and Power 26 (2010) 280-287.

Range of Data:

Temperature [K]	1173	1306
Pressure [atm]	1.68	1.85
Fuel Mole Fraction [%]	0.571	0.571
Oxygen Mole Fraction [%]	4	4
Equivalence Ratio	1	1

Type of Data:

3-Pentanone Table 1: Ignition delay time measurement in argon using sidewall and endwall OH emission near 306 nm (using a time based on extrapolating the maximum signal slope to the intersection with the baseline), and sidewall PZT pressure.

3-Pentanone Table 1:

T _s [K]	P _s [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign Time [μs]
1306	1.68	0.571	4	1	280
1259	1.73	0.571	4	1	545
1209	1.77	0.571	4	1	1022
1188	1.80	0.571	4	1	1337
1173	1.85	0.571	4	1	1626

Butanal

Literature Source of Data:

D. F. Davidson, S. C. Ranganath, K.-Y. Lam, M. Liaw, Z. Hong, R. K. Hanson, *"Ignition Delay Time Measurements of Normal Alkanes and Simple Oxygenates,"* Journal of Propulsion and Power 26 (2010) 280-287.

Range of Data:

Temperature [K]	1188	1551
Pressure [atm]	1.34	2.77
Fuel Mole Fraction [%]	0.73	1.0
Oxygen Mole Fraction [%]	2.75	5.5
Equivalence Ratio	1	2

Type of Data:

Butanal Table 1: Ignition delay time measurement in argon using sidewall and endwall OH emission near 306 nm (using a time based on extrapolating the maximum signal slope to the intersection with the baseline), and sidewall PZT pressure.

Butanal Table 1:

T_5 [K]	P_5 [atm]	Fuel [%]	O_2 [%]	ϕ E.R.	Ign Time [μ s]
1397	1.41	1	5.5	1	156
1296	1.40	1	5.5	1	580
1423	1.37	1	5.5	1	107
1236	1.45	1	5.5	1	1292
1308	1.48	1	5.5	1	454
1209	1.45	1	5.5	1	1836
1264	2.68	1	5.5	1	598
1188	2.77	1	5.5	1	1589
1352	2.51	1	5.5	1	196
1229	2.73	1	5.5	1	1023
1318	2.62	1	5.5	1	306
1388	2.51	1	5.5	1	135
1352	1.46	1	2.75	2	1034
1411	1.39	1	2.75	2	602
1506	1.37	1	2.75	2	237
1551	1.34	1	2.75	2	158
1296	1.47	1	2.75	2	1610
1248	1.78	0.727	4	1	1188
1306	1.75	0.727	4	1	534
1195	1.80	0.727	4	1	2452
1234	1.81	0.727	4	1	1451

Methyl Formate

Literature Source of Data:

W. Ren, A. Farooq, D. F. Davidson, R. K. Hanson, “CO Concentration and Temperature Sensor for Combustion Gases using Quantum-Cascade Laser Absorption near 4.7 Microns,” Applied Physics B 107 (2012) 849-860.

Range of Data:

Temperature [K]	1379	1379
Pressure [atm]	1.67	1.67
Fuel Mole Fraction [%]	0.494	0.494
Oxygen Mole Fraction [%]	0.988	0.988
Equivalence Ratio	1	1

Type of Data:

Methyl Formate Table 1: Ignition delay time measurement in argon using the time to 50% peak of CO mole fraction time-history based on laser absorption measurements.

Methyl Formate Table 1:

T_s [K]	P_s [atm]	Fuel [%]	O_2 [%]	ϕ E.R.	$T[50]$ [μs]
1379	1.67	0.494	0.988	1	300

Methyl Butanoate

Literature Source of Data:

D. F. Davidson, S. C. Ranganath, K.-Y. Lam, M. Liaw, Z. Hong, R. K. Hanson, *"Ignition Delay Time Measurements of Normal Alkanes and Simple Oxygenates,"* Journal of Propulsion and Power 26 (2010) 280-287.

Range of Data:

Temperature [K]	1244	1446
Pressure [atm]	1.67	1.84
Fuel Mole Fraction [%]	0.308	0.615
Oxygen Mole Fraction [%]	4	4
Equivalence Ratio	0.5	1

Type of Data:

Methyl Butanoate Table 1: Ignition delay time measurement in argon using sidewall and endwall OH emission near 306 nm (using a time based on extrapolating the maximum signal slope to the intersection with the baseline), and sidewall PZT pressure.

Methyl Butanoate Table 1:

T_5 [K]	P_5 [atm]	Fuel [%]	O_2 [%]	ϕ E.R.	Ign Time [μ s]
1299	1.77	0.615	4	1	898
1358	1.71	0.615	4	1	440
1385	1.67	0.615	4	1	306
1244	1.80	0.615	4	1	2036
1323	1.74	0.615	4	1	703
1282	1.73	0.615	4	1	1089
1353	1.70	0.615	4	1	433
1373	1.65	0.615	4	1	307
1446	1.69	0.615	4	1	132
1262	1.74	0.615	4	1	1778
1330	1.76	0.615	4	1	607
1403	1.70	0.308	4	0.5	153
1326	1.72	0.308	4	0.5	491
1287	1.80	0.308	4	0.5	926
1238	1.84	0.308	4	0.5	1903
1264	1.81	0.308	4	0.5	1200
1308	1.77	0.308	4	0.5	706

Methyl Decanoate

Literature Source of Data:

D. R. Haylett, D. F. Davidson, R. K. Hanson, "Ignition Delay Times of Low-Vapor-Pressure Fuels Measured Using an Aerosol Shock Tube," Combustion and Flame 159 (2012) 552-561.

D. R. Haylett, "The Development and Application of Aerosol Shock Tube Methods for the Study of Low-Vapor-Pressure Fuels," Ph.D. Thesis, Mechanical Engineering Department, Stanford University, March 2011.

http://hanson.stanford.edu/dissertations/Haylett_2011.pdf

Range of Data:

Temperature [K]	1208	1308
Pressure [atm]	7.6	8.0
Fuel Mole Fraction [%]	0.16	0.18
Oxygen Mole Fraction [%]	21	21
Equivalence Ratio	0.12	0.17

Type of Data:

Methyl Decanoate Table 1: Ignition delay time measurements using sidewall pressure, and confirmed with CH* emission near 430 nm, from **Aerosol Shock Tube experiments**.

Methyl Decanoate Table 1:

T _s [K]	P _s [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign Time [μs]
1302	7.67	0.161	21	0.15	298
1308	7.57	0.101	21	0.09	486
1185	7.84	0.179	21	0.17	1385
1255	8.03	0.126	21	0.12	631
1208	7.90	0.182	21	0.17	831

Butanol Isomers

Literature Source of Data:

I. L. R. Bec, Y. Zhu, D. F. Davidson, R. K. Hanson, , “*Shock Tube Measurements of Ignition Delay Times for the Butanol Isomers using the Constrained-Reaction-Volume Strategy*,” Submitted to International Journal of Chemical Kinetics March 2014.

Y. Zhu, D. F. Davidson, R. K. Hanson, “*1-Butanol Ignition Delay Times at Low Temperatures: An Application of the Constrained-Reaction-Volume Strategy*,” Combustion and Flame 161 (2014) 634-643.

Range of Data:

Temperature [K]	777	1095
Pressure [atm]	15	45
Fuel Mole Fraction [%]	3	6.5
Oxygen Mole Fraction [%]	7	41
Equivalence Ratio	0.5	3.3

Type of Data:

Butanol Isomers Table 1: Ignition delay times in air measured using the **CONSTRAINED REACTION VOLUME STRATEGY** and conventional filling strategies. Ignition delay times measured using OH* sidewall emission near 306 nm. Measurement performed for four isomers: n-butanol, sec-butanol, iso-butanol, and tert-butanol.

Butanol Isomers Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign Time [μs]
Iso-Butanol, Conventional Strategy					
895	15.3	3.35	20.30	0.99	2968
847	15.7	3.41	20.30	1.01	3622
971	22.6	3.43	20.30	1.02	893
871	19.0	3.41	20.30	1.01	3175
824	17.8	3.34	20.30	0.99	3837
819	20.8	3.36	20.30	0.99	7041
794	21.3	3.44	20.30	1.02	6822
994	21.5	3.36	20.30	0.99	1010
775	22.7	3.49	20.30	1.03	7071
1069	22.9	3.43	20.30	1.01	373
798	23.6	3.48	20.30	1.03	6704
Iso-Butanol, CRV Strategy					
844	16.8	3.03	12.80	1.42	11209
845	16.0	2.82	11.60	1.46	11262
846	16.8	2.84	11.40	1.50	11611
861	15.9	2.97	12.60	1.42	7721
875	15.5	3.10	12.80	1.45	6160
900	15.5	3.26	14.00	1.40	4957
922	15.0	3.47	14.40	1.45	3502
969	22.4	3.36	13.80	1.46	1750
994	19.4	3.32	15.80	1.26	1378
997	20.1	3.18	15.20	1.25	1407
1016	18.5	3.30	18.50	1.07	931
1051	18.2	3.38	19.30	1.05	538
961	28.1	3.20	12.80	1.50	1685
922	28.0	3.12	12.40	1.51	2684
912	28.5	3.02	12.00	1.51	2684
880	29.4	3.59	14.20	1.52	4079
889	16.7	1.41	14.00	0.61	8643
891	15.2	1.45	15.20	0.57	7768
905	15.6	1.40	13.20	0.63	6042
924	16.4	1.35	13.00	0.62	5073
930	16.4	1.41	13.80	0.61	5002
932	15.9	1.49	16.10	0.56	4795
943	15.1	1.57	15.90	0.59	4446
950	13.6	1.45	14.80	0.59	3695
950	13.7	1.60	16.30	0.59	3536
971	18.8	1.36	15.00	0.54	2939
973	17.4	1.33	14.20	0.56	2897
987	18.1	1.35	16.10	0.50	2596
1032	14.8	1.67	20.00	0.50	1508

T_s [K]	P_s [atm]	Fuel [%]	O_2 [%]	ϕ E.R.	Ign Time [μs]
Sec-Butanol, CRV Strategy					
828	15.6	3.00	11.60	1.54	13797
868	20.7	3.10	11.20	1.68	7473
832	20.1	3.10	11.00	1.71	14122
864	18.9	3.50	12.40	1.71	6157
926	17.0	3.90	15.20	1.55	2884
908	18.9	3.50	14.00	1.51	4246
954	14.9	3.30	15.80	1.25	2290
1062	17.0	3.70	20.30	1.08	572
1019	16.0	3.20	18.50	1.04	1204
1043	16.9	3.20	17.70	1.08	889
1020	17.6	3.80	19.10	1.21	1018
923	18.9	3.40	13.60	1.52	3836
918	19.9	3.30	13.40	1.49	3323
Tert-Butanol, CRV Strategy					
955	19.7	3.30	13.60	1.46	5646
969	19.2	3.60	16.20	1.33	3559
998	18.4	3.60	17.70	1.22	2856
1045	21.0	2.90	16.60	1.05	1971
1095	19.2	3.50	20.30	1.04	1157
1071	15.1	3.60	20.30	1.06	1504
1073	17.5	3.60	20.30	1.06	1388

T _s [K]	P _s [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign Time [μs]
n-Butanol, CRV Strategy					
1043	16.2	3.38	16.20	1.25	504
1012	22.8	3.38	16.20	1.25	672
959	23.0	3.38	14.20	1.43	1551
913	22.8	3.38	14.20	1.43	2847
882	17.5	3.38	14.20	1.43	5272
869	17.4	3.38	14.20	1.43	5702
858	17.2	3.38	14.20	1.43	7026
850	17.3	3.38	12.20	1.67	8000
848	18.1	3.38	12.20	1.67	8038
841	19.1	3.38	12.20	1.67	8204
839	19.0	3.38	12.20	1.67	9901
820	19.0	3.38	12.20	1.67	11594
922	17.4	1.72	14.40	0.72	4238
908	17.1	1.72	14.40	0.72	4954
902	17.1	1.72	14.40	0.72	4756
883	17.3	1.72	14.40	0.72	6889
882	21.5	1.72	14.40	0.72	5414
880	21.4	1.72	14.40	0.72	5844
874	22.4	1.72	14.40	0.72	6619
856	17.5	1.72	14.40	0.72	9267
847	22.4	6.54	11.80	3.34	3405
830	24.0	6.54	11.80	3.34	5521
827	17.9	6.54	11.80	3.34	6693
806	23.8	6.54	11.80	3.34	7679
879	17.3	3.38	28.40	0.71	2086
874	17.6	3.38	28.40	0.71	2609
851	17.5	3.38	28.40	0.71	3769
833	16.4	3.38	24.40	0.83	3890
822	19.0	3.38	24.40	0.83	4960
800	19.9	3.38	24.40	0.83	6259
777	19.6	3.38	24.40	0.83	10230
990	14.7	3.38	7.14	2.84	2171
980	16.8	3.38	7.14	2.84	2066
905	16.5	3.38	7.14	2.84	4728
890	16.5	3.38	7.14	2.84	5382
884	17.3	3.38	7.14	2.84	5810
862	17.4	3.38	7.14	2.84	7925
855	17.0	3.38	7.14	2.84	11048
851	18.4	3.38	7.14	2.84	9168
829	18.3	3.38	6.12	3.31	12278

T _s [K]	P _s [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign Time [μs]
n-Butanol, Conventional Strategy					
1121	21.7	3.38	20.3	1	140
1047	16.3	3.38	20.3	1	514
1042	21.2	3.38	20.3	1	379
1014	22.2	3.38	20.3	1	572
982	22.4	3.38	20.3	1	851
939	21.5	3.38	20.3	1	1489
917	21.8	3.38	20.3	1	2134
876	15.6	3.38	20.3	1	2740
867	16.2	3.38	20.3	1	3144
831	16.4	3.38	20.3	1	3844
829	17.7	3.38	20.3	1	3722
812	22.6	3.38	20.3	1	4044
809	17.9	3.38	20.3	1	4747
792	18.5	3.38	20.3	1	6212
776	23.9	3.38	20.3	1	8183
1093	19.8	1.72	20.6	0.5	418
1025	20.7	1.72	20.6	0.5	1103
966	21.3	1.72	20.6	0.5	2079
931	15.1	6.54	19.6	2	1505
902	20.3	6.54	19.6	2	1809
875	21.1	6.54	19.6	2	2373
802	17.3	6.54	19.6	2	5440
778	18.1	6.54	19.6	2	6196
749	18.6	6.54	19.6	2	9704
976	23.1	3.38	40.6	0.5	424
933	23.2	3.38	40.6	0.5	677
1063	21.3	3.38	10.2	2	566
1040	21.8	3.38	10.2	2	633
1007	22.2	3.38	10.2	2	970
982	23.1	3.38	10.2	2	1485
945	34.1	3.38	20.3	1	825
883	34.3	3.38	20.3	1	1670
815	34.8	3.38	20.3	1	3113
798	38.3	3.38	20.3	1	4841
762	41.6	3.38	20.3	1	8704
728	41.0	3.38	20.3	1	8571
716	45.0	3.38	20.3	1	9385

Butanol Isomers

Literature Source of Data:

I. Stranic, D. P. Chase, J. T. Harmon, S. Yang, D. F. Davidson, R. K. Hanson. *"Shock Tube Measurements of Ignition Delay Times for the Butanol Isomers,"* Combustion and Flame 159 (2012) 516-527.

I. Stranic, D. P. Chase, J. T. Harmon, S. Yang, D. F. Davidson, R. K. Hanson. *"Shock Tube Measurements of Ignition Delay Times for the Butanol Isomers,"* 23rd International Colloquium on the Dynamics of Explosions and Reacting Systems (2011), Irvine CA, United States.

I. Stranic, D. P. Chase, J. T. Harmon, S. Yang, D. F. Davidson, R. K. Hanson. *"Shock Tube Measurements of Ignition Delay Times for the Butanol Isomers,"* 7th US National Combustion Meeting, March 20-23, 2011, Atlanta GA.

M. R. Harper, W. H. Green, K. M. Van Geem, B. W. Weber, C.-J. Sung, I. Stranic, D. F. Davidson, R. K. Davidson, *"Combustion of Butanol Isomers: Reaction Pathways at Elevated Pressures from Low-to-High Temperatures,"* Paper 938, 7th International Conference on Chemical Kinetics, MIT Boston, July 2011.

Range of Data:

Temperature [K]	1050	1500
Pressure [atm]	1	45
Fuel Mole Fraction [%]	0.3	1
Oxygen Mole Fraction [%]	3	4.5
Equivalence Ratio	0.5	1

Type of Data:

Butanol Isomers Table 1: Ignition delay times in argon measured for all four isomers: n-butanol, sec-butanol, iso-butanol, and tert-butanol. For measurements at pressures below 4 atm endwall OH* emission data was used to infer the ignition delay time. For measurements at pressures above 10 atm sidewall OH* emission data was used to infer the ignition delay time. In both cases, ignition delay time was defined as the time between the arrival of the reflected shock wave at the observation port and the extrapolation of the maximum slope of the emission signal to the baseline.

T _s [K]	P _s [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign Time [μs]
n-Butanol					
1274	1.51	0.667	4	1	1013
1420	1.38	0.667	4	1	207
1275	1.38	0.667	4	1	993
1344	1.42	0.667	4	1	468
1450	1.36	0.667	4	1	158
1320	1.47	0.667	4	1	575
1432	1.47	0.667	4	1	172
1380	1.51	0.667	4	1	296
1458	1.34	0.667	4	1	148
1332	1.73	0.667	4	1	430
1384	1.71	0.667	4	1	259
1405	1.68	0.333	4	0.5	153
1339	1.76	0.333	4	0.5	328
1289	1.78	0.333	4	0.5	574
1245	1.86	0.333	4	0.5	1057
1413	1.75	0.333	4	0.5	134
1336	2.78	0.667	4	1	312
1374	2.97	0.667	4	1	213
1291	3.05	0.667	4	1	506
1247	3.19	0.667	4	1	832
1217	3.15	0.667	4	1	1133
1199	3.18	0.667	4	1	1382
1425	2.88	0.667	4	1	128
1301	1.01	0.500	3	1	1324
1418	0.97	0.500	3	1	326
1355	0.99	0.500	3	1	628
1505	0.93	0.500	3	1	151
1381	1.17	0.500	3	1	401
1495	1.16	0.500	3	1	141
1230	0.95	0.750	4.5	1	1560
1388	0.93	0.750	4.5	1	293
1312	0.96	0.750	4.5	1	653
1534	0.91	0.750	4.5	1	90
1166	18.24	0.450	4	0.60-0.75	702
1186	18.30	0.450	4	0.60-0.75	528
1233	17.95	0.450	4	0.60-0.75	297
1137	18.47	0.450	4	0.60-0.75	962
1087	18.63	0.450	4	0.60-0.75	1750
1272	17.41	0.450	4	0.60-0.75	197
1121	18.75	0.450	4	0.60-0.75	1157
1354	17.58	0.450	4	0.60-0.75	73
1311	17.71	0.450	4	0.60-0.75	120
1177	17.58	0.450	4	0.60-0.75	628

1065	43.17	0.580	4	0.75-1.0	1141
1135	38.93	0.580	4	0.75-1.0	532
1118	43.04	0.580	4	0.75-1.0	612
1270	41.40	0.580	4	0.75-1.0	111
1199	43.33	0.580	4	0.75-1.0	226
1111	39.61	0.580	4	0.75-1.0	700
1054	46.01	0.580	4	0.75-1.0	1221
1280	43.71	0.580	4	0.75-1.0	92
1205	44.72	0.580	4	0.75-1.0	207
1198	40.92	0.580	4	0.75-1.0	252
1139	40.77	0.580	4	0.75-1.0	493
1144	40.78	0.580	4	0.75-1.0	466
1175	42.27	0.580	4	0.75-1.0	320
1205	44.02	0.580	4	0.75-1.0	213
1207	44.06	0.580	4	0.75-1.0	216
1164	45.14	0.580	4	0.75-1.0	330
1239	18.92	0.167	4	0.2-0.3	317
1156	19.74	0.167	4	0.2-0.3	959
1321	18.65	0.167	4	0.2-0.3	106
1302	19.34	0.167	4	0.2-0.3	133
1197	19.98	0.167	4	0.2-0.3	527
1246	18.49	0.167	4	0.2-0.3	289
1138	20.19	0.167	4	0.2-0.3	1119
1302	18.58	0.167	4	0.2-0.3	138
1240	18.56	0.167	4	0.2-0.3	308
1314	19.50	0.167	4	0.2-0.3	111
1317	17.94	0.667	4	1	118
1245	17.73	0.667	4	1	283
1224	18.51	0.667	4	1	341
1268	41.12	0.667	4	1	113
1211	44.95	0.667	4	1	196
1169	45.12	0.667	4	1	314
1263	18.71	0.667	4	1	235
1183	18.77	0.667	4	1	723
1316	18.56	0.667	4	1	113
1220	18.91	0.667	4	1	434

T _s [K]	P _s [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign Time [μs]
sec-Butanol					
1360	1.59	0.667	4	1	564
1434	1.54	0.667	4	1	263
1503	1.44	0.667	4	1	138
1306	1.58	0.667	4	1	1066
1549	1.42	0.667	4	1	90
1409	1.55	0.667	4	1	323
1465	1.47	0.667	4	1	180
1321	1.48	0.667	4	1	936
1339	1.48	0.667	4	1	699
1406	3.26	0.667	4	1	220
1403	3.19	0.667	4	1	226
1405	3.37	0.667	4	1	217
1385	3.41	0.667	4	1	265
1367	3.53	0.667	4	1	313
1274	3.30	0.667	4	1	830
1281	3.28	0.667	4	1	861
1408	3.30	0.667	4	1	214
1329	3.44	0.667	4	1	457
1308	3.44	0.667	4	1	622
1443	3.23	0.667	4	1	149
1293	3.42	0.667	4	1	721
1294	3.56	0.667	4	1	713
1417	3.05	0.667	4	1	202
1475	3.10	0.667	4	1	113
1260	3.46	0.667	4	1	1011
1385	1.74	0.333	4	0.5	260
1334	1.80	0.333	4	0.5	456
1300	1.81	0.333	4	0.5	705
1269	1.87	0.333	4	0.5	973
1441	1.73	0.333	4	0.5	137
1362	3.34	0.333	4	0.5	238
1468	1.19	0.500	3	1	267
1383	1.23	0.500	3	1	602
1532	1.19	0.500	3	1	141
1399	1.16	0.500	3	1	493
1436	1.17	0.500	3	1	349
1424	1.17	0.500	3	1	404
1331	1.24	0.500	3	1	1094
1309	1.23	1.000	6	1	736
1357	1.20	1.000	6	1	430
1287	1.24	1.000	6	1	1041
1393	1.19	1.000	6	1	292
1425	1.18	1.000	6	1	226

1300	1.27	1.000	6	1	810
1382	1.24	1.000	6	1	330
1192	18.12	0.467	4	0.65-0.75	806
1315	18.40	0.467	4	0.65-0.75	164
1258	18.42	0.467	4	0.65-0.75	337
1125	18.47	0.467	4	0.65-0.75	1697
1242	18.69	0.467	4	0.65-0.75	411
1166	18.51	0.467	4	0.65-0.75	1099
1347	17.99	0.467	4	0.65-0.75	122
1124	43.99	0.617	4	0.80-1.05	780
1176	40.49	0.617	4	0.80-1.05	455
1064	42.40	0.617	4	0.80-1.05	1599
1195	44.26	0.617	4	0.80-1.05	339
1132	37.83	0.617	4	0.80-1.05	820
1242	43.21	0.617	4	0.80-1.05	204
1253	42.34	0.617	4	0.80-1.05	186
1165	45.37	0.617	4	0.80-1.05	472
1136	43.64	0.617	4	0.80-1.05	672
1069	40.75	0.617	4	0.80-1.05	1505
1308	41.31	0.617	4	0.80-1.05	104
1235	18.36	0.667	4	1	483
1292	18.76	0.667	4	1	257
1337	18.59	0.667	4	1	149
1242	15.56	0.667	4	1	457
1193	41.77	0.667	4	1	409
1270	41.93	0.667	4	1	170
1209	42.91	0.667	4	1	334
1235	43.87	0.667	4	1	243
1119	43.77	0.667	4	1	892

T _s [K]	P _s [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign Time [μs]
iso-Butanol					
1433	1.41	0.667	4	1	322
1381	1.51	0.667	4	1	501
1284	1.56	0.667	4	1	1210
1319	1.49	0.667	4	1	898
1349	1.5	0.667	4	1	675
1519	1.37	0.667	4	1	142
1483	1.41	0.667	4	1	189
1486	1.49	0.667	4	1	190
1485	1.51	0.667	4	1	178
1588	1.38	0.667	4	1	87
1376	1.69	0.667	4	1	452
1372	1.6	0.667	4	1	497
1401	1.56	0.667	4	1	371
1437	3.38	0.667	4	1	167
1363	3.27	0.667	4	1	348
1367	3.42	0.667	4	1	343
1343	3.42	0.667	4	1	406
1286	3.3	0.667	4	1	792
1310	3.49	0.667	4	1	528
1493	3.22	0.667	4	1	107
1271	3.41	0.667	4	1	805
1278	3.54	0.667	4	1	751
1240	3.55	0.667	4	1	1032
1410	1.57	0.333	4	0.5	250
1344	1.65	0.333	4	0.5	487
1298	1.72	0.333	4	0.5	827
1265	1.74	0.333	4	0.5	1133
1496	1.56	0.333	4	0.5	103
1438	1.17	0.500	3	1	441
1551	1.13	0.500	3	1	149
1483	1.14	0.500	3	1	293
1493	1.12	0.500	3	1	264
1586	1.18	0.500	3	1	112
1526	1.21	0.500	3	1	188
1249	18.84	0.517	4	0.65-0.90	304
1189	18.86	0.517	4	0.65-0.90	588
1322	18.11	0.517	4	0.65-0.90	147
1106	18.95	0.517	4	0.65-0.90	1395
1186	17.96	0.517	4	0.65-0.90	625
1152	19.13	0.517	4	0.65-0.90	866
1136	18.9	0.517	4	0.65-0.90	1020
1120	19.02	0.517	4	0.65-0.90	1207
1121	19.76	0.517	4	0.65-0.90	1148

1070	19.42	0.517	4	0.65-0.90	1949
1358	17.88	0.517	4	0.65-0.90	95
1271	18.78	0.517	4	0.65-0.90	232
1155	43.88	0.650	4	0.90-1.05	381
1072	40.92	0.650	4	0.90-1.05	1041
1143	44.61	0.650	4	0.90-1.05	423
1166	44.74	0.650	4	0.90-1.05	342
1204	44.72	0.650	4	0.90-1.05	225
1131	45.75	0.650	4	0.90-1.05	481
1086	44.56	0.650	4	0.90-1.05	851
1080	45.19	0.650	4	0.90-1.05	903
1225	46.36	0.650	4	0.90-1.05	175
1131	47.33	0.650	4	0.90-1.05	472
1095	47.88	0.650	4	0.90-1.05	731
1045	48.47	0.650	4	0.90-1.05	1303
1176	42.78	0.650	4	0.90-1.05	332
1022	48.5	0.650	4	0.90-1.05	1840
1257	46.82	0.650	4	0.90-1.05	132
1281	45.44	0.650	4	0.90-1.05	103
1287	18.54	0.667	4	1	234
1208	18.58	0.667	4	1	531
1314	17.84	0.667	4	1	178
1167	19.06	0.667	4	1	770
1140	43.37	0.667	4	1	509
1223	43.55	0.667	4	1	216
1089	43.3	0.667	4	1	848

T _s [K]	P _s [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign Time [μs]
tert-Butanol					
1464	1.47	0.667	4	1	635
1507	1.45	0.667	4	1	370
1540	1.44	0.667	4	1	263
1615	1.39	0.667	4	1	126
1425	1.55	0.667	4	1	995
1593	1.45	0.667	4	1	141
1453	1.67	0.667	4	1	625
1405	1.70	0.667	4	1	1135
1458	1.48	0.667	4	1	640
1534	1.50	0.667	4	1	254
1626	1.59	0.667	4	1	106
1447	1.61	0.667	4	1	673
1503	3.06	0.667	4	1	225
1487	3.10	0.667	4	1	277
1504	3.02	0.667	4	1	248
1513	3.03	0.667	4	1	220
1516	3.00	0.667	4	1	212
1578	3.14	0.667	4	1	106
1469	3.07	0.667	4	1	358
1511	2.97	0.667	4	1	239
1516	2.96	0.667	4	1	213
1527	2.98	0.667	4	1	201
1539	3.01	0.667	4	1	161
1489	3.19	0.667	4	1	278
1451	3.24	0.667	4	1	426
1395	3.17	0.667	4	1	804
1428	3.24	0.667	4	1	532
1385	3.19	0.667	4	1	853
1405	3.31	0.667	4	1	690
1395	3.33	0.667	4	1	784
1459	1.60	0.333	4	0.5	347
1389	1.67	0.333	4	0.5	756
1497	1.55	0.333	4	0.5	212
1562	1.50	0.333	4	0.5	105
1479	1.02	0.500	3	1	756
1515	1.02	0.500	3	1	472
1538	0.99	0.500	3	1	352
1432	1.36	0.500	3	1	1112
1453	1.35	0.500	3	1	893
1463	1.3	0.500	3	1	819
1510	1.28	0.500	3	1	453
1563	1.26	0.500	3	1	226
1568	1.21	0.500	3	1	234

1656	1.26	0.500	3	1	102
1377	18.00	0.567	4	0.75-0.95	247
1310	18.58	0.567	4	0.75-0.95	515
1276	18.97	0.567	4	0.75-0.95	847
1442	18.29	0.567	4	0.75-0.95	123
1316	17.7	0.567	4	0.75-0.95	533
1270	19.72	0.567	4	0.75-0.95	853
1240	19.62	0.567	4	0.75-0.95	1214
1319	18.25	0.567	4	0.75-0.95	498
1225	19.68	0.567	4	0.75-0.95	1462
1471	17.62	0.567	4	0.75-0.95	84
1395	17.81	0.567	4	0.75-0.95	207
1469	17.91	0.567	4	0.75-0.95	87
1205	46.15	0.667	4	0.95-1.05	997
1181	46.69	0.667	4	0.95-1.05	1241
1267	46.81	0.667	4	0.95-1.05	427
1249	44.00	0.667	4	0.95-1.05	575
1310	45.84	0.667	4	0.95-1.05	264
1244	47.14	0.667	4	0.95-1.05	575
1286	42.97	0.667	4	0.95-1.05	377
1329	44.50	0.667	4	0.95-1.05	221
1336	43.60	0.667	4	0.95-1.05	212
1223	44.44	0.667	4	0.95-1.05	768
1300	43.59	0.667	4	0.95-1.05	302
1403	44.92	0.667	4	0.95-1.05	91
1374	45.21	0.667	4	0.95-1.05	124
1533	17.29	0.667	4	1	52
1388	17.20	0.667	4	1	252
1317	18.52	0.667	4	1	545
1464	16.79	0.667	4	1	111
1221	42.28	0.667	4	1	810
1322	41.78	0.667	4	1	269
1288	41.61	0.667	4	1	382
1345	39.93	0.667	4	1	218

Large Methyl Esters

Literature Source of Data:

M. F. Campbell, D. F. Davidson, R. K. Hanson, C. K. Westbrook *"Ignition Delay Times of Methyl Oleate and Methyl Linoleate Behind Reflected Shock Waves,"* Proceedings of the Combustion Institute 34 (2013) 419-425.

M. F. Campbell, D. F. Davidson, R. K. Hanson, *"Ignition Delay Times of Very-Low-Vapor-Pressure Biodiesel Surrogates,"* Fuel 126 (2014) 271-281.

Range of Data:

Temperature [K]	1100	1400
Pressure [atm]	3	8
Fuel Mole Fraction [%]	0.05	0.7
Oxygen Mole Fraction [%]	1	21
Equivalence Ratio	0.3	1.5

Type of Data:

Large Methyl Esters Table 1: Ignition delay time measurements in argon using sidewall pressure, and confirmed with CH* emission near 430 nm, from **Aerosol Shock** Tube experiments for 7 methyl esters. Equivalence ratios are based on the total oxygen content of the gaseous mixture, including the two O-atoms in the FAME.

MO = methyl oleate $C_{19}H_{36}O_2$

ML = methyl linoleate $C_{19}H_{34}O_2$

MD = methyl decanoate $C_{11}H_{22}O_2$

MLA = methyl laurate $C_{13}H_{26}O_2$

MM = methyl myristate $C_{15}H_{30}O_2$

MP = methyl palmitate $C_{17}H_{34}O_2$

MOB = methyl oleate blend = 70% MO, 30% smaller FAMEs

Large Methyl Esters Table 1:

Fuel	T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
ML	1222	6.71	0.170	4.00	1.13	782
ML	1194	6.82	0.168	4.00	1.11	1066
ML	1147	6.72	0.191	4.00	1.25	1797
ML	1143	7.02	0.211	4.00	1.38	1976
ML	1268	6.95	0.222	4.00	1.45	613
ML	1300	6.92	0.213	4.00	1.39	414
ML	1319	6.71	0.214	4.00	1.40	334
ML	1378	6.68	0.213	4.00	1.40	185
ML	1325	6.53	0.221	4.00	1.44	272
ML	1258	6.87	0.202	4.00	1.33	571
ML	1276	6.78	0.128	4.00	0.86	376
ML	1239	6.82	0.112	4.00	0.75	684
ML	1192	6.82	0.110	4.00	0.74	1028
ML	1163	6.93	0.115	4.00	0.77	1232
ML	1141	6.94	0.109	4.00	0.73	2071
ML	1330	6.73	0.104	4.00	0.70	205
ML	1328	6.59	0.101	4.00	0.68	199
ML	1188	7.19	0.313	4.00	2.00	1020
ML	1169	7.25	0.309	4.00	1.98	1369
ML	1219	7.01	0.290	4.00	1.87	833
ML	1239	7.07	0.331	4.00	2.11	788
ML	1128	7.15	0.297	4.00	1.91	2032
ML	1290	6.87	0.372	4.00	2.35	458
ML	1324	7.04	0.364	4.00	2.30	364
ML	1332	6.84	0.353	4.00	2.24	272
ML	1327	6.56	0.119	4.00	0.79	190
ML	1334	6.33	0.112	4.00	0.75	208
ML	1209	6.70	0.097	4.00	0.65	831
ML	1233	6.59	0.116	4.00	0.78	574
ML	1179	3.88	0.293	4.00	1.88	2168
ML	1242	3.63	0.226	4.00	1.47	935
ML	1233	3.68	0.218	4.00	1.43	987
ML	1185	3.69	0.239	4.00	1.55	2465
ML	1320	3.45	0.202	4.00	1.33	378
ML	1297	3.40	0.161	4.00	1.07	445
ML	1350	3.27	0.162	4.00	1.07	246
ML	1334	3.49	0.236	4.00	1.54	371
ML	1258	3.59	0.188	4.00	1.23	746

Fuel	T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
MO	1242	6.74	0.171	4.00	1.15	556
MO	1248	6.78	0.174	4.00	1.17	575
MO	1187	6.98	0.322	4.00	2.09	1139
MO	1320	6.69	0.175	4.00	1.18	285
MO	1284	6.90	0.160	4.00	1.08	381
MO	1240	6.69	0.232	4.00	1.54	622
MO	1256	6.66	0.221	4.00	1.47	492
MO	1339	6.62	0.195	4.00	1.30	232
MO	1212	6.95	0.211	4.00	1.41	840
MO	1314	6.84	0.206	4.00	1.38	319
MO	1189	6.87	0.200	4.00	1.34	1110
MO	1162	6.75	0.192	4.00	1.29	1296
MO	1182	7.00	0.189	4.00	1.27	1409
MO	1137	6.75	0.179	4.00	1.20	2137
MO	1332	6.47	0.219	4.00	1.46	233
MO	1314	6.68	0.117	4.00	0.80	193
MO	1242	6.84	0.104	4.00	0.71	544
MO	1188	6.78	0.102	4.00	0.70	869
MO	1152	6.70	0.090	4.00	0.62	1519
MO	1316	6.84	0.345	4.00	2.23	417
MO	1229	6.82	0.310	4.00	2.02	730
MO	1203	6.97	0.296	4.00	1.93	998
MO	1159	7.25	0.373	4.00	2.39	1372
MO	1131	6.94	0.257	4.00	1.69	2533
MO	1115	6.73	0.260	4.00	1.71	2187
MO	1143	6.84	0.284	4.00	1.86	2006
MO	1276	6.82	0.325	4.00	2.11	552
MO	1286	3.48	0.286	4.00	1.87	601
MO	1370	3.42	0.212	4.00	1.41	196
MO	1277	3.54	0.202	4.00	1.35	645
MO	1196	3.50	0.195	4.00	1.30	2129
MO	1220	3.47	0.206	4.00	1.38	1311
MO	1337	3.42	0.233	4.00	1.54	404
MO	1272	6.48	0.112	4.00	0.76	334
MO	1160	6.76	0.110	4.00	0.75	1625
MO	1316	6.59	0.322	4.00	2.10	353
MO	1301	6.74	0.304	4.00	1.98	381
MO	1257	6.72	0.308	4.00	2.01	644
MO	1277	6.67	0.300	4.00	1.96	528
MO	1255	3.43	0.219	4.00	1.46	728
MO	1231	3.40	0.188	4.00	1.26	1367
MO	1320	3.32	0.198	4.00	1.33	432
MO	1355	3.37	0.201	4.00	1.34	216

Fuel	T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
MP	1242	6.32	0.046	4	0.29	467
MP	1305	6.43	0.042	4	0.27	221
MP	1311	6.60	0.048	4	0.30	269
MP	1273	6.56	0.056	4	0.35	378
MP	1267	6.59	0.057	4	0.36	361
MP	1247	6.64	0.056	4	0.35	503
MP	1213	6.55	0.054	4	0.34	705
MP	1220	6.65	0.048	4	0.31	646
MP	1205	6.66	0.057	4	0.36	786
MP	1191	6.73	0.047	4	0.30	962
MP	1180	3.31	0.126	4	0.78	2199
MP	1240	3.36	0.098	4	0.61	627
MP	1228	3.42	0.109	4	0.68	1037
MP	1257	3.38	0.117	4	0.73	592
MP	1251	3.21	0.109	4	0.68	481
MP	1280	3.30	0.130	4	0.81	394
MP	1309	3.34	0.111	4	0.69	245
MP	1238	3.51	0.114	4	0.71	740
MP	1196	3.44	0.110	4	0.69	1535

Fuel	T _s [K]	P _s [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
MLA	1194	6.61	0.27	4	1.24	1660
MLA	1230	6.65	0.27	4	1.24	970
MLA	1247	6.64	0.286	4	1.30	747
MLA	1249	6.47	0.273	4	1.25	749
MLA	1197	6.64	0.267	4	1.22	1159
MLA	1171	6.63	0.298	4	1.36	1707
MLA	1281	6.63	0.295	4	1.34	526
MLA	1309	6.65	0.295	4	1.35	483
MLA	1332	6.75	0.312	4	1.41	407
MLA	1354	6.69	0.304	4	1.38	284
MLA	1332	6.60	0.27	4	1.24	328
MLA	1264	6.42	0.171	4	0.80	599
MLA	1250	6.56	0.141	4	0.67	754
MLA	1218	6.65	0.156	4	0.74	1105
MLA	1163	6.70	0.166	4	0.78	2139
MLA	1321	6.67	0.172	4	0.80	253
MLA	1287	6.51	0.159	4	0.75	391
MLA	1300	6.66	0.172	4	0.81	369
MLA	1179	6.69	0.177	4	0.83	1768
MLA	1217	3.59	0.266	4	1.22	1352
MLA	1254	3.54	0.22	4	1.02	1166
MLA	1282	3.43	0.203	4	0.94	509
MLA	1253	3.51	0.262	4	1.20	1065
MLA	1278	3.54	0.262	4	1.20	608
MLA	1241	6.73	0.159	4	0.75	736
MLA	1274	3.43	0.265	4	1.21	592
MLA	1297	3.51	0.316	4	1.43	605
MLA	1351	3.61	0.318	4	1.44	272

Fuel	T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
MM	1212	6.67	0.199	4	1.07	1016
MM	1214	6.71	0.182	4	0.98	1013
MM	1275	6.66	0.163	4	0.88	379
MM	1241	6.69	0.115	4	0.63	653
MM	1222	6.68	0.097	4	0.54	732
MM	1192	6.72	0.119	4	0.65	1606
MM	1194	6.77	0.156	4	0.85	1085
MM	1198	6.60	0.125	4	0.68	999
MM	1162	6.68	0.127	4	0.69	1807
MM	1221	6.64	0.134	4	0.73	914
MM	1292	6.71	0.151	4	0.82	387
MM	1255	6.62	0.128	4	0.70	410
MM	1257	6.54	0.136	4	0.74	509
MM	1278	6.45	0.151	4	0.82	353
MM	1241	6.82	0.172	4	0.93	679
MM	1214	6.71	0.219	4	1.17	898
MM	1251	7.00	0.246	4	1.31	828
MM	1258	6.84	0.239	4	1.27	654
MM	1254	6.55	0.224	4	1.20	629
MM	1295	6.86	0.251	4	1.33	388
MM	1270	6.47	0.255	4	1.35	519
MM	1322	6.69	0.250	4	1.33	362
MM	1357	6.61	0.206	4	1.11	203
MM	1317	6.61	0.125	4	0.68	209
MM	1214	6.71	0.204	4	1.09	790
MM	1194	6.66	0.201	4	1.08	1188
MM	1195	6.76	0.186	4	1.00	1293
MM	1162	6.71	0.214	4	1.15	1987
MM	1231	3.37	0.136	4	0.74	1101
MM	1297	3.34	0.160	4	0.87	489
MM	1313	3.27	0.125	4	0.68	291
MM	1312	3.25	0.137	4	0.74	391
MM	1260	3.27	0.144	4	0.78	536
MM	1244	3.33	0.167	4	0.90	794
MM	1225	3.29	0.153	4	0.83	935
MM	1207	3.28	0.149	4	0.81	1552
MM	1332	3.23	0.130	4	0.71	219
MM	1334	3.27	0.127	4	0.69	208

Fuel	T _s [K]	P _s [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
MOB	1205	6.57	0.050	4	0.32	827
MOB	1158	6.62	0.065	4	0.42	1653
MOB	1257	6.63	0.065	4	0.42	512
MOB	1209	6.55	0.086	4	0.55	907
MOB	1176	6.63	0.092	4	0.59	1285
MOB	1170	6.68	0.085	4	0.54	1377
MOB	1280	6.85	0.127	4	0.80	443
MOB	1313	6.97	0.158	4	0.99	321
MOB	1338	6.78	0.169	4	1.05	248
MOB	1338	6.50	0.183	4	1.14	267
MOB	1360	6.51	0.224	4	1.38	217
MOB	1242	6.58	0.078	4	0.50	650
MOB	1141	6.78	0.186	4	1.16	1768
MOB	1212	7.07	0.138	4	0.87	925
MOB	1210	6.95	0.186	4	1.16	874
MOB	1331	6.82	0.101	4	0.64	216
MOB	1345	6.55	0.109	4	0.69	192
MOB	1286	6.82	0.086	4	0.55	354
MOB	1200	6.88	0.188	4	1.17	1013
MOB	1271	6.74	0.173	4	1.08	530
MOB	1314	6.67	0.211	4	1.30	325
MOB	1306	6.77	0.181	4	1.13	395
MOB	1235	6.91	0.207	4	1.28	754
MOB	1191	6.88	0.172	4	1.07	1193
MOB	1234	6.73	0.176	4	1.10	705
MOB	1309	6.64	0.231	4	1.42	388
MOB	1175	6.86	0.188	4	1.17	1204
MOB	1181	6.97	0.213	4	1.32	1360
MOB	1234	6.91	0.100	4	0.63	837
MOB	1256	6.88	0.107	4	0.68	642
MOB	1288	6.59	0.105	4	0.67	440
MOB	1358	6.54	0.067	4	0.43	166
MOB	1225	6.92	0.098	4	0.62	836

Fuel	T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
MD	1335	6.64	0.042	1	0.66	494
MD	1368	6.56	0.042	1	0.67	214
MD	1388	6.42	0.052	1	0.81	225
MD	1331	6.86	0.046	1	0.72	559
MD	1278	6.74	0.041	1	0.66	673
MD	1262	6.93	0.041	1	0.65	1077
MD	1213	7.06	0.462	21	0.36	285
MD	1169	7.01	0.372	21	0.29	642
MD	1102	7.12	0.427	21	0.33	1259
MD	1059	7.22	0.460	21	0.36	2226
MD	1162	7.09	0.515	21	0.40	621
MD	1113	7.07	0.468	21	0.36	995
MD	1069	7.14	0.779	21	0.60	1164
MD	1026	7.15	0.725	21	0.55	1944
MD	1062	7.20	0.740	21	0.57	1343

Nitrogen-containing Fuels

TMEDA (Tetramethylethylenediamine)

Literature Source of Data:

D. F. Davidson, R. K. Hanson, Unpublished data from "Spray and Combustion of Gelled Hypergolic Propellants for Future Rocket and Missile Engines," ARO/MURI Contract No. W911NF-08-1-0124, Dr. R. Anthenien: Contract Monitor, Mechanical Engineering Department, Stanford University (August 2011).

Range of Data:

Temperature [K]	1051	1301
Pressure [atm]	14.25	41.76
Fuel Mole Fraction [%]	0.2	0.4
Oxygen Mole Fraction [%]	4	4
Equivalence Ratio	0.5	2.0

Type of Data:

TMEDA Table 1: Ignition delay time measurement from using sidewall PZT pressure and CH* emission near 431 nm.

TMEDA Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1204	31.84	0.4	4	1	205
1133	33.85	0.4	4	1	416
1194	38.71	0.4	4	1	201
1249	41.76	0.4	4	1	98
1077	15.63	0.4	4	1	1339
1160	14.81	0.4	4	1	534
1278	14.50	0.4	4	1	122
1107	14.80	0.4	4	1	965
1233	14.94	0.4	4	1	216
1207	15.19	0.4	4	1	306
1055	15.41	0.4	4	1	1787
1243	13.52	0.4	4	1	197
1203	15.11	0.2	4	0.5	322
1137	15.31	0.2	4	0.5	855
1145	15.30	0.2	4	0.5	776
1186	15.09	0.2	4	0.5	442
1199	15.20	0.2	4	0.5	335
1301	14.25	0.2	4	0.5	89
1272	14.41	0.2	4	0.5	134
1221	14.80	0.2	4	0.5	283
1217	14.79	0.4	4	2	281
1164	15.31	0.4	4	2	456
1129	15.70	0.4	4	2	638
1290	14.88	0.4	4	2	117
1051	15.39	0.4	4	2	1341
1053	14.88	0.4	4	2	1318
1075	15.14	0.4	4	2	1007

MMH (Monomethylhydrazine)

Literature Source of Data:

D. F. Davidson, R. D. Cook, S. H. Pyun, R. K. Hanson, "MMH Pyrolysis and Oxidation: Species Time-History Measurements behind Reflected Shock Waves," Paper #9, 23rd ICDERS Meeting July 24-29, 2011 Irvine, CA.

R. D. Cook, D. F. Davidson, R. K. Hanson, "Shock Tube Measurements of Species Time-Histories in Monomethyl Hydrazine," In preparation for submission to Combustion and Flame.

Range of Data:

Temperature [K]	1136	1287
Pressure [atm]	1.5	1.6
Fuel Mole Fraction [%]	1	1
Oxygen Mole Fraction [%]	2.5	2.5
Equivalence Ratio	1	1

Type of Data:

MMH Table 1: Ignition delay time measurement in argon from using species time-history measurements of NH₂ (time to complete removal) OH (time to 50% of peak), NH₃ (time to complete removal), and CH₄ (time to complete removal).

MMH Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Species	Ign. Time [μs]
1136	1.63	1	2.5	1	NH ₂	1790
1200	1.59	1	2.5	1	NH ₂	975
1287	1.51	1	2.5	1	NH ₂	416
1156	1.66	1	2.5	1	OH	1313
1180	1.59	1	2.5	1	OH	1081
1279	1.48	1	2.5	1	OH	406
1147	1.5	1	2.5	1	CH ₄	1410
1203	1.5	1	2.5	1	CH ₄	950
1292	1.5	1	2.5	1	CH ₄	301
1239	1.61	1	2.5	1	NH ₃	1087

Morpholine

Literature Source of Data:

S. Li, D. F. Davidson, R. K. Hanson, N. J. Labbe, P. R. Westmoreland, P. Oßwald, K. Kohse-Höinghaus, "*Shock Tube Measurements and Model Development for Morpholine Pyrolysis and Oxidation at High Pressures*," Combustion and Flame 160 (2013) 1559-1571.

Range of Data:

Temperature [K]	910	1197
Pressure [atm]	12.5	29.2
Fuel Mole Fraction [%]	0.70	3.65
Oxygen Mole Fraction [%]	4.0	21
Equivalence Ratio	0.5	2

Type of Data:

Morpholine Table 1: Ignition delay time measurement in argon from using sidewall PZT pressure. The six points between 866 and 899 K were not published in the paper and have significantly larger dP_5/dt values.

Morpholine Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
910	14.71	3.65	21	1.0	1350
915	15.76	3.65	21	1.0	1130
925	15.83	3.65	21	1.0	1070
938	14.66	3.65	21	1.0	1000
966	14.79	3.65	21	1.0	636
1009	14.04	3.65	21	1.0	374
1062	13.77	3.65	21	1.0	197
1097	13.02	3.65	21	1.0	115
921	13.79	1.83	21	0.5	1970
983	13.94	1.83	21	0.5	1030
1023	13.37	1.83	21	0.5	547
1110	13.19	1.83	21	0.5	192
1168	12.53	1.83	21	0.5	97
875	16.45	7.30	21	2.0	969
901	16.60	7.30	21	2.0	775
935	16.29	7.30	21	2.0	562
965	15.25	7.30	21	2.0	402
994	16.13	7.30	21	2.0	223
1041	15.08	7.30	21	2.0	136
1027	16.50	3.65	4	1.0	1150
1042	16.64	0.70	4	1.0	845
1091	16.30	0.70	4	1.0	430
1139	15.94	0.70	4	1.0	244
1197	15.56	0.70	4	1.0	145
932	25.21	3.65	21	1.0	625
955	29.18	3.65	21	1.0	389
986	26.70	3.65	21	1.0	279
994	27.02	3.65	21	1.0	241
1046	27.05	3.65	21	1.0	117
1074	26.36	3.65	21	1.0	82
866	14.64	7.30	21	0.5	5680
869	13.76	7.30	21	0.5	4600
870	15.42	7.30	21	0.5	5480
878	14.66	7.30	21	0.5	4270
882	14.97	7.30	21	0.5	3260
899	14.70	7.30	21	0.5	2310

Dimethylamine

Literature Source of Data:

S. Li, D. F. Davidson, R. K. Hanson, “*Shock Tube Study of Dimethylamine Oxidation*,” Submitted for publication, Proceedings of the Combustion Institute 35, November 2013.

Range of Data:

Temperature [K]	1211	1504
Pressure [atm]	0.82	2.97
Fuel Mole Fraction [%]	500ppm	2.13
Oxygen Mole Fraction [%]	0.1875	4
Equivalence Ratio	0.5	2.0

Type of Data:

Dimethylamine Table 1: Ignition delay time measurement in argon using sidewall PZT pressure. The ignition delay times of the two low-concentration points (1417 and 1504 K) are derived from OH species time-history measurements (time to 50% of peak).

Dimethylamine Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1371	0.88	1.067	4	1	485
1450	0.84	1.067	4	1	234
1282	0.93	1.067	4	1	1314
1493	0.82	1.067	4	1	153
1398	1.10	1.067	4	1	308
1457	1.39	1.067	4	1	146
1383	1.53	1.067	4	1	300
1232	1.49	1.067	4	1	1439
1244	1.49	1.067	4	1	1255
1300	1.50	1.067	4	1	650
1352	1.69	1.067	4	1	343
1459	2.57	1.067	4	1	94
1408	2.71	1.067	4	1	150
1334	2.86	1.067	4	1	327
1270	2.96	1.067	4	1	667
1211	2.97	1.067	4	1	1192
1267	1.61	0.533	4	0.5	854
1181	1.71	0.533	4	0.5	2152
1363	1.54	0.533	4	0.5	277
1433	1.50	0.533	4	0.5	131
1296	1.52	2.13	4	2	986
1222	1.56	2.13	4	2	1786
1346	1.43	2.13	4	2	694
1440	1.42	2.13	4	2	304
1498	1.40	2.13	4	2	195
1417	2.20	500ppm	0.1875	1	516
1504	2.10	500ppm	0.1875	1	249

Ethylamine

Literature Source of Data:

S. Li, D. F. Davidson, R. K. Hanson, “*Shock Tube Study of Ethylamine Pyrolysis and Oxidation*,” Paper 07RK-0075, 8th U.S. Combustion Meeting, University of Utah, May 19-22, 2013.

S. Li, D. F. Davidson, R. K. Hanson, “*Shock Tube Study of Ethylamine Pyrolysis and Oxidation*,” Submitted to Combustion and Flame, November 2013.

Range of Data:

Temperature [K]	1200	1448
Pressure [atm]	0.85	2.0
Fuel Mole Fraction [%]		
Oxygen Mole Fraction [%]	0.2	4.0
Equivalence Ratio	0.75	1.25

Type of Data:

Ethylamine Table 1: Ignition delay time measurement in argon using sidewall PZT pressure. The ignition delay times based on the species time-history measurements are derived from NH₂ time to complete removal and OH time to 50% of peak.

Ethylamine Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	Ign. Time [μs]
1273	1.57	1.33	3.975	1.25	1425
1272	1.32	1.33	3.975	1.25	1458
1395	1.31	1.33	3.975	1.25	415
1344	1.28	1.33	3.975	1.25	709
1221	1.38	1.33	3.975	1.25	2685
1428	1.46	0.80	3.975	0.75	177
1279	1.37	0.80	3.975	0.75	1057
1179	1.33	0.80	3.975	0.75	5762
1267	1.20	0.80	3.975	0.75	1838
1282	1.29	0.80	3.975	0.75	1304
1341	1.24	0.80	3.975	0.75	703
1382	1.22	0.80	3.975	0.75	347
1373	1.53	1.06	3.975	1.0	408
1285	1.45	1.06	3.975	1.0	1210
1258	1.48	1.06	3.975	1.0	1597
1309	1.36	1.06	3.975	1.0	769
1305	1.35	1.06	3.975	1.0	850
1324	1.29	1.06	3.975	1.0	777
1448	1.49	1.06	3.975	1.0	187
1301	0.87	1.06	3.975	1.0	1612
1349	0.88	1.06	3.975	1.0	867
1399	0.86	1.06	3.975	1.0	514
1437	0.85	1.06	3.975	1.0	284
1269	1.99	1.06	3.975	1.0	936
1252	1.91	1.06	3.975	1.0	1190
1322	1.88	1.06	3.975	1.0	529
1413	2.06	1.06	3.975	1.0	221
1259	2.02	1.06	3.975	1.0	1120
1217	1.97	1.06	3.975	1.0	1900
1323	1.93	1.06	3.975	1.0	602
1558	2.05	0.21	0.8	1.0	149
1487	2.07	0.21	0.8	1.0	261
1441	2.09	0.21	0.8	1.0	396
1399	1.93	0.05	0.2	1.0	756
1503	1.99	0.05	0.2	1.0	281
1546	2.10	0.05	0.2	1.0	211
1593	2.07	0.05	0.2	1.0	139

Ethylamine Table 1 continued:

NH₂ Species Time-Histories					
1558	2.05	2000	0.80	0.81	149
1487	2.07	2000	0.80	0.81	258
1441	2.09	2000	0.80	0.81	390
OH Species Time-Histories					
1593	2.1	500	0.1625	1	159
1503	2.0	500	0.1625	1	362
1399	1.9	500	0.1625	1	1042

IDT Measurements up to June 2019

Small Fuels

Hydrogen Impurities

J. Urzay, N. Kseib, D.F. Davidson, G. Iaccarino and R.K. Hanson, "Uncertainty-quantification Analysis of the Effects of Residual Impurities on Hydrogen-Oxygen Ignition in Shock Tubes," *Comb. and Flame* **161** 1-15 (2014) [DOI 10.1016/j.combustflame.2013.08.012](https://doi.org/10.1016/j.combustflame.2013.08.012)

This paper is a review of impurities in shock tube experiments and provides an estimate of the level of shock tube impurities found in most of the experiments performed at Stanford.

Hydrogen

Literature Source of Data:

J. Shao, R. Choudhary, A. Susa, D. F. Davidson, R. K. Hanson, "Shock Tube Study of the Rate Constants for $H + O_2 + M \rightarrow HO_2 + M$ ($M = Ar, H_2O, CO_2, N_2$) at Elevated Pressures," *Proceedings of the Combustion Institute* **37** 145-152 (2019) DOI [10.1016/j.proci.2018.05.077](https://doi.org/10.1016/j.proci.2018.05.077)

Range of Data:

Temperature [K]	91179	1376
Pressure [atm]	11	33
Fuel Mole Fraction [%]	3	3
Oxygen Mole Fraction [%]	1.5	1.5
Equivalence Ratio	1	1

Type of Data:

Hydrogen Table 1: Ignition delay time measurement using sidewall pressure. Data was used to determine the rate of $H+O_2+M=HO_2+M$. Data tables are taken from the Supplementary material in this paper.

Hydrogen Table 1:

Table S1: Measured reaction rate coefficients for $M=Ar$

Mixture	Temperature (K)	Pressure (atm)	IDT (μm)
3%H ₂ /1.5%O ₂ /Ar	1249	32.5	77
3%H ₂ /1.5%O ₂ /Ar	1240	32.4	101
3%H ₂ /1.5%O ₂ /Ar	1255	31	60
3%H ₂ /1.5%O ₂ /Ar	1262	31.5	47
3%H ₂ /1.5%O ₂ /Ar	1184	16.8	133
3%H ₂ /1.5%O ₂ /Ar	1196	16.9	90
3%H ₂ /1.5%O ₂ /Ar	1212	16.9	59
3%H ₂ /1.5%O ₂ /Ar	1196	17.4	97
3%H ₂ /1.5%O ₂ /Ar	1224	16.8	48
3%H ₂ /1.5%O ₂ /Ar	1214	17.1	60
3%H ₂ /1.5%O ₂ /Ar	1230	28.7	102

Table S2. Measured reaction rate coefficients for M=N₂

Mixture	Temperature (K)	Pressure (atm)	IDT (μs)
3%H ₂ /1.5%O ₂ /N ₂	1228	12.6	64
3%H ₂ /1.5%O ₂ /N ₂	1201	12.7	126
3%H ₂ /1.5%O ₂ /N ₂	1199	13	132
3%H ₂ /1.5%O ₂ /N ₂	1225	12.6	69
3%H ₂ /1.5%O ₂ /N ₂	1201	13.1	125

Table S3. Measured reaction rate coefficients for M=H₂O

Mixture	Temperature (K)	Pressure (atm)	IDT (μm)
3%H ₂ /1.5%O ₂ /4.8%H ₂ O/Ar	1264	15.7	86
3%H ₂ /1.5%O ₂ /3.9%H ₂ O/Ar	1307	16.6	26
3%H ₂ /1.5%O ₂ /4.4%H ₂ O/Ar	1264	16.8	87
3%H ₂ /1.5%O ₂ /9.7%H ₂ O/Ar	1288	16	105
3%H ₂ /1.5%O ₂ /8.6%H ₂ O/Ar	1270	16.4	168
3%H ₂ /1.5%O ₂ /9.1%H ₂ O/Ar	1344	15.5	32
3%H ₂ /1.5%O ₂ /8.7%H ₂ O/Ar	1315	15.4	60
3%H ₂ /1.5%O ₂ /9.0%H ₂ O/Ar	1329	15	48
3%H ₂ /1.5%O ₂ /9.1%H ₂ O/Ar	1320.5	15.6	58
3%H ₂ /1.5%O ₂ /13.8%H ₂ O/Ar	1376	15.1	37
3%H ₂ /1.5%O ₂ /13.0%H ₂ O/Ar	1282	15.5	195

Table S4. Measured reaction rate coefficients for M=CO₂

Mixture	Temperature (K)	Pressure (atm)	IDT (μm)
3%H ₂ /1.5%O ₂ /20%CO ₂ /Ar	1223	19	217
3%H ₂ /1.5%O ₂ /20%CO ₂ /Ar	1267.5	18.5	93
3%H ₂ /1.5%O ₂ /20%CO ₂ /Ar	1203	17.4	281
3%H ₂ /1.5%O ₂ /20%CO ₂ /Ar	1235	18.4	178
3%H ₂ /1.5%O ₂ /20%CO ₂ /Ar	1189	11.9	221
3%H ₂ /1.5%O ₂ /20%CO ₂ /Ar	1185.5	12.3	236
3%H ₂ /1.5%O ₂ /20%CO ₂ /Ar	1191	11.5	210
3%H ₂ /1.5%O ₂ /20%CO ₂ /Ar	1228	11.7	118
3%H ₂ /1.5%O ₂ /20%CO ₂ /Ar	1185	12.1	234
3%H ₂ /1.5%O ₂ /20%CO ₂ /Ar	1224	16.4	191
3%H ₂ /1.5%O ₂ /20%CO ₂ /Ar	1238	16.1	144
3%H ₂ /1.5%O ₂ /10%CO ₂ /Ar	1180	11.7	189
3%H ₂ /1.5%O ₂ /10%CO ₂ /Ar	1179	11.3	173

Hydrogen

Literature Source of Data:

J. Shao, R. Choudhary, D. F. Davidson and R. K. Hanson, "Ignition Delay Times of Methane and Hydrogen Highly Diluted in Carbon Dioxide at High Pressures up to 300 atm," *Proceedings of the Combustion Institute* **37** 4555-4562 (2019) DOI [10.1016/j.proci.2018.08.002](https://doi.org/10.1016/j.proci.2018.08.002)

J.K. Shao, D.F. Davidson and R.K Hanson, "Ignition Delay Times of Methane and Hydrogen Highly Diluted in Carbon Dioxide," International Symposium on Shock Waves 31, Nagoya, Japan, July 2017.

Range of Data:

Temperature [K]	1083	1291
Pressure [atm]	37	311
Fuel Mole Fraction [%]	5	10
Oxygen Mole Fraction [%]	10	5
Equivalence Ratio	0.25	1

Type of Data:

Hydrogen Table 2: Ignition delay time measurement using sidewall pressure, and OH* emission near 306 nm. Lower pressure data is from University of Central Florida. Data tables are from the supplementary material in the paper.

Hydrogen Table 2:

10%H₂/5%O₂/balance CO₂ phi =1

Data from Stanford University

T	1000/T	P	IDT	Phase
K	1/K	Atm	μσ	
1182	0.846	110.92	176	S
1291	0.775	103.4	36	S
1161	0.861	111	227	S
1154	0.867	113	258	S
1087	0.92	275.4	446	S
1143	0.875	311	135	S
1141	0.876	244.5	147	S
1083	0.923	251.4	366	S

5%H₂/10%O₂/balance CO₂ phi =0.25

Data from University of Central Florida

T	1000/T	P	IDT	Phase
K	1/K	Atm	μσ	
1201	0.83264	39.287	294	G
1221	0.819	37.055	216	G
1270	0.7874	37.587	90	G
1170	0.8547	40.067	354	G
1224	0.81699	38.204	180	G

* G represents gas; S represents super-critical fluid;

Methane

Literature Source of Data:

J. Shao, R. Choudhary, D. F. Davidson and R. K. Hanson, "Ignition Delay Times of Methane and Hydrogen Highly Diluted in Carbon Dioxide at High Pressures up to 300 atm," *Proceedings of the Combustion Institute* **37** 4555-4562 (2019) DOI [10.1016/j.proci.2018.08.002](https://doi.org/10.1016/j.proci.2018.08.002)

J.K. Shao, D.F. Davidson and R.K Hanson, "Ignition Delay Times of Methane and Hydrogen Highly Diluted in Carbon Dioxide," International Symposium on Shock Waves 31, Nagoya, Japan, July 2017.

Range of Data:

Temperature [K]	1055	1578
Pressure [atm]	27	245
Fuel Mole Fraction [%]	3.91	7.5
Oxygen Mole Fraction [%]	9.92	15
Equivalence Ratio	1	1

Type of Data:

Methane Table 1: Ignition delay time measurement in CO₂ using sidewall pressure, and OH* emission near 306 nm. 35 atm data is from University of Central Florida. Data tables are from the supplementary material in the paper.

Methane Table 1:

7.5%CH₄/15%O₂/balance CO₂

35 atm data is from University of Central Florida

T	1000/T	P	IDT	Phase
K	1/K	Atm	μσ	
1346	0.743	34.4	478	G
1434	0.697	35.4	202	G
1421	0.704	32.8	206	G
1374	0.728	27.1	450	G
1265	0.791	109.4	427	S
1277	0.783	111.8	367	S
1362	0.734	103.6	156	S
1411	0.709	100.5	85	S
1100	0.909	285.5	275	S
1045	0.957	259.7	636	S
1087	0.92	249.4	359	S
1055	0.948	245.2	518	S

3.91%CH₄/9.92%O₂/CO₂ (typo in E.R. in publication)

35 atm data is from University of Central Florida

T	1000/T	P	IDT	
1480	0.676	32.2	289	G
1507	0.664	27.5	189	G
1447	0.691	35.7	311	G
1578	0.634	30.3	57	G
1428	0.7	73.1	242	G
1380	0.725	74.5	408	G
1352	0.74	75.1	541	G
1453	0.688	72.2	202	G
1359	0.736	78.6	748	G
1144	0.874	262.7	673	S
1086	0.921	269.4	1098	S
1082	0.924	266.8	1098	S

* G represents gas; S represents super-critical fluid;

Methane

Literature Source of Data:

J. Shao, D.F. Davidson, R.K. Hanson, A Shock Tube Study of Ignition Delay Times in Diluted Methane, Ethylene, Propene and their Blends at Elevated Pressures,” *Fuel* **225** 370-380 (2018) DOI [10.1016/j.fuel.2018.03.146](https://doi.org/10.1016/j.fuel.2018.03.146)

Range of Data:

Temperature [K]	1420	1782
Pressure [atm]	14	52
Fuel Mole Fraction [%]	1.96	3.85
Oxygen Mole Fraction [%]	3.92	3.85
Equivalence Ratio	1	2

Type of Data:

Methane Table 2: Ignition delay time measurement in 4%O₂/argon buffer using sidewall pressure, OH* emission near 306 nm, and fuel absorption at 3.39 microns. Data tables are from the paper appendix.

Methane Table 2

Mixture #	T5 (K)	P5 (atm)	Ign_P (μs)	Ign_OH* (μs)	Ign_3.39um (μs)
<i>Mixture 1, CH₄, phi = 1</i>					
1	1752	13.5	128	108	119
1	1746	13.8	122	110	121
1	1731	13.8	144	134	147
1	1730	13.5	159	140	144
1	1730	13.7	168	143	158
1	1651	13.9	265	254	264
1	1634	14.6	298	286	295
1	1613	14.4	319	307	315
1	1616	14.6	318	309	316
1	1617	14.2	342	332	340
1	1576	14.9	416	413	436
1	1461	14.9	1494	1454	1463

Mixture #	T5 (K)	P5 (atm)	Ign_P (μ s)	Ign_OH* (μ s)	Ign_3.39um (μ s)
1	1420	15.5	2241	2260	2310
1	1663	54.2	107	93	96
1	1587	51.7	213	191	205
1	1537	57.5	286	268	277
1	1534	51.1	309	293	297
1	1507	52.9	372	347	365
1	1488	51.5	417	399	405
1	1453	53.8	565	547	563
1	1437	58.3	602	597	610
<i>Mixture 2, CH₄, phi = 2</i>					
2	1782	13.3	134	117	126
2	1782	13.3	134	117	126
2	1769	13.3	145	128	143
2	1749	13.3	180	163	170
2	1739	13.3	193	172	187
2	1728	13.4	206	189	195
2	1704	13.5	229	215	222
2	1657	13.6	307	309	316
2	1630	13.8	359	385	393
2	1627	13.8	393	392	402
2	1600	13.9	461	469	487
2	1504	14.3	1259	1205	1276
2	1481	14.4	1471	1526	1660
2	1464	14.8	1717	1731	1895

Methane/Ethylene Blend

Literature Source of Data:

J. Shao, D.F. Davidson, R.K. Hanson, A Shock Tube Study of Ignition Delay Times in Diluted Methane, Ethylene, Propene and their Blends at Elevated Pressures,” *Fuel* **225** 370-380 (2018) DOI [10.1016/j.fuel.2018.03.146](https://doi.org/10.1016/j.fuel.2018.03.146)

Range of Data:

Temperature [K]	1133	1340
Pressure [atm]	16	16
Fuel Mole Fraction [%]	1.5	1.5
Oxygen Mole Fraction [%]	3.94	3.94
Equivalence Ratio	1	1

Type of Data:

Methane/Ethylene Blend Table 1: Ignition delay time measurement in 4%O₂/argon buffer using OH* emission near 306 nm. Data table is from the paper appendix. The exact test mixture mole fractions are 0.54% CH₄, 0.96% C₂H₄, 3.94% O₂, balance argon.

Methane/Ethylene Blend Table 1

<i>Mixture 6, 36% CH₄/64% C₂H₄, phi = 1</i>			
Mix	T [K]	P [atm]	IDT [μ s]
6	1340	15.3	108
6	1276	15.2	258
6	1212	15.7	600
6	1165	16.1	1073
6	1133	16.3	1669
6	1238	15.4	425

Ethylene

Literature Source of Data:

J. Shao, D.F. Davidson, R.K. Hanson, A Shock Tube Study of Ignition Delay Times in Diluted Methane, Ethylene, Propene and their Blends at Elevated Pressures," *Fuel* **225** 370-380 (2018) DOI [10.1016/j.fuel.2018.03.146](https://doi.org/10.1016/j.fuel.2018.03.146)

Range of Data:

Temperature [K]	1095	1317
Pressure [atm]	16	60
Fuel Mole Fraction [%]	1.32	2.66
Oxygen Mole Fraction [%]	3.95	3.95
Equivalence Ratio	1	2

Type of Data:

Ethylene Table 1: Ignition delay time measurements in argon using sidewall PZT pressure and OH* emission near 306 nm for two mixtures described in the next table.

Mix #	Components	% C ₂ H ₄	% O ₂	Phi
3	C ₂ H ₄ /O ₂ /Ar	1.32	3.95	1
4	C ₂ H ₄ /O ₂ /Ar	2.66	3.95	2

Ethylene Table 1

Mixture 3, C ₂ H ₄ , phi = 1				
#	T [K]	P [atm]	IDT(P) [μs]	IDT(OH*) [μs]
3	1317	15.7		91
3	1303	15.8		116
3	1295	15.8		127
3	1275	15.5		182
3	1271	15.8	216	184
3	1269	15.8	212	193
3	1267	15.8	223	185
3	1231	16.1	332	322
3	1195	16.3	510	499

3	1186	15.9	550	540	
3	1164	16.4	704	699	
3	1137	16.6	1080	1056	
3	1132	16.7	1141	1120	
3	1311	59.7	69	61	
3	1281	59.6	107	92	
3	1261	60.1	123	121	
3	1263	60.3	135	125	
3	1230	61.7	184	177	
3	1188	61.8	319	317	
3	1156	62.7	452	454	
3	1117	61.9	705	698	
3	1109	63.0	762	755	
3	1095	63.3	913	921	
<i>Mixture 4, C₂H₄, phi = 2</i>					
4	1268	15.7	214	211	
4	1232	15.7	302	321	
4	1213	16.0	380	390	
4	1206	16.0	395	416	
4	1191	16.0	470	496	
4	1185	16.0	480	528	
4	1180	16.1	532	548	
4	1159	16.4	669	689	
4	1135	16.2	881	912	
4	1122	16.4	994	1070	

Ethylene/Propene Blends

Literature Source of Data:

J. Shao, D.F. Davidson, R.K. Hanson, A Shock Tube Study of Ignition Delay Times in Diluted Methane, Ethylene, Propene and their Blends at Elevated Pressures,” *Fuel* **225** 370-380 (2018) DOI [10.1016/j.fuel.2018.03.146](https://doi.org/10.1016/j.fuel.2018.03.146)

P. Gokulakrishnan, C.C. Fuller, M. Klassen, Y. Zhu, D.F. Davidson, R.K. Hanson and B. Kiel, “Ignition of Light Hydrocarbon Mixtures Relevant to Thermal Cracking of Jet Fuels,” AIAA Sci. & Tech. Forum, San Diego, Jan. 2016 DOI [10.2514/6.2016-0661](https://doi.org/10.2514/6.2016-0661)

Range of Data:

Temperature [K]	1206	1329
Pressure [atm]	15	15
Fuel Mole Fraction [%]	1.06	1.17
Oxygen Mole Fraction [%]	3.95	3.96
Equivalence Ratio	1	1

Type of Data:

Table 1: Ethylene/propene mixtures dilute in 4% O₂/argon. Ignition delay time measurements in argon using sidewall PZT pressure and OH* emission near 306 nm. Gokulakrishnan et al. (2016) provided a preliminary presentation of this data. Data table taken from paper Appendix A.

Ethylene/propene Blends Table 1

Mixture	T [K]	P [atm]	IDT (P) [μ s]	IDT(OH*) [μ s]
<i>Mixture 7 in paper: 25% C₂H₄/75% C₃H₆, phi = 1</i>				
7	1324	14.8	748	737
<i>Mixture 8 in paper: 50% C₂H₄/50% C₃H₆, phi = 1</i>				
8	1329	14.9	543	537
8	1212	15.6	2439	2408
<i>Mixture 9 in paper: 75% C₂H₄/25% C₃H₆, phi = 1</i>				
9	1326	14.8	335	327
9	1206	15.5	1042	1035

Propene

Literature Source of Data:

J. Shao, D.F. Davidson, R.K. Hanson, A Shock Tube Study of Ignition Delay Times in Diluted Methane, Ethylene, Propene and their Blends at Elevated Pressures,” *Fuel* **225** 370-380 (2018) DOI [10.1016/j.fuel.2018.03.146](https://doi.org/10.1016/j.fuel.2018.03.146)

Range of Data:

Temperature [K]	1255	1488
Pressure [atm]	16	16
Fuel Mole Fraction [%]	0.88	0.88
Oxygen Mole Fraction [%]	3.96	3.96
Equivalence Ratio	1	1

Type of Data:

Propene Table 1: Ignition delay time measurement in argon using side wall pressure and OH* emission near 306 nm. This data is part of a study of small alkenes.

Propene Table 1:

<i>Mixture 5, C₃H₆/4% O₂/Argon, phi = 1</i>					
Mix	T [K]	P [atm]	IDT(P) [μ s]	IDT(OH*) [μ s]	
5	1327	14.9	860	822	
5	1305	15.6	1195	1341	
5	1255	15.7	2917	3022	
5	1273	15.5	2196	2190	
5	1375	15.1	555	533	
5	1395	14.9	452	452	
5	1431	14.7	349	336	
5	1488	14.5	236	207	

Normal Alkanes

n-Heptane

Literature Source of Data:

M.F. Campbell, S. Wang, C.S. Goldenstein, R.M. Spearrin, A.M. Tulgestke, L.T. Zaczek, D.F. Davidson and R.K. Hanson, "Constrained Reaction Volume Shock Tube Study of *n*-Heptane Oxidation: Ignition Delay Times, Multi-species Time-histories and Temperature Profiles," *Proc. Comb. Inst.* **35** 231-239 (2015) [DOI 10.1016/j.proci.2014.05.001](https://doi.org/10.1016/j.proci.2014.05.001)

Range of Data:

Temperature [K]	651	823
Pressure [atm]	6.1	7.4
Fuel Mole Fraction [%]	1.02	1.02
Oxygen Mole Fraction [%]	15	15
Equivalence Ratio	0.75	0.75

Type of Data:

n-Heptane Table 1: Ignition delay time measurement in argon/5%CO₂ based on OH* emission at 306 nm and extrapolating the fuel removal processes for both 1st and 2nd stage ignition. Measurements were made using conventional and CRV filling strategies.

n-Heptane Table 1:

Temp	Press	Fuel	O2	Ar	CO2	E.R.	Filling	1st IDT	2nd IDT
K	atm	MF	MF	MF	MF			μs	μs
721	6.7	0.00986	0.1485	0.7911	0.0505	0.73	CRV	6391	24463
777	6.2	0.00988	0.1485	0.7911	0.0505	0.73	CRV	4032	20456
746	6.4	0.01069	0.1484	0.7905	0.0505	0.79	CRV	4108	27925
808	6.4	0.00974	0.1485	0.7912	0.0505	0.72	CRV	4961	23263
713	6.7	0.00993	0.1485	0.7911	0.0505	0.74	CRV	6650	27586
690	6.6	0.01005	0.1485	0.7910	0.0505	0.74	CRV	9845	29360
803	6.1	0.00968	0.1485	0.7913	0.0505	0.72	CRV	6124	33520
817	6.2	0.00981	0.1485	0.7912	0.0505	0.73	CRV		47965
660	6.6	0.00987	0.1485	0.7911	0.0505	0.73	CRV	21531	37730
758	6.2	0.01003	0.1485	0.7910	0.0505	0.74	CRV	3412	33808
1223	6.3	0.01019	0.1485	0.7909	0.0505	0.75	CRV		437
652	6.6	0.00967	0.1485	0.7913	0.0505	0.72	CRV	29476	45867
651	7.0	0.00991	0.1485	0.7911	0.0505	0.73	CRV	41105	

672	6.7	0.00825	0.1490	0.7922	0.0506	0.61	Full	17797	44157
721	6.7	0.00834	0.1490	0.7921	0.0506	0.62	Full	5731	34859
765	6.4	0.01051	0.1484	0.7906	0.0505	0.78	Full	3846	18340
779	6.4	0.01051	0.1484	0.7906	0.0505	0.78	CRV	4027	30204
705	6.6	0.00972	0.1485	0.7912	0.0505	0.72	CRV	8114	26816
703	6.5	0.00985	0.1485	0.7911	0.0505	0.73	CRV	8182	26832
766	6.4	0.00984	0.1485	0.7911	0.0505	0.73	CRV	3761	28719
763	6.3	0.00968	0.1485	0.7913	0.0505	0.72	CRV	3691	28418
670	6.8	0.00980	0.1485	0.7912	0.0505	0.73	CRV	18222	35618
673	6.9	0.00980	0.1485	0.7912	0.0505	0.73	CRV	16950	34799
770	6.4	0.00976	0.1485	0.7912	0.0505	0.72	CRV	3462	32373
673	6.9	0.00991	0.1485	0.7911	0.0505	0.73	CRV	16755	32557
703	6.5	0.00990	0.1485	0.7911	0.0505	0.73	CRV	7852	25037
669	6.8	0.00988	0.1485	0.7911	0.0505	0.73	CRV	17643	33630
667	6.7	0.01010	0.1485	0.7909	0.0505	0.75	Full	23082	35189
704	6.6	0.01046	0.1484	0.7906	0.0505	0.77	Full	8831	22741
814	6.3	0.01092	0.1484	0.7903	0.0504	0.81	Full	6082	23197
1099	7.4	0.00815	0.1489	0.7924	0.0506	0.60	CRV		1851
1134	7.1	0.00820	0.1489	0.7923	0.0506	0.61	CRV		1153
1194	7.2	0.00825	0.1489	0.7923	0.0506	0.61	CRV		545
1072	7.3	0.00821	0.1489	0.7923	0.0506	0.61	CRV		2701
1173	7.2	0.00837	0.1488	0.7922	0.0506	0.62	Full		664
1116	7.5	0.00842	0.1488	0.7921	0.0506	0.62	Full		1968
1127	7.2	0.00847	0.1488	0.7921	0.0506	0.63	Full		1285
1068	7.2	0.00863	0.1488	0.7920	0.0506	0.64	Full		3090

***n*-Heptane**

Literature Source of Data:

M.F. Campbell, T. Parise, A.M. Tulgestke, R.M. Spearrin, D.F. Davidson and R.K. Hanson, "Strategies for Obtaining Long Constant-pressure Test Times in Shock Tubes," *Shock Waves* **25** (6) 651-665 (2015) [DOI 10.1007/s00193-015-0596-x](https://doi.org/10.1007/s00193-015-0596-x)

Range of Data:

Temperature [K]	648	648
Pressure [atm]	4.6	4.6
Fuel Mole Fraction [%]	0.95	0.95
Oxygen Mole Fraction [%]	21	21
Equivalence Ratio	1.0	1.0

Type of Data:

n-Heptane Table 2: Ignition delay time measurement in O₂/argon based on OH* emission at 306 nm and extrapolating the fuel removal processes for both 1st and 2nd stage ignition.

n-Heptane Table 2:

Temp	Press	Fuel	O ₂	Ar	E.R.	1st IDT	2nd IDT
K	atm	MF	MF	MF		ms	ms
648	4.6	0.0095	0.21	0.79	0.5	48.0	65.8

***n*-Heptane**

Literature Source of Data:

J. Shao, R. Choudhary, Y. Peng, D.F. Davidson and R.K. Hanson, "High-speed Imaging of Homogeneous and Inhomogeneous Ignition in a High-pressure Shock Tube," Paper 2019-1771, Aerospace Sciences Meeting, Jan. 2019. <https://arc.aiaa.org/doi/pdf/10.2514/6.2019-1771>

Range of Data:

Temperature [K]	799	1156
Pressure [atm]	16	16
Fuel Mole Fraction [%]	0.95	1.9
Oxygen Mole Fraction [%]	4	21
Equivalence Ratio	0.5	1.0

Type of Data:

n-Heptane Table 3: Imaging data of n-heptane ignition in air. Ignition delay time measurement based on pressure, high-speed imaging, and laser absorption of fuel removal. IDT values given are approximate from paper figures.

n-Heptane Table 3:

Temp	Press	Fuel	O2	Balance	E.R.	1st IDT	2nd IDT
K	atm	MF	MF			ms	ms
1156	15.6	0.0036	0.04	Argon	1		1.6
876	16	0.95	0.21	Nitrogen	0.5	1	4.4
799	16	1.9	0.21	Nitrogen	1	0.7	1.2
1045	17	1.56	0.21	Argon	0.82	inhomo	inhomo

***n*-Heptane**

Literature Source of Data:

MF Campbell, S Wang, DF Davidson, RK Hanson, "Shock tube study of normal heptane first-stage ignition near 3.5 atm," *Combustion and Flame* **198** 376-392 (2018) DOI [0.1016/j.combustflame.2018.08.008](https://doi.org/10.1016/j.combustflame.2018.08.008)

Range of Data:

Temperature [K]	664	1208
Pressure [atm]	2	4
Fuel Mole Fraction [%]	0.8	0.8
Oxygen Mole Fraction [%]	21	21
Equivalence Ratio	0.4	0.4

Type of Data:

n-Heptane Table 4: Ignition delay time measurement in argon using sidewall PZT pressure, OH* emission near 306 nm, and lasers absorption of fuel removal at 3.39 microns. Data in table is taken from the Supplementary Material in the paper Table S1.

n-Heptane Table 4:

Shock Number	Initial T_5 [K]	Initial P_5 [atm]	x_{Ar}	x_{O_2}	$x_{C_7H_{16}}$	ϕ	Filling Method	τ_{ign} [μ s]	τ_1 [μ s]
789	736	3.95	0.78362	0.20830	0.00808	0.43	CRV	-	5496
791	784	3.86	0.78308	0.20816	0.00876	0.46	CRV	-	5881
793	703	3.98	0.78329	0.20822	0.00849	0.45	CRV	-	10477
800	665	4.13	0.78382	0.20836	0.00783	0.41	CRV	-	27868
803	671	4.03	0.78371	0.20833	0.00796	0.42	CRV	-	23705
805	676	3.92	0.78328	0.20821	0.00851	0.45	CRV	-	19420
806	685	3.93	0.78288	0.20811	0.00901	0.48	CRV	-	16125
808	693	4.00	0.78220	0.20793	0.00987	0.52	CRV	-	13146
809	750	3.73	0.78210	0.20790	0.00999	0.53	CRV	-	5110
812	746	3.74	0.78154	0.20775	0.01070	0.57	CRV	-	5266
814	767	3.65	0.78360	0.20830	0.00810	0.43	CRV	-	5124
816	754	3.64	0.78420	0.20846	0.00734	0.39	CRV	-	5117
817	767	3.60	0.78339	0.20824	0.00836	0.44	CRV	-	5347
819	723	3.66	0.78319	0.20819	0.00863	0.46	CRV	-	6816
826	789	3.49	0.78334	0.20823	0.00843	0.45	CRV	-	8613
790	745	4.03	0.78367	0.20832	0.00802	0.42	Conventional	-	5086
792	792	3.93	0.78291	0.20812	0.00897	0.47	Conventional	-	6604
794	701	3.97	0.78423	0.20847	0.00730	0.39	Conventional	-	11887
795	705	4.15	0.78240	0.20798	0.00962	0.51	Conventional	-	10469
801	664	4.11	0.78373	0.20833	0.00794	0.42	Conventional	-	29613
804	668	3.99	0.78357	0.20829	0.00814	0.43	Conventional	-	23289
807	693	3.99	0.78322	0.20820	0.00858	0.45	Conventional	-	13491

818	765	3.58	0.78339	0.20824	0.00837	0.44	Conventional	-	5220
821	723	3.72	0.78327	0.20821	0.00852	0.45	Conventional	-	7243
873	1093	2.60	0.78271	0.20806	0.00923	0.49	CRV	3695	-
872	1095	2.59	0.78294	0.20812	0.00894	0.47	CRV	3909	-
854	1111	2.85	0.78242	0.20798	0.00960	0.51	CRV	3191	-
853	1122	2.93	0.78185	0.20783	0.01031	0.55	CRV	2417	-
741	1131	2.57	0.78228	0.20795	0.00978	0.52	CRV	2077	-
876	1159	2.50	0.78287	0.20811	0.00902	0.48	CRV	1966	-
874	1166	2.51	0.78285	0.20810	0.00905	0.48	CRV	1245	-
856	1168	2.82	0.78457	0.20856	0.00688	0.36	CRV	1386	-
744	1173	2.60	0.78305	0.20815	0.00879	0.46	CRV	1149	-
743	1182	2.57	0.78277	0.20808	0.00915	0.48	CRV	930	-
855	1195	2.86	0.78340	0.20824	0.00836	0.44	CRV	776	-
742	1226	2.36	0.78554	0.20882	0.00564	0.30	CRV	772	-
756	1105	2.60	0.78252	0.20801	0.00947	0.50	Conventional	3021	-
857	1132	2.87	0.78357	0.20829	0.00814	0.43	Conventional	2097	-
755	1136	2.43	0.78184	0.20783	0.01032	0.55	Conventional	1735	-
757	1208	2.29	0.78261	0.20804	0.00935	0.49	Conventional	737	-

***n*-Heptane**

Literature Source of Data:

JK Shao, R Choudhary, Y Peng, DF Davidson, RK Hanson, "A shock tube study of *n*-heptane, iso-octane, *n*-dodecane and iso-octane/*n*-dodecane blends oxidation at elevated pressures and intermediate temperatures," *Fuel* **243** 541-553 (2019) DOI [10.1016/j.fuel.2019.01.152](https://doi.org/10.1016/j.fuel.2019.01.152)

Range of Data:

Temperature [K]		
Pressure [atm]		
Fuel Mole Fraction [%]		
Oxygen Mole Fraction [%]		
Equivalence Ratio		

Type of Data:

n-Heptane Table 5: Ignition delay time measurement in air using pressure, OH* emission near 306 nm, and laser absorption of fuel. Data table is taken from the paper.

n-Heptane Table 5:

T5 (K)	P5 (atm)	phi	Ign_P (μs)	Ign_OH* (μs)	Ign_3.39 μm (μs)	Ign_1st (μs)
n-heptane/air, P5 ~ 28 atm, phi = 1, Conventional Shock Tube Filling Strategy						
751	31.5	0.90	1165	1152	1158	941
756	30.5	0.92	958	941	945	726
763	30.3	1.01	1087	1082	1078	884
800	29.1	0.99	829	816.8	818	472
819	28.1	0.98	729	716.4	722	306
834	27.4	0.99	806	796.8	800	307
868	28.0	1.01	844	839.6	842	230
884	26.6	0.98	1014	1010	1012	221
890	27.2	1.01	1018	1016	1018	226
892	28.3	1.12	981	987	981	246
895	26.2	1.00	1144	1126	1140	297
920	26.8	1.02	1216	1199	1178	

T5 (K)	P5 (atm)	phi	Ign_P (μs)	Ign_OH* (μs)	Ign_3.39 μm (μs)	Ign_1st (μs)
924	26.8	0.98	1362	1349	1338	
947	25.5	1.00	1280	1281	1276	
978	25.8	0.98	1075	1072	1075	
986	28.3	1.00	690	701	690	
992	27.8	1.07	973	962	964	
1021	26.4	0.92	517	514.4	517	
1028	27.2	0.96	709	710.8	709	
1043	28.5	1.01	508.8	520	508.8	
1061	24.7	0.90	523	515.6	516	
1072	26.7	1.06	425	414	416	
1116	28.7	1.00	312	296	304	
1119	23.8	0.98	254	254.8	247	
1127	24.3	0.98	229	218	220	

T5 (K)	P5 (atm)	phi	Ign_P (μs)	Ign_OH* (μs)	Ign_3.39 μm (μs)	Ign_1st(μs)
n-heptane/air, P5 ~ 28 atm, phi = 1, CRV Shock Tube Filling Strategy						
768	29.6	0.96	988	978.4	980	705
793	29.0	0.89	915	908.4	908	490
793	29.3	0.91	879	868	870	491
796	29.0	0.83	961	954.8	960	477
828	28.5	0.92	907	899.6	900	387
880	28.5	0.94	1074	1061	1069	296
905	28.0	0.92	1395	1378	1387	221

***n*-Dodecane**

Literature Source of Data:

JK Shao, R Choudhary, Y Peng, DF Davidson, RK Hanson, "A shock tube study of n-heptane, iso-octane, n-dodecane and iso-octane/n-dodecane blends oxidation at elevated pressures and intermediate temperatures," *Fuel* **243** 541-553 (2019) DOI [10.1016/j.fuel.2019.01.152](https://doi.org/10.1016/j.fuel.2019.01.152)

Range of Data:

Temperature [K]		
Pressure [atm]		
Fuel Mole Fraction [%]		
Oxygen Mole Fraction [%]		
Equivalence Ratio		

Type of Data:

n-Dodecane Table 1: Ignition delay time measurement in air using pressure, OH* emission near 306 nm, and laser absorption of fuel at 3.39 microns. Data table is taken from the paper.

n-Dodecane Table 1

T5 (K)	P5 (atm)	phi	Ign_P (μs)	Ign_OH* (μs)	Ign_3.39 μm (μs)	Ign_1st (μs)
n-dodecane/air, P5 ~ 28 atm, phi = 1						
715	18.5	0.86	2641		2641	1814
718	18.4	0.93	2324		2304	1654
719	18.7	1.04	2120		2129	1504
726	17.9	0.80	2048		2037	1307
729	17.7	0.85	2047		2029	1218
735	17.5	0.84	1794	1791	1794	1003
740	18.2	0.97	1365	1357	1366	632
740	17.5	0.92	1706	1705	1708	897
756	16.9	0.96	1280	1280	1280	595.6
769	17.4	0.92	1275	1271	1275	489
770	17.2	0.88	1097	1089	1094	413
772	16.7	0.90	1186	1182	1183	455

T5 (K)	P5 (atm)	phi	Ign_P (μs)	Ign_OH* (μs)	Ign_3.39 μm (μs)	Ign_1st (μs)
776	17.7	0.90	1331	1328	1333	446
778	17.2	0.97	1287	1288	1285	431.2
781	16.9	0.93	1062	1056	1052	305.2
783	16.9	0.96	1218	1229	1213	360
785	17.4	0.95	1295	1288	1290	376
790	16.3	0.97	1150	1152	1141	340
791	17.4	0.97	1267	1271	1263	364
793	17.2	1.01	1238	1226	1233	302
799	17.1	0.95	1173	1170	1167	295
799	17.4	0.98	1220	1228	1213	292
829	16.8	0.86	1550	1595	1538	247
840	16.5	0.86	1704	1744	1698	220
852	16.6	0.97	1534	1532	1534	200
883	15.6	0.95	2061	2051	2062	241
884	16.3	0.95	1987	1970	1952	208
886	16.1	0.93	2018	2122	2018	194
901	15.7	0.95	2302	2257	2283	
906	16.0	0.96	2130	2109	2117	
923	15.5	0.94	2575	2540	2602	
949	15.5	0.93	2080	2031	2065	
981	15.4	0.94	1948	1935	1946	
1008	14.8	0.91	1566	1556	1562	
1031	14.2	0.93	1239	1230	1232	
1080	14.0	0.88	683	678	683	
1115	13.5	0.90	484	480	483	
1161	15.7	0.95	257	241	246	
1199	13.3	0.90	203	195	193	
1288	12.9	0.87	69.2	60	60.8	
n-dodecane/air, P5 = 50–70 atm, phi = 1						
833	68.4	0.70	358	370	406	

T5 (K)	P5 (atm)	phi	lgn_P (μs)	lgn_OH* (μs)	lgn_3.39 μm (μs)	lgn_1st (μs)
843	65.3	0.79	332	338	357	
878	51.0	0.75	293	329	306	
925	57.1	0.74	255	248	255	
990	58.6	0.78	258	243	260	
1021	57.3	0.92	185	175	184	
1067	52.9	1.02	97	87	97	

***n*-Dodecane/Nitric Oxide**

Literature Source of Data:

J.K. Shao, Y. Zhu, D.F. Davidson and R.K Hanson, "Shock Tube Study of the Effect of NO Addition on Ignition Delay Times of n-Dodecane/Air Mixtures," International Symposium on Shock Waves 31, Nagoya, Japan, July 2017.

Range of Data:

Temperature [K]	750	1083
Pressure [atm]	30	30
Fuel Mole Fraction [%]	1.13	1.13
Oxygen Mole Fraction [%]	21	21
Equivalence Ratio	1	1

Type of Data:

Table 1: n-Dodecane/Air/Nitric Oxide mixtures with ppm NO added. IDT measurements in air based on OH* emission at 306 nm, side wall pressure and laser absorption of fuel. Figure 5 data in the paper was normalized to 30 atm.

n-Dodecane/Air/Nitric Oxide Table 1

NO add	P	T	IDT
ppm	atm	K	ms
0	30	1083	294
0	30	1036	405
0	30	949	511
0	30	931	488
0	30	918	487
0	30	907	435
0	30	864	399
0	30	815	438
0	30	777	667
0	30	750	1125
500	30	1058	200
500	30	1008	356
500	30	932	385
500	30	876	380
1000	30	1052	197
1000	30	1003	280
1000	30	931	334
1000	30	872	364
1000	30	778	589

Branched Alkanes

iso-Octane

Literature Source of Data:

JK Shao, R Choudhary, Y Peng, DF Davidson, RK Hanson, "A shock tube study of n-heptane, iso-octane, n-dodecane and iso-octane/n-dodecane blends oxidation at elevated pressures and intermediate temperatures," *Fuel* **243** 541-553 (2019) DOI [10.1016/j.fuel.2019.01.152](https://doi.org/10.1016/j.fuel.2019.01.152)

Range of Data:

Temperature [K]		
Pressure [atm]		
Fuel Mole Fraction [%]		
Oxygen Mole Fraction [%]		
Equivalence Ratio		

Type of Data:

Iso-Octane Table 1: Ignition delay time measurement in air using pressure, OH* emission near 306 nm, and laser absorption of fuel. Data table is taken from the paper.

Iso-Octane Table 1

T5 (K)	P5 (atm)	phi	Ign_P (μs)	Ign_OH* (μs)	Ign_3.39	Ign_1st
<i>iso-octane/air</i> , P5 ~ 28 atm, phi = 1 Conventional Shock Tube Filling Strategy						
934	28	1.07	2166	2159	2142	
947	28.1	1.06	1512	1497	1503	
951	28.9	1.05	1595	1590	1584	
954	28	1.05	1594	1571	1586	
955	28.4	1.05	1635	1634	1625	
978	28	1.02	1426	1411	1412	
983	28.4	1.02	1424	1412	1416	
993	27.8	1.01	1197	1195	1188	
1013	26.8	0.99		1036	1035	
1020	27.3	0.98	826	818	814	
1031	27.1	0.97	945	930	934	

T5 (K)	P5 (atm)	phi	Ign_P (μs)	Ign_OH* (μs)	Ign_3.39	Ign_1st
1034	28.2	0.97	908	892	891	
1070	26.4	0.93	573	569	563	
1108	26.5	0.9	351	342	347	
1138	25	0.88		240	239	
1144	25.6	0.87	230	218	224	
1160	25	0.86	201	181	193	
1185	25	0.84	144	134	144	
<i>iso</i> -octane/air, P5 = 45–70 atm, phi = 1 Conventional Shock Tube Filling Strategy						
665	41.9	1.02	6644		6646	
687	45.2	0.91	4441		4442	3288
716	68.3	0.7	4686	4680	4684	3063
717	70	0.79	3692	3707	3707	2644
726	67.4	0.85	2989	2986	2993	1753
761	40.7	0.97	3834		3834	1178
770	43.4	1.1	2413		2413	973
790	50.1	1.02	2592		2595	768
800	47.8	1.04	2594		2595	619
830	53.1	1.01	2102		2102	525
852	51.5	1.01	1944		1943	
893	53.1	1.02	1613		1613	
901	51.6	1.01	1575		1576	
918	61.3	0.91	1966	1990	1984	
966	55.3	1	807	796	797	
1010	45.8	0.99	578	567	562	
1019	49.7	0.93	532.4	525	520	
1022	58.8	1	478	470	464	
1034	52	0.97	394	385	382	
1037	48.5	0.98	372	367	360	
1039	49.1	0.98	425	415	418	
1047	50.6	1.05	357	350	350	

T5 (K)	P5 (atm)	phi	Ign_P (μs)	Ign_OH* (μs)	Ign_3.39	Ign_1st
1058	51.4	0.98	344	334	334	
1061	51.9	1	324	312	314	
1062	52	1	311	301	304	
1062	51.9	0.99	262	257	254	
1070	52.6	0.99	292	281	284	
1071	52	0.95	288	280	282	
1079	53.9	0.99	257	252	247	
1080	53.6	0.99	256	250	246	
1081	53.8	0.99	252	242	245	
1087	54.1	0.96	222	215	216	
1097	55.6	0.98	213	206	202	

T5 (K)	P5 (atm)	phi	Ign_OH* (μs)	Ign_3.39 μm (μs)
<i>iso</i> -octane/air, P5 ~ 28 atm, phi = 1, CRV Shock Tube Filling Strategy				
914	27.7	0.91	2084	2102
938	27.1	0.87	1627	1609
940	27.3	0.88	1771	1769
943	27.9	0.91	1680	1669
944	27.4	0.85	1545	1666
969	26.7	0.91	1430	1416
970	27.4	0.98	1299	1266
977	27.5	0.93	1344	1319
993	27.0	0.91	1269	1334
994	27.0	0.90	1312	1221
1011	26.0	0.84	1163	1153

iso-Octane/n-Dodecane Blend

Literature Source of Data:

JK Shao, R Choudhary, Y Peng, DF Davidson, RK Hanson, "A shock tube study of n-heptane, iso-octane, n-dodecane and iso-octane/n-dodecane blends oxidation at elevated pressures and intermediate temperatures," *Fuel* **243** 541-553 (2019) DOI [10.1016/j.fuel.2019.01.152](https://doi.org/10.1016/j.fuel.2019.01.152)

Range of Data:

Temperature [K]		
Pressure [atm]		
Fuel Mole Fraction [%]		
Oxygen Mole Fraction [%]		
Equivalence Ratio		

Type of Data:

Iso-Octane/n-Dodecane blend Table 1: Ignition delay time measurement in air using pressure, OH* emission near 306 nm, and laser absorption of fuel. Data table is taken from the paper.

Iso-Octane/n-Dodecane blend Table 1

T5 (K)	P5 (atm)	phi	Ign_P (μs)	Ign_OH* (μs)	Ign_3.39 μm (μs)
50% iso-octane/50% n-dodecane/air, P5 ~ 17 atm, phi = 1					
748	21.6	1.02	1373	1373	1373
771	21.6	1.01	1347	1333	1342
789	20.2	1.04	1412	1405	1406
815	20.2	1.02	1540	1547	1540
835	19.9	1.06	1585	1601	1582
866	17.7	1.06	2125	2109	2106
882	16.5	1.05	2464	2421	2437
885	13.7	1.03	2832	2825	2817
904	14.6	1.06	2501	2507	2473
904	15.3	1.04	2473	2452	2471
906	13.4	1.04	2379	2361	2353
918	14.6	1.03	2362	2336	2338

T5 (K)	P5 (atm)	phi	Ign_P (μs)	Ign_OH* (μs)	Ign_3.39 μm (μs)
929	14.9	1.02	1904	1895	1901
938	12.7	1.02	2113	2095	2095
998	15.9	1.01	1250	1237	1242
1113	14.3	0.96	517	510	516
1149	12.4	0.98	334	322	331

2,7-Dimethyl Octane

Literature Source of Data:

S. Li, S.M. Sarathy, D.F. Davidson, R.K. Hanson and C.K. Westbrook, "Shock Tube and Modeling Study of 2,7-dimethyloctane Pyrolysis and Oxidation," *Comb. and Flame* **162** 2296-2306 (2015) [DOI 10.1016/j.combustflame.2015.01.027](https://doi.org/10.1016/j.combustflame.2015.01.027)

Range of Data:

Temperature [K]	666	1216
Pressure [atm]	12	27
Fuel Mole Fraction [%]	0.68	1.35
Oxygen Mole Fraction [%]	21	21
Equivalence Ratio	0.5	1.0

Type of Data:

2,7-Dimethyl Octane Table 1: Ignition delay time measurement in air using the onset of sidewall OH* emission near 306 nm as well as from side wall pressure measurements.

2,7-Dimethyl Octane Table 1

E.R.	<i>T</i>	<i>P</i>	IDT
	K	atm	ms
<i>Conventional filling</i>			
1	666	17.9	7903
1	688	18.7	4269
1	715	16.1	2421
1	754	16.4	2083
1	820	21.1	1884
1	893	18.5	2000
1	906	14.8	1928
1	919	17.1	1895
1	937	14.5	1812
1	954	13.7	1615
1	981	16	1359
1	1007	13.6	1484
1	1012	13.9	1391
1	1067	13.2	1050
1	1097	13.1	746
1	1148	13	376

1	1191	12.5	247
0.5	712	18.7	4359
0.5	827	20.8	2859
0.5	937	17.6	2220
0.5	1033	13.5	1685
0.5	1076	13.4	1162
0.5	1125	13.2	653
0.5	1159	12.8	432
0.5	1216	12.2	222
<i>CRV filling</i>			
1	704	27.2	5980
1	713	27	5050
1	723	16.6	5160
1	729	16.2	5295
1	759	16.2	4617
1	762	20.5	3244
1	776	22	3318
1	791	21.7	3380
1	803	20.2	3970
1	834	18.7	3960
1	868	16.8	4005
1	880	12.7	4575
1	922	13.5	3890
1	927	13.3	3479
1	930	13.5	3558
1	986	13.1	2697
1	1021	11.6	1939
1	1064	11	1135
0.5	771	20.8	5150
0.5	887	18.4	4744
0.5	984	16.8	2700

Cyclic Fuels

JP-10

Literature Source of Data:

Y. Tao, R. Xu, K. Wang, J. Shao, S.E. Johnson, A. Movaghar, X. Han, J-W. Park, T. Lu, K. Brezinsky, F.N. Egolfopoulos, D.F. Davidson, R.K. Hanson, C.T. Bowman and H. Wang, "A Physics-based Approach to Modeling Real-fuel Combustion Chemistry—III. Reaction Kinetic Model of JP10," *Combustion and Flame* **198** 466-476 (2018) DOI [10.1016/j.combustflame.2018.08.022](https://doi.org/10.1016/j.combustflame.2018.08.022)

Range of Data:

Temperature [K]	1045	1671
Pressure [atm]	1	16
Fuel Mole Fraction [%]	0.2	3
Oxygen Mole Fraction [%]	1.4	21
Equivalence Ratio	0.5	2

Type of Data:

JP-10 Table 1: Ignition delay time measurement in O₂/nitrogen and O₂/argon using pressure and OH* emission near 306 nm. Data are from the supplementary material in the paper. Data identified as [1] is from D.F. Davidson, D.C. Horning, J.T. Herbon, R.K. Hanson, Shock tube measurements of JP-10 ignition, Proceedings of the Combustion Institute 28 (2000) 1687-1692.

JP-10 Table 1:

Ref.	T5 [K]	P5 [atm]	IDT [μ s]	JP-10 [%]	O2 [%]	Balance
	1208	15.9	156	1.44	20.7	N2
	1045	15.9	679	2.72	20.4	N2
	1190	14.2	235	3.06	20.4	N2
	1049	16.0	907	0.723	20.87	N2
	1230	14.2	151	0.723	20.8	N2
[1]	1671	1.05	63	0.2	2.99	Ar
[1]	1406	3.30	854	0.2	2.82	Ar
[1]	1580	3.17	105	0.2	2.76	Ar
[1]	1378	6.17	846	0.2	2.89	Ar
[1]	1490	6.40	228	0.2	2.76	Ar
[1]	1474	3.05	148	0.2	5.54	Ar
[1]	1482	3.05	982	0.2	1.39	Ar

Distillate Fuels

Distillate Fuels

Literature Source of Data:

D.F. Davidson, Y.Y. Zhu, J. Shao and R.K. Hanson, "Ignition Delay Time Correlations for Distillate Fuels," *Fuel* **187** 26–32 (2017) [DOI 10.1016/j.fuel.2016.09.047](https://doi.org/10.1016/j.fuel.2016.09.047)

J. Shao, Y. Zhu, S. Wang, D.F. Davidson, R. K. Hanson, "A Shock Tube Study of Jet Fuel Pyrolysis and Ignition at Elevated Pressures and Temperatures," *Fuel* **226** 338-344 (2018) DOI [10.1016/j.fuel.2018.04.028](https://doi.org/10.1016/j.fuel.2018.04.028)

Range of Data:

Temperature [K]	1000	1400
Pressure [atm]	6	60
Fuel Mole Fraction [%]		
Oxygen Mole Fraction [%]	4	21
Equivalence Ratio	0.4	2.1

Type of Data:

IDT measurements for a series of distillate fuels (given in the accompanying table) are provided. IDT measurements derived from OH* emission or sidewall pressure measurements. Table 1 are IDT values in 21%O₂/argon (airgon), Table 2 are IDT values in air. The same data may also exist in other Stanford publications, including the HyChem series of papers.

FUEL/identifier	Equivalent molecular formula	Military POSF# (when available)
JP-8 (A1)	C10.8H21.8	10264
Jet A (A2)	C11.4H22.1	10325
JP-5 (A3)	C11.9H22.6	10289
RP-2 (R4)	C12.0H24.1	7688
RP-2 (R5)	C12.6H25.7	5433
Kerosene (K6)	C11.5H25.0	
Kerosene (K7)	C12.0H23.0	
DF-2 (D8)	C13.7H23.0	
DF-2 (D9)	C13.6H25.9	12407
Gasoline (G10)	C7.4H13.7	

Fuel in Argon Table 1

Ignition Delay Times for Fuels in Argon				
Fuel	t_{ign} (us)	T (K)	P (atm)	Phi
Mixture is Fuel in 21%O₂ balance Argon				
A1	112	1220	16.10	1.31
A1	257	1120	17.90	1.53
A1	149	1188	17.00	1.42
A1	729	1054	17.26	0.60
A1	299	1140	17.30	0.84
A1	296	1105	18.10	1.08
A1	288	1112	18.20	1.05
A1	275	1108	17.90	1.17
A1	190	1107	34.11	1.06
A1	187	1092	33.85	0.97
A1	250	1080	34.59	0.94
A1	418	1022	35.95	1.10
A2	303	1153	15.80	0.77
A2	452	1072	16.60	1.12
A2	481	1051	17.05	1.17
A2	152	1209	15.70	0.74
A2	344	1117	16.40	1.07
A2	438	1073	16.66	1.12
A2	417	1110	15.80	0.76
A2	274	1126	16.35	1.25
A2	314	1112	16.60	1.31
A2	611	1051	16.80	0.80
A3	367	1116	16.10	0.97
A3	458	1061	17.20	1.25
A3	160	1194	15.80	0.80
A3	506	1053	17.10	1.04
A3	444	1085	16.90	0.87
A3	416	1082	17.20	1.10
Mixture is Fuel in 4%O₂ balance Argon				
A1	532	1286	11.96	1.02
A1	1309	1205	12.40	1.04
A1	783	1244	12.26	1.05
A1	2498	1155	12.55	1.04
A1	1806	1181	12.42	1.07

A1	483	1290	12.22	1.11
A1	2711	1154	12.35	1.02
A1	770	1240	12.66	1.01
A1	2039	1173	12.87	1.01
A1	365	1317	15.12	1.11
A1	232	1374	14.80	1.02
A1	530	1274	15.42	1.11
A1	790	1229	15.56	1.14
A1	1050	1198	15.83	1.12
A1	2200	1154	15.98	1.13
A1	653	1231	12.54	0.43
A1	1279	1193	12.65	0.49
A1	1403	1182	12.82	0.49
A1	2120	1157	13.00	0.50
A1	379	1284	12.24	0.49
A1	215	1334	12.16	0.48
A1	1715	1186	12.59	2.14
A1	942	1243	12.27	2.07
A1	403	1324	11.87	2.14
A1	1967	1169	12.40	2.11
A1	873	1248	12.10	2.08
A1	400	1330	11.61	2.10
A1	822	1110	53.80	1.25
A1	599	1147	50.09	1.21
A1	253	1235	49.28	1.21
A1	1573	1067	51.99	1.05
A1	2613	1030	52.21	1.06
A2	800	1239	12.81	1.03
A2	1673	1190	12.98	1.05
A2	523	1285	12.39	0.99
A2	2711	1145	13.16	1.05
A2	267	1356	12.31	1.02
A2	170	1411	11.78	0.95
A2	363	1318	15.02	0.97
A2	217	1373	14.76	0.93
A2	497	1278	15.33	1.02
A2	760	1229	15.38	1.04
A2	2094	1161	15.93	0.99
A2	702	1264	12.65	1.00

A2	1535	1199	12.95	1.15
A3	1754	1185	12.84	1.02
A3	2623	1147	13.11	1.05
A3	800	1242	12.75	1.04
A3	530	1287	12.44	1.02
A3	320	1342	12.06	0.95
A3	237	1386	11.82	0.90
A3	2275	1146	15.70	1.11
A3	1240	1181	15.45	1.08
A3	815	1219	15.26	1.15
A3	507	1278	15.17	0.94
A3	370	1313	14.95	1.06
A3	230	1374	14.99	1.16
A2	164	1483	1.28	0.88
A2	431	1421	1.34	0.82
A2	1185	1338	1.35	1.03
A2	1795	1296	1.38	1.12
A2	1020	1356	1.33	1.11
A2	164	1483	1.28	0.88
A2	431	1421	1.34	0.82
A2	1185	1338	1.35	1.03
A2	1795	1296	1.38	1.12
A2	1020	1356	1.33	1.11
D9	312	1350	13.07	1.09
D9	545	1265	13.59	0.81
D9	47	1495	12.14	0.87
D9	1441	1196	13.51	1.00
D9	577	1286	13.00	0.91
D9	1544	1193	13.77	0.82
D9	1020	1217	13.39	0.97
D9	102	1422	12.64	0.89
D9	192	1382	12.83	1.04
D9	118	1438	14.58	1.05
D9	276	1338	15.40	1.07
R4	566	1296	12.51	1.13
R4	853	1248	12.87	1.04
R4	260	1379	12.35	1.09
R4	192	1410	12.26	1.04

R4	1007	1217	13.03	1.05
R4	1228	1209	12.96	0.90
R4	1093	1216	13.07	0.90
R4	1065	1216	12.99	1.06
R4	1978	1181	13.28	1.06
R4	1532	1202	13.24	1.05
R5	174	1508	1.16	1.01
R5	41	1608	1.22	0.92
R5	230	1476	1.21	0.91
R5	157	1512	1.26	1.15
R5	382	1442	1.21	0.89
R5	671	1385	1.28	0.93
R5	88	1572	1.23	0.92
R5	1883	1284	1.43	1.02
R4	61	1610	1.09	0.89
R4	99	1563	1.15	0.92
R4	465	1438	1.08	1.10
R4	515	1436	1.14	1.13
R4	350	1484	1.12	0.96
R4	231	1537	1.09	1.46
R4	53	1694	1.14	1.35

Fuel in Air Table 2

Ignition Delay Times for Fuels in Air				
Fuel	tign (us)	T (K)	P (atm)	Phi
A1	343	1169	11.25	0.99
A1	793	1103	11.89	0.87
A1	1293	1050	12.80	0.94
A1	1593	1034	11.35	1.21
A1	355	1174	10.45	0.86
A1	357	1162	10.54	0.96
A1	1385	1052	11.59	0.97
A1	792	1101	11.25	1.03
A1	233	1201	10.57	1.02
A1	125	1256	10.34	0.37
A1	430	1161	11.20	0.42
A1	510	1148	10.90	0.41
A1	1300	1087	11.42	0.40

A1	800	1120	11.28	0.41
A1	255	1200	10.57	0.38
A1	1545	1008	14.80	0.85
A1	517	1109	14.10	0.99
A1	133	1247	13.00	0.97
A1	180	1107	34.11	1.06
A1	245	1080	34.59	0.94
A1	415	1022	35.95	1.10
A1	125	1256	10.34	0.37
A1	430	1161	11.20	0.42
A1	510	1148	10.90	0.41
A1	1300	1087	11.42	0.40
A1	800	1120	11.28	0.41
A1	255	1200	10.57	0.38
A2	296	1166	11.61	1.18
A2	1645	1024	12.50	0.87
A2	718	1109	9.86	1.11
A2	1440	1031	11.05	1.13
A2	115	1253	9.51	1.06
A2	210	1208	10.24	1.07
A2	565	1118	11.03	1.16
A2	930	1074	11.65	0.97
A2	1221	1011	15.20	1.04
A2	715	1072	14.40	1.12
A3	1381	1024	12.63	1.12
A3	640	1118	11.99	0.86
A3	134	1229	10.93	1.19
A3	199	1092	30.63	1.26
A3	517	1029	32.36	0.91
A3	1452	999	15.30	1.19
A3	498	1087	14.60	1.22
A3	1285	1015	14.90	1.19
A3	377	1138	14.00	1.30
D9	786	1072	11.95	0.94

D9	511	1107	13.09	0.97
D9	241	1171	12.98	1.13
D9	222	1183	12.34	0.88
D9	119	1234	11.71	1.11
D9	269	1015	56.99	0.95
D9	241	1020	55.42	1.05
D9	138	1052	56.35	1.11
D9	123	1060	53.99	1.07
G10	360	1072	27.84	1.03
G10	258	1112	22.92	1.04
G10	83	1178	26.27	1.05
G10	1101	1028	14.64	1.03
G10	685	1081	14.31	1.04
G10	343	1133	13.24	1.02
G10	168	1200	13.19	1.01
K6	245	1006	59.30	1.00
K6	177	1054	57.20	0.99
K6	102	1106	52.80	0.98
K6	58	1128	54.30	1.05
K6	619	1019	25.10	1.14
K6	471	1064	25.30	1.02
K6	363	1103	24.20	0.87
K6	916	1090	12.50	1.05
K7	324	1015	59.70	1.11
K7	406	1000	61.60	1.14
K7	350	1019	59.50	1.03
K7	219	1061	55.50	1.11
K7	803	1016	26.50	0.93
K7	785	1023	23.30	0.89
K7	498	1055	22.60	1.00
K7	304	1098	21.40	0.97
K7	230	1124	21.40	0.91
D8	432	1018	54.60	0.95
D8	264	1051	55.00	1.05
D8	332	1028	56.50	1.09

D8	150	1074	59.30	1.01
R4	349	1152	10.80	1.09
R4	1703	1010	11.99	1.18
R4	806	1091	11.24	0.94
R4	144	1233	10.24	0.97
R4	1089	1056	11.64	1.02
R5	1277	1030	11.85	0.97
R5	309	1174	10.65	0.92
R5	1037	1084	11.21	0.96
R5	426	1153	10.75	1.09
R5	249	1196	10.47	1.04
R5	723	1100	11.36	1.18

Jet Fuels and Rocket Propellants with HyChem Modeling

Literature Source of Data:

Y. Zhu, R.K. Hanson and D.F. Davidson, "Shock Tube/ Laser Absorption Measurements of Jet Fuel Pyrolysis and Oxidation," 53rd AIAA Aerospace Sciences Meeting, Kissimmee, FL January 2015 [DOI 10.2514/6.2015-1158](https://doi.org/10.2514/6.2015-1158)

D.F. Davidson, Y. Zhu, S. Wang, T. Parise, R. Sur, and R.K. Hanson, "Shock Tube Measurements of Jet and Rocket Fuels," AIAA Sci. & Tech. Forum, San Diego, January 2016 [DOI 10.2514/6.2016-0178](https://doi.org/10.2514/6.2016-0178)

K. Wang, R. Xu, T. Parise, J.K. Shao, D.J. Lee, A. Movaghar, D.F. Davidson, R.K. Hanson, H. Wang, C.T. Bowman and F.N. Egolfopoulos, "Combustion Kinetics of Conventional and Alternative Jet Fuels using a Hybrid Chemistry (HyChem) Approach," 10th U.S. National Combustion Meeting, April 2017.

K. Wang, R. Xu, T. Parise, J.K. Shao, D.F. Davidson, R.K. Hanson, H. Wang, C.T. Bowman, "Evaluation of a Hybrid Chemistry Approach for Combustion of Blended Petroleum and Bio-derived Jet Fuels," 10th U.S. National Combustion Meeting, April 2017.

J.K. Shao, D.F. Davidson and R.K. Hanson, "Shock Tube Study of Jet Fuel Pyrolysis and Ignition at Elevated Pressure," 10th U.S. National Combustion Meeting, April 2017.

R. Xu, D. Chen, K. Wang, Y. Tao, J.K. Shao, T. Parise, Y. Zhe, S. Wang, R. Zhao, D.J. Lee, F.N. Egolfopoulos, D.F. Davidson, R.K. Hanson, C.T. Bowman and H. Wang, "HyChem Model for Petroleum-Derived Jet Fuels," 10th U.S. National Combustion Meeting, April 2017.

Zhu et al. (2015), Davidson et al. (2016), Wang et al., Shao et al. and Xu et al. (2017) were preliminary presentations of the jet fuel experiments and modeling covered in the later HyChem papers.

HyChem Papers

H. Wang, R. Xu, J. Wang, C. T. Bowman, R. K. Hanson, D. F. Davidson, K. Brezinsky, F. N. Egolfopoulos, "A Physics-based Approach to Modeling Real-fuel Combustion Chemistry-I. Evidence from Experiments and Thermodynamics, Chemical Kinetics and Statistical Considerations," *Combustion and Flame* **193** 502-519 (2018) DOI [10.1016/j.combustflame.2018.03.019](https://doi.org/10.1016/j.combustflame.2018.03.019)

R. Xu, K. Wang, S. Banerjee, J. Shao, T. Parise, Y. Zhu, S. Wang, A. Movaghar, D. J. Lee, R. Zhao, X. Han, Y. Gao, T. Lu, K. Brezinsky, F.N. Egolfopoulos, D. F. Davidson, R. K. Hanson, C. T. Bowman, H. Wang, "A Physics-based Approach to

Modeling Real-fuel Combustion Chemistry-II. Reaction Kinetics Models of Jet and Rocket Fuels,” *Combustion and Flame* **193** 520-537 (2018) DOI <https://doi.org/10.1016/j.combustflame.2018.03.021>

Y. Tao, R. Xu, K. Wang, J. Shao, S.E. Johnson, A. Movaghar, X. Han, J-W. Park, T. Lu, K. Brezinsky, F.N. Egolfopoulos, D.F. Davidson, R.K. Hanson, C.T. Bowman and H. Wang, “A Physics-based Approach to Modeling Real-fuel Combustion Chemistry—III. Reaction Kinetic Model of JP10,” *Combustion and Flame* 198 466-476 (2018) DOI <https://doi.org/10.1016/j.combustflame.2018.08.022>

K. Wang, R. Xu, T. Parise, J. Shao, A. Movaghar, D. J. Lee, J. Part, Y. Gao, T. Lu, F. Egolfopoulos, D. F. Davidson, R. K. Hanson, C. T. Bowman, H. Wang, “A Physics-based Approach to Modeling Real-fuel Combustion Chemistry – IV. HyChem Modeling of combustion Kinetics of a Bio-derived Jet Fuel and its Blends with a Conventional Jet A,” *Combustion and Flame* 198 477-489 (2018) DOI <https://doi.org/10.1016/j.combustflame.2018.07.012>

Range of Data:

Temperature [K]	1000	1400
Pressure [atm]	1	60
Fuel Mole Fraction [%]		
Oxygen Mole Fraction [%]	4	21
Equivalence Ratio	0.4	2

Type of Data:

Jet and Rocket Fuel IDT data found in H. Wang et al. (2018) and R. Xu et al. (2018) can be found in this database under distillate fuels: D.F. Davidson, Y.Y. Zhu, J. Shao and R.K. Hanson, “Ignition Delay Time Correlations for Distillate Fuels,” *Fuel* **187** 26–32 (2017) DOI [DOI 10.1016/j.fuel.2016.09.047](https://doi.org/10.1016/j.fuel.2016.09.047)

JP-10 IDT data found in Y. Tao et al. (2018) can be found in this volume and in volume 1 of this database: D.F. Davidson, D.C. Horning, J.T. Herbon, R.K. Hanson, “shock tube measurements of JP-10 ignition,” *Proceedings of the Combustion Institute* 28, 1687-1692 (2000) DOI [https://doi.org/10.1016/S0082-0784\(00\)80568-8](https://doi.org/10.1016/S0082-0784(00)80568-8)

Cat A and Cat C fuel descriptions can be found in the table below.

Table 1: Binary Mixtures of A2 and C1 Fuels. This data supports the K. Wang et al. (2018) paper discussions. Data was digitized from Figure 12 in that paper. Fuel mixtures are by liquid volume, not mole fraction.

Table 2: IDT datasets for A1, A2 and Cat C1 fuels extending to 700 K.

Table 3: Normalized Cat A1, A2 and A3 data. Data normalized to 12 atm and E.R.=1 using the following pressure scaling: for $T > 1000\text{K}$, P^{-1} ; for $T < 1000$, $P^{(-1.76+0.76*T/1000)}$, and the following E.R. Scaling: ϕ^{-1} .

Table 4: Normalized Cat C1 to C6 data. Data normalized to 12 atm and E.R.=1 using the following pressure scaling: for $T > 1000\text{K}$, P^{-1} ; for $T < 1000$, $P^{(-1.76+0.76*T/1000)}$, and the following E.R. Scaling: ϕ^{-1} . This data was acquired for the NJFCP FAA research program and was not published in a journal. Note there are two sets of C2 data, one set with the heated shock tube walls at 80C and one set with the walls at 110C. IDTs are effectively the same.

Fuel Description Table

Fuel/Solvent Mixture	POSF Number	Composition % volume	Description
A-1	10264	Petroleum JP-8	Low flash, viscosity, and aromatics
A-2	10325	Petroleum Jet A	Nominal jet fuel
A-3	10289	Petroleum JP-5	High flash, viscosity, and aromatics
C-1	11498 12368 12384	Gevo ATJ, Highly branched C12 and C16 paraffins	Low DCN, unusual boiling range
C-2	11813 12223	16% tri-methylbenzene + 84% C14 iso-paraffins	Chemically-asymmetric boiling range
C-3	12341 12363	64% A-3 + 36% farnesane	High viscosity fuel, at viscosity limit for jet fuel at -20°C
C-4	12344 12489	60% C9-12 iso-paraffins, 40% C-1	Low DCN, conventional, wide boiling range
C-5	12345 12713 12789 12816	73% C10 iso-paraffins + 27% tri-methylbenzene	Flat boiling range
C-6		High cycloalkane content	High cyclo-paraffins

Binary Mixtures of A2 and C1 Fuels Table 1

Fuel Mix	T [K]	P [atm]	E.R.	1000/T	IDT us
80%A2/20%C1	1425	0.56	1.00	0.70181	204
	1383	0.56	1.00	0.72332	326
	1346	0.56	1.00	0.74321	444
	1299	0.56	1.00	0.76955	656
	1278	0.56	1.00	0.78217	793
50%A2/50%C1	1415	0.56	1.00	0.70693	289
	1389	0.56	1.00	0.71993	386
	1352	0.56	1.00	0.73943	512
	1349	0.56	1.00	0.74149	598
	1312	0.56	1.00	0.762	914
	1281	0.56	1.00	0.7805	1459
	1293	0.56	1.00	0.77366	1610
A2	1463	0.56	1.00	0.68331	130
	1404	0.56	1.00	0.71238	197
	1348	0.56	1.00	0.74182	332
	1354	0.56	1.00	0.73877	454
	1338	0.56	1.00	0.74727	435
	1335	0.56	1.00	0.74933	538
	1323	0.56	1.00	0.75583	470
	1308	0.56	1.00	0.76472	480
	1308	0.56	1.00	0.76439	714
	1268	0.56	1.00	0.78834	793
	1293	0.56	1.00	0.77366	1045
	1243	0.56	1.00	0.80478	1123
	1238	0.56	1.00	0.80784	1543
	1204	0.56	1.00	0.83079	2048
C1	1460	0.56	1.00	0.68503	222
	1439	0.56	1.00	0.69493	244
	1425	0.56	1.00	0.70181	305
	1404	0.56	1.00	0.71238	417
	1400	0.56	1.00	0.7141	542
	1375	0.56	1.00	0.72744	602
	1352	0.56	1.00	0.73977	960

Complete IDT Data for Cat A1, A2, C1 Fuels Table 2

Fuel	tign (us)	T (K)	P (atm)	Phi	Xfuel	XO2	XN2
Cat A1							
10264	343	1165	11.18	0.99	0.0127	0.2073	0.7800
10264	793	1099	11.82	0.87	0.0111	0.2077	0.7812
10264	1293	1046	12.72	0.94	0.0120	0.2075	0.7806
10264	1593	1030	11.26	1.21	0.0154	0.2068	0.7779
10264	2395	982	12.53	0.88	0.0112	0.2076	0.7811
10264	1964	959	13.84	0.95	0.0121	0.2075	0.7805
10264	3120	887	11.16	0.89	0.0114	0.2076	0.7810
10264	3572	861	10.48	0.96	0.0122	0.2074	0.7803
10264	2459	810	15.57	0.97	0.0124	0.2074	0.7802
10264	2120	778	18.08	1.01	0.0129	0.2073	0.7798
10264	3194	777	12.26	1.10	0.0140	0.2071	0.7790
10264	3119	740	13.91	1.00	0.0128	0.2073	0.7799
10264	791	957	35.50	1.10	0.0140	0.2071	0.7789
10264	886	861	32.01	1.04	0.0133	0.2072	0.7795
10264	777	834	39.21	1.03	0.0132	0.2072	0.7796
10264	1237	757	35.45	1.00	0.0128	0.2073	0.7799
Cat A2							
10325	296	1164	11.50	1.18	0.0145	0.2070	0.7785
10325	589	1124	12.01	0.83	0.0103	0.2079	0.7819
10325	710	1110	11.63	0.81	0.0099	0.2079	0.7822
10325	1645	1022	12.42	0.87	0.0108	0.2077	0.7815
10325	1908	919	12.56	0.99	0.0122	0.2074	0.7804
10325	3485	907	8.84	0.89	0.0109	0.2077	0.7814
10325	3816	856	9.94	0.82	0.0101	0.2079	0.7821
10325	4433	834	9.39	0.87	0.0107	0.2078	0.7816
10325	3723	827	13.94	0.95	0.0117	0.2076	0.7808
10325	3216	824	10.33	1.30	0.0160	0.2066	0.7774
10325	3043	813	18.48	1.12	0.0138	0.2071	0.7791
10325	3644	805	10.54	1.34	0.0165	0.2065	0.7770
10325	3149	776	15.57	0.90	0.0110	0.2077	0.7813
10325	3100	758	15.18	1.10	0.0135	0.2072	0.7793
10325	4845	733	12.83	0.81	0.0100	0.2079	0.7821
10325	3397	728	14.54	0.92	0.0114	0.2076	0.7810
Cat C1							
11498	1050	1048	13.04	0.92	0.0098	0.2079	0.7822
11498	434	1111	11.93	0.95	0.0102	0.2079	0.7820
11498	160	1197	11.87	0.97	0.0104	0.2078	0.7818

11498	2921	931	13.35	1.03	0.0110	0.2077	0.7813
11498	2454	894	18.14	1.07	0.0115	0.2076	0.7810
11498	3918	858	16.10	1.02	0.0109	0.2077	0.7814
11498	4664	839	13.85	1.09	0.0117	0.2076	0.7808
11498	2234	888	17.85	1.12	0.0120	0.2075	0.7805
11498	4030	860	15.76	1.08	0.0116	0.2076	0.7809
11498	5605	822	14.52	1.09	0.0117	0.2076	0.7808
11498	855	1053	12.70	1.05	0.0112	0.2076	0.7811
11498	1061	1033	13.21	1.14	0.0122	0.2074	0.7804
11498	1127	1032	12.55	0.97	0.0104	0.2078	0.7818
11498	1207	1017	13.41	1.09	0.0117	0.2076	0.7808
11498	1504	1010	13.48	1.05	0.0112	0.2076	0.7811
11498	413	1104	12.64	1.07	0.0115	0.2076	0.7810

Table 3: Normalized Cat A1, A2 and A3 data Table 3.

T [K]	IDT [μ s]
A1	10264
1165	318
1099	679
1046	1285
1030	1803
982	2202
959	2157
887	2578
861	2950
810	3222
778	3466
777	3585
740	3724
A2	10325
1164	284
1124	589
1110	688
1022	1703
907	2511
856	3097
834	3363
824	3528
776	3824
758	4502

733	4263
728	3953
A3	10289
1022	1601
1116	540
1227	142
758	6142
861	5028
899	3771
711	6866
749	5555

Table 4: Normalized Cat C1 to C6 data Table 4.

C1	11498
1048	1047
1111	411
1197	154
1053	950
1033	1331
1032	1143
1017	1468
1010	1773
1104	465
929	2954
931	3367
894	4105
858	5522
839	5974
888	3851
860	5888
822	7587
C2-80C	10223
1160	423
1100	847
1281	90
1236	162
1056	952
1021	1790
765	4213
769	3737

840	3300
905	2304
934	2484
799	3571
1085	876
1205	204
974	2259
796	3505
759	3951
794	3926
769	4506
724	5242
708	6181
C2-110C	10223
1085	876
1205	204
974	2259
796	3505
759	3951
794	3926
769	4506
724	5242
708	6181
C3	12341
1104	545
1211	189
1104	491
1034	1400
1022	1472
986	2290
979	2256
785	4189
798	4123
797	4028
768	3883
755	4371
873	3226
756	3972
724	4273
724	4236
720	4180

C4	12344
1076	621
1151	238
1235	103
1044	1013
988	2231
1010	1709
843	3448
789	4925
824	3570
C5	12345
1086	975
1041	1703
1146	507
1246	172
768	3757
835	3366
975	1878
779	3894
734	4654
730	5020
C6	10279-2
1108	585
1171	284
1274	72
1148	376
1078	858
1043	1321
764	4224
732	4462
809	3775
857	2432
860	2491
904	2308
964	1816
781	4313
761	4245

Gasoline

Literature Source of Data:

D. F. Davidson, J. K. Shao, R. Choudhary, M. Mehl, N. Obrecht and R. K. Hanson, "Ignition Delay Time Measurements and Modeling for Gasoline at Very High Pressures," Proceedings of the Combustion Institute 37 4885-4892 (2019) [DOI 10.1016/j.proci.2018.08.032](https://doi.org/10.1016/j.proci.2018.08.032)

Range of Data:

Temperature [K]	705	1174
Pressure [atm]	30	260
Fuel Mole Fraction [%]	~1.9	~1.9
Oxygen Mole Fraction [%]	21	21
Equivalence Ratio	0.7	1.1

Type of Data:

Gasoline Table 1: Ignition delay time measurement in synthetic air from heated high pressure shock tube experiments using sidewall PZT pressure and OH* emission near 306 nm. Data tables are taken from the supplementary material of the paper. The composition of the four fuels is given in the following table also taken from the paper.

Composition of liquid fuels A, B, C and D in mass%.

Fuel composition	A	B	C	D
Olefins	9	19	17	17
Aromatics	2	81	36	29
N-Paraffins	3		2	2
Iso-Paraffins	81		34	32
Napthenes	5		11	12
Oxygenates				8
Total	100	100	100	100
MW	103.4	90.2	97	90.1
RON	101	104	105	106.5
MON	94	91	92	91.5

Gasoline Table 1

Fuel A					
T (K)	1000/T	P (atm)	E.R.	IDT 2nd (μs)	IDT_1st (μs)
705	1.4184	70.2	1.02	3722	
706	1.4167	70.2	1.02	3758	
710	1.4083	71.4	0.93	4160	
716	1.3971	73.5	1.04	3518	
721	1.3864	68.7	0.93	2930	
724	1.3812	72.4	0.76	4578	3586
785	1.2739	69.0	0.87	1755	800
800	1.2500	68.9	0.87	1722	625
812	1.2315	67.5	0.86	1792	600
854	1.1710	70.2	0.86	1525	
868	1.1520	68.2	0.86	1503	
892	1.1212	68.6	0.90	1250	
898	1.1136	84.0	0.90	1031	
906	1.1037	65.0	0.86	1160	
953	1.0490	84.8	1.03	506	
1003	0.9966	82.6	0.93	311	
1034	0.9671	80.4	0.91	233	
1055	0.9479	62.6	0.87	229	
1061	0.9422	53.6	1.04	230	
723	1.3831	143.5	0.83	1637	
760	1.3158	152.0	0.92	817	
761	1.3141	155.0	0.97	767	
853	1.1723	134.7	0.60	579	
790	1.2663	185.4	0.66	638	
871	1.1481	247.7	0.75	187	130
895	1.1179	222.8	0.70	214	160
Fuel B					
T (K)	1000/T	P (atm)	E.R.	IDT 2nd (μs)	IDT_1st (μs)
763	1.3106	65.4	0.95	5200	3446
764	1.3089	57.5	0.92	5800	3356
775	1.2903	58.4	0.89	4500	2345
776	1.2887	51.1	0.93	5200	3026
844	1.1845	53.6	0.91	2501	2105
904	1.1066	64.3	0.93	1515	
944	1.0597	57.5	0.94	1190	
948	1.0549	57.0	0.83	1503	

971	1.0300	75.9	0.85	912	
1083	0.9235	60.1	0.92	278	
1108	0.9024	72.2	0.95	200	
1174	0.8516	64.3	1.01	78	
772	1.2953	143.0	0.80	4906	1944
789	1.2674	128.0	0.90	3240	1590
797	1.2547	157.5	0.73	4101	1935
834	1.1990	119.0	0.57	1984	1500
854	1.1710	137.0	0.92	1642	825
920	1.0870	138.0	0.89	816	530
786	1.2723	176.3	0.76	4423	2627
824	1.2142	202.5	0.82	2665	1904
Fuel C					
T (K)	1000/T	P (atm)	E.R.	IDT 2nd (μs)	IDT_1st (μs)
725	1.3797	67.5	0.890	5286	3771
742	1.3469	64.9	0.920	4171	2881
752	1.3302	58.6	0.900	4802	3018
774	1.2914	68.7	0.870	3574	2809
791	1.2645	69.2	0.870	2849	2316
821	1.2183	68.3	0.880	2360	1568
837	1.1954	67.3	0.870	1928	1401
880	1.1361	83.9	0.890	1456	1095
906	1.1033	83.1	0.860	1105	--
950	1.0528	56.5	0.880	776	
998	1.0015	78.5	0.890	302	--
1060	0.9438	77.8	0.860	153	--
795	1.2579	120.3	0.660	3364	2375
840	1.1907	109.9	0.750	3050	1250
936	1.0684	104.5	0.830	462	600
947	1.0561	119.9	0.750	526	--
948	1.0548	99.8	0.820	417	--
1012	0.9878	111.4	0.790	189	--
1078	0.9281	112.8	0.790	72	--
1111	0.9002	108.7	0.830	39	--
732	1.3661	131.0	0.880	4644	
797	1.2547	131.1	0.925	1700	1166
802	1.2469	152.0	0.670	2737	
809	1.2361	156.0	0.680	2149	1300
810	1.2346	130.2	0.770	2333	1500
838	1.1933	143.5	0.770	1566	765
878	1.1387	152.5	0.670	1432	845

880	1.1366	134.6	0.700	1356	872
883	1.1319	149.9	0.740	1214	902
900	1.1105	147.8	0.620	1228	--
987	1.0130	134.4	0.780	279	--
992	1.0077	132.0	0.850	227	--
998	1.0019	133.7	0.850	230	--
787	1.2707	185.1	0.770	1934	1550
796	1.2563	190.0	0.770	1493	1000
803	1.2453	203.9	0.950	959	700
831	1.2034	214.0	0.770	884	480
851	1.1751	259.0	0.880	510	
883	1.1331	256.7	0.770	615	
Fuel D					
T (K)	1000/T	P (atm)	E.R.	IDT 2nd (μs)	IDT_1st (μs)
716	1.3967	66.8	0.89	4432	3352
763	1.3110	61.5	0.84	3500	2442
783	1.2771	61.3	0.88	3807	2163
830	1.2051	78.5	0.85	1761	1388
859	1.1641	89.5	1.00	1418	1150
872	1.1468	69.7	0.89	1263	
874	1.1442	64.6	0.87	1366	1200
904	1.1062	88.8	1.03	901	600
943	1.0605	69.4	0.87	670	
1042	0.9601	80.1	0.84	167	--
740	1.3514	142.6	1.00	3550	1550
740	1.3514	143.0	1.00	3784	1459
741	1.3495	142.6	0.98	3667	
742	1.3477	149.0	0.72	3403	
746	1.3401	142.9	0.92	3814	
754	1.3263	138.4	0.67	3875	
811	1.2324	150.2	0.91	1100	
838	1.1933	135.6	0.80	1807	891
954	1.0479	163.5	0.89	386	
782	1.2796	189.5	0.83	1470	
817	1.2244	215.4	0.89	988	600
Fuel D					
T (K)	1000/T	P (atm)	E.R.	IDT 2nd (μs)	IDT_1st (μs)
706	1.4164	60.2	0.874		4400
709	1.4099	63.5	0.816	6844	4352
749	1.3355	58.3	0.843	4137	3059

760	1.3154	68.9	0.805	3319	2690
820	1.2200	84.0	0.753	2491	2000
837	1.2012	79.9	0.853	1658	1314
842	1.1871	69.5	0.840	1726	1520
898	1.1139	74.0	0.833	1112	792
920	1.0867	65.8	0.909	860	--
951	1.0518	79.9	0.849	528	--
951	1.0515	79.8	0.842	634	--
952	1.0510	63.9	0.843	576	--
955	1.0470	64.5	0.838	586	--
959	1.0433	81.4	0.827	431	--
1000	1.0000	66.0	0.879	304	--
1103	0.9066	74.7	0.873	52	
746	1.3405	142.7	0.914	4077	2877
746	1.3401	156.1	0.781	1750	1140
747	1.3396	145.4	0.980	3363	1100
753	1.3280	148.2	0.950	2700	
758	1.3196	150.8	0.727	2350	800
758	1.3186	113.7	0.890	3506	2755
775	1.2908	147.5	0.635	3386	833
840	1.1912	115.9	0.874	1997	1159
876	1.1421	133.2	0.963	1125	
819	1.2218	230.3	0.760	1277	940
872	1.1469	257.8	0.728	582	
Fuel C					
T (K)	1000/T	P (atm)	E.R.	IDT 2nd (ms)	IDT_1st (ms)
954	1.0482	32.8	0.85	1418	
973	1.4099	32.8	0.85	1100	
Fuel D					
T (K)	1000/T	P (atm)	E.R.	IDT 2nd (ms)	IDT_1st (ms)
934	1.0707	32.2	0.85	1632	
976	1.0246	32.4	0.85	967	
926	1.0799	32.9	0.85	1596	
942	1.0616	34.3	0.85	1359	

Jet Fuel

Literature Source of Data:

Y. Wang, D.F. Davidson and R.K. Hanson, "A New Method of Predicting Derived Cetane Number for Hydrocarbon Fuels," *Fuel* **241** 319-326 (2019) DOI [10.1016/j.fuel.2018.12.027](https://doi.org/10.1016/j.fuel.2018.12.027)

Y. Wang, Y. Cao, D.F. Davidson and R.K. Hanson, "Ignition Delay Time Measurements for Distillate and synthetic Jet Fuels," Paper 2019-2248, Aerospace Sciences Meeting, Jan. 2019. <https://arc.aiaa.org/doi/pdf/10.2514/6.2019-2248>

Range of Data:

Temperature [K]	1250	1475
Pressure [atm]	4	4
Fuel Mole Fraction [%]		
Oxygen Mole Fraction [%]	4	4
Equivalence Ratio	1	1

Type of Data:

Jet Fuel Table 1: Ignition delay time measurement of various jet fuels in synthetic air using sidewall PZT pressure and OH* emission near 306 nm. In these experiments the fuel was mixed in the shock tube. T(shock tube) = 20-25 C. Oxidizer bottle was 4% O₂ in Argon. Detailed properties of the individual fuels are given in Table A that was taken from the Y. Wang et al. Fuel paper.

Fuel/Solvent Mixture	POSF Number	Composition % volume	Description
A-1	10264	Petroleum JP-8	Low flash, viscosity, and aromatics
A-2	10325	Petroleum Jet A	Nominal jet fuel
A-3	10289	Petroleum JP-5	High flash, viscosity, and aromatics
Geographical 1 (A-4)	12784	Petroleum Jet A	
Geographical 2 (A-5)	12831	Petroleum Jet A	
Geographical 3 (A-6)	12843	Petroleum Jet A	
Geographical 4 (A-7)	12905	Petroleum JP-5	

Fuel/Solvent Mixture	POSF Number	Composition % volume	Description
Geographical 5 (A-8)	12906	Petroleum Jet A	
CN30	13197	46 wt% pentamethylheptane, remainder methyl- and dimethyl- iso-paraffins	
CN35	13198	39 wt% pentamethylheptane, remainder methyl- and dimethyl- iso-paraffins	
CN40	13199	20 wt.% pentamethylheptane, remainder methyl- and dimethyl- iso-paraffins	
CN45	13200	14 wt% pentamethylheptane, remainder methyl- and dimethyl- iso-paraffins	
CN50	13201		
CN55	13202		
C-1	11498 12368 12384	Gevo ATJ, Highly branched C12 and C16 paraffins	Low DCN, unusual boiling range
C-2	11813 12223	16% tri-methylbenzene + 84% C14 iso-paraffins	Chemically-asymmetric boiling range
C-3	12341 12363	64% A-3 + 36% farnesane	High viscosity fuel, at viscosity limit for jet fuel at -20 °C
C-4	12344 12489	60% C9-12 iso-paraffins, 40% C-1	Low DCN, conventional, wide boiling range
C-5	12345 12713 12789 12816	73% C10 iso-paraffins + 27% tri-methylbenzene	Flat boiling range
C-6		High cycloalkane content	High cyclo-paraffins
C-7	12925	75% RP-2, 23% A-3, 2% decalin	High cyclo-paraffins
C-8	12923	Jet A + Exxon aromatic blend	High (maximum allowable) aromatics
C-9	12933	80% R-8 HEFA, 20% n-C12	High DCN

Jet Fuel Table 1:

Fuel	T5 [K]	P5 [atm]	Phi	idt OH [us]	idt P [us]	idt laser [us]
POSF10325 A2	1326	3.79	0.94	791	813	806
POSF10325 A2	1327	3.88	0.91	810	808	829
POSF10325 A2	1419	3.83	1.00	294	280	309
POSF10325 A2	1286	4.14	0.99	1196	1216	1229
POSF12843	1380	3.72	0.93	399	413	410
POSF12843	1325	3.79	1.04	851	846	883
POSF12843	1275	3.86	1.06	1539	1568	1539
POSF12843	1415	3.65	0.93	325	308	339
POSF12906	1430	3.67	0.97	265	252	283
POSF12906	1372	3.76	1.06	498	472	510
POSF12906	1282	3.82	0.97	1249	1286	1264
POSF12906	1322	3.78	1.03	928	950	993
POSF12784	1341	3.73	0.95	764	785	792
POSF12784	1422	3.57	1.02	311	298	323
POSF12831	1362	3.74	1.10	573	555	586
POSF12831	1319	3.82	1.10	940	977	974
POSF12784	1293	3.92	0.98	1324	1349	1373
POSF12784	1399	3.76	1.02	411	391	426
POSF12831	1279	3.92	1.02	1536	1575	1579
POSF12831	1425	3.67	1.10	310	319	319
POSF12905	1374	3.71	1.06	566	565	581
POSF12905	1300	3.78	1.04	1169	1209	1189
POSF12905	1412	3.59	1.18	360	370	376
POSF12905	1275	3.91	1.04	1611	1660	1628
POSF12905	1431	3.62	1.00	269	288	278
POSF13202 CN55	1268	3.94	0.98	1254	1260	1267
POSF13202 CN55	1239	4.03	1.00	1885	1897	1915
POSF13202 CN55	1316	3.86	1.06	822	851	817
POSF13202 CN55	1400	3.68	1.03	333	324	353
POSF13197 CN30	1319	3.82	1.04	1503	1522	1510
POSF13197 CN30	1410	3.68	1.03	639	630	647
POSF13197 CN30	1282	3.94	0.95	2142	2135	2138
POSF13197 CN30	1373	3.81	1.04	965	996	972
POSF13197 CN30	1475	3.58	0.99	323	323	326
POSF13200 CN45	1327	3.92	1.01	994	1004	1013
POSF13200 CN45	1287	4.03	0.97	1321	1355	1345
POSF13200 CN45	1381	3.78	0.96	522	533	546
POSF13200 CN45	1454	3.70	0.96	263	251	273
POSF13198 CN35	1322	3.84	0.90	1351	1344	1362

POSF13198 CN35	1434	3.65	1.07	388	404	406
POSF13198 CN35	1274	3.98	1.00	1983	2010	1994
POSF13198 CN35	1375	3.75	1.00	730	695	761
POSF13199 CN40	1341	3.96	0.90	867	876	890
POSF13199 CN40	1279	4.11	0.95	1778	1817	1799
POSF13199 CN40	1407	3.93	1.00	464	515	509
POSF13199 CN40	1477	3.75	0.97	243	247	256
POSF13201 CN50	1287	4.15	1.00	1150	1156	
POSF13201 CN50	1384	3.88	1.02	419	430	443
POSF13201 CN50	1352	4.06	1.10	490	483	504
POSF13201 CN50	1250	4.16	0.93	1572	1595	1594
C4	1295	4.11	1.00	1711		
C4	1365	3.88	0.88	966		
C4	1385	3.65	1.05	708		

Species Measurement to December 2005

Small Fuels

Methane

Literature Source of Data:

E. J. Chang, “*Shock Tube Experiments for the Development and Validation of Models of Hydrocarbon Combustion*,” M. Eng. Thesis, (also published as HTGL Report No. T-320), Mechanical Engineering Department, Stanford University, Stanford CA (1995).

M. Frenklach, H. Wang, M. Goldenberg, G. P. Smith, D. M. Golden, C. T. Bowman, R. K. Hanson, W. C. Gardiner, V. Lissianski, “*GRI-Mech – An Optimized Detailed Chemical Reaction Mechanism for Methane Combustion*,” Topical Report GRI-95/0058, Gas Research Institute (1995).

Range of Data:

Temperature [K]	1933	2470
Pressure [atm]	0.37	1.22
Fuel Mole Fraction [%]	0.1	0.3
Oxygen Mole Fraction [%]	0.1	1.21
Equivalence Ratio	0.5	4

Type of Data:

Methane Table 1: CO₂ species concentration time-history IR laser absorption measurements at 4.200 μm (2380.72 cm^{-1}) in argon.

Methane Table 2: CH₃ species concentration time-history UV laser absorption measurements at 216 nm in argon.

Methane Table 3: OH species concentration time-history UV laser absorption measurements at 306.7 nm (32606.56 cm^{-1}) in argon.

Formaldehyde pyrolysis data are reported in Chang (1995), but are not included here.

Methane Table 1:

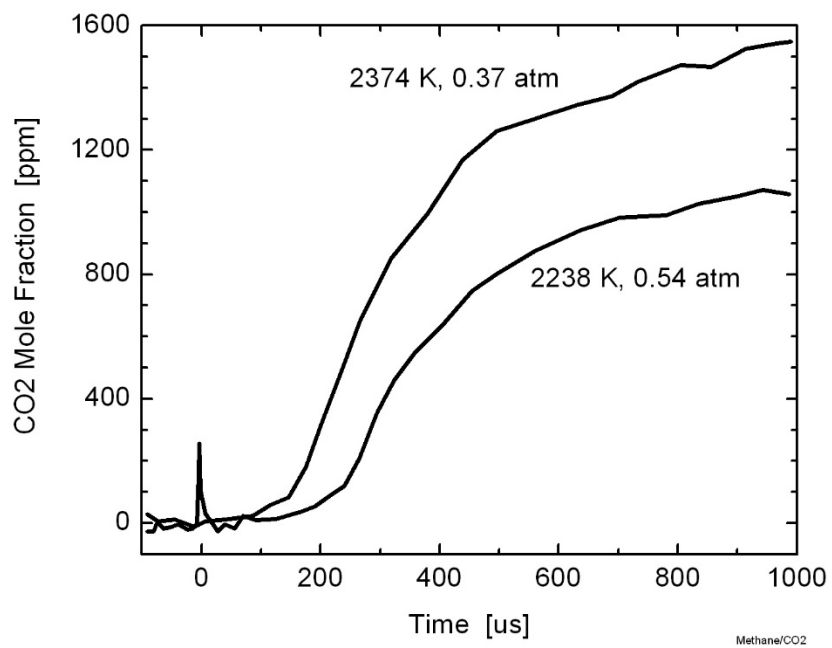
T ₂ [K]	P ₂ [atm]	Methane [ppm]	Oxygen [ppm]	φ (E.R.)	CO ₂ @1ms [ppm]	Plateau CO ₂ [ppm]	¼ Plat [μs]
2164	0.395	3007	6055	1	941	1980	600
2238	0.541	3007	6055	1	1057	1828	340
2317	0.541	3007	6055	1	1123	1588	235
2470	0.512	3007	6055	1	998	1144	141
2140	0.616	3005	12079	0.5	1764	2661	304
2298	0.559	3005	12079	0.5	1599	2270	193
2374	0.370	3005	12079	0.5	1532	1937	238
2408	0.496	3005	12079	0.5	1477	1913	155

Methane Table 2:

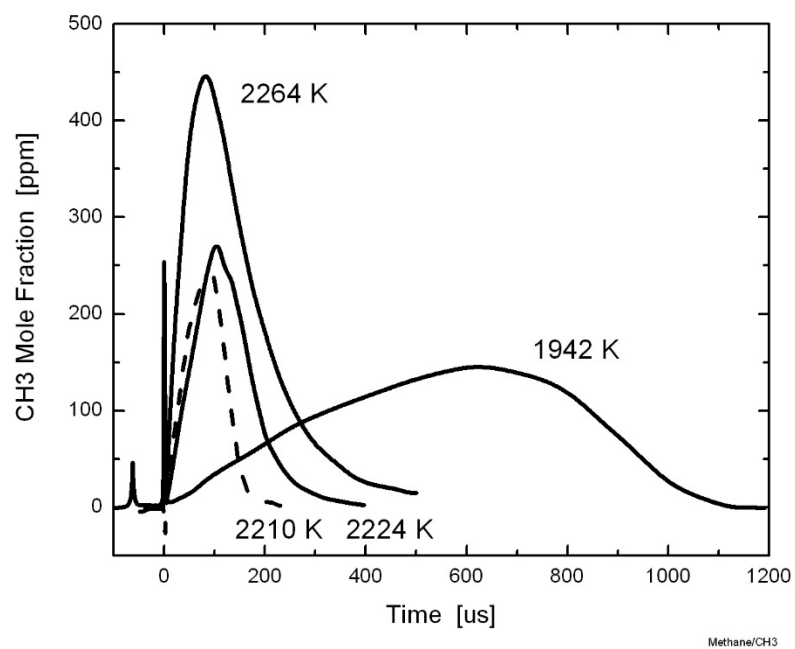
T ₅ [K]	P ₅ [atm]	Methane [ppm]	Oxygen [ppm]	φ (E.R.)	Peak CH ₃ [ppm]	Time [μs]
1951	1.066	2012	1000	4	236	530
2075	1.011	2012	1000	4	337	256
2264	1.024	2012	1000	4	451	83
2401	0.939	2012	1000	4	550	42
1942	1.092	1000	2011	0.99	145	625
1989	1.090	994	2021	0.98	171	443
1997	1.032	994	2021	0.98	187	281
2053	1.016	994	2021	0.98	203	230
2169	0.990	1000	2011	0.99	216	129
2224	1.039	994	2021	0.98	270	109
2338	1.219	994	2021	0.98	342	34
2503	0.924	1000	2011	0.99	325	35
1933	1.099	1000	4034	0.5	134	508
2210	1.006	1000	4034	0.5	240	89
2460	0.910	1000	4034	0.5	333	27.5

Methane Table 3:

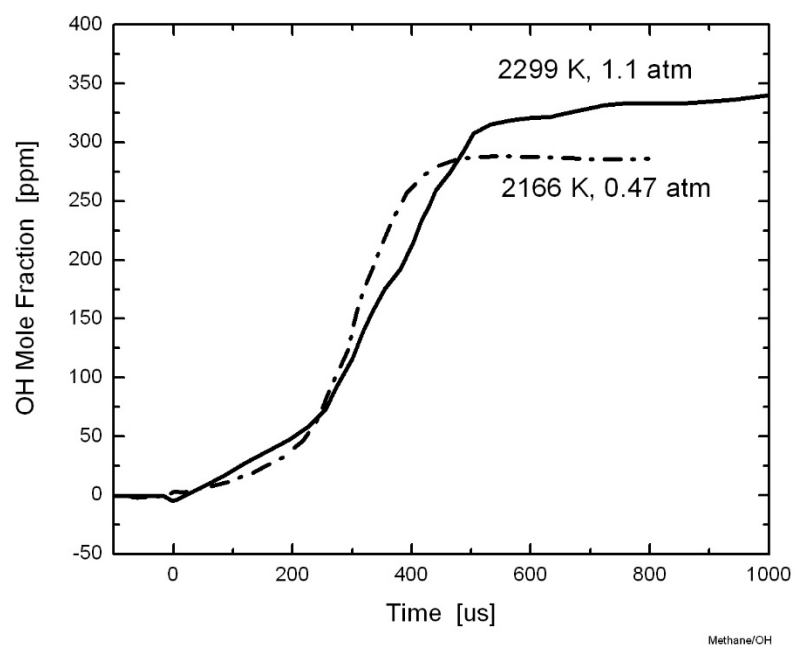
T ₅ or T ₂ [K]	P ₅ or P ₂ [atm]	Methane [ppm]	Oxygen [ppm]	φ (E.R.)	Max OH [ppm]	½ Max [μs]
2056	1.036	997	1996	1	254	632
2145	1.063	1002	2000	1	295	287
2166	1.098	1002	2000	1	290	305
2190	1.130	997	1996	1	303	230
2216	1.081	1002	2000	1	321	195
2169	0.497	1007	2014	1	286	590
2223	0.348	1007	2014	1	304	650
2279	0.490	1007	2014	1	323	403
2299	0.472	1007	2014	1	336	355
2623	0.413	1007	2014	1	429	140



Methane Figure 1.



Methane Figure 2.



Methane Figure 3.

Methane

Literature Source of Data:

D. Woiki, M. Votsmeier, D. F. Davidson, R. K. Hanson, C. T. Bowman, “CH-Radical Concentration Measurements in Fuel-Rich CH₄/O₂/Ar and CH₄/O₂/NO/Ar Mixtures Behind Shock Waves,” Combustion and Flame 113: 624-626 (1998).

Range of Data:

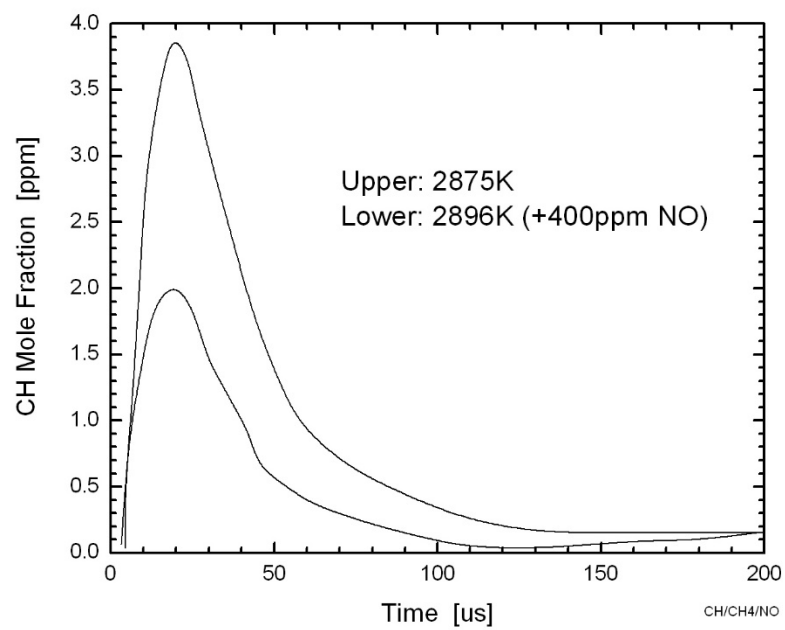
Temperature [K]	2644	3254
Pressure [atm]	1.7	1.8
Fuel Mole Fraction [ppm]	80	80
Oxygen Mole Fraction [ppm]	100	100
Nitric Oxide Mole Fraction [ppm]	0	400
Equivalence Ratio	0.53	1.6

Type of Data:

Methane Table 7: CH species concentration time-history measurements in argon using laser absorption at 431 nm. Data was derived from a digitization of Fig. 2 of Woiki et al. (1998).

Methane Table 7:

T ₅ [K]	P ₅ [atm]	Methane [ppm]	Oxygen [ppm]	NO [ppm]	φ (E.R.)	CH Peak [ppm]
3038	1.7	80	100	0	1.6	6.1
3018	1.7	80	100	0	1.6	6.4
2875	1.8	80	100	0	1.6	3.9
2773	1.7	80	100	0	1.6	3.0
2773	1.7	80	100	0	1.6	2.8
2656	1.7	80	100	0	1.6	1.8
2644	1.7	80	100	0	1.6	1.9
3254	1.7	80	100	400	0.53	4.9
3212	1.7	80	100	400	0.53	4.0
2970	1.7	80	100	400	0.53	2.9
2896	1.8	80	100	400	0.53	2.0
2806	1.7	80	100	400	0.53	1.6
2687	1.7	80	100	400	0.53	1.2



Methane Figure 7.

Methane

Literature Source of Data:

E. L. Petersen, M. Rohrig, D. F. Davidson, R. K. Hanson, C. T. Bowman, “*High Pressure Methane Oxidation Behind Reflected Shock Waves*,” Proceedings of the Combustion Institute 26: 799-806 (1996).

E. L. Petersen, D. F. Davidson, M. Rohrig, R. K. Hanson, C. T. Bowman, “*A Shock Tube Study of High Pressure Methane Oxidation*,” Paper 95F-153, Western States Section/The Combustion Institute Fall Meeting, Stanford (1995).

E. L. Petersen, D. F. Davidson, M. Rohrig, R. K. Hanson, “*Shock-Induced Ignition of High-Pressure H_2 - O_2 -Ar and CH_4 - O_2 -Ar Mixtures*,” AIAA 95-3113, 31st AIAA/ASME/SAE/ASEE Joint Propulsion Conference and Exhibit, San Diego (1995).

Range of Data:

Temperature [K]	1408	2043
Pressure [atm]	9.2	86.8
Fuel Mole Fraction [%]	0.25	3.4
Oxygen Mole Fraction [%]	0.14	6.7
Equivalence Ratio	0.5	4.0

Type of Data:

Methane Table 8: OH species concentration time-history measurements in argon using laser absorption at 306.7 nm.

Methane Table 8: High pressure ignition delay time data in argon using same definition as found in Methane Table 4 of Volume 1.

Methane Table 9: CH_4 species concentration time-history induction time and end time are determined from the interception points (times) of a linear fit through the decaying CH_4 signal and the initial and final signal levels respectively.

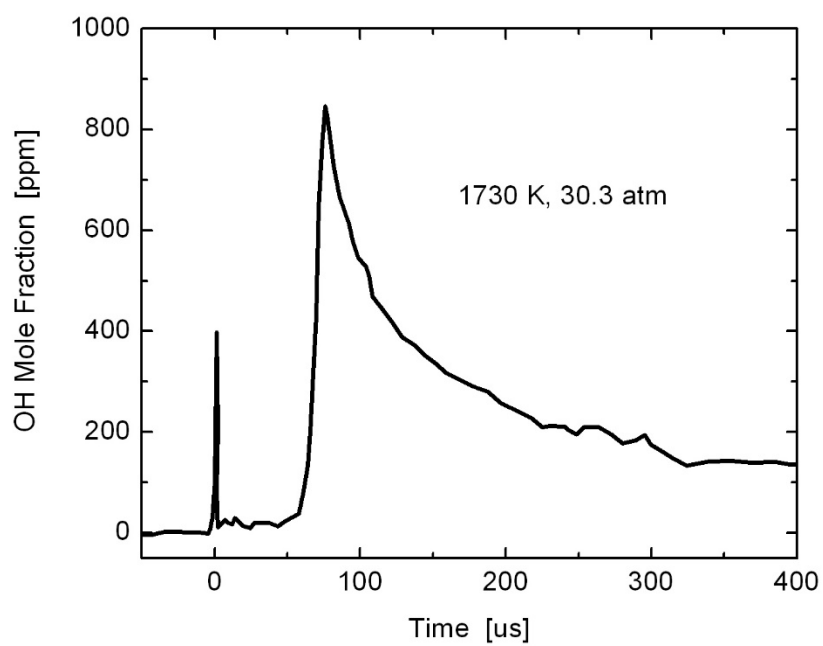
Methane Table 9: High pressure ignition delay time data using same definition as found in Methane Table 4 of Volume 1.

Methane Table 8:

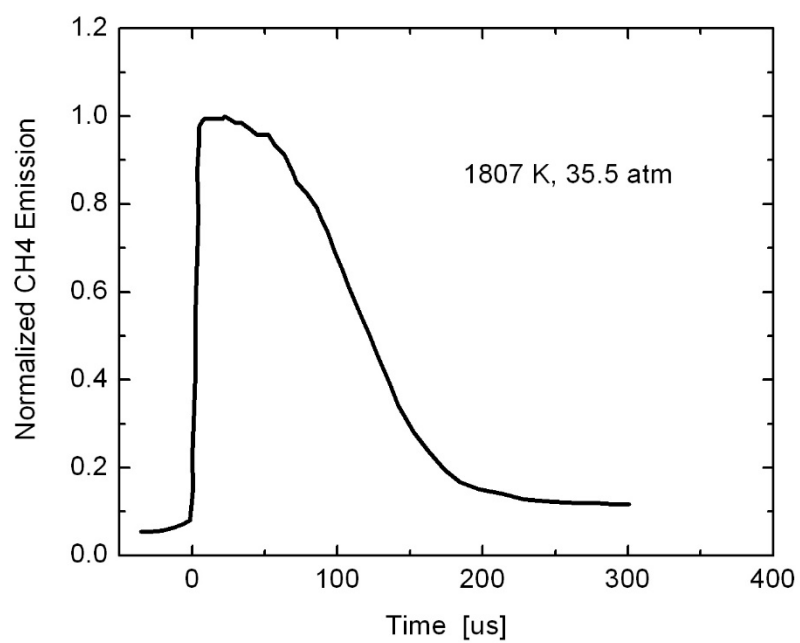
T ₅ [K]	P ₅ [atm]	Methane [%]	O ₂ [%]	ϕ □ □ (E.R.)	OH First Rise [μs]	OH Peak [μs]	Ignition (Pressure) [μs]
1680	85.3	0.25	1	0.5	92	118	
1770	31.2	0.25	1	0.5	62	89	
1730	30.3	0.5	2	0.5	64	73	60
1659	80.9	0.5	2	0.5	75	82	74
1665	81.2	0.5	2	0.5			62
1830	9.8	0.5	2	0.5	68	98	66
1853	10.5	0.5	2	0.5	51	75	47
1706	83.9	0.28	0.56	1		200	160
1811	31.3	0.28	0.56	1		128	
1687	30.5	0.28	0.56	1		401	
1761	36.8	0.28	0.56	1		177	
1821	79.4	0.28	0.56	1		74	
1778	79.1	0.56	1.14	1	47	53	46
1957	9.7	0.56	1.14	1	40	58	
2043	9.4	0.56	1.14	1	26	41	
1534	30.1	3.4	6.7	1			206
1535	61.9	3.4	6.7	1			113
1633	31.7	3.4	6.7	1			51
1584	34.1	3.4	6.7	1			96
1496	34.4	3.4	6.7	1			229
1408	34.6	3.4	6.7	1			554
1408	34.6	3.4	6.7	1			554

Methane Table 9:

T ₅ [K]	P ₅ [atm]	Methane [%]	O ₂ [%]	$\phi_{\square\square}$ (E.R.)	CH ₄ Induction [μs]	CH ₄ End [μs]	Ignition (Pressure) [μs]
1673	84.4	0.25	1	0.5	95	127	
1590	86.8	0.25	1	0.5	209	278	
1671	32.8	0.25	1	0.5	142	184	140
1627	31.7	1	4	0.5	88	95	92
1532	32.3	1	4	0.5	227	274	237
1512	34.5	1	4	0.5	314	346	317
1426	34.1	1	4	0.5	725	810	747
1739	10.1	1	4	0.5	82	94	74
1988	30.2	0.25	0.25	2	14	63	
2031	9.3	0.25	0.25	2	20	97	
1710	39.5	0.5	0.5	2	169	491	
1896	77.1	0.5	0.5	2	15	56	
1926	9.4	0.5	0.5	2	63	144	
1887	9.4	0.5	0.5	2	81	178	
1807	35.5	0.5	0.5	2	63	165	
1425	36.6	5	5	2	783		785
1579	34.6	5	5	2	173		165
1536	33.7	5	5	2	376		352
1455	33.9	5	5	2	671		645
1555	30.4	5	5	2	268		271
1633	29.1	5	5	2	118		110
1635	10.2	5	5	2	333		336
1744	9.8	5	5	2	110		114
1478	60.3	5	5	2	321		307
1706	80.9	0.25	0.14	3.6	174		
1742	29.9	0.25	0.14	3.6	163		
1977	32.3	0.25	0.14	3.6	16		
1692	79.9	0.25	0.14	3.6	163		
1949	9.5	0.25	0.14	3.6	53	266	
1802	9.2	0.25	0.14	3.6	238	910	
1690	85.7	0.5	0.25	4	177	702	
1818	31.4	0.5	0.25	4	60	283	



Methane Figure 8.



Methane Figure 9.

Ethane

Literature Source of Data:

E. J. Chang, "Shock Tube Experiments for the Development and Validation of Models of Hydrocarbon Combustion," M. Eng. Thesis, (also published as HTGL Report No. T-320), Mechanical Engineering Department, Stanford University, Stanford CA (1995).

M. Frenklach, H. Wang, M. Goldenberg, G. P. Smith, D. M. Golden, C. T. Bowman, R. K. Hanson, W. C. Gardiner, V. Lissianski, "GRI-Mech – An Optimized Detailed Chemical Reaction Mechanism for Methane Combustion," Topical Report GRI-95/0058, Gas Research Institute (1995).

Range of Data:

Temperature [K]	1717	2497
Pressure [atm]	0.54	1.23
Fuel Mole Fraction [ppm]	247	2012
Oxygen Mole Fraction [ppm]	1055	7062
Methane Mole Fraction [ppm]	0	503
Equivalence Ratio	0.98	1.01

Type of Data:

Ethane Table 1: CO₂ species concentration time-history IR laser absorption measurements at 4.200 μm (2380.72 cm^{-1}) in argon. Low pressure experiments were performed behind incident shock waves.

Ethane Table 2: CH₃ species concentration time-history UV laser absorption measurements at 216 nm in argon. Includes mixtures with added methane.

Ethane Table 3: OH species concentration time-history UV laser absorption measurements at 306.7 nm (32606.56 cm^{-1}) in argon.

Ethane Table 1:

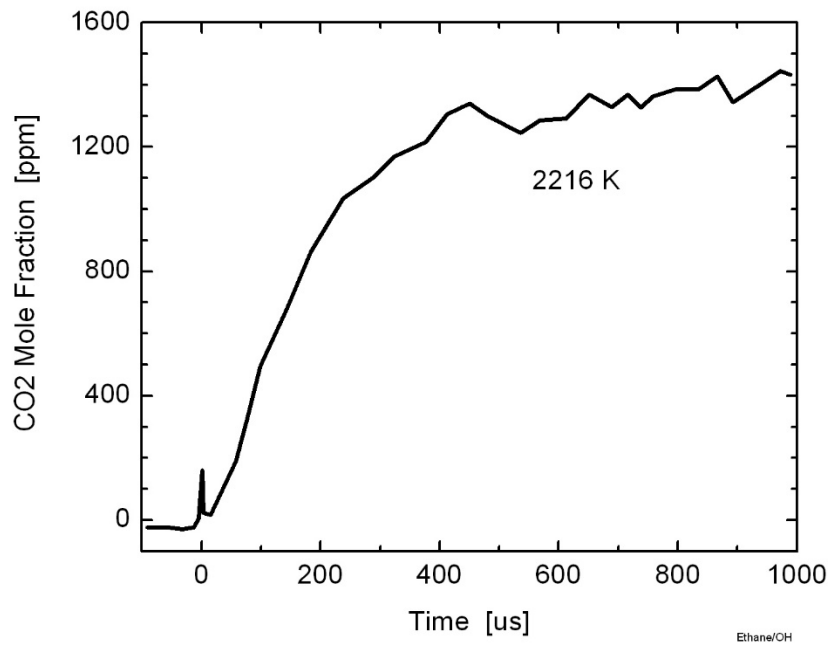
T2 [K]	P2 [atm]	Ethane [ppm]	Oxygen [ppm]	ϕ (E.R.)	CO ₂ @1ms [ppm]	Plateau CO ₂ [ppm]	¼ Plat [μs]
2216	0.62	2012	7062	1.0	1455	2568	133
2497	0.537	2012	7062	1.0	1123	1475	68

Ethane Table 2:

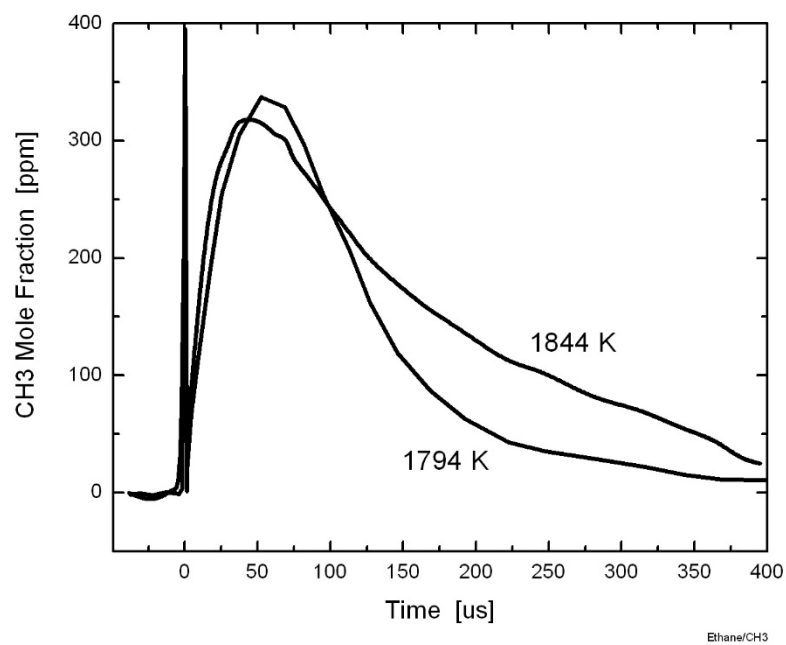
T ₅ [K]	P ₅ [atm]	Ethane [ppm]	Methane [ppm]	Oxygen [ppm]	φ (E.R.)	Peak CH ₃ [ppm]	Time [μs]
1794	1.166	295	0	1055	0.98	342	59
1717	1.201	247	503	1855	1.01	80	256
1759	1.163	247	503	1855	1.01	55	302
1785	1.181	247	503	1855	1.01	67	297
1844	1.233	247	503	1855	1.01	48	321

Ethane Table 3:

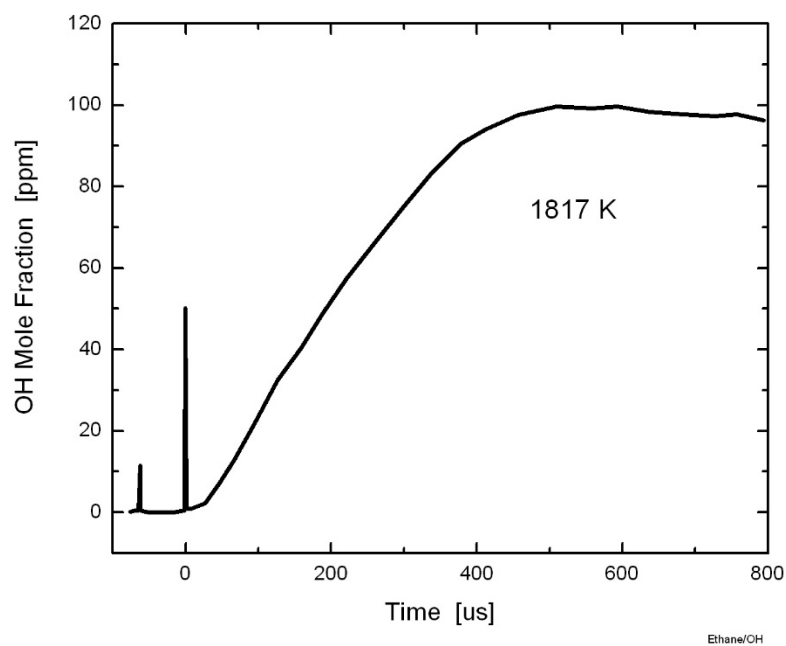
T ₅ [K]	P ₅ [atm]	Ethane [ppm]	Oxygen [ppm]	φ (E.R.)	Max OH [ppm]	½ Max [μs]
1757	1.179	300	1052	1.0	94	250
1817	1.209	300	1052	1.0	100	195
2012	1.183	300	1052	1.0	135	115



Ethane Figure 1.



Ethane Figure 2.



Ethane Figure 3.

Ethane

Literature Source of Data:

M. Rohrig, E. L. Petersen, D. F. Davidson, R. K. Hanson, C. T. Bowman, *"Measurement of the Rate Coefficient of the Reaction $\text{CH} + \text{O}_2 = \text{Products}$ in the Temperature Range 2200 to 2600 K,"* International Journal Chemical Kinetics 29: 781-789 (1997).

Range of Data:

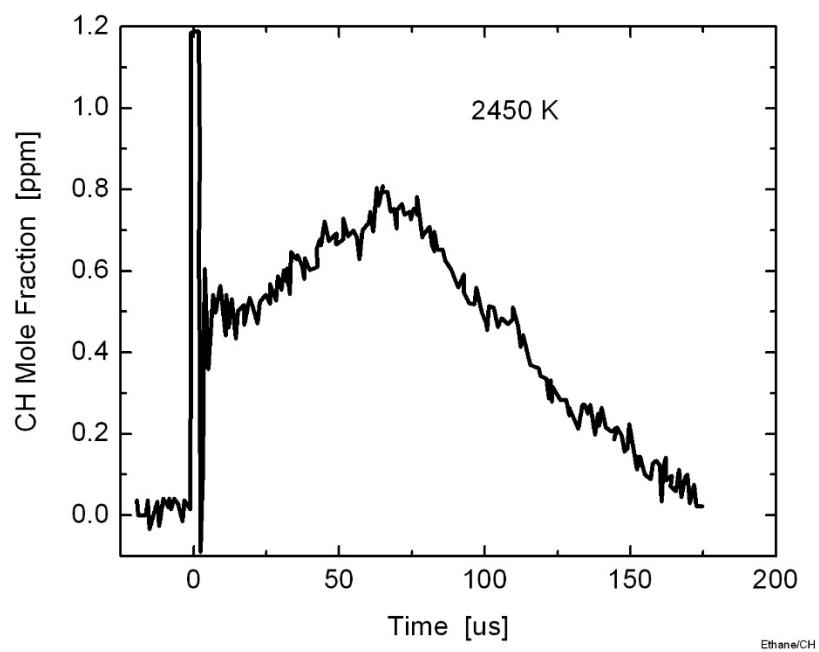
Temperature [K]	2121	2450
Pressure [atm]	1.1	1.2
Fuel Mole Fraction [%]	100	150
Oxygen Mole Fraction [%]	972	1002
Equivalence Ratio	0.35	0.54

Type of Data:

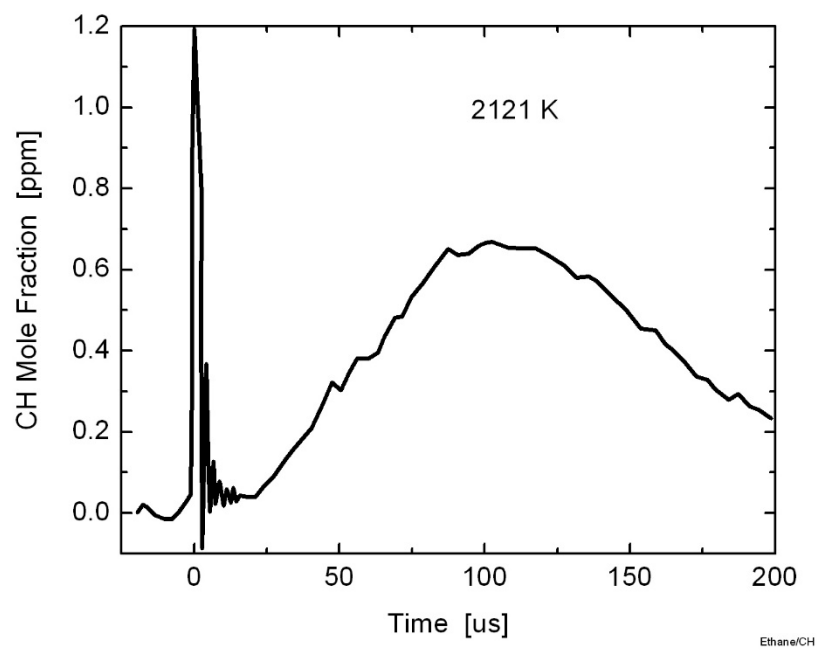
Ethane Table 4: CH species concentration time-history measurements based on laser absorption measurements at 431 nm in argon. CH data was derived from a digitization of Figs. 1 and 5 of Rohrig et al. (1997).

Ethane Table 4:

T ₅ [K]	P ₅ [atm]	Ethane [ppm]	Oxygen [ppm]	ϕ (E.R.)	1 st Plateau [ppm]	Peak CH [ppm]	Peak [μs]
2450	1.2	100	1002	0.35	0.5	7.5	70
2121	1.1	150	972	0.54		6.5	100



Ethane Figure 4a.



Ethane Figure 4b.

Normal Alkanes

Propane

Literature Source of Data:

D. F. Davidson, J. T. Herbon, D. C. Horning, R. K. Hanson, "*OH Concentration Time-histories in n-Alkane Oxidation*," International Journal of Chemical Kinetics 33: 775-783 (2001).

Range of Data:

Temperature [K]	1497	1687
Pressure [atm]	2.13	2.24
Fuel Mole Fraction [ppm]	500	1500
Oxygen Mole Fraction [ppm]	2500	7500
Equivalence Ratio	1	1

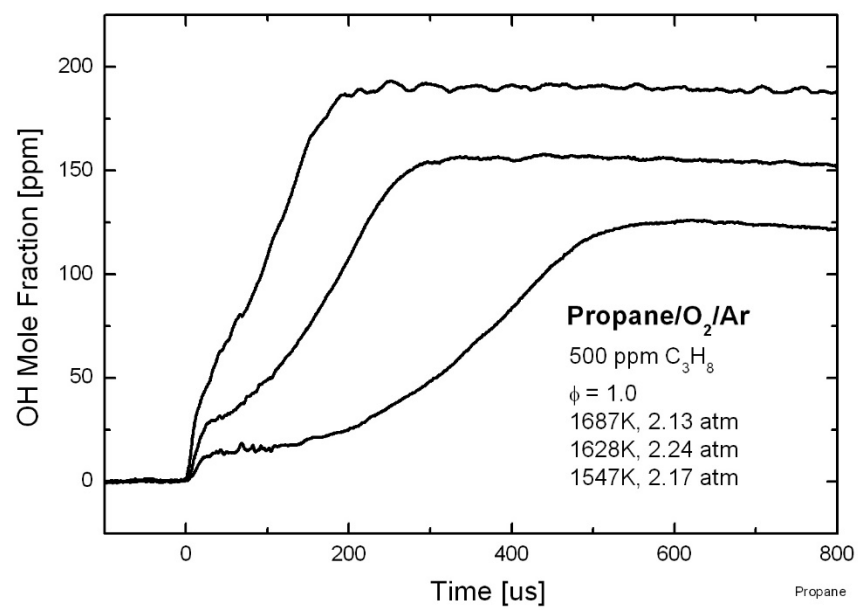
Type of Data:

Propane Table 1: OH concentration time-history measurements using laser absorption at 306.7 nm. $T_1(90\%)$ and $T_2(50\%)$ are the time for the OH mole fraction to reach 90% of the initial plateau level and 50% of the final plateau level respectively. For those conditions where no 1st plateau is indicated, no distinct plateau existed because of the rapid rise to the final plateau level. Carrier gas is argon.

Other unpublished data exist from this study.

Propane Table 1:

T_5 [K]	P_5 [atm]	Propane [ppm]	O ₂ [ppm]	Absorption Coefficient [atm ⁻¹ cm ⁻¹]	$T_1(90\%)$ [μs]	1 st Plateau [ppm]	$T_2(50\%)$ [μs]	2 nd Plateau [ppm]
1547	2.17	500	2500	146	28	14	344	129
1628	2.24	500	2500	130	25	30	156	157
1687	2.13	500	2500	127			89	190



Propane Figure 1.

***n*-Butane**

Literature Source of Data:

D. F. Davidson, J. T. Herbon, D. C. Horning, R. K. Hanson, "*OH Concentration Time-histories in n-Alkane Oxidation*," International Journal of Chemical Kinetics 33: 775-783 (2001).

Range of Data:

Temperature [K]	1531	1761
Pressure [atm]	2.02	2.06
Fuel Mole Fraction [ppm]	500	500
Oxygen Mole Fraction [ppm]	3250	3250
Equivalence Ratio	1	1

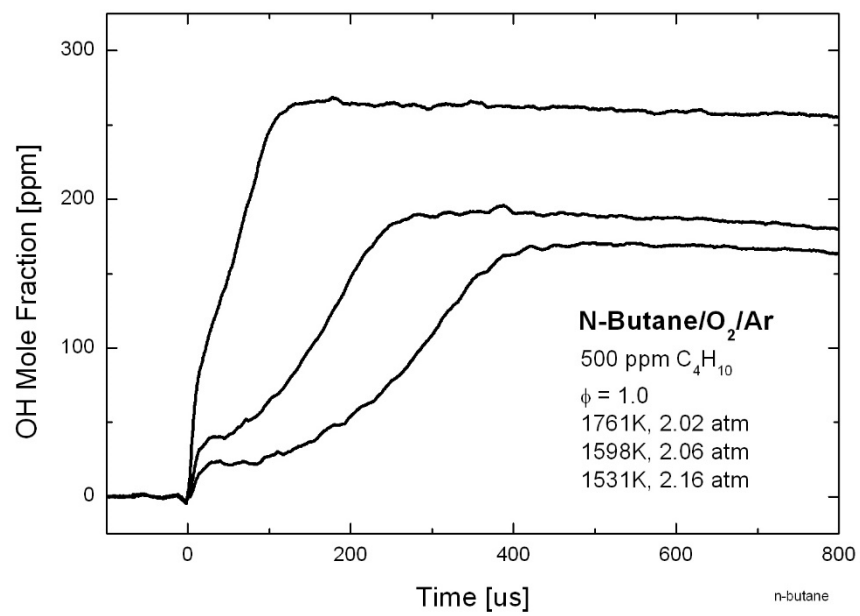
Type of Data:

Butane Table 1: OH concentration time-history measurements using laser absorption at 306.7 nm. $T_1(90\%)$ and $T_2(50\%)$ are the time for the OH mole fraction to reach 90% of the initial plateau level and 50% of the final plateau level respectively. For those conditions where no 1st plateau is indicated, no distinct plateau existed because of the rapid rise to the final plateau level. Carrier gas is argon.

Other unpublished data exist from this study.

n-Butane Table 1:

T_5 [K]	P_5 [atm]	n- Butane [ppm]	O ₂ [ppm]	Absorption Coefficient [atm ⁻¹ cm ⁻¹].	$T_1(90\%)$ [μ s]	1 st Plateau [ppm]	$T_2(50\%)$ [μ s]	2 nd Plateau [ppm]
1531	2.16	500	3250	147	24	23	260	175
1598	2.06	500	3250	144	21	40	144	192
1761	2.02	500	3250	121			40	266



n-Butane Figure 1.

***n*-Heptane**

Literature Source of Data:

D. F. Davidson, J. T. Herbon, D. C. Horning, R. K. Hanson, "*OH Concentration Time-histories in n-Alkane Oxidation*," International Journal of Chemical Kinetics 33: 775-783 (2001).

Range of Data:

Temperature [K]	1540	1784
Pressure [atm]	2.04	3.80
Fuel Mole Fraction [ppm]	300	300
Oxygen Mole Fraction [ppm]	3300	3300
Equivalence Ratio	1	1

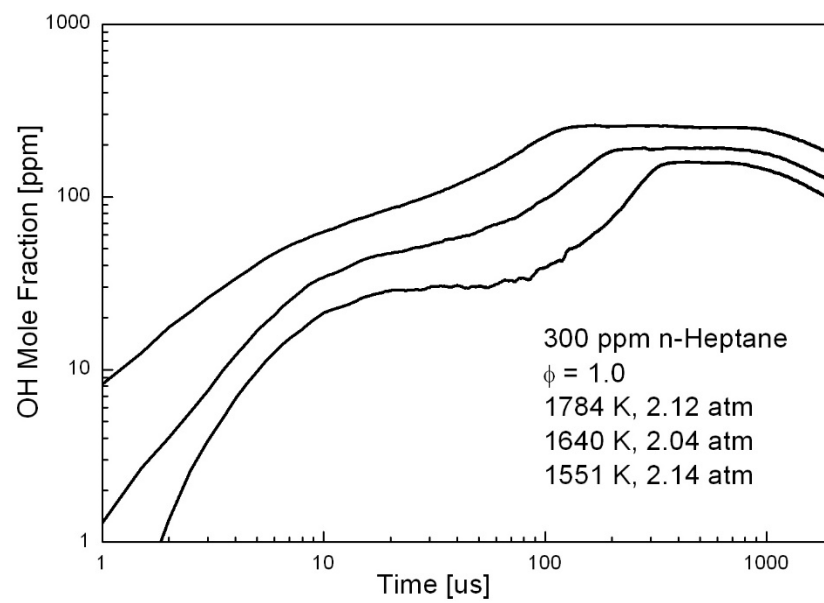
Type of Data:

n-Heptane Table 1: OH concentration time-history measurements using laser absorption at 306.7 nm. $T_1(90\%)$ and $T_2(50\%)$ are the time for the OH mole fraction to reach 90% of the initial plateau level and 50% of the final plateau level respectively. For those conditions where no 1st plateau is indicated, no distinct plateau existed because of the rapid rise to the final plateau level. Carrier gas is argon.

Other unpublished data exist from this study.

n-Heptane Table 1:

T_5 [K]	P_5 [atm]	<i>n</i> - Heptane [ppm]	O ₂ [ppm]	Absorption Coefficient [atm ⁻¹ cm ⁻¹].	$T_1(90\%)$ [μs]	1 st Plateau [ppm]	$T_2(50\%)$ [μs]	2 nd Plateau [ppm]
1551	2.14	300	3300	150	16	30	207	160
1640	2.04	300	3300	139			99	191
1784	2.12	300	3300	118			47	255



n-Heptane Figure 1.

***n*-Heptane**

Literature Source of Data:

D. F. Davidson, M. A. Oehlschlaeger, R. K. Hanson, "*Methyl Concentration Time-Histories during iso-Octane and n-Heptane Oxidation and Pyrolysis*," Proceedings of the Combustion Institute 31: in press (2006).

Range of Data:

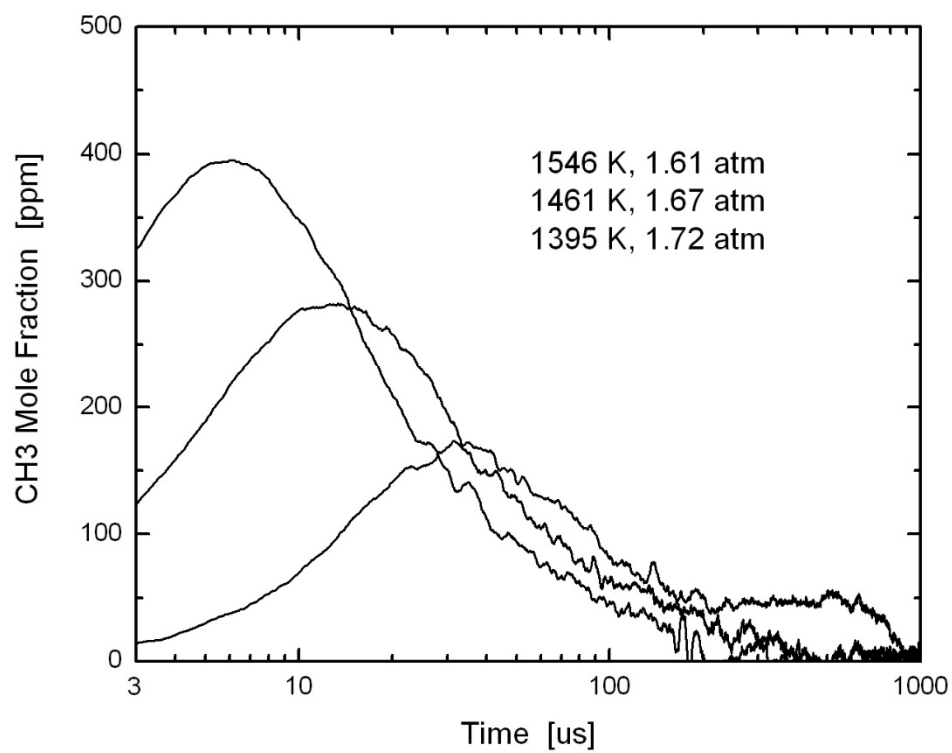
Temperature [K]	1100	1560
Pressure [atm]	1.6	2.0
Fuel Mole Fraction [ppm]	100	500
Oxygen Mole Fraction [%]	0.55	0.625
Equivalence Ratio	1.0	1.0

Type of Data:

n-Heptane Table 2-1: CH₃ concentration time-history measurements in argon using narrow-linewidth ring-dye laser absorption of the peak of the pre-dissociatively broadened CH₃ absorpti0n feature B²A'₁-X²A''₂ near 216 nm. Measurements at three wavelengths, 216.55, 214.97 and 218.70 nm, were deconvolved to separate out the CH₃ absorption from that of a time-dependent interference absorption.

n-Heptane Table 2-1:

T ₅ [K]	P ₅ [atm]	C ₇ H ₁₆ [ppm]	O ₂ [ppm]
1395	1.72	500	5500
1461	1.67	500	5500
1546	1.61	500	5500



n-Heptane Figure 2-1.

n-Decane

Literature Source of Data:

D. F. Davidson, J. T. Herbon, D. C. Horning, R. K. Hanson, "*OH Concentration Time-histories in n-Alkane Oxidation*," International Journal of Chemical Kinetics 33: 775-783 (2001).

Range of Data:

Temperature [K]	1357	1706
Pressure [atm]	2.03	2.28
Fuel Mole Fraction [ppm]	300	2000
Oxygen Mole Fraction [ppm]	3875	3100
Equivalence Ratio	0.8	1.2

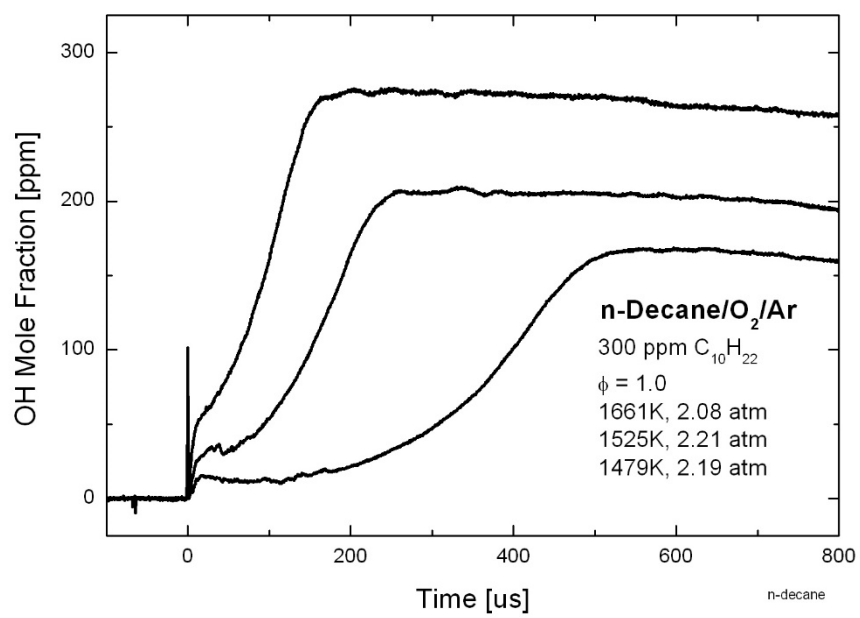
Type of Data:

n-Decane Table 1: OH concentration time-history measurements using laser absorption at 306.7 nm. $T_1(90\%)$ and $T_2(50\%)$ are the time for the OH mole fraction to reach 90% of the initial plateau level and 50% of the final plateau level respectively. For those conditions where no 1st plateau is indicated, no distinct plateau existed because of the rapid rise to the final plateau level. Carrier gas is argon.

Other unpublished data exist from this study.

n-Decane Table 1:

T_5 [K]	P_5 [atm]	n- Decane [ppm]	O ₂ [ppm]	Absorption Coefficient [atm ⁻¹ cm ⁻¹].	$T_1(90\%)$ [μs]	1 st Plateau [ppm]	$T_2(50\%)$ [μs]	2 nd Plateau [ppm]
1479	2.19	300	4650	164	11	14	375	166
1525	2.21	300	4650	140		34	154	206
1661	2.08	300	4650	134			87	273



n-Decane Figure 1.

Branched Alkanes

Iso-Butane

Literature Source of Data:

M. A. Oehlschlaeger, D. F. Davidson, J. T. Herbon, R. K. Hanson, "*Shock Tube Measurements of Branched Alkane Ignition Times and OH Concentration Time-histories*," International Journal of Chemical Kinetics 36: 67-78 (2004).

M. A. Oehlschlaeger, D. F. Davidson, J. T. Herbon, R. K. Hanson, "*Shock Tube Measurements of Branched Alkane Ignition Times and OH Concentration Time-histories*," AIAA paper 2003-0830, 41st AIAA Aerospace Sciences Meeting and Exhibit, Reno NV (2003).

Range of Data:

Temperature [K]	1291	1864
Pressure [atm]	1.33	12.58
Fuel Mole Fraction [%]	0.01	1
Oxygen Mole Fraction [%]	0.065	6.5
Equivalence Ratio	0.25	2.0

Type of Data:

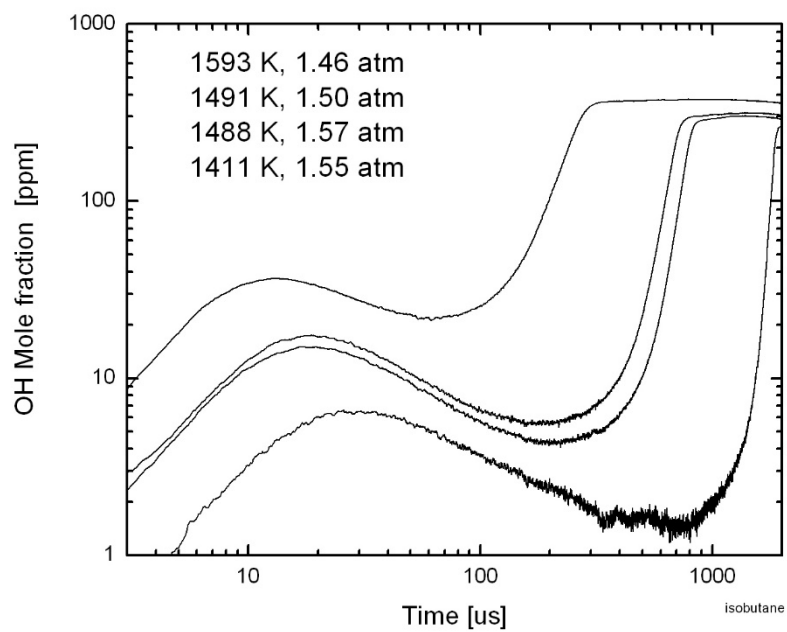
Iso-Butane Table 1: OH concentration time-history measurements in argon using narrow-linewidth ring-dye laser absorption of the R₁(5) line of the OH A-X (0,0) band at 306.7 nm. "Sat." indicates laser transmission of ~0.

Other unpublished data exists from this study.

Iso-Butane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	ϕ E.R.	First Peak [μ s]	First Peak [ppm]	Minimum [μ s]	Minimum [ppm]	50% Peak [μ s]	Peak [ppm]
1366	1.61	0.5	3.25	1	16	5	346	3	1429	Sat.
1426	1.55	0.5	3.25	1	9	13	198	6	680	Sat.
1520	1.49	0.5	3.25	1	7	34	76	17	233	Sat.
1646	1.46	0.5	3.25	1	4	96	20	63	65	Sat.
1339	1.69	0.25	3.25	0.5	27	4	353	2	1128	641
1402	1.62	0.25	3.25	0.5	19	10	156	6	488	646
1488	1.58	0.25	3.25	0.5	11	28	55	19	165	777
1541	1.47	0.25	3.25	0.5	9	45	30	36	93	831
1291	1.67	0.125	3.25	0.25	65	3	381	2	1476	342
1341	1.63	0.125	3.25	0.25	45	9	212	6	735	386
1390	1.56	0.125	3.25	0.25	34	19	130	15	372	466
1474	1.51	0.125	3.25	0.25	26	56	40	55	120	608
1411	1.55	0.1	0.65	1	30	6	550	2	1810	268
1488	1.57	0.1	0.65	1	18	15	212	4	758	303
1491	1.50	0.1	0.65	1	19	17	178	6	662	315
1593	1.46	0.1	0.65	1	13	37	61	22	237	375
1720	1.41	0.1	0.65	1	12	71	20	69	87	470
1444	1.59	0.05	0.325	1	33	7	390	2	1770	143
1546	1.58	0.05	0.325	1	23	17	155	7	572	177
1630	1.49	0.05	0.325	1	18	29	61	23	248	214
1703	1.40	0.05	0.325	1	18	42	31	41	146	242
1603	2.70	0.05	0.325	1	12	21	69	11	255	187
1625	2.50	0.05	0.325	1	12	23	55	14	224	187
1723	2.50	0.05	0.325	1	11	37	24	35	100	255
1545	5.61	0.05	0.325	1	7	9	85	3	368	137
1577	5.03	0.05	0.325	1	7	16	64	5	255	180
1750	4.78	0.05	0.325	1	5	42	12	40	49	254

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	Absorption Coefficient [atm ⁻¹ cm ⁻¹]
1411	1.55	0.1	0.65	187
1488	1.57	0.1	0.65	171
1491	1.50	0.1	0.65	172
1593	1.46	0.1	0.65	154



iso-Butane Figure 1.

Iso-Pentane

Literature Source of Data:

M. A. Oehlschlaeger, D. F. Davidson, J. T. Herbon, R. K. Hanson, "Shock Tube Measurements of Branched Alkane Ignition Times and OH Concentration Time-histories," International Journal of Chemical Kinetics 36: 67-78 (2004).

Range of Data:

Temperature [K]	1300	1726
Pressure [atm]	1.56	5.67
Fuel Mole Fraction [%]	0.025	1.25
Oxygen Mole Fraction [%]	0.8	20
Equivalence Ratio	0.25	2

Type of Data:

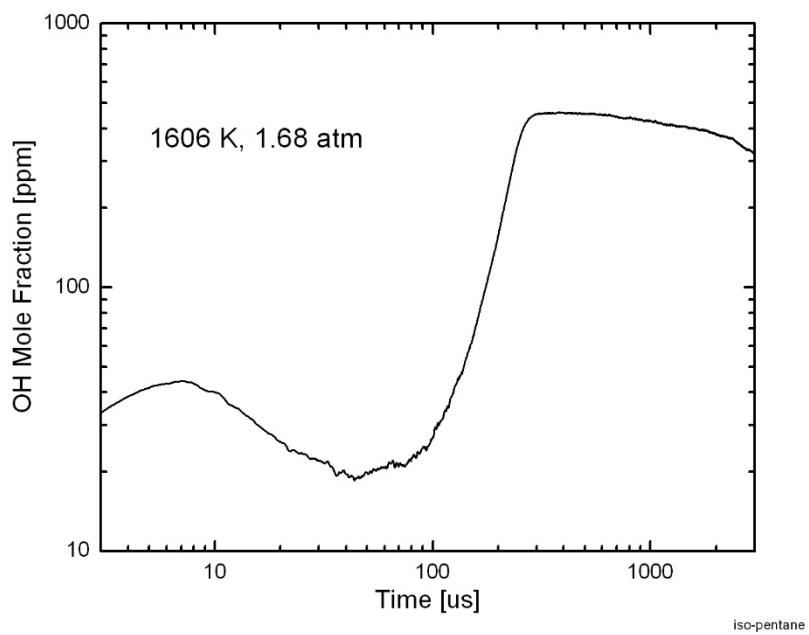
Iso-Pentane Table 1: OH concentration time-history measurements in argon using narrow-linewidth ring-dye laser absorption of the R₁(5) line of the OH A-X (0,0) band at 306.7 nm. "Sat." indicates laser transmission of ~0.

Other unpublished data exists from this study.

Iso-Pentane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	First Peak [μs]	First Peak [ppm]	Minimum [μs]	Minimum [ppm]	50% Peak [μs]	Peak [ppm]
1534	1.68	0.5	4.0	1	7	21	117	12	235	Sat.
1552	1.62	0.5	4.0	1	5	28	113	15	212	Sat.
1413	1.74	0.1	0.8	1	19	9	458	2	1788	344
1517	1.74	0.1	0.8	1	9	26	112	8	551	396
1606	1.68	0.1	0.8	1	7	44	44	19	219	459
1708	1.62	0.1	0.8	1	6	71	22	56	95	571
1609	5.40	0.1	0.8	1	3	18	17	4	183	281
1635	5.67	0.1	0.8	1	3	35	13	7	146	308
1726	5.22	0.1	0.8	1	2	39	10	19	103	355

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	Absorption Coefficient [atm ⁻¹ cm ⁻¹].
1606	1.68	0.1	0.8	145



iso-Pentane Figure 1.

Iso-Octane

Literature Source of Data:

D. F. Davidson, M. A. Oehlschlaeger, J. T. Herbon, R. K. Hanson, “*Shock Tube Measurements of Iso-Octane Ignition Times and OH Concentration Time-histories*,” Proceedings of the Combustion Institute 29: 1295-1301 (2002).

Range of Data:

Temperature [K]	1177	2009
Pressure [atm]	1.10	8.17
Fuel Mole Fraction [%]	0.01	1.0
Oxygen Mole Fraction [%]	0.125	12.5
Equivalence Ratio	0.25	2.0

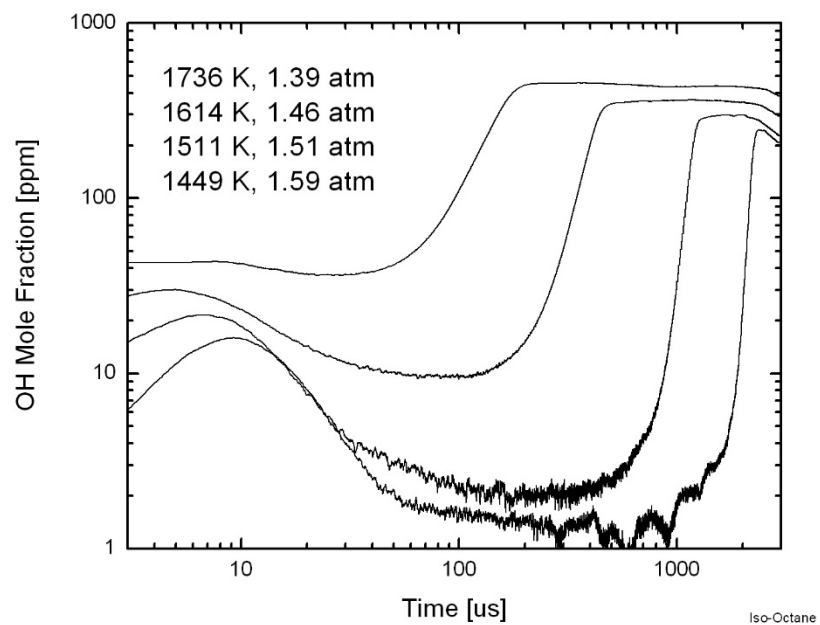
Type of Data:

Iso-Octane Table 1: OH concentration time-history measurements in argon using narrow-linewidth ring-dye laser absorption of the R₁(5) line of the OH A-X (0,0) band at 306.7 nm. “Sat.” indicates laser transmission of ~0.

Other unpublished data exist from this study.

Iso-Octane Table 1:

T ₅ [K]	P ₅ [atm]	Fuel [%]	O ₂ [%]	φ E.R.	First Peak [μs]	First Peak [ppm]	Minimum [μs]	Minimum [ppm]	50% Peak [μs]	Peak [ppm]
1449	1.59	0.05	0.625	1	8	16		1	2182	245
1511	1.51	0.05	0.625	1	6	22		2	1152	288
1614	1.43	0.05	0.625	1	5	30	86	9	388	352
1736	1.40	0.05	0.625	1	5	44	27	36	137	456



iso-Octane Figure 1.

Iso-Octane

Literature Source of Data:

D. F. Davidson, M. A. Oehlschlaeger, R. K. Hanson, "*Methyl Concentration Time-Histories during iso-Octane and n-Heptane Oxidation and Pyrolysis*," Proceedings of the Combustion Institute 31: in press (2006).

Range of Data:

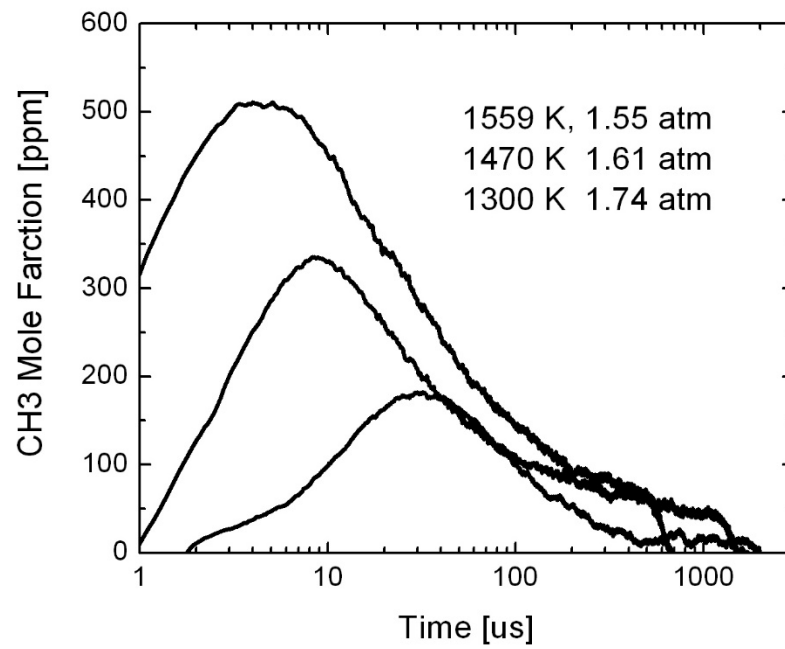
Temperature [K]	1100	1560
Pressure [atm]	1.6	2.0
Fuel Mole Fraction [ppm]	100	500
Oxygen Mole Fraction [%]	0.55	0.625
Equivalence Ratio	1.0	1.0

Type of Data:

Iso-Octane Table 2-1: CH₃ concentration time-history measurements in argon using narrow-linewidth ring-dye laser absorption of the peak of the pre-dissociatively broadened CH₃ absorpti0n feature B²A'₁-X²A''₂ near 216 nm. Measurements at three wavelengths, 216.55, 214.97 and 218.70 nm, were deconvolved to separate out the CH₃ absorption from that of a time-dependent interference absorption.

Iso-Octane Table 2-1:

T ₅ [K]	P ₅ [atm]	C ₈ H ₁₈ [ppm]	O ₂ [ppm]
1300	1.74	500	6250
1470	1.61	500	6250
1559	1.55	500	6250



iso-Octane Figure 2-1.

Iso-Octane

Literature Source of Data:

D. F. Davidson, B. M. Gauthier, R. K. Hanson, “*Shock Tube Ignition Measurements of Iso-Octane/Air and Toluene/Air at High Pressures,*” Proceedings of the Combustion Institute 20: 1175-1182 (2005).

Range of Data:

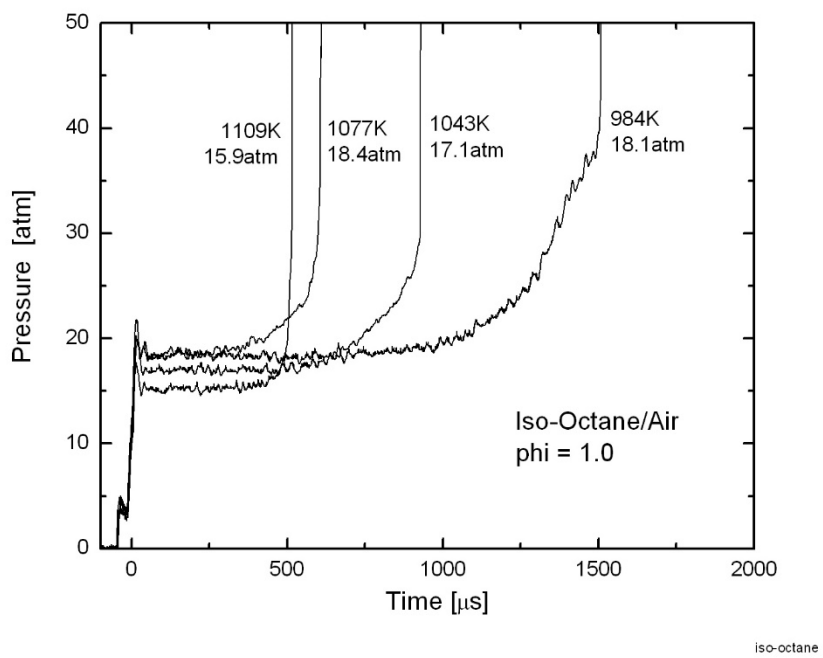
Temperature [K]	855	1159
Pressure [atm]	14.9	59.3
Fuel Mole Fraction [%]	0.829	1.652
Oxygen Mole Fraction [%]	20.8	20.7
Equivalence Ratio	0.5	1.0

Type of Data:

Iso-Octane Table 2-2: Ignition delay time measurement in synthetic air (79% N₂, 21% O₂) based on sidewall PZT pressure measurements and confirmed with CH* (at 431 nm) and OH* (at 306nm) emission measurements.

Iso-Octane Table 2-2:

T ₅ [K]	P ₅ [atm]	Iso-Octane [%]	φ (E.R.)
984	18.1	1.65	1
1043	17.1	1.65	1
1077	18.4	1.65	1
1109	15.9	1.65	1



iso-Octane Figure 2-2.

Cyclo-Alkanes

JP-10

Literature Source of Data:

D. F. Davidson, D. C. Horning, J. T. Herbon, R. K. Hanson, "*Shock Tube Measurements of JP-10 Ignition*," Proceedings of the Combustion Institute 28: 1687-1692 (2000).

Range of Data:

Temperature [K]	1352	1671
Pressure [atm]	1.15	9.20
Fuel Mole Fraction [%]	0.2	0.4
Oxygen Mole Fraction [%]	1.39	5.89
Equivalence Ratio	0.48	2.01

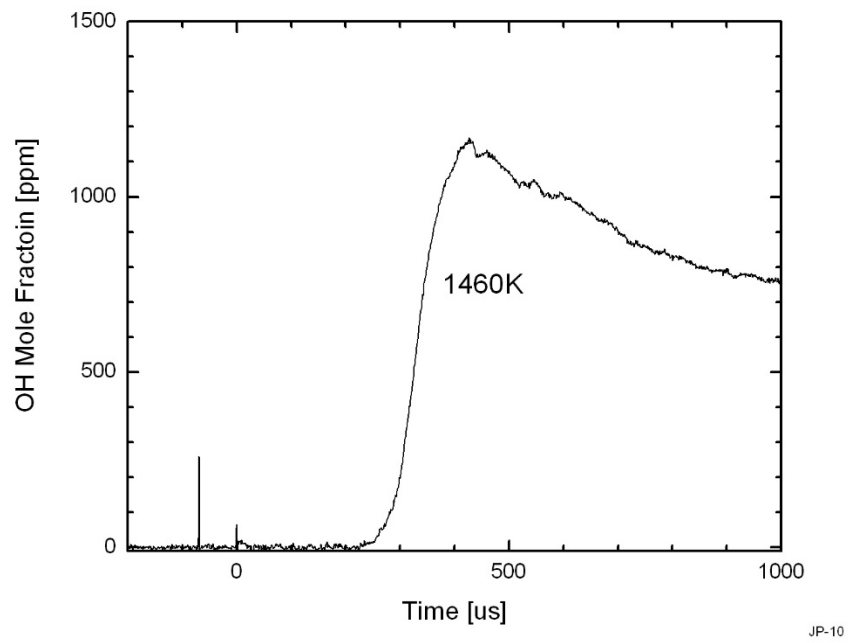
Type of Data:

JP-10 Table 1: OH concentration time-history measurements in argon using narrow-linewidth ring-dye laser absorption of the R₁₂(7) line of the OH A-X (0,0) band near 306.7 nm. The absorption coefficient for 1460K example is $k_{\lambda} = 21.56$ [atm⁻¹cm⁻¹].

Other unpublished data exists.

JP-10 Table 1:

T5 [K]	P5 [atm]	JP-10 [%]	O2 [%]	ϕ (E.R.)	Time-to-peak [μ s]	Peak OH [ppm]
1460	3.02	0.20	2.83	0.99	429	1159



JP-10 Figure 1.

Olefins

1,3-Butadiene

Literature Source of Data:

C. S. Libby, D. F. Davidson, R. K. Hanson, "A Shock Tube Study of the Oxidation of 1,3-Butadiene," AIAA 2004-1322 (2004).

Range of Data:

Temperature [K]	1405	1585
Pressure [atm]	1.449	1.712
Fuel Mole Fraction [ppm]	250	1000
Oxygen Mole Fraction [%]	0.55	0.55
Equivalence Ratio	0.25	1.0

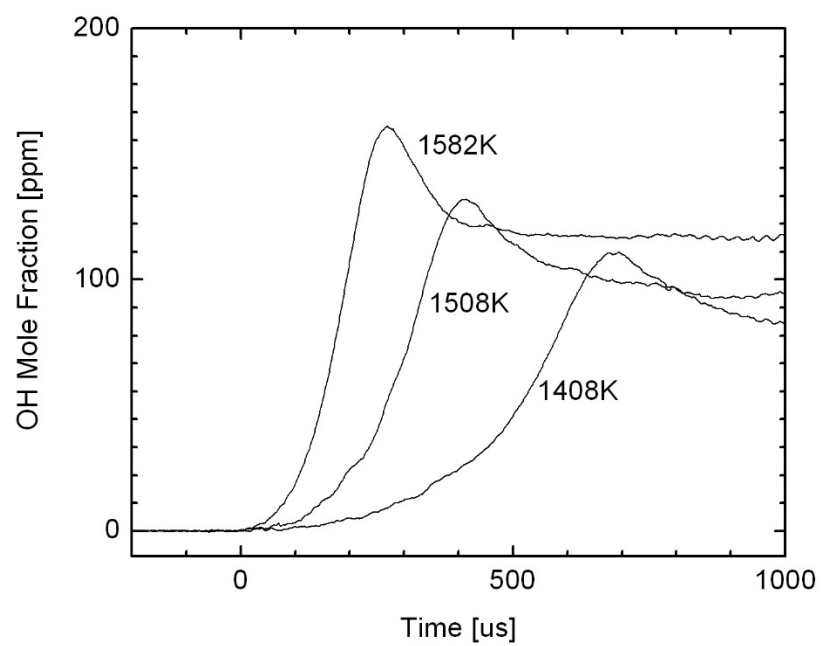
Type of Data:

1,3-Butadiene Table 1: OH concentration time-history measurements in argon using narrow-linewidth ring-dye laser absorption of the peak of the R₁(5) line of the OH A-X (0,0) band at 306.687 nm.

1,3-Butadiene Table 1:

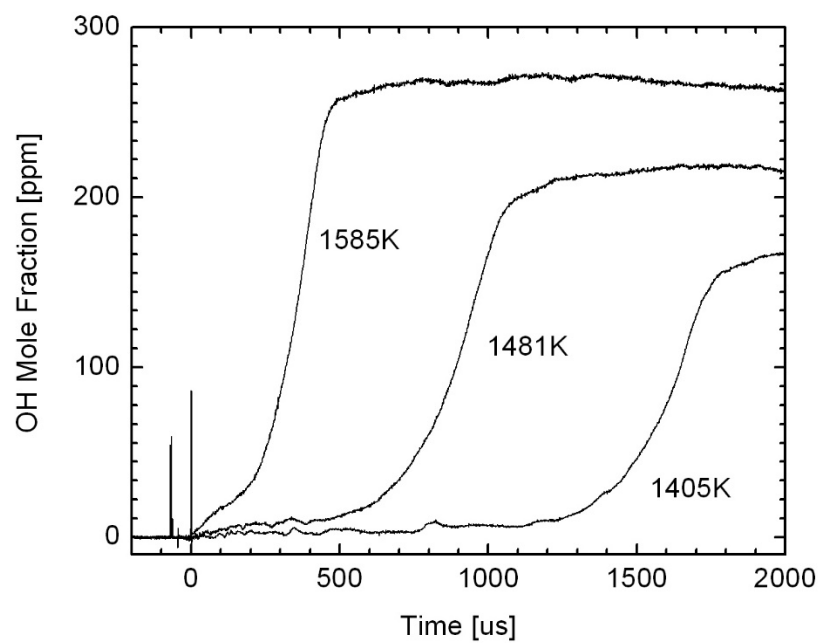
T ₅ [K]	P ₅ [atm]	C ₄ H ₆ [ppm]	φ (E.R.)	1/8 Peak [μs]	1/8 Peak [ppm]	1/4 Peak [μs]	1/4 Peak [ppm]	Plateau [ppm]	50% peak [μs]	50% peak [ppm]	Peak [μs]	Peak [ppm]
1408	1.712	250	0.25	86	1	173	4	111	529	56	690	111
1508	1.688	250	0.25	52	1	103	4	132	295	66	412	132
1582	1.682	250	0.25	34	3	68	8	117	178	81	270	161
1405	1.513	1000	1	287	3.2	574	3.5	175	1624	88	2294	175
1481	1.496	1000	1	207	7.8	414	9.6	218	906	109	1655	218
1585	1.449	1000	1	72	1.7	144	14.5	266	416	129	576	257

T ₅ [K]	P ₅ [atm]	Wavelength [nm]	Pathlength [cm]	Absorption Coefficient [atm ⁻¹ cm ⁻¹]
1408	1.712	306.687	15.24	190.84
1508	1.688	306.687	15.24	169.49
1582	1.682	306.687	15.24	155.33
1405	1.513	306.687	15.24	197.74
1481	1.496	306.687	15.24	180.14
1585	1.449	306.687	15.24	159.84



1,3-Butadiene Figure 1a.

1,3-butadiene



1,3-Butadiene Figure 1b.

1,3-butadiene

Aromatics

Toluene

Literature Source of Data:

D. F. Davidson, B. M. Gauthier, R. K. Hanson, “*Shock Tube Ignition Measurements of Iso-Octane/Air and Toluene/Air at High Pressures,*” Proceedings of the Combustion Institute 20: 1175-1182 (2005).

Range of Data:

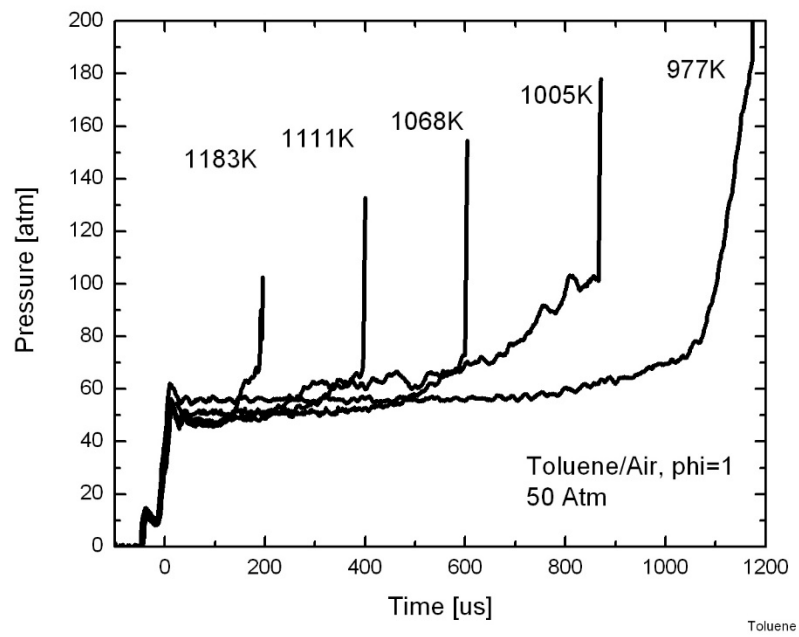
Temperature [K]	971	1269
Pressure [atm]	13.1	54.3
Fuel Mole Fraction [%]	1.15	2.28
Oxygen Mole Fraction [%]	20.8	20.5
Equivalence Ratio	0.5	1.0

Type of Data:

Toluene Table 2-1: Ignition delay time data in air using PZT pressure measurements of the time between the arrival of the reflected shock (the center of the reflected shock bifurcation feature) and the distinct ignition pressure rise (the time of the intersection of the linear extrapolation of the pressure rise with the pre-ignition pressure floor). Similar ignition delay times were recovered from CH and OH emission measurements.

Toluene Table 2-1:

T_5 [K]	P_5 [atm]	Toluene [%]	ϕ (E.R.)
977	54.3	2.28	1
1005	50.4	2.28	1
1068	48.0	2.28	1
1111	46.7	2.28	1
1183	46.2	2.28	1



Toluene Figure 2-1.

Toluene

Literature Source of Data:

V. Vasudevan, D. F. Davidson, R. K. Hanson, "*Shock Tube Measurements of Toluene Ignition Times and OH Concentration Time-histories*," Proceedings of the Combustion Institute 330: 1155-1163 (2005).

Range of Data:

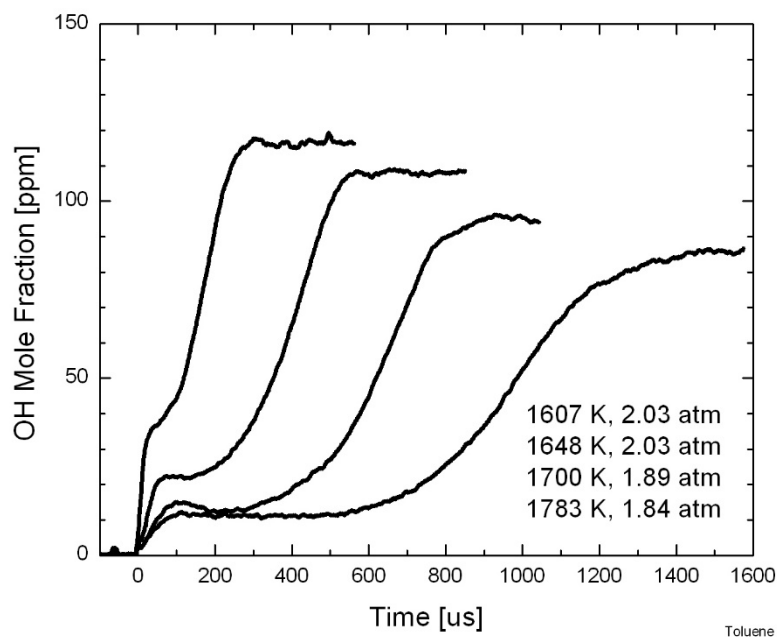
Temperature [K]	1458	1847
Pressure [atm]	1.75	4.54
Fuel Mole Fraction [ppm]	250	5000
Oxygen Mole Fraction [%]	0.225	4.5
Equivalence Ratio	0.5	1.5

Type of Data:

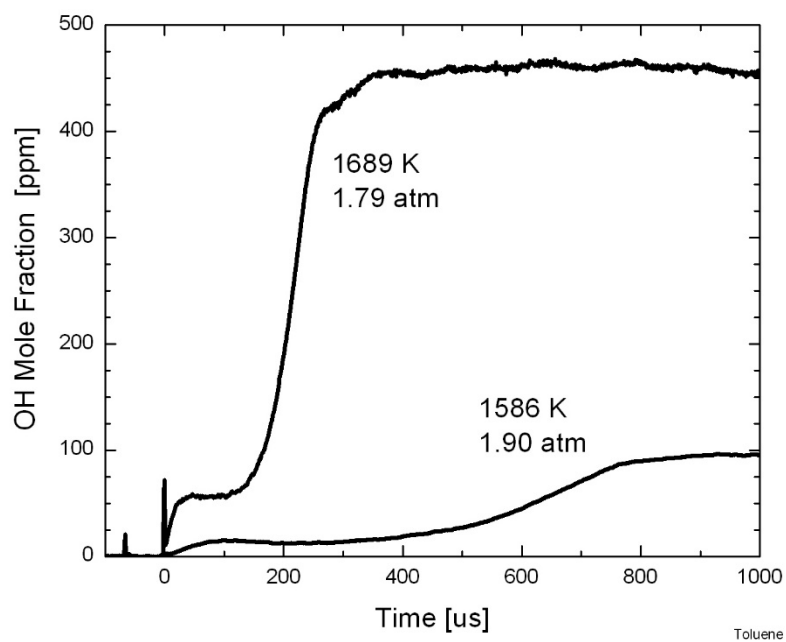
Toluene Table 1: OH concentration time-history measurements in argon using narrow-linewidth ring-dye laser absorption of the R₁(5) line of the OH A-X (0,0) band at 306.7 nm. "n.d." indicates that the plateau was not well defined and "sat." indicated a saturated signal with laser transmission ~0.

Toluene Table 1:

T ₅ [K]	P ₅ [atm]	C ₇ H ₈ [ppm]	O ₂ [%]	φ (E.R.)	First Plateau [μs]	First Plateau [ppm]	Ignition Time [μs]	Peak [ppm]
1607	2.03	250	0.225	1	121	12	949	87
1648	2.03	250	0.225	1	104	15	608	96
1700	1.89	250	0.225	1	64	23	369	109
1783	1.84	250	0.225	1	24	36	136	118
1564	1.95	1000	0.9	1	127	31	1068	324
1586	1.90	1000	0.9	1	97	32	702	377
1614	1.80	1000	0.9	1	64	40	389	388
1689	1.79	1000	0.9	1	43	56	209	460
1527	4.54	1000	0.9	1	n.d.	32	798	306
1541	4.43	1000	0.9	1	n.d.	36	651	318
1697	4.26	1000	0.9	1	37	53	150	421
1458	1.99	1000	1.8	0.5	n.d.	23	1123	424
1504	1.98	1000	1.8	0.5	230	26	725	448
1540	1.96	1000	1.8	0.5	110	31	501	455
1550	1.94	1000	1.8	0.5	54	31	386	501
1666	1.92	1000	1.8	0.5	28	65	153	591
1616	1.82	1000	0.6	1.5	100	35	1090	162
1627	1.92	1000	0.6	1.5	47	36	922	201
1714	1.77	1000	0.6	1.5	26	55	384	259
1847	1.75	1000	0.6	1.5	12	83	143	298
1434	2.03	5000	4.5	1	n.d.	75	1070	sat
1454	1.66	5000	4.5	1	n.d.	87	750	sat.
1618	1.88	5000	4.5	1	n.d.	177	130	sat.
1635	1.83	5000	4.5	1	n.d.	182	107	sat.



Toluene Figure 1a.



Toluene Figure 1b.

Species Measurement to January 2014

Hydrocarbon Fuels

Toluene

M. A. Oehlschlaeger, D. F. Davidson, R. K. Hanson, "*Investigation of the reaction of toluene with molecular oxygen in shock heated gases,*" Combustion and Flame 147 (2006) 195-208.

Benzyl radicals measured at 266 nm.

M. A. Oehlschlaeger, D. F. Davidson, R. K. Hanson, "*Experimental investigation of toluene+H = benzyl+H₂ at high temperatures,*" Journal of Physical Chemistry A 110 (2006) 9867-9873.

Benzyl radical measured at 266 nm

M. A. Oehlschlaeger, D. F. Davidson, R. K. Hanson, "*High-temperature thermal decomposition of benzyl radicals,*" Journal of Physical Chemistry A 110 (2006) 6649-6653.

Benzyl radical measured at 266 nm

M. A. Oehlschlaeger, D. F. Davidson, R. K. Hanson, "*Thermal decomposition of toluene: overall rate and branching ratio,*" Proceedings of the Combustion Institute 31 (2007) 211-219.

Benzyl radical measured at 266 nm

n-Heptane, n-Nonane, n-Butane

D. F. Davidson, M. A. Oehlschlaeger, R. K. Hanson, "*Methyl concentration time-histories during iso-octane and n-heptane oxidation and pyrolysis,*" Proceedings of the Combustion Institute 31 (2007) 321-328.

Methyl radicals measured at 216 nm.

Z. Hong, D. F. Davidson, R. K. Hanson, "*Shock tube studies of soot formation in heptane and heptane/DME mixtures,*" AIAA paper 2007-774, 43th AIAA Reno 2007.

Soot measured using extinction at 670 nm and emission at 633 nm.

Z. Hong, D. F. Davidson, R. K. Hanson, "*Shock tube studies of soot formation in rich heptane/oxygen mixtures with DME/acetone/butanal additives,*" Paper 07F-34, Fall Meeting, Western States Section/Combustion Institute, Sandia, Livermore, October 2007.

Soot measured using extinction at 670 nm and emission at 633 nm.

H. Li, A. Farooq, R. D. Cook, D. F. Davidson, J. B. Jeffries, R. K. Hanson, "A diode laser absorption sensor for rapid measurements of temperature and water vapor in a shock tube," Paper 2660 26th International Symposium on Shock Waves, July 2007, Germany.

Temperature and water measured near 1400 nm.

H. Li, Z. C. Owens, D. F. Davidson, R. K. Hanson, "A simple reactive gasdynamic model for the computation of gas temperature and species concentrations behind reflected shock waves," International Journal of Chemical Kinetics 40 (2008) 189-198.

Temperature and water measured near 1400 nm.

A. Farooq, J. B. Jeffries, R. K. Hanson, "CO₂ concentration and temperature sensor for combustion gases using diode-laser absorption near 2.7 microns," Applied Physics B 90 (2008) 619-628.

Temperature and CO₂ measured near 2700 nm.

A. E. Klingbeil, J. M. Porter, J. B. Jeffries, R. K. Hanson, "Two-wavelength mid-IR absorption diagnostic for simultaneous measurement of temperature and hydrocarbon fuel concentration," Proceedings of the Combustion Institute 32 (2009) 821-829.

Temperature and fuel measured near 3400 nm.

Z. Hong, D. F. Davidson, S. S. Vasu, R. K. Hanson, "The effect of oxygenates on soot formation in rich heptane mixtures: a shock tube study," Fuel 88 (2009) 1901-1906.

Soot measured using extinction at 670 nm and emission at 633 nm.

D. F. Davidson, Z. Hong, G. L. Pilla, A. Farooq, R. D. Cook, R. K. Hanson, "Species time-history measurements during n-heptane oxidation behind reflected shock waves," Paper 31F1, 6th U.S. Combustion Meeting, Ann Arbor MI 2009.

OH radicals measured at 306 nm.

CO₂ measured at 2700 nm.

H₂O measured at 2500 nm.

S. S. Vasu, D. F. Davidson, R. K. Hanson, "OH time-histories during oxidation of n-heptane and methylcyclohexane at high pressures and temperatures," Combustion and Flame 156 (2009) 736-749.

OH radicals measured at 306 nm.

A. Farooq, J. B. Jeffries, R. K. Hanson, "Sensitive detection of temperature behind reflected shock waves using wavelength modulation of CO₂ near 2.7 microns," Applied Physics B96 (2009) 161-173.

Temperature and CO₂ measured near 2700 nm.

D. F. Davidson, Z. Hong, G. L. Pilla, A. Farooq, R. D. Cook, R. K. Hanson, "*Multi-species time-history measurements during n-heptane oxidation behind reflected shock waves*," Combustion and Flame 157 (2010) 1899-1905.

OH radicals measured at 306 nm.

CO₂ measured at 2700 nm.

H₂O measured at 2500 nm.

C₂H₄ measured at 10.53 microns.

n-heptane measured at 3.39 microns.

D. F. Davidson, Z. Hong, G. L. Pilla, A. Farooq, R. D. Cook, R. K. Hanson, "*Multi-species measurements behind reflected shock waves in hydrocarbons using laser absorption*," 48th AIAA Aerospace Sciences Meeting, 4-7 January 2010, Paper AIAA 2010-198.

OH radicals measured at 306 nm.

CO₂ measured at 2700 nm.

H₂O measured at 2500 nm.

C₂H₄ measured at 10.53 microns.

G. L. Pilla, D. F. Davidson, R. K. Hanson, "*Shock tube/laser absorption measurements of ethylene time-histories during ethylene and n-heptane pyrolysis*," Proceedings of the Combustion Institute 33 (2011) 333-340.

C₂H₄ measured at 10.53 microns.

S. H. Pyun, J. Cho, D. F. Davidson, R. K. Hanson, "*Interference-free mid-IR laser absorption detection of methane*," Measurement Science Technology 22 (2011) 025303 (9pp).

CH₄ measured at 3.4 microns.

W. Ren, D. F. Davidson, R. K. Hanson, "*Shock tube measurements of ethylene concentration time-histories during ethylene, n-heptane and methylcyclohexane pyrolysis*," 7th US National Technical Meeting of the Combustion Institute, March 20-23, 2011.

C₂H₄ measured at 10.53 microns.

G. A. Pang, R. K. Hanson, D. M. Golden, C. T. Bowman, "*High-temperature measurements of the rate constants for reactions of OH with a series of large normal alkanes: n-pentane, n-heptane and n-nonane*," Z. Phys. Chem. 225 (2011) 1157-1178.

OH radicals measured at 306 nm.

S. H. Pyun, W. Ren, D. F. Davidson, R. K. Hanson, "*Methane and ethylene time-history measurements in n-butane and n-heptane pyrolysis behind reflected shock waves*," Fuel 108 (2013) 557-564.

CH₄ measured at 3.4 microns.

C₂H₄ measured at 10.53 & 10.67 microns.

Iso-octane, Iso-cetane

D. F. Davidson, M. A. Oehlschlaeger, R. K. Hanson, "*Methyl concentration time-histories during iso-octane and n-heptane oxidation and pyrolysis*," Proceedings of the Combustion Institute 31 (2007) 321-328.

Methyl radicals measured at 216 nm.

S. H. Pyun, J. M. Porter, J. B. Jeffries, R. K. Hanson, J. C. Montoya, M. G. Allen, K. R. Sholes, "Two-color-absorption sensor for time-resolved measurements of gasoline concentration and temperature," Applied Optics 48 (2009) 6492-6500.

Fuel measured near 3370 nm.

M. E. MacDonald, W. Ren, Y. Zhu, D. F. Davidson, R. K. Hanson, "*Fuel and ethylene measurements during n-dodecane, methylcyclohexane, and iso-cetane decomposition in shock tubes*," Fuel 103 (2013) 1060-1068; see also AIAA Joint Propulsion Conference, July 31-August 3, 2011, San Diego CA..

C₂H₄ measured at 10.53 microns.

Fuel measured at 3.39 microns.

Ethane

V. Vasudevan, R. K. Hanson, C. T. Bowman, D M. Golden, D. F. Davidson, "*High-temperature shock tube measurements of methyl radical decomposition*," Journal of Physical Chemistry A111 (2007) 4062-4072..

CH radicals measured at 431 nm.

OH radicals measured at 306 nm.

V. Vasudevan, R. K. Hanson, C. T. Bowman, D M. Golden, D. F. Davidson, "*Shock tube study of the reaction of CH with N₂: overall rate and branching ratio*," Journal of Physical Chemistry A111 (2007) 11818-11830.

CH radicals measured at 431 nm.

NCN radicals measured at 329 nm.

Methyl Iodide

V. Vasudevan, R. K. Hanson, C. T. Bowman, D M. Golden, D. F. Davidson, "*High-temperature shock tube measurements of methyl radical decomposition*," Journal of Physical Chemistry A111 (2007) 4062-4072.

CH radicals measured at 431 nm.

OH radicals measured at 306 nm.

V. Vasudevan, R. D. Cook, R. K. Hanson, C. T. Bowman, D M. Golden, "*High-temperature shock tube study of the reactions of CH₃+OH = Products and CH₃OH+Ar = Products*," International Journal of Chemical Kinetics 40 (2008) 488-495.

OH radicals measured at 306 nm.

Ethylene, Propene, Butene, Cyclohexene, 1,3-Butadiene

S. S. Vasu, Z. Hong, D. F. Davidson, R. K. Hanson, D. M. Golden, “*Shock tube/laser absorption measurements of the reaction of OH with ethylene and propene*,” Journal of Physical Chemistry A114 (2010) 11529-11537.

OH radicals measured at 306 nm.

G. L. Pilla, D. F. Davidson, R. K. Hanson, “*shock tube/laser absorption measurements of ethylene time-histories during ethylene and n-heptane pyrolysis*,” Proceedings of the Combustion Institute 33 (2011) 333-340.

C₂H₄ measured at 10.53 microns.

S. S. Vasu, L. K. Huynh, D. F. Davidson, R. K. Hanson, D. M. Golden, “*Reactions of OH with butene isomers: measurements of the overall rates and a theoretical study*,” Journal Physical Chemistry A115 (2011) 2549-2556.

OH measured at 306 nm.

W. Ren, D. F. Davidson, R. K. Hanson, “*Shock tube measurements of ethylene concentration time-histories during ethylene, n-heptane and methylcyclohexane pyrolysis*,” 7th US National Technical Meeting of the Combustion Institute, March 20-23, 2011.

C₂H₄ measured at 10.53 microns.

W. Ren, D. F. Davidson, R. K. Hanson, “*IR laser absorption diagnostic for C₂H₄ in shock tube kinetics studies*,” International Journal of Chemical Kinetics 44 (2012) 423-432.

C₂H₄ measured at 10.53 microns.

I. Stranic, D. F. Davidson, R. K. Hanson, “*Shock tube measurements of the rate constant for the reaction cyclohexene = ethylene + 1,3-butadiene*,” Chemical Physics Letters 584 (2013) 18-23.

C₂H₄ measured at 10.53 microns.

S. S. Vasu, J. Zador, D. F. Davidson, R. K. Hanson, D. M. Golden, J. A. Miller, “*High-temperature measurements and a theoretical study of the reaction of OH with 1,3-butadiene*,” Journal of Physical Chemistry A114 (2010) 8312-8318.

OH radicals measured at 306 nm.

n-Dodecane, n-Hexadecane

A. E. Klingbeil, J. B. Jeffries, D. F. Davidson, R. K. Hanson, “*Two-wavelength mid-IR diagnostic for temperature and n-dodecane concentration in an aerosol shock tube*,” Applied Physics B 93 (2008) 627-638.

Temperature and n-dodecane measured at 3409 and 3432 nm.

M. E. MacDonald, D. F. Davidson, R. K. Hanson, "*Decomposition rate measurements of RP-1, RP-2, n-dodecane, and RP-1 with fuel stabilizers*," AIAA Paper 2008-4766, 44th AIAA/JPC meeting Hartford CT July 2008.

Fuel measured at 3390 nm.

S. S. Vasu, D. F. Davidson, Z. Hong, V. Vasudevan, R. K. Hanson, "*n-Dodecane oxidation at high-pressures: measurements of ignition delay times and OH concentration time-histories*," Proceedings of the Combustion Institute 32 (2009) 173-180.

OH radicals measured at 306 nm.

D. F. Davidson, Z. Hong, G. L. Pilla, A. Farooq, R. D. Cook, R. K. Hanson, "*Multi-species time-history measurements during n-dodecane oxidation behind reflected shock waves*," Proceedings of the Combustion Institute 33 (2011) 151-157.

OH radicals measured at 306 nm.

CO₂ measured at 2700 nm.

H₂O measured at 2500 nm.

C₂H₄ measured at 10.53 microns.

n-dodecane at 3.39 microns

M. E. MacDonald, D. F. Davidson, R. K. Hanson, "*Decomposition measurements of RP-1, RP-2, JP-7, n-dodecane, and tetrahydroquinoline in shock tubes*," Journal of Propulsion and Power 27 (5) (2011) 981-989.

Fuel measured at 3390 nm.

M. E. MacDonald, W. Ren, D. F. Davidson, R. K. Hanson, "*Fuel and ethylene measurements during n-dodecane, methylcyclohexane, and iso-cetane decomposition in shock tubes*," AIAA Joint Propulsion Conference, July 31-August 3, 2011, San Diego CA.

C₂H₄ measured at 10.53 microns.

Fuel measured at 3.39 microns.

M. E. MacDonald, W. Ren, Y. Zhu, D. F. Davidson, R. K. Hanson, "*Fuel and ethylene measurements during n-dodecane, methylcyclohexane, and iso-cetane decomposition in shock tubes*," Fuel 103 (2013) 1060-1068; see also AIAA Joint Propulsion Conference, July 31-August 3, 2011, San Diego CA..

C₂H₄ measured at 10.53 microns.

Fuel measured at 3.39 microns.

D. R. Haylett, R. D. Cook, D. F. Davidson, R. K. Hanson, "*OH and C₂H₄ species time-histories during hexadecane and diesel ignition behind reflected shock waves*," Proceedings of the Combustion Institute 33 (2011) 167-173.

OH radicals measured at 306 nm.

C₂H₄ measured at 10.53 microns.

D. R. Haylett, D. F. Davidson, R. D. Cook, Z. Hong, W. Ren, S. H. Pyun, R. K. Hanson, "*Multi-species time-histories during hexadecane oxidation behind reflected shock waves*," Proceedings of the Combustion Institute 34 (2013) 369-376.

OH radicals measured at 306 nm.

C₂H₄ measured at 10.53 microns.

CH₄ measured at 3.4 microns.

CO₂ measured at 2.7 microns.

H₂O measured at 2.55 microns.

Methylcyclohexane, Cyclohexane, n-Butylcyclohexane

S. S. Vasu, D. F. Davidson, R. K. Hanson, "*OH time-history absorption measurements at high pressures and temperatures behind reflected shocks during methylcyclohexane oxidation*," Paper 08S-13, Spring Meeting Western States Section/Combustion Institute, Los Angeles March 2008.

OH radicals measured at 306 nm.

S. S. Vasu, D. F. Davidson, R. K. Hanson, "*OH time-histories during oxidation of n-heptane and methylcyclohexane at high pressures and temperatures*," Combustion and Flame 156 (2009) 736-749.

OH radicals measured at 306 nm.

Z. Hong, K.-Y. Lam, D. F. Davidson, R. K. Hanson, "*A comparative study of the oxidation characteristics of cyclohexane, methylcyclohexane, and n-butylcyclohexane at high temperatures*," Combustion and Flame 158 (2011) 1456-1468.

OH radicals measured at 306 nm.

H₂O measured at 2.55 microns.

W. Ren, D. F. Davidson, R. K. Hanson, "*Shock tube measurements of ethylene concentration time-histories during ethylene, n-heptane and methylcyclohexane pyrolysis*," 7th US National Technical Meeting of the Combustion Institute, March 20-23, 2011.

C₂H₄ measured at 10.53 microns.

M. E. MacDonald, W. Ren, Y. Zhu, D. F. Davidson, R. K. Hanson, "*Fuel and ethylene measurements during n-dodecane, methylcyclohexane, and iso-cetane decomposition in shock tubes*," Fuel 103 (2013) 1060-1068; see also AIAA Joint Propulsion Conference, July 31-August 3, 2011, San Diego CA..

C₂H₄ measured at 10.53 microns.

Fuel measured at 3.39 microns.

Diesel DF-2

D. R. Haylett, R. D. Cook, D. F. Davidson, R. K. Hanson, "*OH and C₂H₄ species time-histories during hexadecane and diesel ignition behind reflected shock waves*," Proceedings of the Combustion Institute 33 (2011) 167-173.

OH radicals measured at 306 nm.

C₂H₄ measured at 10.53 microns.

RP and JP Fuels

M. E. MacDonald, D. F. Davidson, R. K. Hanson, "*Decomposition rate measurements of RP-1, RP-2, n-dodecane, and RP-1 with fuel stabilizers*," AIAA Paper 2008-4766, 44th AIAA/JPC meeting Hartford CT July 2008.

Fuel measured at 3390 nm.

D. F. Davidson, M. E. MacDonald, G. Pilla, R. K. Hanson, "*Shock Tube/Laser Absorption Studies of the Decomposition of Fuel Surrogates*," 57th JANNAF Propulsion Meeting (JPM)/5th Liquid Propulsion Subcommittee (LPS) May 3-7, 2010.

Fuel measured at 3.39 microns.

C₂H₄ measured at 10.53 microns.

M. E. MacDonald, D. F. Davidson, R. K. Hanson, "*Decomposition measurements of RP-1, RP-2, JP-7, n-dodecane, and tetrahydroquinoline in shock tubes*," Journal of Propulsion and Power 27 (5) (2011) 981-989.

Fuel measured at 3390 nm.

D. F. Davidson, S. Li, K.-Y. Lam, J. T. Harmon, R. K. Hanson, "*Shock tube/laser absorption measurements of JP-8 ignition delay times and multi-species time-histories*," Paper #2127, JANNAF Propulsion Meeting, December 2011, Huntsville AB.

OH radicals measured at 307 nm.

C₂H₄ measured at 10.53 microns.

D. F. Davidson, W. Ren, D. P. Chase, S. F. Bellimoria, R. K. Hanson, "*Shock tube/laser absorption database for the oxidation of HiFire JP-7 surrogate fuel*," Paper #2127, JANNAF Propulsion Meeting, December 2011, Huntsville AB. See also D. F. Davidson, W. Ren, R. K. Hanson, "*Experimental database for development of a HiFire JP-7 surrogate fuel mechanism*," Paper AIAA-2012-0620, 50th AIAA Aerospace Sciences Meeting, Tennessee, January 2012.

C₂H₄ measured at 10.53 microns.

D. F. Davidson, M. E. MacDonald, R. K. Hanson, "*RP-fuel and surrogate pyrolysis: multi-species time-history measurements using shock tube/laser absorption*," Paper #2129, JANNAF Propulsion Meeting, December 2011, Huntsville AB.

C₂H₄ measured at 10.53 microns.

Fuel measured at 3.39 microns.

M. E. MacDonald, D. F. Davidson, R. K. Hanson, W. J. Pitz, M. Hehl, C. K. Westbrook, *“Formulation of an RP-1 surrogate from shock tube measurements of fuel and ethylene time-histories,”* Fuel 103 (2013) 1051-1059.

C₂H₄ measured at 10.53 microns.

Fuel measured at 3.39 microns.

Nitrogen-Bearing Fuels

Azomethane

V. Vasudevan, R. D. Cook, R. K. Hanson, C. T. Bowman, D M. Golden, *“High-temperature shock tube study of the reactions of CH₃+OH = Products and CH₃OH+Ar = Products,”* International Journal of Chemical Kinetics 40 (2008) 488-495.

OH radicals measured at 306 nm.

Ethylamine

S. Li, D. F. Davidson, R. K. Hanson, *“Shock tube study of ethylamine pyrolysis and oxidation,”* 8th U.S. National Combustion Meeting, WSS/CI University of Utah, May 19-22, 2013.

NH₂ radicals measured at 597 nm.

OH radicals measured at 307 nm.

Monomethyl Hydrazine

D. F. Davidson, R. D. Cook, S. H. Pyun, R. K. Hanson, *“MMH pyrolysis and oxidation: species time-history measurements behind reflected shock waves,”* International Colloquium on Detonations and Energetic Reactive Systems 23, Paper #9, July 24-29, 2011.

NH₂ measured at 597 nm.

OH radicals measured at 307 nm.

CH₄ measured at 3.4 microns.

NH₃ measured at 10.22 microns

MMH measured at 9.6 microns

R. D. Cook, S. H. Pyun, J. Cho, D. F. Davidson, R. K. Hanson, *“Shock tube measurements of species time-histories in monomethyl hydrazine pyrolysis,”* Combustion and Flame 158 (2011) 790-795.

NH₂ measured at 597 nm.

NH₃ measured at 10.22 microns

MMH measured at 9.6 microns

R. D. Cook, D. F. Davidson, R. K. Hanson, "*Shock tube measurements of species time-histories in monomethyl hydrazine oxidation*," was not submitted to a journal, results limited to ARO annual reports; contact Dr. David Davidson for further information.

NH₂ measured at 597 nm.

OH radicals measured at 307 nm.

Morpholine

S. Li, D. F. Davidson, R. K. Hanson, N. J. Labbe, P. R. Westmoreland, P. Oßwald, K. Kohse-Höinghaus, "*Shock tube measurements and model development for morpholine pyrolysis and oxidation at high pressures*," Combustion and Flame 160 (2013) 1559-1571.

Morpholine measured at 3.39 microns

Oxygenated Fuels

Dimethyl Ether

R. D. Cook, D. F. Davidson, R. K. Hanson, “*Measurements of ignition delay times and OH species concentrations in DME/O₂/Ar mixtures*,” Paper 2970P 26th International Symposium on Shock Waves, July 2007, Germany
OH radicals measured at 306 nm.

R. D. Cook, D. F. Davidson, R. K. Hanson, “*Shock tube measurements of ignition delay times and OH time-histories in dimethyl ether oxidation*,” Proceedings of the Combustion Institute 32 (2009) 189-196.
OH radicals measured at 306 nm.

R. D. Cook, D. F. Davidson, R. K. Hanson, “*Rate Measurements of DME decomposition and DME+OH at elevated temperatures*,” Paper 1P08, 6th U.S. Combustion Meeting, Ann Arbor MI 2009.
OH radicals measured at 306 nm.

R. D. Cook, D. F. Davidson, R. K. Hanson, “*High-temperature shock tube measurements of dimethyl ether decomposition and the reaction of dimethyl ether with OH*,” Journal of Physical Chemistry A 113 (2009) 9974-9980.
OH radicals measured at 306 nm.

S. H. Pyun, W. Ren, K.-Y. Lam, D. F. Davidson, R. K. Hanson, “*Shock tube measurements of methane, ethylene and carbon monoxide time-histories in DME pyrolysis*,” Combustion and Flame 160 (2013) 747-754.
CO measured at 4.6 microns.
CH₄ measured at 3.4 microns.
C₂H₄ measured at 10.53 & 10.67 microns.

Acetic Anhydride

V. Vasudevan, R. K. Hanson, C. T. Bowman, D M. Golden, D. F. Davidson, “*Shock tube study of the reaction of CH with N₂: overall rate and branching ratio*,” Journal of Physical Chemistry A 111 (2007) 11818-11830.
CH radicals measured at 431 nm.

Methanol

V. Vasudevan, R. D. Cook, R. K. Hanson, C. T. Bowman, D M. Golden, “*High-temperature shock tube study of the reactions of CH₃+OH = Products and CH₃OH+Ar = Products*,” International Journal of Chemical Kinetics 40 (2008) 488-495.
OH radicals measured at 306 nm.

W. Ren, E. Dames, D. Hyland, D. F. Davidson, R. K. Hanson, "Shock tube study of methanol, methyl formate pyrolysis: CH₃OH and CO time-history measurements," Combustion and Flame 160 (2013) 2669-2679.

methanol measured at 9.67 microns.

CO measured at 4.56 microns.

Butanol Isomers

S. S. Vasu, D. F. Davidson, R. K. Hanson, D M. Golden, "Measurements of the reaction of OH with n-butanol at high-temperatures," Chemical Physics Letters 497 (2010) 26-29.

OH radicals measured at 306 nm.

R. D. Cook, R. Sur, I. Stranic, D. F. Davidson, R. K. Hanson, "Multi-species laser measurements of n-butanol pyrolysis behind reflected shock waves," International Symposium on Shock Waves 28, paper #2462, July 2011. See also R. D. Cook, R. Sur, I. Stranic, D. F. Davidson, R. K. Hanson, "Multi-species laser measurements of n-butanol pyrolysis behind reflected shock waves," 7th International Colloquium on Chemical Kinetics, MIT, July 10-14, 2011.

OH radicals measured at 306 nm.

H₂O measured at 2.5 microns.

CH₂O measured at 306.7 nm.

G. A. Pang, R. K. Hanson, D. M. Golden, C. T. Bowman, "Rate constant measurements for the overall reaction of OH+1-butanol = products from 900 to 1200 K," Journal of Physical Chemistry A 116 (2012) 2475-2483.

OH radicals measured at 306 nm.

G. A. Pang, R. K. Hanson, D. M. Golden, C. T. Bowman, "Experimental determination of the high-temperature rate constant for the reaction of OH with sec-butanol," Journal of Physical Chemistry A 116 (2012) 9607-9613.

OH radicals measured at 306 nm.

R. D. Cook, D. F. Davidson, R. K. Hanson, "Multi-species laser measurements of n-butanol pyrolysis behind reflected shock waves," International Journal of Chemical Kinetics 44 (2012) 303-311.

OH radicals measured at 306 nm.

H₂O measured at 2.5 microns.

CH₂O/CH₃CHO measured at 306.8 nm.

G. A. Pang, R. K. Hanson, D. M. Golden, C. T. Bowman, "High-temperature rate constant determination for the reaction of OH with iso-butanol," Journal of Physical Chemistry A 116 (2012) 4720-4725.

OH radicals measured at 306 nm.

I. Stranic, S. H. Pyun, D. F. Davidson, R. K. Hanson, *"Multi-species measurements in 1-butanol pyrolysis behind reflected shock waves,"* Combustion and Flame 159 (2012) 3242-3250.

OH radicals measured at 306 nm.

H₂O measured at 2.55 microns.

C₂H₄ measured at 10.53 & 10.67 microns.

CO measured at 4.56 microns.

CH₄ measured at 3.4 microns.

I. Stranic, S. H. Pyun, D. F. Davidson, R. K. Hanson, *"Multi-species measurements in 2-butanol and i-butanol pyrolysis behind reflected shock waves,"* Combustion and Flame 160 (2013) 1012-1019.

OH radicals measured at 306 nm.

H₂O measured at 2.55 microns.

C₂H₄ measured at 10.53 & 10.67 microns.

CO measured at 4.56 microns.

CH₄ measured at 3.4 microns.

I. Stranic, G. A. Pang, D. F. Davidson, R. K. Hanson, D. M. Golden, C. T. Bowman, *"Shock tube measurements of the tert-butanol + OH reaction rate and the tert-C₄H₈OH radical β -scission branching ratio using isotopic labeling,"* Journal of Physical Chemistry A117 (2013) 4777-4784. See also 8th U.S. National Combustion Meeting, WSS/CI University of Utah, May 19-22, 2013, Paper #0014.

OH radicals measured at 306 nm.

Methyl Esters

A. Farooq, D. F. Davidson, R. K. Hanson, L. K. Huynh, A. Violi, *"An experimental and computational study of methyl ester decomposition pathways using shock tubes,"* Proceedings of the Combustion Institute 32 (2009) 247-253.

CO₂ measured near 2700 nm.

W. Ren, A. Farooq, D. F. Davidson, R. K. Hanson, *"CO concentration and temperature sensor for combustion gases using quantum-cascade laser absorption near 4.7 microns,"* Proceedings of the Combustion Institute 32 (2009) 247-253.

CO measured near 4.7 microns.

K.-Y. Lam, D. F. Davidson, R. K. Hanson, *"High-temperature measurements of OH with small methyl esters: methyl formate, methyl acetate, methyl propanoate, and methyl butanoate,"* Journal of Physical Chemistry A116 (2012) 12229-12241.

OH radicals measured at 307 nm.

W. Ren, K.-Y. Lam, S. H. Pyun, A. Farooq, D. F. Davidson, R. K. Hanson, *"Shock tube/laser absorption studies of the decomposition of methyl formate,"* Proceedings of the Combustion Institute 34 (2013) 453-461.

CO measured at 4.6 microns.
CH₂O measured at 306 nm.
CH₄ measured at 3.4 microns.

W. Ren, E. Dames, D. Hyland, D. F. Davidson, R. K. Hanson, "*Shock tube study of methanol, methyl formate pyrolysis: CH₃OH and CO time-history measurements*," Combustion and Flame 160 (2013) 2669-2679.

methanol measured at 9.67 microns.
CO measured at 4.56 microns.

W. Ren, R. M. Spearrin, D. F. Davidson, R. K. Hanson, "*Thermal decomposition of C3-C5 ethyl esters: CO, CO₂ and H₂O time-history measurements behind reflected shock waves*," 8th U.S. National Combustion Meeting, WSS/CI, University of Utah, May 19-23, 2013.

CO measured at 4.6 microns.
CO₂ measured at 4.2 microns.
H₂O measured at 2.55 microns.

Hydrogen Peroxide, Hydrogen, Peroxyl Radicals

Z. Hong, A. Farooq, E. A. Barbour, D. F. Davidson, R. K. Hanson, "*Hydrogen peroxide decomposition rate: a shock tube study using tunable laser absorption of H₂O at 2.5 microns*," Paper 09F65, Western States Section/Combustion Institute Fall Meeting (2009).

H₂O measured at 2550 nm.

Z. Hong, A. Farooq, E. A. Barbour, D. F. Davidson, R. K. Hanson, "*Hydrogen peroxide decomposition rate: a shock tube study using tunable laser absorption of H₂O at 2.5 microns*," Journal of Physical Chemistry A 113 (2009) 12919-12925.

H₂O measured at 2550 nm.

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H. Ning, J. Wu, L. Ma, W. Ren, D.F. Davidson and R.K. Hanson, "Chemical Kinetic Modeling and Shock Tube Study of Methyl Propanoate Decomposition," *Comb. and Flame* **184** 30-40 (2017) [DOI 10.1016/j.combustflame.2017.06.001](https://doi.org/10.1016/j.combustflame.2017.06.001)

Water, Hydrogen, Tert-butyl Hydroperoxide

C.S. Goldenstein, R.M. Spearrin, I.A. Schultz, J.B. Jeffries and R.K. Hanson, "Wavelength Modulation Spectroscopy near 1.4 μm for Measurements of H₂O and Temperature in High-pressure and -temperature Gases," *Meas. Sci. and Tech.* **25** 1-9 (2014) [DOI 10.1088/0957-0233/25/5/055101](https://doi.org/10.1088/0957-0233/25/5/055101)

S. Wang, S. Li, D.F. Davidson and R.K. Hanson, "Shock Tube Measurement of the High-temperature Rate Constant for OH + CH₃ $\rightarrow$ Products," *J. Phys. Chem. A* **119** (33) 8799–8805 (2015) [DOI 10.1021/acs.jpca.5b05725](https://doi.org/10.1021/acs.jpca.5b05725)

J. Shao, R. Choudhary, A. Susa, D. F. Davidson, R. K. Hanson, "Shock Tube Study of the Rate Constants for H + O₂ + M $\rightarrow$ HO₂ + M (M = Ar, H₂O, CO₂, N₂) at Elevated Pressures," *Proceedings of the Combustion Institute* **37**, 145-152 (2019). <https://doi.org/10.1016/j.proci.2018.05.077>

J.J. Girard, R. Choudhary and R.K. Hanson, "Collisional-induced Broadening and Shift Parameters of OH with Ar and N₂ near 308.6 nm, Measured at T = 1300 – 2000K and P = 20 – 100 atm," *JQSRT* **221**, 194-201, (2018). <https://doi.org/10.1016/j.jqsrt.2018.10.007>

Acetone and Higher Ketones

E. Dames, K.Y. Lam, D.F. Davidson and R.K. Hanson, "An Improved Kinetic Mechanism for 3-Pentanone Pyrolysis and Oxidation using Multispecies Time Histories in Shock Tubes" *Comb. and Flame* **161** 1135-1145 (2014) [DOI 10.1016/j.combustflame.2013.11.010](https://doi.org/10.1016/j.combustflame.2013.11.010)

K. Sun, S. Wang, R. Sur, X. Chao, J.B. Jeffries and R.K. Hanson, "Time-resolved in situ Detection of CO in a Shock Tube using Cavity-enhanced Absorption Spectroscopy with a Quantum-cascade Laser near 4.6 μm ," *Optics Express* **22** (20) 24559-24565 (2014) [DOI 10.1364/OE.22.024559](https://doi.org/10.1364/OE.22.024559)

S. Wang, K. Sun, D.F. Davidson, J.B. Jeffries and R.K. Hanson, "Shock Tube Measurement of Acetone Dissociation using Cavity-enhanced Absorption Spectroscopy of CO," *J. Phys. Chem. A* **119** 7257-7262 (2015) [DOI 10.1021/jp511642a](https://doi.org/10.1021/jp511642a)

C. Hemken, U. Burke, K.Y. Lam, D. F. Davidson, R. K. Hanson, K. A. Heufer and K. Kohse-Hoinghaus, "Towards a Better Understanding of 2-butanone Oxidation: Detailed Species Measurements and Kinetic Modeling," *Comb. and Flame* **184** 195-207 (2017) [DOI 10.1016/j.combustflame.2017.06.001](https://doi.org/10.1016/j.combustflame.2017.06.001)

Formaldehyde and Higher Aldehydes

S. Wang, D.F. Davidson and R.K. Hanson, "High Temperature Measurements for the Rate Constants of C₁-C₄ Aldehydes with OH in a Shock Tube," *Proc. Comb. Inst.* **35** 473-480 (2015) [DOI 10.1016/j.proci.2014.06.112](https://doi.org/10.1016/j.proci.2014.06.112)

S. Wang, E. Dames, D.F Davidson and R.K. Hanson, "Reaction Rate Constant of CH₂O + H=HCO + H₂ Revisited: A Combined Study of Direct Shock Tube Measurement and Transition State Theory Calculation," *J. Phys. Chem. A* **118** 10201-10209 (2014) [DOI 10.1021/jp5085795](https://doi.org/10.1021/jp5085795)

S. Wang, D.F. Davidson and R.K. Hanson, "Rate Constants of Long, Branched, and Unsaturated Aldehydes with OH at Elevated Temperatures," 36th Symp. (Int.) on Combustion, Seoul, Korea, 2016 [DOI 10.1016/j.proci.2016.06.017](https://doi.org/10.1016/j.proci.2016.06.017)

S. Wang, D.F. Davidson and R.K. Hanson, "Shock Tube Measurement for the Dissociation Rate Constant of Acetaldehyde using Sensitive CO Diagnostic," *J. Phys. Chem. A* **120** (35) 6895–6901 (2016) [DOI 10.1021/acs.jpca.6b03647](https://doi.org/10.1021/acs.jpca.6b03647)

S. Wang, D.F. Davidson and R.K. Hanson, "A Shock Tube Study of CH₂O Oxidation via Simultaneous Laser Absorption Measurements of OH and CO," International Conference on Chemical Kinetics, Chicago, IL, May 2017.

S. Wang, D.F. Davidson and R.K. Hanson, "A Shock Tube and Laser Absorption Study of CH₂O Oxidation via Simultaneous Measurements of OH and CO," *J. Phys. Chem. A* **121**,45,8561-8568 (2017).
<https://pubs.acs.org/doi/abs/10.1021/acs.jpca.7b09362>

Nitrogen-Bearing Species

Nitric Oxide

R.M. Spearrin, I.A. Schultz, J.B. Jeffries and R.K. Hanson, "Laser Absorption of Nitric Oxide for Thermometry in High-enthalpy Air," *Meas. Sci. and Tech.* **25** 125103 (7pp) (2014) [DOI 10.1088/0957-0233/25/12/125103](https://doi.org/10.1088/0957-0233/25/12/125103)

Ethylamine, Dimethylamine

S. Li, D.F. Davidson and R.K. Hanson, "High-Temperature Measurements of the Reactions of OH with Ethylamine and Dimethylamine," *J. Phys. Chem. A* **118** 70-77 (2014) [DOI 10.1021/jp411141w](https://doi.org/10.1021/jp411141w)

S. Li, D.F. Davidson and R.K. Hanson, "Shock Tube Study Dimethylamine Oxidation," *IJCK* **47** (1) 19-26 (2014) [DOI 10.1002/kin.20888](https://doi.org/10.1002/kin.20888)

S. Li, D.F. Davidson and R.K. Hanson, "Shock Tube Study of Ethylamine Pyrolysis and Oxidation," *Comb and Flame* **161** 2512-2518 (2014) [DOI 10.1016/j.combustflame.2014.04.002](https://doi.org/10.1016/j.combustflame.2014.04.002)

Monomethyl Hydrazine

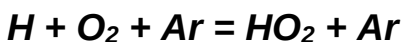
S. Li, D.F. Davidson and R.K. Hanson, "Shock Tube Study of the Pressure Dependence of Monomethyl Hydrazine Pyrolysis," *Comb. and Flame* **161** 16-22 (2014) [DOI 10.1016/j.combustflame.2013.07.023](https://doi.org/10.1016/j.combustflame.2013.07.023)

Azomethane

S. Wang, S. Li, D.F. Davidson and R.K. Hanson, "Shock Tube Measurement of the High-temperature Rate Constant for $\text{OH} + \text{CH}_3 \rightarrow \text{Products}$," *J. Phys. Chem. A* **119** (33) 8799–8805 (2015) [DOI 10.1021/acs.jpca.5b05725](https://doi.org/10.1021/acs.jpca.5b05725)

Reaction Rate Measurements to January 2009

Dissociation/Association Reactions



G. A. Pang, D. F. Davidson, R. K. Hanson, "Experimental Study and Modeling of Shock Tube Ignition Delay Times for Hydrogen-Oxygen-Argon Mixtures at Low Temperatures," Proceedings of the Combustion Institute 32 (2009) 181-188.

For 908-1118 K, 3.0-3.8 atm:

$$k_{\infty} = 1.04 \times 10^{13} T^{0.2} [\text{cm}^3/\text{mol/s}];$$

$$k_0 = 6.99 \times 10^{18} T^{-1.2} [\text{cm}^6/\text{mol}^2/\text{s}], F_{\text{cent}} = 0.7.$$

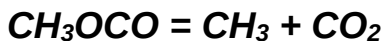
This study provides a preliminary measurement of this rate in this temperature range.



V. Vasudevan, R. D. Cook, R. K. Hanson, C. T. Bowman, D. M. Golden, "High-Temperature Shock Tube Study of the Reactions $CH_3 + OH = \text{Products}$ and $CH_3OH + Ar = \text{Products}$," International Journal of Chemical Kinetics 40 (2008) 488-495.

For 1904-2298 K, 1.21-1.41 atm:

No rate expression given; data table provided. Measurements are in agreement with experiments of N. K. Srinivasan, M.-C. Su, J. V. Michael, J. Phys. Chem. A 111 (2007) 3951-3958 and theory of A. W. Jasper, S. J. Klippenstein, L. B. Harding, B. Ruscis, J. Phys. Chem. A 111 (2007) 3932-3950..

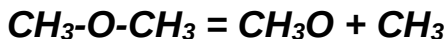


A. Farooq, D. F. Davidson, R. K. Hanson, L. K. Huynh, A. Violi, "An Experimental and Computational Study of Methyl Ester Decomposition Pathways using Shock Tubes," Proceedings of the Combustion Institute 32 (2009) 247-253.

For 1260-1653 K, 1.4-1.7 atm:

$$k = 1.55 \times 10^{12} T^{0.514} \exp(-15182/RT) [1/\text{s}].$$

This study provides a confirmation of the reaction rate suggested by L. K. Huynh, A. Violi, J. Org. Chem. 73 (2008) 94-101.



R. D. Cook, D. F. Davidson, R. K. Hanson, "Shock Tube Measurements of Ignition Delay Times and OH Time-Histories in Dimethyl Ether Oxidation," Proceedings of the Combustion Institute 32 (2009) 189-196.

For 680-1750 K, 1.6-6.6 bar:

$$k = 1.61 \times 10^{79} T^{-18.4} \exp(116438/RT) [1/\text{s}]$$

This rate was inferred based on this and previous studies.

CH₃ + Ar = Products

V. Vasudevan, R. K. Hanson D. M. Golden, C. T. Bowman, D. F. Davidson, "*High-Temperature Shock Tube Measurements of Methyl Radical Decomposition*," Journal Physical Chemistry A 111 (2007) 4062-4072.

For 2706-3527 K, 0.7-4.2 atm:

$k = 3.09 \times 10^{15} \exp(-80871/RT)$ [cm³/mol/s] for the CH+H₂ channel

For 2253-2975 K, 1.1-3.9 atm:

$k = 2.24 \times 10^{15} \exp(-82659/RT)$ [cm³/mol/s] rate for the CH₂+H channel

This study also includes the rate for the CH+H₂ channel.

C₆H₅CH₃ = Products

M. A. Oehlschlaeger, D. F. Davidson, R. K. Hanson, "*Thermal Decomposition of Toluene: Overall Rate and Branching Ratio*," Proceedings of the Combustion Institute 31 (2007) 211-219. See also Western States Section/Combustion Institute Fall 2005 Meeting, October 17-18, 2005, paper 05F-59.

For 1400-1780 K, 1.5 bar:

$k = 2.09 \times 10^{15} \exp(-87510/RT)$ [cm³/mol/s] for the C₆H₅CH₂+H channel.

$k = 2.66 \times 10^{16} \exp(-97880/RT)$ [cm³/mol/s] for the C₆H₅+CH₃ channel.

n-C₇H₁₆ = Products

D. F. Davidson, M. A. Oehlschlaeger, R. K. Hanson, "*Methyl Concentration Time-Histories during Iso-Octane and N-Heptane Oxidation and Pyrolysis*," Proceedings of the Combustion Institute 31 (2007) 321-328. See also Western States Section/Combustion Institute Fall 2005 Meeting, October 17-18, 2005, paper 05F-61.

For 1300-1690 K, 1.8 atm:

$k = 9.00 \times 10^{14} \exp(-67300/RT)$ [1/s]

This study also includes the rate for iso-octane decomposition.

iso-C₈H₁₈ = Products

D. F. Davidson, M. A. Oehlschlaeger, R. K. Hanson, "*Methyl Concentration Time-Histories during Iso-Octane and N-Heptane Oxidation and Pyrolysis*," Proceedings of the Combustion Institute 31 (2007) 321-328. See also Western States Section/Combustion Institute Fall 2005 Meeting, October 17-18, 2005, paper 05F-61.

For 1200-1400 K, 1.7 atm:

$k = 5.20 \times 10^{15} \exp(-67800/RT)$ [1/s]

This study also includes the rate for n-heptane decomposition.

CH₂O + Ar = Products

V. Vasudevan, D. F. Davidson, R. K. Hanson, C. T. Bowman, D. M. Golden, "*High-Temperature Measurements of the Rates of the Reactions CH₂O+Ar=Products and CH₂O+O₂=Products*," Proceedings of the Combustion Institute 31 (2007) 175-183. See also Western States Section/Combustion Institute Fall 2005 Meeting, October 17-18, 2005, paper 05F-40.

For 2258-2687 K, 1.6 atm:

$k = 5.85 \times 10^{14} \exp(-63783/RT)$ [cm³/mol/s] HCO+H channel

$k = 4.64 \times 10^{14} \exp(-57027/RT)$ [cm³/mol/s] H₂+CO channel

This study also includes rate measurements of CH₂O+O₂.

C₆H₅CH₂ = Products

M. A. Oehlschlaeger, D. F. Davidson, R. K. Hanson, "High-Temperature Thermal Decomposition of Benzyl Radicals," Journal Physical Chemistry A 110 (2006) 6649-6653.

For 1430-1730 K, 1.5 bar:

$k = 8.20 \times 10^{14} \exp(-80672/RT)$ [1/s]

This study assumes the products are C₇H₆+H.

(CH₃)₃-CO-OH = (CH₃)₃CO + OH

V. Vasudevan, D. F. Davidson, R. K. Hanson, "Direct Measurements of the Reaction of OH+CH₂O=HCO+H₂O at High Temperatures," International Journal of Chemical Kinetics 37 (2005) 98-109.

For 900-1000 K, 1.6 bar:

$k = 2.50 \times 10^{15} \exp(-43000/RT)$ [cm³/mol/s]

This study also includes a measurement of the reaction of CH₂O+OH.

CO₂ + M = CO + O(³P) + M

M. A. Oehlschlaeger, D. F. Davidson, J. B. Jeffries, R. K. Hanson, "Carbon Dioxide Thermal Decomposition: Observation of Incubation," Z. Phys. Chem. 219 (2005) 555-567.

For 3200-4600 K, 45-100 kPa:

$k = 3.14 \times 10^{14} \exp(-101933/RT)$ [cm³/mol/s]

This study includes a determination of the incubation time for this reaction.

C₂H₆ = 2CH₃

M. A. Oehlschlaeger, D. F. Davidson, R. K. Hanson, "High-Temperature Ethane and Propane Decomposition," Proceedings of the Combustion Institute 30 (2005) 1119-1127. See also Western States Section/Combustion Institute Fall 2003 Meeting October 20-21, 2003, Paper 03F-56.

For 700-1924 K, 0.13-8.4 atm, in argon:

$k_{\infty} = 1.88 \times 10^{50} T^{-9.72} \exp(-107338/RT)$ [1/s];

$k_0 = 3.72 \times 10^{65} T^{-13.14} \exp(-101575/RT)$ [cm³/mol/s];

$F_{\text{cent}} = 0.61 \exp(-T/100K) + 0.39 \exp(-T/1900K) + \exp(-6000K/T)$.

This study includes a determination of the propane decomposition rate also.

C₃H₈ = CH₃ + C₂H₅

M. A. Oehlschlaeger, D. F. Davidson, R. K. Hanson, "High-Temperature Ethane and Propane Decomposition," Proceedings of the Combustion Institute 30 (2005) 1119-1127. See also Western States Section/Combustion Institute Fall 2003 Meeting October 20-21, 2003, Paper 03F-56.

For 600-1653 K, 0.13-8.4 atm in argon:

$$k_{\infty} = 1.29 \times 10^{37} T^{-5.84} \exp(-97383/RT) [1/s];$$

$$k_0 = 5.64 \times 10^{74} T^{-15.74} \exp(-98714/RT) [\text{cm}^3/\text{mol/s}];$$

$$F_{\text{cent}} = 0.69 \exp(-T/50K) + 0.31 \exp(-T/3000K) + \exp(-9000K/T).$$

This study includes a determination of the ethane decomposition rate also.

CH₂O + M = Products

G. Friedrichs, D. F. Davidson, R. K. Hanson, "Validation of a Thermal Decomposition Mechanism of Formaldehyde by Detection of CH₂O and HCO behind Shock Waves," International Journal of Chemical Kinetics 36 (2004) 157-169.

For 1675-2080 K, 1 bar:

$$k = 3.3 \times 10^{39} T^{-6.3} \exp(-100000/RT) [\text{cm}^3/\text{mol/s}] \text{ for the H+HCO channel.}$$

$$k = 3.1 \times 10^{45} T^{-8.0} \exp(-97608/RT) [\text{cm}^3/\text{mol/s}] \text{ for the H}_2\text{+CO channel.}$$

These rates are based on RRKM fits to the current study's data. A reaction rate for CH₂OH+HCO is all provided in this study.

Iso-C₄H₁₀ = CH₃ + i-C₃H₇

M. A. Oehlschlaeger, D. F. Davidson, R. K. Hanson, "High-Temperature Thermal Decomposition of Iso-Butane and n-Butane Behind Shock Waves," Journal Physical Chemistry A 108 (2004) 4247-4253. See also Western States Section/Combustion Institute Fall 2003 Meeting October 20-21, 2003, Paper 03F-56.

For 1320-1560 K, 0.20-8.8 atm, in argon:

$$k_{\infty} = 4.83 \times 10^{16} \exp(-79897/RT) [1/s];$$

$$k_0 = 2.41 \times 10^{14} \exp(-52576/RT) [\text{cm}^3/\text{mol/s}];$$

$$F_{\text{cent}} = 0.75 \exp(-T/750K).$$

This study includes a determination of the n-butane decomposition rate also.

n-C₄H₁₀ = Products

M. A. Oehlschlaeger, D. F. Davidson, R. K. Hanson, "High-Temperature Thermal Decomposition of Iso-Butane and n-Butane Behind Shock Waves," Journal Physical Chemistry A 108 (2004) 4247-4253. See also Western States Section/Combustion Institute Fall 2003 Meeting October 20-21, 2003, Paper 03F-56.

For 1320-1600 K, 0.20-8.8 atm, in argon:

$$k_{\infty} = 4.28 \times 10^{14} \exp(-69903/RT) [1/s];$$

$$k_0 = 5.34 \times 10^{17} \exp(-42956/RT) [\text{cm}^3/\text{mol/s}];$$

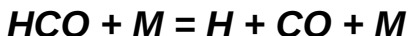
$$F_{\text{cent}} = 0.28 \exp(-T/1500K) \text{ for the CH}_3\text{+n-C}_3\text{H}_7 \text{ channel.}$$

$$k_{\infty} = 2.72 \times 10^{15} \exp(-75605/RT) [1/s];$$

$$k_0 = 4.72 \times 10^{18} \exp(-49576/RT) [\text{cm}^3/\text{mol/s}];$$

$$F_{\text{cent}} = 0.28 \exp(-T/1500K) \text{ for the C}_2\text{H}_5\text{+C}_2\text{H}_5 \text{ channel.}$$

This study includes a determination of the iso-butane decomposition rate also.



G. Friedrichs, J. T. Herbon, D. F. Davidson, R. K. Hanson, "Quantitative Detection of HCO behind Shock Waves: The Thermal Decomposition of HCO," Physical Chemistry Chemical Physics 4 (2002) 5778-5788.

For 835-1230 K and near 1 bar:

$$k = 4.0 \times 10^{13} \exp(-15550/RT) \text{ [cm}^3/\text{mol/s]}.$$

This study includes an RRKM fit to this rate, as well as determinations of the rate of $\text{H} + \text{HCO}$ and $\text{HCO} + \text{HCO}$.



S. Song, D. M. Golden, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of Benzylamine Decomposition: Overall Rate Coefficient and Heat of Formation of the Benzyl Radical," Journal Physical Chemistry A 106 (2002) 6094-6098.

For 1225-1599 K, 1.19-1.47 bar:

$$k = 5.49 \times 10^{14} \exp(-65790/RT) \text{ [cm}^3/\text{mol/s]}.$$

A high-pressure-limit rate is also provided in this study.



R. W. Bates, D. M. Golden, R. K. Hanson, C. T. Bowman, "Experimental Study and Modeling of the Reaction $\text{H} + \text{O}_2 + \text{M} = \text{HO}_2 + \text{M}$ ($\text{M} = \text{Ar}, \text{N}_2, \text{H}_2\text{O}$) at Elevated Pressures and Temperatures between 1050 and 1250 K," Physical Chemistry Chemical Physics 3 (2001) 2337-2342. See also Joint U. S. Sections Technical Meeting, The Combustion Institute, Washington DC, March 1999, Abstract book pp. 629-632.

For 1050-1250 K, 7-152 bar:

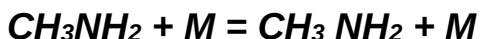
$$k_\infty = 9.04 \times 10^{12} T^{0.2} \text{ [cm}^3/\text{mol/s]};$$

$$k_0 = 6.80 \times 10^{18} T^{-1.2} \text{ [cm}^6/\text{mol}^2/\text{s]} \text{ for Ar:}$$

$$k_0 = 2.65 \times 10^{19} T^{-1.3} \text{ [cm}^6/\text{mol}^2/\text{s]} \text{ for N}_2:$$

$$k_0 = 3.70 \times 10^{19} T^{-1.0} \text{ [cm}^6/\text{mol}^2/\text{s]} \text{ for H}_2\text{O:}$$

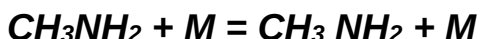
$$F_{\text{cent}} = 0.7 \text{ for Ar and N}_2 \text{ and } F_{\text{cent}} = 0.8 \text{ for H}_2\text{O}.$$



M. Votsmeier, S. Song, D. F. Davidson, R. K. Hanson, "Shock Tube Study of Monomethylamine Thermal Decomposition and NH_2 High Temperature Absorption Coefficient," International Journal of Chemical Kinetics 31 (1999) 323-330.

For 1550-1900 K, 1.6 atm:

$$k = 2.51 \times 10^{15} \exp(-56490/RT) \text{ [cm}^3/\text{mol/s]}.$$

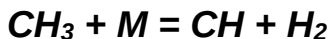


M. Votsmeier, S. Song, D. F. Davidson, R. K. Hanson, "Sensitive Detection of NH_2 in Shock Tube Experiments Using Frequency Modulation Spectroscopy," International Journal of Chemical Kinetics 31 (1999) 323-330. . See also Joint U. S. Sections Technical Meeting, The Combustion Institute, Washington DC, March 1999. Abstract book pp. 645-648.

For 1530-1975 K, 1.3 atm:

$k = 8.17 \times 10^{16} \exp(-61020/RT) [\text{cm}^3/\text{mol/s}]$.

A related study using conventional absorption spectroscopy is found in M. Votsmeier, S. Song, D. F. Davidson, R. K. Hanson, International Journal of Chemical Kinetics 31 (1999) 323-330.



M. Roehrig, E. L. Petersen, D. F. Davidson, R. K. Hanson, C. T. Bowman, "Measurement of the Rate Coefficient of the Reaction $\text{CH} + \text{O}_2 = \text{Products}$ in the Temperature Range 2200 to 2600 K," International Journal of Chemical Kinetics 29 (1997) 781-789.

For 2200-2600 K, 1.3 atm:

$k = 1.00 \times 10^{16} \exp(-85242/RT) [\text{cm}^3/\text{mol/s}]$.

This study also includes a determination of the rate of $\text{CH} + \text{O}_2$ and $\text{CH}_2 + \text{H}$.



M. Roehrig, E. L. Petersen, D. F. Davidson, R. K. Hanson, "A Shock Tube Study of the Pyrolysis of NO_2 ," International Journal of Chemical Kinetics 29 (1997) 483-493. See also Western States Section/The Combustion Institute 1995 Fall Meeting, Stanford University, October 30-31, 1995, paper 95F-156.

For 1350-2100 K, 42-380 atm, for argon:

$k_\infty = 3.98 \times 10^{14} \exp(-71770/RT) [1/\text{s}]$;

$k_0 = 3.98 \times 10^{15} \exp(-60048/RT) [\text{cm}^3/\text{mol/s}]$ for Ar:

This study also includes a determination of the reaction rate of $\text{NO}_2 + \text{NO}_2$.

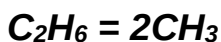


M. Roehrig, E. L. Petersen, D. F. Davidson, R. K. Hanson, "The Pressure Dependence of the Thermal Decomposition of N_2O ," International Journal of Chemical Kinetics 28 (1996) 599-608.

For 1570-3100 K, 0.3-450 atm, for argon:

$k_\infty = 1.26 \times 10^{12} \exp(-56700/RT) [1/\text{s}]$;

$k_0 = 3.98 \times 10^{14} \exp(-62679/RT) [\text{cm}^3/\text{mol/s}]$.



D. F. Davidson, R. K. Hanson, C. T. Bowman, "Communication: Revised Values of the Rate Coefficients of Ethane and Methane Decomposition," International Journal of Chemical Kinetics 27 (1995) 305-308.

For 1450-1945 K, 1 atm, for argon:

$k = 7.94 \times 10^{11} \exp(-60206/RT) [1/\text{s}]$.

This paper provides a reaction rate for methane decomposition as well.



D. F. Davidson, R. K. Hanson, C. T. Bowman, "Communication: Revised Values of the Rate Coefficients of Ethane and Methane Decomposition," International Journal of Chemical Kinetics 27 (1995) 305-308.

For 1785-2325 K, 1 atm, for argon:

$k = 1.04 \times 10^{18} \exp(-96389/RT)$ [1/s].

This paper provides a reaction rate for ethane decomposition as well.



D. F. Davidson, E. L. Petersen, M. Roehrig, R. K. Hanson, "Measurement of the Rate Coefficient of $H+O_2+M=HO_2+M$ for $M=Ar$ and N_2 at High Pressures," Proceedings of the Combustion Institute 26 (1996) 481-488.

For 1260-1375 K, 50-115 atm:

$k_0 = 7.0 \times 10^{17} T^{-0.8}$ [cm⁶/mol²/s] for $M = Ar$;

$k_0 = 2.6 \times 10^{19} T^{-1.24}$ [cm⁶/mol²/s] for $M = N_2$.



M. S. Wooldridge, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of Nitric Acid Decomposition," Proceedings of the 19th International Symposium on Shock Waves, Marseille, France, July 26-30, 1993, Vol. II, pp. 83-88.

For 1120-1960 K, 0.15-1.01 atm:

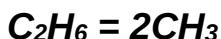
$k = 1.01 \times 10^{16} \exp(-34335/RT)$ [cm³/mol/s].



L. S. Zelson, D. F. Davidson, R. K. Hanson, "V.U.V. Absorption Diagnostic for Shock Tube Kinetics Studies of C_2H_4 ," Journal Quantitative Spectroscopy and Radiative Transfer 52 (1994) 31-43. See also Western States Section/The Combustion Institute 1993 Fall Meeting, Menlo Park CA, October 18-19, 1993, Paper 93-057.

For 1600-2300 K, 1.2 atm:

$k = 9.17 \times 10^{16} \exp(-75069/RT)$ [cm³/mol/s].



D. F. Davidson, M. D. Di Rosa, R. K. Hanson, C. T. Bowman, "A Study of Ethane Decomposition in a Shock Tube using Laser Absorption of CH_3 ," International Journal of Chemical Kinetics 25 (1993) 969-982.

For 1350-2110 K, 0.58-4.4 atm:

Measurement results were corrected in D. F. Davidson, R. K. Hanson, C. T. Bowman, Internat. J. Chem. Kinet. 27 (1995) 305-308.



D. F. Davidson, M. D. Di Rosa, A. Y. Chang, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of Methane Decomposition using Laser Absorption of CH_3 ," Proceedings of the Combustion Institute 24 (1992) 589-596.

For 1790-2325 K, 0.56-3.76 atm:

Measurement results were corrected in D. F. Davidson, R. K. Hanson, C. T. Bowman, Internat. J. Chem. Kinet. 27 (1995) 305-308.



D. F. Davidson, K. Kohse-Hoeinghaus, A. Y. Chang, R. K. Hanson, , "A Pyrolysis Mechanism for Ammonia," International Journal of Chemical Kinetics 22 (1990) 513-535. See also Western States Section/ The Combustion Institute 1989 Fall Meeting, Livermore CA October 23-24, 1989, Paper 89-95.

For 2200-2800 K, 0.8-1.1 atm:

$$k = 2.2 \times 10^{16} \exp(-93470/RT) \text{ [cm}^3/\text{mol/s]}.$$

Reaction rates for $\text{NH}_2 + \text{H}$, $\text{NH}_2 + \text{NH}$, and $\text{NH}_2 + \text{NH}_2$ are also included in this paper.

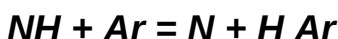


J. D. Mertens, A. Y. Chang, R. K. Hanson, C. T. Bowman, "Reaction Kinetics of NH in the Shock Tube Pyrolysis of HNCO," International Journal of Chemical Kinetics 21 (1989) 1049-1067. See also Western States Section/ The Combustion Institute 1988 Fall Meeting, Dana Point CA October 17-18, 1988, Paper 88-64.

For 1830-3340 K, 0.3-2.1 atm:

$$k = 9.84 \times 10^{15} \exp(-85441/RT) \text{ [cm}^3/\text{mol/s]}.$$

This paper provides reaction rates for $\text{NH} + \text{NH}$ and $\text{NH} + \text{Ar}$ as well.



J. D. Mertens, A. Y. Chang, R. K. Hanson, C. T. Bowman, "Reaction Kinetics of NH in the Shock Tube Pyrolysis of HNCO," International Journal of Chemical Kinetics 21 (1989) 1049-1067. See also Western States Section/ The Combustion Institute 1988 Fall Meeting, Dana Point CA October 17-18, 1988, Paper 88-64.

For 3140-3320 K, 0.3-2.1 atm:

$$k = 2.65 \times 10^{14} \exp(-75506/RT) \text{ [cm}^3/\text{mol/s]}.$$

This paper provides reaction rates for $\text{HNCO} + \text{Ar}$ and $\text{NH} + \text{NH}$ as well.

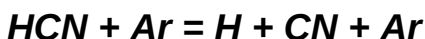


M. Y. Louge, R. K. Hanson, "High Temperature Kinetics of NCO," Combustion and Flame 58 (1984) 291-300.

For 2150-2400 K, 0.6 atm:

$$k = 6.3 \times 10^{16} T^{-0.5} \exp(-47688/RT) \text{ [cm}^3/\text{mol/s]}.$$

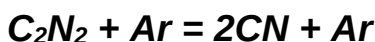
Rates are also presented for $\text{NCO} + \text{H}$ and $\text{NCO} + \text{H}_2$.



A. Szekely, R. K. Hanson, C. T. Bowman, "Thermal Decomposition of Hydrogen Cyanide behind Incident Shock Waves," Journal Physical Chemistry 88 (1984) 666-668.

For 2700-3600 K, 0.6 atm:

$$k = 1.0 \times 10^{16} \exp(-108589/RT) \text{ [cm}^3/\text{mol/s]}.$$



A. Szekely, R. K. Hanson, C. T. Bowman, "Shock Tube Study of the Thermal Decomposition of Hydrogen Cyanogen," Journal Physical Chemistry 80 (1984) 4982-4985.

For 2500-3450 K, 0.23-58 bar:
 $k = 6.3 \times 10^{16} \exp(-99429/RT)$ [cm³/mol/s].



T. R. Roose, R. K. Hanson, C. H Kruger, "*Thermal Decomposition of NH₃ in Shock Waves*," Proceedings of the 12th International Symposium on Shock Tubes and Waves (1980) pp. 476-485.

For 2200-3450 K, 0.14-0.6 atm:

$k = 2.52 \times 10^{16} \exp(-93786/RT)$ [cm³/mol/s].

Rates for the NH+H₂ product channel are also provided in this paper.



W. L Flower, R. K. Hanson, C. H Kruger, "*Experimental Study of Nitric Oxide Decomposition by Reaction with Hydrogen*," Combustion, Science and Technology 15 (1977) 115-128.

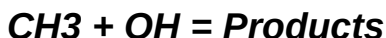
For 1815-3365 K, 0.6 atm:

$k = 1.42 \times 10^{14} \exp(-51280/RT)$ [cm³/mol/s].

A rate for N₂O+O is also presented in this paper.

Bimolecular Reactions

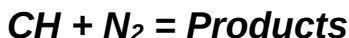
2007-2008



V. Vasudevan, R. D. Cook, R. K. Hanson, C. T. Bowman, D. M. Golden, "*High-Temperature Shock Tube Study of the Reactions CH₃+OH = Products and CH₃OH+Ar = Products*," International Journal of Chemical Kinetics 40 (2008) 488-495.

For 1081-1426 K, 1.21-4.95 atm:

No rate expression given; data table provided. Measurements are in agreement with theory of A. W. Jasper, S. J. Klippenstein, L. B. Harding, B. Ruscis, J. Phys. Chem. A 111 (2007) 3932-3950.

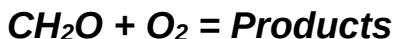


V. Vasudevan, R. K. Hanson, C. T. Bowman, D. M. Golden, D. F. Davidson, "*Shock Tube Study of the Reaction of CH with N₂: Overall Rate and Branching Ratio*," Journal Physical Chemistry A 111 (2007) 11818-11830.

For 1943-3543 K, 0.9-1.4 atm:

$k = 6.03 \times 10^{12} \exp(-22155/RT)$ [cm³/mol/s]

This study includes evidence of the NCN+H channel.



V. Vasudevan, D. F. Davidson, R. K. Hanson, C. T. Bowman, D. M. Golden, "*High-Temperature Measurements of the Rates of the Reactions CH₂O+Ar=Products*

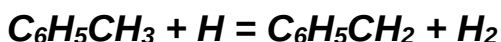
and $\text{CH}_2\text{O} + \text{O}_2 = \text{Products}$,” Proceedings of the Combustion Institute 31 (2007) 175-183. See also Western States Section/Combustion Institute Fall 2005 Meeting, October 17-18, 2005, paper 05F-40.

For 1480-2367 K, 0.91-1.96 atm:

$$k = 5.08 \times 10^{14} \exp(-46297/RT) \text{ [cm}^3/\text{mol/s]}$$

This study also includes rate measurements of $\text{CH}_2\text{O} + \text{Ar}$.

2005-2006

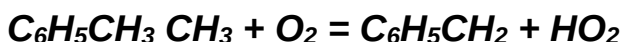


M. A. Oehlschlaeger, D. F. Davidson, R. K. Hanson, “*Experimental Investigation of Toluene + H = Benzyl + H₂ at High Temperatures*,” Journal Physical Chemistry A 110 (2006) 9867-9873.

For 1256-1667 K, 1.7 bar:

$$k = 1.33 \times 10^{15} \exp(-14880/RT) \text{ [cm}^3/\text{mol/s]}$$

This study provides a non-Arrhenius fit to this data and the lower temperature data C. Ellis, M. S. Scott, R. W. Walker, Combust. Flame 132 (2003) 291.

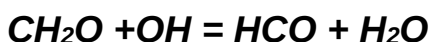


M. A. Oehlschlaeger, D. F. Davidson, R. K. Hanson, “*Investigation of the Reaction of Toluene with Molecular Oxygen in Shock-Heated Gases*,” Combustion and Flame 147 (2006) 195-208.

For 1117-1366 K, 1.7 bar:

$$k = 2.18 \times 10^7 T^{2.5} \exp(-46045/RT) \text{ [cm}^3/\text{mol/s]}$$

This reaction rate determination includes data from the lower temperature study of T. Ingham, R. W. Walker, R. E. Woolford, Proc. Combust. Inst. 25 (1994) 767-774.



V. Vasudevan, D. F. Davidson, R. K. Hanson, “*Direct Measurements of the Reaction of OH + CH₂O = HCO + H₂O at High Temperatures*,” International Journal of Chemical Kinetics 37 (2005) 98-109. See also Joint Meeting of the U.S. Sections of the Combustion Institute, Philadelphia PA, March 22, 2005, Paper C23.

For 934-1670 K, 1.6 bar:

$$k = 7.82 \times 10^7 T^{1.63} \exp(1055/RT) \text{ [cm}^3/\text{mol/s]}$$

This reaction rate determination includes data from the lower temperature study of V. Sivakumaran, D. Holscher, T. J. Dillon, J. N. Crowley, Phys. Chem. Chem. Phys. 5 (2003) 4821-4827.

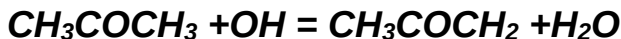


V. Vasudevan, D. F. Davidson, R. K. Hanson, “*High-Temperature Measurements of the Reactions of OH with Toluene and Acetone*,” Journal of Physical Chemistry A 109 (2005) 3352-3359. See also the 25th International Symposium on Shock Waves, Bangalore India, July 2005. See also Joint Meeting of the U.S. Sections of the Combustion Institute, Philadelphia PA, March 22, 2005, Paper C23.

For 570-1389 K, 2.25 bar:

$$k = 1.62 \times 10^{13} \exp(-2770/RT) \text{ [cm}^3\text{/mol/s]}$$

This reaction rate determination includes data from the lower temperature study of F. P. Tully, A. R. Ravishankara, R. L. Thomson, J. M. Nicovich, R. C. Shah, N. M. Kreutter, P. H. Wine, J. Phys. Chem. 85 (1981) 2262-2269. See also International



V. Vasudevan, D. F. Davidson, R. K. Hanson, "High-Temperature Measurements of the Reactions of OH with Toluene and Acetone," Journal of Physical Chemistry A 109 (2005) 3352-3359. See also Joint Meeting of the U.S. Sections of the Combustion Institute, Philadelphia PA, March 22, 2005, Paper C23.

For 982-1300 K, 1.65 bar:

$$k = 2.95 \times 10^{13} \exp(-4564/RT) \text{ [cm}^3\text{/mol/s]}$$

This study also includes a measurement of the rate of toluene+OH.



J. T. Herbon, R. K. Hanson, C. T. Bowman, D. M. Golden, "The Reaction of $\text{CH}_3 + \text{O}_2$: Experimental Determination of the Rate Coefficients for the Product Channels at High Temperatures" Proceedings of the Combustion Institute 30 (2005) 955-963.

For 1590-2430 K, 1.4 atm:

$$k = 6.08 \times 10^7 T^{1.54} \exp(-27828/RT) \text{ [cm}^3\text{/mol/s]} \text{ for the } \text{CH}_3\text{O} + \text{O} \text{ channel.}$$

$$k = 68.6 T^{2.86} \exp(-9768/RT) \text{ [cm}^3\text{/mol/s]} \text{ for the } \text{CH}_2\text{O} + \text{OH} \text{ channel.}$$

2000-2004

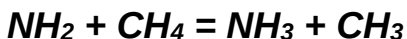


G. Friedrichs, D. F. Davidson, R. K. Hanson, "Validation of a Thermal Decomposition Mechanism of Formaldehyde by Detection of CH_2O and HCO behind Shock Waves," International Journal of Chemical Kinetics 36 (2004) 157-169.

For 1710-2100 K, 1 bar:

$$k = 1.5 \times 10^{13} \text{ [cm}^3\text{/mol/s]}.$$

A reaction rate for $\text{CH}_2\text{O} + \text{M}$ is all provided in this study.



S. Song, D. M. Golden, R. K. Hanson, C. T. Bowman, J. P. Senosianin, C. B. Musgrave, G. Friedrichs, "A Shock Tube Study of the Reaction $\text{NH}_2 + \text{CH}_4 = \text{NH}_3 + \text{CH}_3$ and Comparison with Transition State Theory," International Journal of Chemical Kinetics 35 (2003) 304-309.

For 1591-2084 K, 1.10-1.33 bar:

$$k = 1.47 \times 10^{13} T^{3.01} \exp(-9934/RT) \text{ [cm}^3\text{/mol/s]}.$$

This rate is based on the experiment and TST theory.

$H + HCO = H_2 + CO$

G. Friedrichs, J. T. Herbon, D. F. Davidson, R. K. Hanson, "Quantitative Detection of HCO behind Shock Waves: The Thermal Decomposition of HCO," Physical Chemistry Chemical Physics 4 (2002) 5778-5788.

For 820 K and lower, and pressures lower than 1 bar:

$$k = 1.1 \times 10^{14} \text{ [cm}^3\text{/mol/s]}.$$

This study includes determinations of the rate of HCO+HCO and HCO+M also.

 $HCO + HCO = CH_2O + CO$

G. Friedrichs, J. T. Herbon, D. F. Davidson, R. K. Hanson, "Quantitative Detection of HCO behind Shock Waves: The Thermal Decomposition of HCO," Physical Chemistry Chemical Physics 4 (2002) 5778-5788.

For 820 K and lower, and pressures lower than 1 bar:

$$k = 2.7 \times 10^{13} \text{ [cm}^3\text{/mol/s]}.$$

This study includes determinations of the rate of H+HCO and HCO+M also.

 $H + CH_2O = H_2 + HCO$

G. Friedrichs, D. F. Davidson, R. K. Hanson, "Direct Measurements of the Reaction of $H+CH_2O=H_2+HCO$ Behind Shock Waves by Means of Vis-UV (correction V-UV) Detection of Formaldehyde," International Journal of Chemical Kinetics 34 (2002) 374-386.

For 1510-1969 K, 1.3 atm:

$$k = 6.6 \times 10^{14} \exp(-9313/RT) \text{ [cm}^3\text{/mol/s]}.$$

Title correction: July 18, 2002.

 $NH_2 + NO = \text{Products}$

S. Song, R. K. Hanson, C. T. Bowman, D. M. Golden, "A Shock Tube Study of the Product Branching Ratio of the NH_2+NO Reaction at High Temperatures," Journal Physical Chemistry A 106 (2002) 9233-9235.

For 1826-2159 K, 1.10-1.21 bar:

Table of branching ratio data provided for $NNH+OH$ and N_2+H_2O channels.

 $NH_2 + NO_2 = \text{Products}$

S. Song, D. M. Golden, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of the NH_2+NO_2 Reaction," Proceedings of the Combustion Institute 29 (2002) 2163-2170.

For 1330-1527 K, 1.31-1.49 bar:

$$k = 5.5 \times 10^{12} \text{ [cm}^3\text{/mol/s]}.$$

A branching ratio to the N_2O+H_2O channel is also provided.

 $NH_2 + NO = \text{Products}$

S. Song, R. K. Hanson, C. T. Bowman, D. M. Golden, "Shock Tube Determination of the Overall Rate of $NH_2+NO=\text{Products}$ in the Thermal De-Nox Temperature Window," International Journal of Chemical Kinetics 33 (2001) 715-721. See also

2nd Joint Meeting of the U.S. Section of the Combustion Institute, Oakland CA, March 25-28, 2001, Paper 204.

For 1262-1726 K, 1.14-1.44 bar:

$$k = 6.83 \times 10^{15} T^{-1.203} \exp(211/RT) \text{ [cm}^3\text{/mol/s]}.$$

This reaction rate determination combines the data from this study and our previous higher temperature study S. Song, R. K. Hanson, C. T. Bowman, D. M. Golden, Proc. Comb. Inst. 28 (2000) 2403-2409.

NH₂ + NO = Products

S. Song, R. K. Hanson, C. T. Bowman, D. M. Golden, “*Shock Tube Determination of the Overall Rate of NH₂+NO=Products at High Temperatures*,” Proceedings of the Combustion Institute 28 (2000) 2403-2409.

For 1716-2507 K, 1.0-1.38 bar:

$$k = 2.08 \times 10^{16} T^{-1.34} \text{ [cm}^3\text{/mol/s]}.$$

A lower temperature study of this reaction rate was also performed: S. Song, R. K. Hanson, C. T. Bowman, D. M. Golden, Internat. J. Chem. Kinet. 33 (2001) 715-721.

1995-1999

NH₂ + NO = Products

M. Votsmeier, S. Song, R. K. Hanson, C. T. Bowman, “*A Shock Tube Study of the Product Branching Ratio for the Reaction NH₂+NO Using Frequency-Modulation Detection of NH₂*,” Journal Physical Chemistry A 103 (1999) 1566-1571.

For 1340-1670 K, 1.3 atm:

Table of branching ratio data provided for NNH+OH and N₂+H₂O channels.

CH + O₂ = Products

M. Roehrig, E. L. Petersen, D. F. Davidson, R. K. Hanson, C. T. Bowman, “*Measurement of the Rate Coefficient of the Reaction CH+O₂=Products in the Temperature Range 2200 to 2600 K*,” International Journal of Chemical Kinetics 29 (1997) 781-789.

For 2200-2600 K, 1.3 atm:

$$k = 9.77 \times 10^{13} \text{ [cm}^3\text{/mol/s]}.$$

This study also includes a determination of the rate of CH₃+M and CH₂+H.

CH₂ + H = CH + H₂

M. Roehrig, E. L. Petersen, D. F. Davidson, R. K. Hanson, C. T. Bowman, “*Measurement of the Rate Coefficient of the Reaction CH+O₂=Products in the Temperature Range 2200 to 2600 K*,” International Journal of Chemical Kinetics 29 (1997) 781-789.

For 2200-2600 K, 1.3 atm:

$$k = 1.41 \times 10^{14} \text{ [cm}^3\text{/mol/s]}.$$

This study also includes a determination of the rate of CH₃+M and CH +O₂.

$\text{NO}_2 + \text{NO}_2 = \text{Products}$

M. Roehrig, E. L. Petersen, D. F. Davidson, R. K. Hanson, "A Shock Tube Study of the Pyrolysis of NO_2 ," International Journal of Chemical Kinetics 29 (1997) 483-493.

For 1350-2100 K, 42-380 atm:

$k = 2.00 \times 10^{12} \exp(-25120/RT)$ [$\text{cm}^3/\text{mol/s}$] for the $2\text{NO} + \text{O}_2$ channel;

$k = 1.00 \times 10^{13} \exp(-25837/RT)$ [$\text{cm}^3/\text{mol/s}$] for the $\text{NO}_3 + \text{NO}$ channel.

This study also includes a determination of the reaction rate of $\text{NO}_2 + \text{M}$.

$\text{CH}_3 + \text{CH}_3 = \text{Products}$

D. F. Davidson, M. D. Di Rosa, E. J. Chang, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of Methyl-Methyl Reactions between 1200 and 2400 K," International Journal of Chemical Kinetics 27 (1995) 1179-1196.

For 1570-1780 K, 0.46-1.52 atm:

$k = 2.4 \times 10^{13} \exp(-12875/RT)$ [$\text{cm}^3/\text{mol/s}$] for the $\text{C}_2\text{H}_5 + \text{H}$ channel.

k is less than 1×10^{11} [$\text{cm}^3/\text{mol/s}$] for the $\text{C}_2\text{H}_4 + \text{H}_2$ channel.

This paper also provides a reaction rate for $\text{CH}_3 + \text{H}$.

$\text{CH}_3 + \text{H} = {}^1\text{CH}_2 + \text{H}_2$

D. F. Davidson, M. D. Di Rosa, E. J. Chang, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of Methyl-Methyl Reactions between 1200 and 2400 K," International Journal of Chemical Kinetics 27 (1995) 1179-1196.

For 1800-2200 K, 1.79-2.20 atm:

$k = 1.3 \times 10^{14}$ [$\text{cm}^3/\text{mol/s}$] for the reverse reaction rate ${}^1\text{CH}_3 + \text{H}_2 = \text{CH}_3 + \text{H}$.

This paper also provides a reaction rate for $\text{CH}_3 + \text{CH}_3 = \text{Products}$.

$\text{HCN} + \text{OH} = \text{CN} + \text{H}_2\text{O}$

S. T. Wooldridge, R. K. Hanson, C. T. Bowman, "Simultaneous Laser Absorption Measurements of $\text{CN} + \text{OH}$ in a Shock Tube Study of $\text{HCN} + \text{OH} = \text{Products}$," International Journal of Chemical Kinetics 27 (1995) 1075-1087.

For 1120-1960 K, 0.15-1.01 atm:

$k = 3.90 \times 10^6 T^{1.83} \exp(-10290/RT)$ [$\text{cm}^3/\text{mol/s}$].

$\text{CN} + \text{OH} = \text{Products}$

S. T. Wooldridge, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of the Reactions of CN with HCN , OH and H_2 using CN and OH Laser Absorption," International Journal of Chemical Kinetics 28 (1996) 245-258.

For 1250-1860 K, 0.19-0.35 atm:

$k = 4.0 \times 10^{13}$ [$\text{cm}^3/\text{mol/s}$].

This study also includes reaction rates for $\text{CN} + \text{HCN}$ and $\text{CN} + \text{H}_2$.

$\text{CN} + \text{HCN} = \text{C}_2\text{N}_2 + \text{H}$

S. T. Wooldridge, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of the Reactions of CN with HCN , OH and H_2 using CN and OH Laser Absorption," International Journal of Chemical Kinetics 28 (1996) 245-258.

For 940-1860 K, 0.19-0.35 atm:

$k = 1.51 \times 10^7 T^{1.71} \exp(-1530/RT)$ [cm³/mol/s] (reaction rate of D. L. Yang, T. Yu, M. C. Lin, C. F. Melius, J. Chem. Phys. 97 (1992) 222.)

This study also includes reaction rates for CN+OH and CN+H₂.

CN + H₂ = HCN + H

S. T. Wooldridge, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of the Reactions of CN with HCN, OH and H₂ using CN and OH Laser Absorption," International Journal of Chemical Kinetics 28 (1996) 245-258.

For 940-1860 K, 0.19-0.35 atm:

$k = 2.95 \times 10^5 T^{2.45} \exp(-2237/RT)$ [cm³/mol/s].

This study also includes reaction rates for CN+HCN and CN+OH.

CN + H₂ = HCN + H

S. T. Wooldridge, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of the Reactions of CN with HCN, OH and H₂ using CN and OH Laser Absorption," International Journal of Chemical Kinetics 28 (1996) 245-258.

For 940-1860 K, 0.19-0.35 atm:

$k = 2.95 \times 10^5 T^{2.45} \exp(-2237/RT)$ [cm³/mol/s].

This study also includes reaction rates for CN+HCN and CN+OH.

HNCO + OH = Products

M. S. Wooldridge, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of CO+OH=CO₂+H and HNCO+OH=Products via Simultaneous Laser Absorption Measurements of OH and CO₂," International Journal of Chemical Kinetics 28 (1996) 361-372.

For 620-1860 K, 0.149-0.247 atm:

$k = 3.63 \times 10^7 T^{1.5} \exp(-3594/RT)$ [cm³/mol/s].

This study also includes reaction rates for CO+OH.

CO + OH = CO₂ + H

M. S. Wooldridge, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of CO+OH=CO₂+H and HNCO+OH=Products via Simultaneous Laser Absorption Measurements of OH and CO₂," International Journal of Chemical Kinetics 28 (1996) 361-372.

For 1155-1432 K, 0.243-0.490 atm:

$k = 2.12 \times 10^{12} \exp(-5226/RT)$ [cm³/mol/s].

This study also includes reaction rates for HNCO+OH.

1990-1994

CN + NO₂ = Products

S. T. Wooldridge, J. D. Mertens, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of the Reactions of CN and NCO with NO₂," Proceedings of the Combustion

Institute 25 (1994) 983-991. See also Western States Section/The Combustion Institute 1993 Fall Meeting, Menlo Park CA, October 18-19, 1993, paper 93-111.

For 1000-1600 K, 1.4 atm:

$$k = 1.59 \times 10^{13} \exp(-1133/RT) \text{ [cm}^3/\text{mol/s]}.$$

This study also includes reaction rates for NCO+NO₂.

NCO + NO₂ = Products

S. T. Wooldridge, J. D. Mertens, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of the Reactions of CN and NCO with NO₂," Proceedings of the Combustion Institute 25 (1994) 983-991. See also Western States Section/The Combustion Institute 1993 Fall Meeting, Menlo Park CA, October 18-19, 1993, paper 93-111.

For 1250 K, 1.4 atm:

$$k = 4.5 \times 10^{12} \text{ [cm}^3/\text{mol/s]}.$$

This study also includes reaction rates for CN+NO₂.

H + O₂ = OH + O

C.-L. Yu, M. Frenklach, D. A. Masten, R. K. Hanson, C. T. Bowman, "Reexamination of Shock Tube Measurements of the Rate Coefficient of H+O₂=OH+O," Journal Physical Chemistry 98 (1994) 4770-4771.

For 1800-2500 K, 1.4 atm:

$$k = 8.3 \times 10^{13} \exp(-14425/RT) \text{ [cm}^3/\text{mol/s]}.$$

This is a review article.

CO + OH = CO₂ + H

M. S. Wooldridge, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of the CO+OH=CO₂+H Reaction," Proceedings of the Combustion Institute 25 (1994) 741-748.

For 1090-2370 K, 0.19-0.82 atm:

$$k = 2.12 \times 10^{12} \exp(-5226/RT) \text{ [cm}^3/\text{mol/s]}.$$

OH + OH = H₂O + O

M. S. Wooldridge, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of the OH+OH=H₂O+O Reaction," International Journal of Chemical Kinetics 26 (1994) 389-401. See also Western States Section/The Combustion Institute 1992 Fall Meeting, Berkeley CA, October 12-13, 1992, paper 92-93.

For 1050-2380 K, 0.18-0.60 atm:

$$k = 9.40 \times 10^{-5} T^{4.77} \exp(-9080/RT) \text{ [cm}^3/\text{mol/s]}.$$

O + HNCO = Products

J. D. Mertens, A. Y. Chang, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of Reactions of Atomic Oxygen with Isocyanic Acid," International Journal of Chemical Kinetics 24 (1992) 279-295.

For 2120-3190 K, 0.25-1.11 atm:

$$k = 1.9 \times 10^{13} \exp(-14207/RT) \text{ [cm}^3/\text{mol/s]} \text{ for the NH+CO}_2 \text{ channel.}$$

$$k = 1.4 \times 10^{14} \exp(-20466/RT) \text{ [cm}^3/\text{mol/s]} \text{ for the OH+NCO channel.}$$

This study also includes reaction rates for HNCO+OH.

OH + HNCO = Products

J. D. Mertens, A. Y. Chang, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of Reactions of Atomic Oxygen with Isocyanic Acid," International Journal of Chemical Kinetics 24 (1992) 279-295.

For 2120-2350 K, 0.25-1.11 atm:

An upper limit for this rate was determined to be $k = 1.5 \times 10^{12}$ [cm³/mol/s].

This study also includes reaction rates for HNCO+O.

C(³P) + H₂ = CH + H

A. J. Dean, D. F. Davidson, R. K. Hanson, "A Shock Tube Study of Reactions of C Atoms with H₂ and O₂ using Excimer Photolysis of C₃O₂ and C Atom Atomic Resonance Absorption Spectroscopy," Journal Physical Chemistry 95 (1991) 183-191.

For 1504-2042 K, 0.61-0.83 atm:

$k = 4.0 \times 10^{14} \exp(-23247/RT)$ [cm³/mol/s].

This study also includes rates for C+O₂ and C+C₃O₂.

C(³P) + O₂ = CO + O

A. J. Dean, D. F. Davidson, R. K. Hanson, "A Shock Tube Study of Reactions of C Atoms with H₂ and O₂ using Excimer Photolysis of C₃O₂ and C Atom Atomic Resonance Absorption Spectroscopy," Journal Physical Chemistry 95 (1991) 183-191.

For 1517-4185 K, 0.64-1.21 atm:

$k = 1.2 \times 10^{14} \exp(-3994/RT)$ [cm³/mol/s].

This study also includes rates for C+H₂ and C+C₃O₂.

C(³P) + C₃O₂ = Products

A. J. Dean, D. F. Davidson, R. K. Hanson, "A Shock Tube Study of Reactions of C Atoms with H₂ and O₂ using Excimer Photolysis of C₃O₂ and C Atom Atomic Resonance Absorption Spectroscopy," Journal Physical Chemistry 95 (1991) 183-191.

For 1450-1900 K, 0.6-1.2 atm:

$k = 2.4 \times 10^{19} T^{-1.75}$ [cm³/mol/s].

This study also includes rates for C+O₂ and C+H₂.

CN + O = CO + N

D. F. Davidson, A. J. Dean, M. D. DiRosa, R. K. Hanson, C. T. Bowman, "Shock Tube Measurements of the Reactions of CN with O and O₂," International Journal of Chemical Kinetics 23 (1991) 1035-1050. See also The Combustion Institute/Western States Section 1990 Fall Meeting, La Jolla CA, October 15-16, 1990, Paper 90-13.

For 3000-4500 K, 0.45-0.90 atm:

$k = 7.7 \times 10^{13}$ [cm³/mol/s].

This study also includes reaction rates for CN+O₂.

CN + O₂ = NCO + O

D. F. Davidson, A. J. Dean, M. D. DiRosa, R. K. Hanson, C. T. Bowman, "Shock Tube Measurements of the Reactions of CN with O and O₂," International Journal of Chemical Kinetics 23 (1991) 1035-1050. See also The Combustion Institute/Western States Section 1990 Fall Meeting, La Jolla CA, October 15-16, 1990, Paper 90-13.

For 2700-3800 K, 0.62-1.05 atm:

$$k = 1.0 \times 10^{13} \text{ [cm}^3\text{/mol/s]}.$$

This study also includes reaction rates for CN+O.

N₂O + O = Products

D. F. Davidson, M. D. DiRosa, A. Y. Chang, R. K. Hanson, "Shock Tube Measurements of the Major Product Channels of N₂O+O," Proceedings of the 18th International Symposium on Shock Waves, Sendai, Japan, July 21-26, 1991, pp. 813-818.

For 1680-3340 K, 0.7-2.0 atm:

$$k = 2.9 \times 10^{13} \exp(-23149/RT) \text{ [cm}^3\text{/mol/s]} \text{ for the NO+NO channel.}$$

$$k = 1.4 \times 10^{12} \exp(-10809/RT) \text{ [cm}^3\text{/mol/s]} \text{ for the N}_2\text{+O}_2 \text{ channel.}$$

C(³P) + NO = Products

A. J. Dean, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of Reactions of C Atoms and CH with NO including Product Channel Measurements," Journal Physical Chemistry 95 (1991) 3180-3189.

For 1550-4050 K, 0.5-1.0 atm:

$$k = 4.8 \times 10^{13} \text{ [cm}^3\text{/mol/s]}.$$

This study also includes branching ratio data for the two product channels CH+O and CO+N, and a determination of the CH+NO rate.

CH + NO = Products

A. J. Dean, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of Reactions of C Atoms and CH with NO including Product Channel Measurements," Journal Physical Chemistry 95 (1991) 3180-3189.

For 1550-4050 K, 0.5-1.0 atm:

$$k = 1.0 \times 10^{14} \text{ [cm}^3\text{/mol/s]}.$$

This study also includes branching ratio data for the three product channels including HCN+O, NH+CO and OH+CN, and a determination of the C+NO rate.

H + HNCO = NH₂ + CO

J. D. Mertens, K. Kohse-Hoeinghaus, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of H+HNCO=NH₂+CO," International Journal of Chemical Kinetics 23 (1991) 655-668.

For 2340-3270 K, 0.33-0.49 atm:

$$k = 2.1 \times 10^{14} \exp(-16889/RT) \text{ [cm}^3\text{/mol/s]}.$$

An upper limit for $\text{NH}_2 + \text{HNCO}$ is also included in this paper.

$\text{NH}_2 + \text{HNCO} = \text{NH}_3 + \text{NCO}$

J. D. Mertens, K. Kohse-Hoeinghaus, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of $\text{H} + \text{HNCO} = \text{NH}_2 + \text{CO}$," International Journal of Chemical Kinetics 23 (1991) 655-668.

For 2340-2680 K, 0.33-0.49 atm:

Upper limit for $k = 5.0 \times 10^{11}$ [$\text{cm}^3/\text{mol/s}$].

A reaction rate for $\text{H} + \text{HNCO}$ is also included in this paper.

$\text{NH} + \text{NO} = \text{Products}$

J. D. Mertens, A. Y. Chang, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of the Reactions of NH with NO , O_2 and O ," International Journal of Chemical Kinetics 23 (1991) 173-196. See also Western States Section/ The Combustion Institute 1989 Fall Meeting, Livermore CA October 23-24, 1989, Paper 89-96.

For 2220-3350 K, 0.1-1.0 atm:

$k = 1.7 \times 10^{14} \exp(-12716/RT)$ [$\text{cm}^3/\text{mol/s}$].

Reaction rates for $\text{NH} + \text{O}_2$ and $\text{NH} + \text{O}$ are also included in this paper.

$\text{NH} + \text{O}_2 = \text{Products}$

J. D. Mertens, A. Y. Chang, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of the Reactions of NH with NO , O_2 and O ," International Journal of Chemical Kinetics 23 (1991) 173-196. See also Western States Section/ The Combustion Institute 1989 Fall Meeting, Livermore CA October 23-24, 1989, Paper 89-96.

For 2220-3350 K, 0.1-1.0 atm:

$k = 3.5 \times 10^{13} \exp(-17088/RT)$ [$\text{cm}^3/\text{mol/s}$].

Reaction rates for $\text{NH} + \text{NO}$ and $\text{NH} + \text{O}$ are also included in this paper.

$\text{NH} + \text{O} = \text{Products}$

J. D. Mertens, A. Y. Chang, R. K. Hanson, C. T. Bowman, "A Shock Tube Study of the Reactions of NH with NO , O_2 and O ," International Journal of Chemical Kinetics 23 (1991) 173-196. See also Western States Section/ The Combustion Institute 1989 Fall Meeting, Livermore CA October 23-24, 1989, Paper 89-96.

For 2220-3350 K, 0.1-1.0 atm:

$k = 9.2 \times 10^{13}$ [$\text{cm}^3/\text{mol/s}$].

Reaction rates for $\text{NH} + \text{O}_2$ and $\text{NH} + \text{NO}$ are also included in this paper.

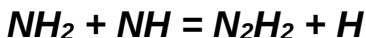
$\text{NH}_2 + \text{H} = \text{NH} + \text{H}_2$

D. F. Davidson, K. Kohse-Hoeinghaus, A. Y. Chang, R. K. Hanson, , "A Pyrolysis Mechanism for Ammonia," International Journal of Chemical Kinetics 22 (1990) 513-535. See also Western States Section/ The Combustion Institute 1989 Fall Meeting, Livermore CA October 23-24, 1989, Paper 89-95.

For 2200-2800 K, 0.8-1.1 atm:

$k = 4.0 \times 10^{13} \exp(-7252/RT)$ [$\text{cm}^3/\text{mol/s}$].

Reaction rates for $\text{NH}_2 + \text{NH}$, $\text{NH}_2 + \text{NH}_2$ and $\text{NH}_3 + \text{M}$ are also included in this paper.

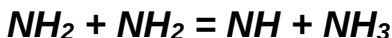


D. F. Davidson, K. Kohse-Hoeinghaus, A. Y. Chang, R. K. Hanson, , "A Pyrolysis Mechanism for Ammonia," International Journal of Chemical Kinetics 22 (1990) 513-535. See also Western States Section/ The Combustion Institute 1989 Fall Meeting, Livermore CA October 23-24, 1989, Paper 89-95.

For 2200-2800 K, 0.8-1.1 atm:

$$k = 1.5 \times 10^{15} T^{-0.5} [\text{cm}^3/\text{mol/s}].$$

Reaction rates for $\text{NH}_2 + \text{H}$, $\text{NH}_2 + \text{NH}_2$ and $\text{NH}_3 + \text{M}$ are also included in this paper.

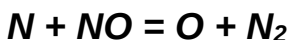


D. F. Davidson, K. Kohse-Hoeinghaus, A. Y. Chang, R. K. Hanson, , "A Pyrolysis Mechanism for Ammonia," International Journal of Chemical Kinetics 22 (1990) 513-535. See also Western States Section/ The Combustion Institute 1989 Fall Meeting, Livermore CA October 23-24, 1989, Paper 89-95.

For 2200-2800 K, 0.8-1.1 atm:

$$k = 5.0 \times 10^{13} \exp(-19870/RT) [\text{cm}^3/\text{mol/s}].$$

Reaction rates for $\text{NH}_2 + \text{NH}$, $\text{NH}_2 + \text{H}_2$ and $\text{NH}_3 + \text{M}$ are also included in this paper.



D. F. Davidson, R. K. Hanson, , "High Temperature Reaction Rate Coefficients Derived from N-ARAS Measurements and Excimer Photolysis of NO," International Journal of Chemical Kinetics 22 (1990) 843-861. See also AIP Conference Proceedings 208, 17th International Symposium on Shock Waves and Shock Tubes, Bethlehem PA 1989, pp. 525-530.

For 1400-3500 K, 0.6 atm:

$$k = 4.29 \times 10^{13} \exp(-1564/RT) [\text{cm}^3/\text{mol/s}].$$

A reaction rate for $\text{N} + \text{H}_2$ is also included in this paper.

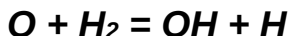


D. F. Davidson, R. K. Hanson, , "High Temperature Reaction Rate Coefficients Derived from N-ARAS Measurements and Excimer Photolysis of NO," International Journal of Chemical Kinetics 22 (1990) 843-861. See also AIP Conference Proceedings 208, 17th International Symposium on Shock Waves and Shock Tubes, Bethlehem PA 1989, pp. 525-530.

For 1400-3500 K, 0.6 atm:

$$k = 1.60 \times 10^{14} \exp(-63663/RT) [\text{cm}^3/\text{mol/s}].$$

A reaction rate for $\text{N} + \text{NO}$ is also included in this paper.



D. F. Davidson, R. K. Hanson, "A Direct Comparison of Shock Tube Photolysis and Pyrolysis Methods in the Determination of the Rate Coefficient for $\text{O} + \text{H}_2 = \text{OH} + \text{H}$," Combustion and Flame 82 (1990) 445-447.

For 2120-2750 K, 0.5 atm:

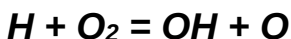
$$k = 8.13 \times 10^{14} \exp(-18955/RT) [\text{cm}^3/\text{mol/s}].$$



D. F. Davidson, R. K. Hanson, "Shock Tube Measurements of the Rate Coefficient for $N+CH_3=H_2CN+H$ using N-Atom ARAS and Excimer Photolysis of NO," Proceedings of the Combustion Institute 23 (1990) 267-273.

For 1600-2000 K, 0.5 atm:

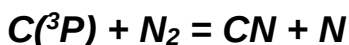
$$k = 7.1 \times 10^{13} \text{ [cm}^3\text{/mol/s]}.$$



D. A. Mastern, R. K. Hanson, C. T. Bowman, "Shock Tube Study of the Reaction of $H+O_2=OH+O$ Using OH Laser Absorption," Journal Physical Chemistry 94 (1990) 7119-7128. See also Western States Section/ The Combustion Institute 1989 Fall Meeting, Livermore CA October 23-24, 1989, Paper 89-97.

For 1450-3370 K, 0.35-2.5 atm:

$$k = 9.33 \times 10^{13} \exp(-14800/RT) \text{ [cm}^3\text{/mol/s]}.$$



A. J. Dean, R. K. Hanson, C. T. Bowman, "High Temperature Shock Tube Study of Reactions of CH and C-Atoms with N_2 ," Proceedings of the Combustion Institute 23 (1990) 267-273.

For 2660-4660 K, 0.5-1.0 atm:

$$k = 6.3 \times 10^{13} \exp(-46018/RT) \text{ [cm}^3\text{/mol/s]}.$$

This paper includes a rate for $CH+N_2$ as well.



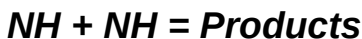
A. J. Dean, R. K. Hanson, C. T. Bowman, "High Temperature Shock Tube Study of Reactions of CH and C-Atoms with N_2 ," Proceedings of the Combustion Institute 23 (1990) 267-273.

For 2500-3800 K, 0.6-1.0 atm:

$$k = 4.4 \times 10^{12} \exp(-21976/RT) \text{ [cm}^3\text{/mol/s]}.$$

This paper includes a rate for $CH+N_2$ as well.

1986-1989

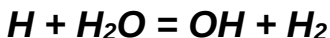


J. D. Mertens, A. Y. Chang, R. K. Hanson, C. T. Bowman, "Reaction Kinetics of NH in the Shock Tube Pyrolysis of HNCO," International Journal of Chemical Kinetics 21 (1989) 1049-1067. See also Western States Section/ The Combustion Institute 1988 Fall Meeting, Dana Point CA October 17-18, 1988, Paper 88-64.

For 2070-2730 K, 0.3-2.1 atm:

$$k = 5.1 \times 10^{13} \text{ [cm}^3\text{/mol/s]}.$$

This paper provides reaction rates for $HNCO+M$ and $NH+Ar$ as well.

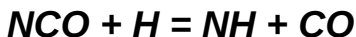


D. F. Davidson, A. Y. Chang, R. K. Hanson, "Laser Photolysis Shock Tube for Combustion Kinetics Studies," Proceedings of the Combustion Institute 22 (1988) 1877-1885.

For 1600-2500 K, 1.2 atm:

$$k = 2.4 \times 10^{14} \exp(-21360/RT) [\text{cm}^3/\text{mol/s}].$$

1974-1985

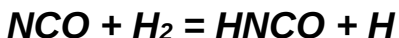


M. Y. Louge, R. K. Hanson, "High Temperature Kinetics of NCO," Combustion and Flame 58 (1984) 291-300.

For 1490 K, 0.6 atm:

$$k = 5.37 \times 10^{13} [\text{cm}^3/\text{mol/s}].$$

Rates are also presented for NCO+M and NCO+H₂.

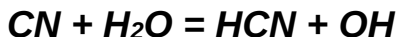


M. Y. Louge, R. K. Hanson, "High Temperature Kinetics of NCO," Combustion and Flame 58 (1984) 291-300.

For 1490 K, 0.6 atm:

$$k = 1.25 \times 10^{12} [\text{cm}^3/\text{mol/s}].$$

Rates are also presented for NCO+M and NCO+H.



A. Szekely, R. K. Hanson, C. T. Bowman, "High Temperature Determination of the Rate Coefficient for the Reaction $\text{H}_2\text{O} + \text{CN} = \text{HCN} + \text{OH}$," International Journal of Chemical Kinetics 16 (1984) 1609-1621.

For 2460-2840 K, 0.4-0.6 atm:

$$k = 2.3 \times 10^{13} \exp(-13312/RT) [\text{cm}^3/\text{mol/s}].$$

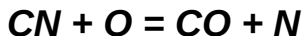


M. Louge, R. K. Hanson, "Shock Tube Study of Cyanogen Oxidation Kinetics," International Journal of Chemical Kinetics 16 (1984) 231-250.

For 2000 K, 0.6 atm:

$$k = 5.0 \times 10^{11} [\text{cm}^3/\text{mol/s}].$$

Rates are also presented for CN+O.



M. Louge, R. K. Hanson, "Shock Tube Study of Cyanogen Oxidation Kinetics," International Journal of Chemical Kinetics 16 (1984) 231-250.

For 2000 K, 0.6 atm:

$$k = 1.8 \times 10^{13} [\text{cm}^3/\text{mol/s}].$$

Rates are also presented for C₂N₂+O.

HCN + O = Products

A. Szekely, R. K. Hanson, C. T. Bowman, "*Shock Tube Study of the Reaction between Hydrogen Cyanide and Atomic Oxygen*," Proceedings of the Combustion Institute 20 (1984) 647-654.

For 1800-2600 K, 0.6 atm:

$k = 2.2 \times 10^{13} \exp(-15379/RT)$ [cm³/mol/s] for the NH+CO channel.

k is less than 5.0×10^{13} [cm³/mol/s] for the CN+OH channel.

NCO + O = CO + NO

M. Y. Louge, R. K. Hanson, "*Shock Tube Study of NCO Kinetics*," Proceedings of the Combustion Institute 20 (1984) 665-672.

For 1450 K, 0.6 atm:

$k = 5.63 \times 10^{13}$.

A rate for HCN+O is also presented in this paper.

HCN + O = NCO + H

M. Y. Louge, R. K. Hanson, "*Shock Tube Study of NCO Kinetics*," Proceedings of the Combustion Institute 20 (1984) 665-672.

For 1440 K, 0.6 atm:

$k = 5.01 \times 10^{11}$.

A rate for NCO+O is also presented in this paper.

CN + H₂ = HCN + H

A. Szekely, R. K. Hanson, C. T. Bowman, "*High Temperature Determination of the Rate Coefficient for the Reaction H₂+CN=H+HCN*," International Journal of Chemical Kinetics 15 (1983) 915-923.

For 2700-3500 K, 0.4-0.6 atm:

$k = 7.5 \times 10^{13}$ [cm³/mol/s].

CN + HCN = C₂N₂ + H

A. Szekely, R. K. Hanson, C. T. Bowman, "*High Temperature Determination of the Rate Coefficient for the Reaction H₂+CN=H+HCN*," International Journal of Chemical Kinetics 15 (1983) 1237-1241.

For 2720-3070 K, 0.4-0.6 atm:

$k = 1.0 \times 10^{13}$ [cm³/mol/s].

NH₂ + NO = Products

T. R. Roose, R. K. Hanson, C. H. Kruger, "*A Shock Tube Study of the Decomposition of NO in the Presence of NH₃*," Proceedings of the Combustion Institute 18 (1981) 853-862.

For 1680-2850 K, 0.6 atm:

$k = 3.0 \times 10^{13} \exp(-15697/RT)$ [cm³/mol/s] for all products.

$k = 7.0 \times 10^{13} \exp(-27818/RT)$ [cm³/mol/s] for the N₂O+H₂ channel.

Rates for NH+NO are also provided in this paper.

NH + NO = Products

T. R. Roose, R. K. Hanson, C. H Kruger, "A Shock Tube Study of the Decomposition of NO in the Presence of NH₃," Proceedings of the Combustion Institute 18 (1981) 853-862. See also Proceedings of the 11th International Symposium on Shock Tubes and Waves (1978) pp. 245-253.

For 1760-2850 K, 0.6 atm:

$$k = 8.0 \times 10^{13} \exp(-29407/RT) \text{ [cm}^3/\text{mol/s]}.$$

Rates for NH₂+NO are also provided in this paper.

NO + H = N + OH

W. L Flower, R. K. Hanson, C. H Kruger, "Experimental Study of Nitric Oxide Decomposition by Reaction with Hydrogen," Combustion, Science and Technology 15 (1977) 115-128. See also the Proceedings of the Combustion Institute 15 (1975) 823-832.

For 2400-4200 K, 0.6 atm:

$$k = 2.22 \times 10^{14} \exp(-50500/RT) \text{ [cm}^3/\text{mol/s]}.$$

N₂O + O = 2NO

W. L Flower, R. K. Hanson, C. H Kruger, "Experimental Study of Nitric Oxide Decomposition by Reaction with Hydrogen," Combustion, Science and Technology 15 (1977) 115-128.

For 1815-3365 K, 0.6 atm:

$$k = 6.23 \times 10^{13} \exp(-24540/RT) \text{ [cm}^3/\text{mol/s]}.$$

A rate for N₂O+M is also presented in this paper.

N₂ + O = NO + N

J. P. Monat, R. K. Hanson, C. H Kruger, "Shock Tube Determination of the Rate Coefficient for the Reaction N₂+O=NO+N," Proceedings of the Combustion Institute 15 (1975) 543-550.

For 2384-3850 K, 0.6 atm:

$$k = 1.84 \times 10^{14} \exp(-76250/RT) \text{ [cm}^3/\text{mol/s]}.$$

O + NO = N + O₂

R. K. Hanson, W. L. Flower, C. H Kruger, "Determination of the Rate Constant for the Reaction O+NO=N+O₂," Combustion Science and Technology 9 (1974) 79-86.

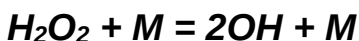
For 2500-4100 K, 0.6 atm:

$$k = 2.36 \times 10^9 T \exp(-38640/RT) \text{ [cm}^3/\text{mol/s]}.$$

Reaction Rate Measurements to May 2014

Dissociation/Association Reactions

2009



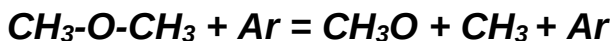
Z. Hong, A. Farooq, E. A. Barbour, D. F. Davidson, R. K. Hanson, "Hydrogen peroxide decomposition rate: a shock tube study using tunable laser absorption near 2.5 μm ," Journal of Physical Chemistry A113 (2009) 12919-12925.

Z. Hong, A. Farooq, E. A. Barbour, D. F. Davidson, R. K. Hanson, "Hydrogen peroxide decomposition rate: a shock tube study using tunable laser absorption near 2.5 μm ," Paper 09F65, Western States Section/Combustion Institute Fall 2009 Meeting, October 2009.

For 1000-1200 K, 0.9-3.2 atm:

$$k_0 = 10^{15.97} \exp(-21220 \text{ K/T}) [\text{cc/mol/s}] \text{ for } \text{M}=\text{Ar}$$

Experiments in N_2 , show a collision efficiency of 0.67



R. D. Cook, D. F. Davidson, R. K. Hanson, "High-temperature shock tube measurements of dimethyl ether decomposition and the reaction of dimethyl ether with OH," Journal of Physical Chemistry A113 (2009) 9974-9980.

R. D. Cook, D. F. Davidson, R. K. Hanson, "Rate measurements of DME decomposition and DME + OH at elevated temperatures," Paper 1P08, 6th U.S. Combustion Meeting, Ann Arbor MI May 2009.

For 1349-1790 K, 0.66-11.5 atm:

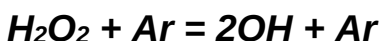
$$k_{\text{infinity}} = 4.38 \times 10^{21} \text{ T}^{-1.57} \exp(-42220 \text{ K/T}) [1/\text{s}]$$

$$k_0 = 7.52 \times 10^{15} \exp(-21537 \text{ K/T}) [\text{cc/mol/s}]$$

$$F_{\text{cent}} = 0.454 \exp(-T/2510)$$

This rate is an RRKM fit to the measured data. The rate of DME+OH was also measured.

2010



Z. Hong, R. D. Cook, D. F. Davidson, R. K. Hanson, "A shock tube study of OH + $\text{H}_2\text{O}_2 = \text{H}_2\text{O} + \text{HO}_2$ and $\text{H}_2\text{O}_2 + \text{M} = 2\text{OH} + \text{M}$ using laser absorption of H_2O and OH," Journal Physical Chemistry A114 (2010) 5718-5727.

For 1020-1460 K, 1.0-1.8 atm:

$$k = 9.5 \times 10^{15} \exp(-21250 \text{ K/T}) [\text{cm}^3/\text{mol/s}] \text{ over this T and P range.}$$

This study also includes a measurement of $\text{OH} + \text{H}_2\text{O} = \text{H}_2\text{O} + \text{HO}_2$.

Butanal = products

D. F. Davidson, S. C. Ranganath, K.-Y. Lam, M. Liaw, Z. Hong, R. K. Hanson, *"Ignition delay time measurements of normal alkanes and simple oxygenates,"* Journal of Physical Chemistry A114 (2010) 11529-11537.

For 1188-1551 K, 1.4-2.7 atm:

$$k = 3 \times 10^{16} \exp(-82000/RT) \text{ [1/s]},$$

Rate expressions for the two decomposition channels ($\text{C}_2\text{H}_5 + \text{CH}_2\text{CHO}$ and $\text{nC}_3\text{H}_7 + \text{HCO}$) were estimated to be equal that given above.

Tert-butyl-hydroperoxide = products

S. S. Vasu, Z. Hong, D. F. Davidson, R. K. Hanson, D. M. Golden, *"Shock tube/laser absorption measurements of the reaction rate of OH with ethylene and propene,"* Journal of Physical Chemistry A114 (2010) 11529-11537.

For 400-1050 K, 0.6-2.6 atm:

$$k = 8.13 \times 10^{-12} T^{7.83} \exp(-14598 \text{ K}/T) \text{ [1/s]},$$

This expression is derived from a fit of the current and lower T data.

Measurements of $\text{OH} + \text{ethylene}$ and $\text{OH} + \text{propene}$ were also performed.

2011

H+O₂+CO₂ = HO₂ + CO₂

S. S. Vasu, D. F. Davidson, R. K. Hanson, *"Shock tube study of syngas ignition in rich CO₂ mixtures and determination of the rate of H+O₂+CO₂=HO₂+CO₂,"* Energy and Fuel 25 (2011) 990-997.

For 1300 K, 8 atm:

$$k_0 = 4.2 \times 10^{18} T^{-0.86} \text{ [cm}^6\text{/mol}^2\text{/s]},$$

Measurements are in agreement with earlier recommendations from the GRI-Mech v3.0 natural gas mechanism.

H₂O₂ + M = 2OH + M

Z. Hong, D. F. Davidson, R. K. Hanson, *"An improved H₂/O₂ mechanism based on recent shock tube/laser absorption measurements,"* Combustion and Flame 158 (2011) 633-644.

For 950-3000 K:

This study provides a summary of recent rate constant measurements in the H_2/O_2 system and a validated detailed reaction mechanism with these rates.

TBHP = OH + tert-butoxy

G. A. Pang, R. K. Hanson, D. M. Golden, C. T. Bowman, "High-temperature measurements of the rate constants for reactions of OH with a series of large normal alkanes: *n*-pentane, *n*-heptane, and *n*-nonane," *Z. Phys. Chem.* 225 (2011) 1157-1178.

For 870-1360 K, 2 atm:

$$k(\text{TBHP decomposition}) = 3.57 \times 10^{13} \exp(-18000 \text{ K/T}) [1/\text{s}]$$

The paper includes a discussion of TBHP (tertbutylhydroperoxide) kinetics.

***n*-dodecane = products**

M. E. Macdonald, D. F. Davidson, R. K. Hanson, "Decomposition measurements of RP-1, RP-2, JP-7, *n*-dodecane and tetrahydroquinoline in shock tubes," *Journal of Propulsion and Power* 27 (2011) 981-989.

For 1214-1286 K, 5-44 atm:

Rate constant data are given in text. Measurements made by fitting the JetSurF 1.0 mechanism to data.

MMH = CH₃NH + NH₂

R. D. Cook, S. H. Pyun, J. Cho, D. F. Davidson, R. K. Hanson, "Shock tube measurements of species time-histories in monomethylhydrazine pyrolysis," *Combustion and Flame* 158 (2011) 790-795.

For 900-1300 K, 2 atm:

$$k = 1.50 \times 10^{58} T^{-12.84} \exp(-39580 \text{ K/T}) [1/\text{s}]$$

The rate for the CH₃ decomposition channel is given as 0-20% of the NH₂ channel resulting in an overall unimolecular decomposition rate for MMH of

$$k = 1.64 \times 10^{58} T^{-12.84} \exp(-39580 \text{ K/T}) [1/\text{s}]$$

This study also includes measurements of MMH+NH₂.

2012

H₂O₂ + M = 2OH + M

Z. Hong, D. F. Davidson, K.-Y. Lam, R. K. Hanson, "A shock tube study of the rate constants of HO₂ and CH₃ reactions," *Combustion and Flame* 159 (2012) 3007-3013.

For 1054-1249 K, 3.5 atm:

The rate measurements were determined from species time-history measurements of OH, H₂O and HO₂, and are given in tabular form in the paper along with rate constant for CH₃ +HO₂.



W. Ren, D. F. Davidson, R. K. Hanson, "IR laser absorption diagnostic for C_2H_4 in shock tube kinetics studies" Combustion and Flame 158 (2011) 633-644.

For 1519-1895 K, 2-3 atm:

$$k = 2.6 \times 10^{16} \exp(-34130 \text{ K/T}) [\text{cc/mol/s}]$$

This study provides ethylene time-histories as well.

3-pentanone = products

K.-Y. Lam, W. Ren, Z. Hong, D. F. Davidson, R. K. Hanson, "Shock tube measurements of 3-pentanone pyrolysis and oxidation" Combustion and Flame 159 (2012) 3251-3263.

For 1070-1330 K, 1.6 atm:

$$k = 4.383 \times 10^{49} T^{-10} \exp(-44780 \text{ K/T}) [1/\text{s}]$$

This study provides a rate constant for the association reaction $C_2H_4 + CO + M = CH_3CHCO + M$.

2013

Methyl formate = products

W. Ren, K.-Y. Lam, S. H. Pyun, A. Farooq, D. F. Davidson, R. K. Hanson, "Shock tube/laser absorption studies of the decomposition of methyl formate," Proceedings of the Combustion Institute 34 (2013) 453-461.

For 1202-1607 K, 1.4-1.7 atm:

$$k(\text{CH}_3\text{OH} + \text{CO}) = 1.1 \times 10^{13} \exp(-29556 \text{ K/T}) [1/\text{s}]$$

$$k(\text{CH}_2\text{O} + \text{CH}_2\text{O}) = 2.6 \times 10^{12} \exp(-32052 \text{ K/T}) [1/\text{s}]$$

$$k(\text{CH}_4 + \text{CO}_2) = 4.4 \times 10^{11} \exp(-29078 \text{ K/T}) [1/\text{s}]$$

This study provides species time-history measurements of all the major O-atom carriers in the MF decomposition system.

Acetone = $CH_3 + CH_3CO$

2-butanone = products

K.-Y. Lam, W. Ren, S. H. Pyun, A. Farooq, D. F. Davidson, R. K. Hanson, "Multi-species time-history measurements during high-temperature acetone and 2-butanone pyrolysis," Proceedings of the Combustion Institute 34 (2013) 607-615.

For 1100-1600 K, 1.6 atm:

$$k(\text{acetone}) = 2.46 \times 10^{14} \exp(-69300/\text{RT}) [1/\text{s}] \text{ at } 1.6 \text{ atm}$$

$$k(2\text{-butanone}) = 6.08 \times 10^{13} \exp(-63100/\text{RT}) [1/\text{s}] \text{ at } 1.5 \text{ atm}$$

This study provides a suggested rate constant for methyl ketene decomposition as well.

Trioxane = 3 CH₂O

S. Wang, D. F. Davidson, R. K. Hanson, "High-temperature laser absorption diagnostics for CH₂O and CH₃CHO and their application to shock tube kinetics studies," Combustion and Flame 160 (2013) 1930-1938.

For 869-1037 K, 2.1 atm:

$$k = 3.58 \times 10^{12} \exp(-18590 \text{ K/T}) [1/\text{s}] \text{ at } 2.1 \text{ atm}$$

This study provides species time-history measurements of aldehydes during pyrolysis.

Cyclohexene = C₂H₄ + 1,3-butadiene

I. Stranic, D. F. Davidson, R. K. Hanson, "Shock tube measurements of the rate constant for the reaction cyclohexene = ethylene + 1,3-butadiene," Chemical Physics Letters 584 (2013) 18-23.

For 950-1300 K, 0.8-3.7 atm:

$$k = 4.84 \times 10^{14} \exp(-31900 \text{ K/T}) [1/\text{s}]$$

The study provides a very accurate determination of the rate constant used in chemical thermometers in shock tube experiments..

CH₂OH = CH₂O + H

CH₃O = CH₂O + H

E. E. Dames, D. M. Golden, "Master equation modeling of the unimolecular decompositions of hydroxymethyl (CH₂OH) and methoxy (CH₃O) radicals to formaldehyde (CH₂O) + H," Journal of Physical Chemistry A 117 (2013) 7686-7696.

For 600-1800 K, 0.01-100 atm:

This study provides Troe-format-fitted expressions for the two title reactions.

2014

Ethylamine = C₂H₅ + NH₂

Ethylamine = CH₂NH₂ + CH₃

S. Li, D. F. Davidson, R. K. Hanson, "Shock tube study of ethylamine pyrolysis and oxidation," Accepted by Combustion and Flame April 2014.

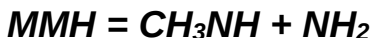
S. Li, D. F. Davidson, R. K. Hanson, K. Moshhammer, K. Kohse-Höinghaus, "Shock tube study of ethylamine pyrolysis and oxidation," Paper 070RK-0075, 8th U.S. National Combustion Meeting, University of Utah, May 19-22, 2013.

For 1220-1448 K, 0.85-2.0 atm:

$$k(\text{C}_2\text{H}_5 + \text{NH}_2) = 4.49 \times 10^{12} \exp(-63158 / \text{RT}) [1/\text{s}]$$

$$k(\text{CH}_2\text{NH}_2 + \text{CH}_3) = 5.58 \times 10^{12} \exp(-61826 / \text{RT}) [1/\text{s}]$$

The study includes NH₂ and OH species time-history measurements from oxidation and pyrolysis experiments.



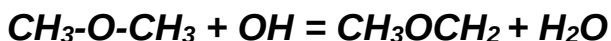
S. Li, D. F. Davidson, R. K. Hanson, “Shock tube study of the pressure dependence of monomethylhydrazine,” Combustion and Flame 161 (2014) 16-22.

For 1100-1400 K, 0.3-5 atm:

Tabulated data for this reaction are given in the paper. This study also includes a rate constant for $\text{NHNH}_2 + \text{H} = 2\text{NH}_2$.

Bimolecular Reactions

2009



R. D. Cook, D. F. Davidson, R. K. Hanson, “High-temperature shock tube measurements of dimethyl ether decomposition and the reaction of dimethyl ether with OH,” Journal of Physical Chemistry A113 (2009) 9974-9980.

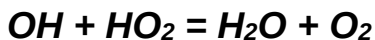
R. D. Cook, D. F. Davidson, R. K. Hanson, “Rate measurements of DME decomposition and DME + OH at elevated temperatures,” Paper 1P08, 6th U.S. Combustion Meeting, Ann Arbor MI May 2009.

For 923-1423 K, 1.65 atm:

$$k = 6.32 \times 10^6 T^2 \exp(328 \text{ K}/T) \text{ [cc/mol/s]}$$

This rate from Curran et al. was found to be in excellent agreement with this measurement. The rate of DME+Ar was also measured.

2010



Z. Hong, S. S. Vasu, D. F. Davidson, R. K. Hanson, “Experimental study of the rate of $\text{OH} + \text{HO}_2 = \text{H}_2\text{O} + \text{O}_2$ at high temperatures using the reverse reaction,” Journal Physical Chemistry A114 (2010) 5520-5525.

For 1600-2200 K, 1.7 atm:

$$k = 3.3 \times 10^{13} \text{ [cm}^3\text{/mol/s]}$$

This study provides a non-Arrhenius fit to this data and the lower temperature data of Keyser 1988.

OH + H₂O₂ = H₂O + HO₂

Z. Hong, R. D. Cook, D. F. Davidson, R. K. Hanson, "A shock tube study of OH + H₂O₂ = H₂O + HO₂ and H₂O₂ + M = 2OH + M using laser absorption of H₂O and OH," Journal Physical Chemistry A114 (2010) 5718-5727.

For 1020-1460 K, 1.0-1.8 atm:

$k = 4.6 \times 10^{13} \exp(-2630 \text{ K/T}) [\text{cm}^3/\text{mol/s}]$

This study also includes a measurement of H₂O₂+M.

OH + n-butanol = products

S. S. Vasu, D. F. Davidson, R. K. Hanson, D. M. Golden, "Measurements of the reaction of OH with n-butanol at high-temperatures," Chemical Physics Letters 497 (2010) 26-29.

For 1017-1269 K, 2.25 atm:

Data are given in tabular format. Discussion about the individual branching channels is included.

OH + ethylene = products***OH + propene = products***

S. S. Vasu, Z. Hong, D. F. Davidson, R. K. Hanson, D. M. Golden, "Shock tube/laser absorption measurements of the reaction rate of OH with ethylene and propene," Journal of Physical Chemistry A114 (2010) 11529-11537.

For 973-1438 K, 2-10 atm:

$k = 2.23 \times 10^4 T^{2.745} \exp(-1115 \text{ K/T}) [\text{cc/mol/s}]$, +ethylene, 600-2000K

$k = 1.94 \times 10^6 T^{2.229} \exp(-540 \text{ K/T}) [\text{cc/mol/s}]$, +propene, 700-1700K

These expressions are TST calculations fit to the current and lower T data.

A measurement of TBHP (tertbutylhydroperoxide) = products was also performed.

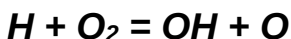
OH + 1,3-butadiene = products

S. S. Vasu, J. Zador, D. F. Davidson, R. K. Hanson, D. M. Golden, "High-temperature measurements and a theoretical study of the reaction of OH with 1,3-butadiene," Journal of Physical Chemistry A114 (2010) 8312-8318.

For 1011-1406 K, 2.2 atm:

Data is given in tabular format. Theoretical double Arrhenius expressions are given for certain channels as well.

2011

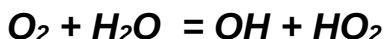
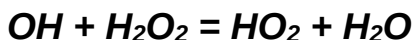
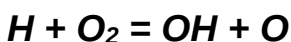


Z. Hong, D. F. Davidson, E. A. Barbour, R. K. Hanson, "A new shock tube study of $H + O_2 = OH + O$ reaction rate using tunable diode laser absorption of H_2O near $2.5 \mu m$," Proceedings of the Combustion Institute 33 (2011) 309-316.

For 1100-1530 K, 2 atm:

$$k = 1.126 \times 10^{14} \exp(-7805 \text{ K/T}) [\text{cm}^3/\text{mol/s}]$$

This study provides another rate constant expression over a wider temperature range using the data of Masten et al. (1990).



Z. Hong, D. F. Davidson, R. K. Hanson, "An improved H_2/O_2 mechanism based on recent shock tube/laser absorption measurements," Combustion and Flame 158 (2011) 633-644.

For 950-3000 K:

This study provides a summary of recent rate constant measurements in the H_2/O_2 system and a validated detailed reaction mechanism with these rates. Improved values for $H_2O_2 + M$ are included as well.



S. S. Vasu, L. K. Huynh, D. F. Davidson, R. K. Hanson, D. M. Golden, "Reactions of OH with butene isomers: measurements of the overall rates and a theoretical study," Journal of Physical Chemistry A 115 (2011) 2549-2556.

For 880-1341 K, 2.2 atm:

Data is given in tabular format. Theoretical discussions about the rate constants for 1-butene, trans-2-butene and cis-2-butene are also given.



G. A. Pang, R. K. Hanson, D. M. Golden, C. T. Bowman, "High-temperature measurements of the rate constants for reactions of OH with a series of large normal alkanes: *n*-pentane, *n*-heptane, and *n*-nonane," Z. Phys. Chem. 225 (2011) 1157-1178.

For 870-1360 K, 2 atm:

$k(\text{OH} + n\text{-pentane}) = 1.27 \times 10^{14} \exp(-2038 \text{ K/T}) [\text{cc/mol/s}]$

$k(\text{OH} + n\text{-heptane}) = 1.47 \times 10^{14} \exp(-1804 \text{ K/T}) [\text{cc/mol/s}]$

$k(\text{OH} + n\text{-nonane}) = 1.91 \times 10^{14} \exp(-1801 \text{ K/T}) [\text{cc/mol/s}]$

$k(\text{OH} + \text{CH}_3) = 1.65 \times 10^{14} [\text{cc/mol/s}]$

The paper includes a discussion of TBHP (tertbutylhydroperoxide) kinetics.



R. D. Cook, S. H. Pyun, J. Cho, D. F. Davidson, R. K. Hanson, "Shock tube measurements of species time-histories in monomethyl hydrazine pyrolysis," Combustion and Flame 158 (2011) 790-795.

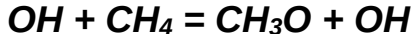
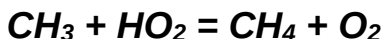
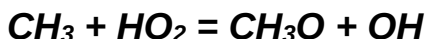
For 900-1300 K, 2 atm:

$k = 3.70 \times 10^{14} \exp(-2620 \text{ K/T}) [1/\text{s}]$

The recommended rate constant for the reaction $\text{CH}_3\text{NNH} + \text{NH}_2 = \text{CH}_3\text{NN} + \text{NH}_3$ is higher than the value proposed by Sun et al.

This study also include measurements of MMH decomposition.

2012



Z. Hong, D. F. Davidson, K.-Y. Lam, R. K. Hanson, "A shock tube study of the rate constants of HO_2 and CH_3 reactions," Combustion and Flame 159 (2012) 3007-3013.

For 1054-1249 K, 3.5 atm:

$k(\text{CH}_3\text{O} + \text{OH}) = 6.8 \times 10^{12} [\text{cc/mol/s}]$

$k(\text{CH}_4 + \text{O}_2) = 4.4 \times 10^{12} [\text{cc/mol/s}]$

$k(\text{OH} + \text{CH}_4)$ = Given in tabular form in paper

The rate measurements were determined from species time-history measurements of OH, H_2O and HO_2 .



G. A. Pang, R. K. Hanson, D. M. Golden, C. T. Bowman, "Rate constant measurements for the overall reaction of $\text{OH} + 1\text{-butanol} = \text{products}$ from 900 to 1200 K," Journal of Physical Chemistry A 116 (2011) 2475-2483.

For 900-1200 K, 1-2 atm:

$k = 1.95 \times 10^{14} \exp(-2505 \text{ K/T}) [\text{cc/mol/s}]$

The paper includes a discussion of the high-pressure decomposition rate constants for $\text{C}_4\text{H}_9\text{O}$.

OH + sec-butanol = products

G. A. Pang, R. K. Hanson, D. M. Golden, C. T. Bowman, "Experimental determination of the high-temperature rate constant for the reaction of OH with sec-butanol," Journal of Physical Chemistry A116 (2011) 9607-9613.

For 887-1178 K, 1 atm:

$$k = 4.2 \times 10^{13} \exp(-1550 \text{ K/T}) [\text{cc/mol/s}]$$

The paper includes a three-parameter fit using previous lower temperature data covering 263-1178K.

OH + iso-butanol = products

G. A. Pang, R. K. Hanson, D. M. Golden, C. T. Bowman, "High-temperature rate constant determination for the reaction of OH with iso-butanol," Journal of Physical Chemistry A116 (2012) 4720-4725.

For 907-1147 K, 1 atm:

The paper includes a rate for ($k_{\text{overall}} - k_{\text{beta}}$) and a recommendation for the overall rate constant.

OH + small methyl esters = products

K.-Y. Lam, D. F. Davidson, R. K. Hanson, "High-temperature measurements of the reactions of OH with small methyl esters: methyl formate, methyl acetate, methyl propanoate, and methyl butanoate," Journal of Physical Chemistry A116 (2012) 12229-12241.

For 876-1371 K, 1.5 atm:

$$k(\text{OH} + \text{methyl formate}) = 2.56 \times 10^{13} \exp(-2026 \text{ K/T}) [\text{cc/mol/s}]$$

$$k(\text{OH} + \text{methyl acetate}) = 3.59 \times 10^{13} \exp(-2438 \text{ K/T}) [\text{cc/mol/s}]$$

$$k(\text{OH} + \text{methyl propanoate}) = 6.65 \times 10^{13} \exp(-2539 \text{ K/T}) [\text{cc/mol/s}]$$

$$k(\text{OH} + \text{methyl butanoate}) = 1.13 \times 10^{14} \exp(-2515 \text{ K/T}) [\text{cc/mol/s}]$$

The paper includes a discussion of the use of the Kwok and Atkinson structure-activity relations to predict these rate constants.

OH + ketones = products

K.-Y. Lam, D. F. Davidson, R. K. Hanson, "High-temperature measurements of the reactions of OH with a series of ketones: acetone, 2-butanone, 3-pentanone, and 2-pentanone," Journal of Physical Chemistry A116 (2013) 5549-5559.

For 870-1360 K, 1-2 atm:

$$k(\text{OH} + \text{acetone}) = 3.30 \times 10^{13} \exp(-2437 \text{ K/T}) [\text{cc/mol/s}]$$

$$k(\text{OH} + 2\text{-butanone}) = 6.35 \times 10^{13} \exp(-2270 \text{ K/T}) [\text{cc/mol/s}]$$

$$k(\text{OH} + 3\text{-pentanone}) = 9.29 \times 10^{13} \exp(-2361 \text{ K/T}) [\text{cc/mol/s}]$$

$$k(\text{OH} + 2\text{-pentanone}) = 7.06 \times 10^{13} \exp(-2020 \text{ K/T}) [\text{cc/mol/s}]$$

The paper also includes a discussion of the use of the Kwok and Atkinson structure-activity relations to predict these rate constants.

2013

H + iso-butanol = products

I. Stranic, S. H. Pyun, D. F. Davidson, R. K. Hanson, “*Multi-species measurements in 2-butanol and i-butanol pyrolysis*,” Combustion and Flame 160 (2013) 1012-1019.

For 1270-1640 K, 1.3-1.9 atm:

The paper provides recommended rate constants for the various channels of the H+isobutanol reactions, which when implemented improved the simulations of the Sarathy et al. (2012) mechanism.

OH + tert-butanol = products

I. Stranic, G. A. Pang, R. K. Hanson, D. M. Golden, C. T. Bowman, “*Shock tube measurements of the tert-butanol + OH reaction rate and the tert-C₄H₈OH radical β -scission branching ratio using isotopic labeling*,” Journal of Physical Chemistry A117 (2013) 4777-4784.

I. Stranic, G. A. Pang, R. K. Hanson, D. M. Golden, C. T. Bowman, “*Shock tube measurements of the tert-butanol + OH reaction rate and the tert-C₄H₈OH radical β -scission branching ratio using isotopic labeling*,” Paper 0014, 8th U.S. National Combustion Meeting, University of Utah, May 19-22, 2013.

For 900-1200 K, 1.1 atm:

$$k = 7.48 \times 10^{13} \exp(-2501 \text{ K/T}) [\text{cc/mol/s}]$$

The paper includes rates for the different H-atoms abstraction sites using isotopic measurements.

OH + H₂ = H₂O + H

K.-Y. Lam, D. F. Davidson, R. K. Hanson, “*A shock tube study of H₂ + OH = H₂O + H using OH laser absorption*,” International Journal of Chemical Kinetics (2013) 363-373.

For 902-1518 K, 1.15-1.52 atm:

$$k = 4.38 \times 10^{13} \exp(-3518 \text{ K/T}) [\text{cc/mol/s}]$$

This study provides a high accuracy measurement of this reaction rate constant.

OH + HO₂ = H₂O + O₂

HO₂ + HO₂ = H₂O₂ + O₂

Z. Hong, K.-Y. Lam, R. Sur, S. Wang, D. F. Davidson, R. K. Hanson, “*On the rate constants of OH + HO₂ and HO₂ + HO₂: a comprehensive study of H₂O₂ thermal decomposition using multi-species laser absorption*,” Proceedings of the Combustion Institute (2013) 565-571.

For 1072-12830 K, 1.7 atm:

$$k(\text{OH}+\text{HO}_2) = 7.0 \times 10^{12} \exp(550 \text{ K/T}) + 4.5 \times 10^{14} \exp(-5500 \text{ K/T}) [\text{cm}^3/\text{mol/s}]$$

$$k(\text{HO}_2+\text{HO}_2) = 1.0 \times 10^{14} \exp(-5556 \text{ K/T}) - 1.9 \times 10^{11} \exp(709 \text{ K/T}) [\text{cm}^3/\text{mol/s}]$$

This rate constant for HO_2+HO_2 is unchanged from Kappel et al (2002).

2014

OH + methanol = products

L. T. Zaczek, K.-Y. Lam, D. F. Davidson, R. K. Hanson, "A shock tube study of $\text{CH}_3\text{OH} + \text{OH} = \text{products}$ using OH laser absorption," Accepted by the Proceedings of the Combustion Institute, April 2014.

For 961-1231 K, 1.2-1.5 atm:

$$k = 5.71 \times 10^4 T^{2.62} \exp(343 \text{ K/T}) [\text{cc}/\text{mol/s}]$$

This three-parameter fit uses the combined data from Hess and Tully (1989, Jimenez et al (2002), Dillion et al. (2004) and the current study.

OH + ethanol = products

I. Stranic, G. A. Pang, R. K. Hanson, D. M. Golden, C. T. Bowman, "Shock tube measurements of the rate constant for the reaction ethanol + OH," Journal of Physical Chemistry A 118 (2014) 822-828.

For 900-1270 K, 1.1 atm:

$$k = 5.07 \times 10^5 T^{2.31} \exp(608 \text{ K/T}) [\text{cc}/\text{mol/s}]$$

This measurement used isotopic ethanol to isolate the different H-atom abstraction sites.

OH + small aldehydes = products

S. Wang, D. F. Davidson, R. K. Hanson, "High temperature measurements for the rate constants of $\text{C}_1\text{-C}_4$ aldehydes with OH in a shock tube," Accepted by the Proceedings of the Combustion Institute, April 2014.

For 950-1400 K, 1-2 atm:

$$k(\text{CH}_2\text{O}) = 1.02 \times 10^7 T^{1.92} \exp(779 \text{ K/T}) [\text{cc}/\text{mol/s}]$$

$$k(\text{CH}_3\text{CHO}) = 4.32 \times 10^6 T^{2.02} \exp(716 \text{ K/T}) [\text{cc}/\text{mol/s}]$$

$$k(\text{C}_2\text{H}_5\text{CHO}) = 5.94 \times 10^6 T^{1.98} \exp(823 \text{ K/T}) [\text{cc}/\text{mol/s}]$$

$$k(\text{nC}_3\text{H}_7\text{CHO}) = 5.36 \times 10^6 T^{2.06} \exp(658 \text{ K/T}) [\text{cc}/\text{mol/s}]$$

This study provides the first measurements for the aldehydes acetaldehyde, propionaldehyde and n-butyraldehyde at high temperatures.

OH + ethylamine = products

OH + dimethylamine = products

S. Li, D. F. Davidson, R. K. Hanson, “*High-temperature measurements of the reaction of OH with ethylamine and dimethylamine,*” Submitted to Journal of Physical Chemistry A, November 2013.

For 901-1368 K, 1.2 atm:

$$k(\text{ethylamine}) = 1.10 \times 10^7 T^{1.93} \exp(1450 \text{ K}/T) \text{ [cc/mol/s]}$$

$$k(\text{dimethylamine}) = 2.26 \times 10^4 T^{2.69} \exp(1797 \text{ K}/T) \text{ [cc/mol/s]}$$

This study is in good agreement with VTST calculations by Galano and Alverrez-Idaboy (2008).

NHNH₂ + H = 2NH₂

S. Li, D. F. Davidson, R. K. Hanson, “*Shock tube study of the pressure dependence of monomethylhydrazine,*” Combustion and Flame 161 (2014) 16-22.

For 1100-1400 K, 0.3-5 atm:

$$k = 1.5 \times 10^{14} \exp(-755 \text{ K}/T) \text{ [cc/mol/s]}$$

This study also includes a rate constant for MMH = CH₃NH + NH₂.

Reaction Rate Measurements to June 2019

Dissociation/Association Reactions

2015

Acetone = Products

S. Wang, K. Sun, D.F. Davidson, J.B. Jeffries and R.K. Hanson, "Shock Tube Measurement of Acetone Dissociation using Cavity-enhanced Absorption Spectroscopy of CO," J. Phys. Chem. A 119 7257-7262 (2015) [DOI 10.1021/jp511642a](https://doi.org/10.1021/jp511642a)

For 1004-1494 K, 1.6 atm

$$k = 4.39 \times 10^{55} T^{-11.394} \exp(-52140/T) \pm 24\% \text{ s}^{-1}$$

2016

CH₄ + Ar = CH₃ + H + Ar

S. Wang, D.F. Davidson and R.K. Hanson, "Improved Shock Tube Measurement of the CH₄ + Ar = CH₃ + H + Ar Rate Constant using UV Cavity-enhanced Absorption Spectroscopy of CH₃," J. Phys. Chem. A 120 (28) 5427–5434 (2016) [DOI 10.1021/acs.jpca.6b02572](https://doi.org/10.1021/acs.jpca.6b02572)

For 1487-1866 K and near 1.7 atm

$$k_d = 3.7 \times 10^{16} \exp(-42200/T) \text{ cm}^3/\text{mol/s}$$

Acetaldehyde = CH₃ + HCO

S. Wang, D.F. Davidson and R.K. Hanson, "Shock Tube Measurement for the Dissociation Rate Constant of Acetaldehyde using Sensitive CO Diagnostic," J. Phys. Chem. A 120 (35) 6895–6901 (2016) [DOI 10.1021/acs.jpca.6b03647](https://doi.org/10.1021/acs.jpca.6b03647)

For 1377-1571 K

$$k_d = 5.5 \times 10^{12} \exp(-32900/T) \text{ s}^{-1} \text{ (0.34 atm)}$$

For 1273-1618 K

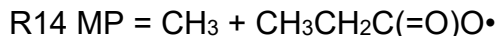
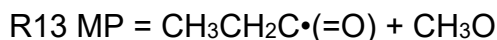
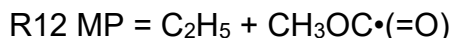
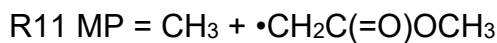
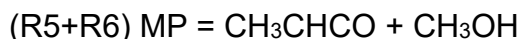
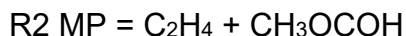
$$k_d = 1.1 \times 10^{14} \exp(-36700/T) \text{ s}^{-1} \text{ (1.6 atm)}$$

2017

Methyl Propanoate = Products

H. Ning, J. Wu, L. Ma, W. Ren, D.F. Davidson and R.K Hanson, "Chemical Kinetic Modeling and Shock Tube Study of Methyl Propanoate Decomposition," *Comb. and Flame* 184 30-40 (2017) [DOI 10.1016/j.combustflame.2017.06.001](https://doi.org/10.1016/j.combustflame.2017.06.001)

Calculated rate constants (1000-2000 K) for the following reactions are given in the paper.



Methylcyclohexane = Products

S. Wang, T. Parise, S.E. Johnson, D.F. Davidson and R.K. Hanson, "A New Diagnostic for Hydrocarbon Fuels using 3.41-micron Diode Laser Absorption," *Comb. and Flame* 186 129-139 [DOI 10.1016/j.combustflame.2017.07.026](https://doi.org/10.1016/j.combustflame.2017.07.026)

For 1260–1400 K and near 1.5 atm:

$$k_d = 3.3 \times 10^{15} \exp(-38000 \text{ K}/T) \text{ s}^{-1} (+ 28\%/-34\%)$$

2019

H + O₂ + M = HO₂ + M

J. Shao, R. Choudhary, A. Susa, D. F. Davidson, R. K. Hanson, "Shock Tube Study of the Rate Constants for H + O₂ + M -> HO₂ + M (M = Ar, H₂O, CO₂, N₂) at Elevated Pressures," *Proceedings of the Combustion Institute* 37, 145-152 (2019). <https://doi.org/10.1016/j.proci.2018.05.077>

For pressures of 12-33 atm, at temperatures near 1200 K:

$$k_0 [\text{M} = \text{Ar}] = 2.6 \times 10^{-32} (T/300)^{-1.2} \text{ cm}^6 \text{ molecule}^{-2} \text{ s}^{-1} (\pm 12\%)$$

$$k_0 [\text{M} = \text{N}_2] = 3.6 \times 10^{-32} (T/300)^{-1.2} \text{ cm}^6 \text{ molecule}^{-2} \text{ s}^{-1} (\pm 12\%)$$

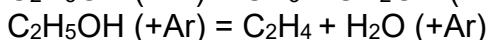
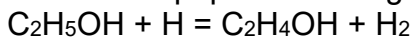
$$k_0 [\text{M} = \text{CO}_2] = 13.0 \times 10^{-32} (T/300)^{-1.2} \text{ cm}^6 \text{ molecule}^{-2} \text{ s}^{-1} (\pm 24\%)$$

$$k_0 [M = \text{H}_2\text{O}] = 59.8 \times 10^{-32} (T/300)^{-1.2} \text{ cm}^6 \text{ molecule}^{-2} \text{ s}^{-1} (\pm 19\%)$$

Ethanol = Products

Rishav Choudhary, Yuzhe Peng, Jiankun Shao, David F. Davidson, Ronald K. Hanson, "Multi-species time history measurements during ethanol pyrolysis behind reflected shock waves," 11th U. S. National Combustion Meeting, Organized by the Western States Section of the Combustion Institute, March 24–27, 2019, Pasadena, California

An archival paper extending this paper is in preparation.



H + O₂ + M = HO₂ + M

R. Choudhary, J. J. Girard, Y. Peng, J. Shao, D. F. Davidson, R. K. Hanson, "Measurement of the reaction rate of H + O₂ + M -> HO₂ + M for M = Ar, N₂, CO₂ at high temperature with a sensitive OH direct absorption diagnostic," submitted to Combustion and Flame 2018

For temperatures of 1450 to 2000 K, at pressures of 10-22 atm:

$$K [M=\text{Ar}] = 2.659 \times 10^{19} T^{-1.36} \text{ cm}^6 \text{ mol}^{-2} \text{ s}^{-1}$$

$$K [M=\text{N}_2] = 2.55 \times 10^{21} T^{-1.95} \text{ cm}^6 \text{ mol}^{-2} \text{ s}^{-1}$$

$$K [M=\text{CO}_2] = 2.233 \times 10^{18} T^{-0.79} \text{ cm}^6 \text{ mol}^{-2} \text{ s}^{-1}$$

Bimolecular Reactions

2014

CH₂O + H = HCO + H₂

S. Wang, E. Dames, D.F Davidson and R.K. Hanson, "Reaction Rate Constant of CH₂O + H=HCO + H₂ Revisited: A Combined Study of Direct Shock Tube Measurement and Transition State Theory Calculation," J. Phys. Chem. A 118 10201-10209 (2014) [DOI 10.1021/jp5085795](https://doi.org/10.1021/jp5085795)

For 1304-2006 K, near 1 atm

$$k = 1.97 \times 10^{11} T^{1.06} \exp(-3818/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$$

For 200-3000 K

$$k = 5.86 \times 10^3 T^{3.13} \exp(-762/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$$

2015

CH₃ + OH = Products

S. Wang, S. Li, D.F. Davidson and R.K. Hanson, "Shock Tube Measurement of the High-temperature Rate Constant for OH + CH₃ → Products," J. Phys. Chem. A 119 (33) 8799–8805 (2015) DOI [10.1021/acs.jpca.5b05725](https://doi.org/10.1021/acs.jpca.5b05725)

For 1214–1933 K and 0.35–1.37 atm
 $k = 41.9 T^{3.15} \exp(5270/T) \text{ cm}^3/\text{mol/s}$

2016

Aldehydes + OH = Products

S. Wang, D.F. Davidson and R.K. Hanson, "Rate Constants of Long, Branched, and Unsaturated Aldehydes with OH at Elevated Temperatures," 36th Symp. (Int.) on Combustion, Seoul, Korea, 2016 DOI [10.1016/j.proci.2016.06.017](https://doi.org/10.1016/j.proci.2016.06.017)

For 976–1346 K and pressures near 0.6 atm, rates are provided for the following aldehydes + OH : isobutyraldehyde (i-C₃H₇CHO), *n*-valeraldehyde (*n*-C₄H₉CHO), isovaleraldehyde (i-C₄H₉CHO), *trans*-2-pentenal (C₂H₅CH=CHCHO), trimethylacetaldehyde ((CH₃)₃CCHO) and benzaldehyde (C₆H₅CHO).

$$k(\text{iC}_3\text{H}_7\text{CHO}) = 7.70 \times 10^{13} \exp(-1700 \text{ K}/T)$$

$$k(\text{nC}_4\text{H}_9\text{CHO}) = 1.03 \times 10^{14} \exp(-1730 \text{ K}/T)$$

$$k(\text{iC}_4\text{H}_9\text{CHO}) = 1.02 \times 10^{14} \exp(-1750 \text{ K}/T)$$

$$k(\text{CH}_3)_3\text{CCHO} = 7.20 \times 10^{13} \exp(-1670 \text{ K}/T)$$

$$k(\text{C}_2\text{H}_5\text{CHCHCHO}) = 6.93 \times 10^{13} \exp(-1340 \text{ K}/T)$$

$$k(\text{C}_6\text{H}_5\text{CHO}) = 3.10 \times 10^{13} \exp(-1380 \text{ K}/T)$$

2017

H + O₂ = O + OH

OH + CO = CO₂ + H

S. Wang, D.F. Davidson and R.K. Hanson, "A Shock Tube and Laser Absorption Study of CH₂O Oxidation via Simultaneous Measurements of OH and CO," J. Phys. Chem. A 121,45,8561–8568 (2017).

<https://pubs.acs.org/doi/abs/10.1021/acs.jpca.7b09362>

For 1428 to 1685 K:

$\text{H} + \text{O}_2 = \text{O} + \text{OH}$: $k = 8.04 \times 10^{13} \exp(-7370 \text{ K}/T) \text{ cc/mol/s}$ (A high accuracy measurement of this rate.)

$\text{OH} + \text{CO} = \text{CO}_2 + \text{H}$: $k = 1.90 \times 10^{12} \exp(-2760 \text{ K}/T) \text{ cc/mol/s}$

2019

JP-10 + OH = Products

L.T. Zaczek and R.K. Hanson, "Direct Measurement of the JP-10+OH=Products Reaction Rate in Shock Tube Experiments," AIAA Paper 2019-2367, Aerospace Sciences Meeting, January 2019. <https://arc.aiaa.org/doi/pdf/10.2514/6.2019-2367>

An archival version of this paper with a fuller, revised dataset is in preparation.

C₂H₄ + H = C₂H₃ + H₂

J. Shao, R. Choudhary, Y. Peng, D.F. Davidson and R.K. Hanson, "Shock Tube Measurement of the C₂H₄ + H -> C₂H₃ + H₂ Rate Constant," J. Phys. Chem. A 2019, 123, 1, 15-20. [DOI: 10.1021/acs.jpca.8b09452](https://doi.org/10.1021/acs.jpca.8b09452)

For 1619-1948 K, near 10 atm:

$k = 3.9 \times 10^{22} T^{3.62} \exp(-5670/T) \text{ cc/molecule/s.}$

C₂H₅OH + H = C₂H₄OH + H₂

Rishav Choudhary, Yuzhe Peng, Jiankun Shao, David F. Davidson, Ronald K. Hanson, "Multi-species time history measurements during ethanol pyrolysis behind reflected shock waves," 11th U. S. National Combustion Meeting, Organized by the Western States Section of the Combustion Institute, March 24–27, 2019, Pasadena, California

An archival paper extending this paper is in preparation.

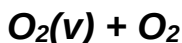
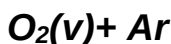
$\text{C}_2\text{H}_5\text{OH} + \text{H} = \text{C}_2\text{H}_4\text{OH} + \text{H}_2$

$\text{C}_2\text{H}_5\text{OH} (+\text{Ar}) = \text{CH}_3 + \text{CH}_2\text{OH} (+\text{Ar})$

$\text{C}_2\text{H}_5\text{OH} (+\text{Ar}) = \text{C}_2\text{H}_4 + \text{H}_2\text{O} (+\text{Ar}).$

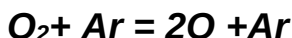
High-Temperature O₂ Kinetics Reaction Rates

2016



Kyle G. Owen, David F. Davidson, and Ronald K. Hanson. "Oxygen Vibrational Relaxation Times: Shock Tube/Laser Absorption Measurements", *Journal of Thermophysics and Heat Transfer*, Vol. 30, No. 4 (2016), pp. 791-798. <https://doi.org/10.2514/1.T4505>

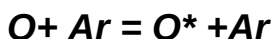
Vibrational Relaxation time for 1000 to 4000 K and pressures of 0.2 to 1 atm. Values are in excellent agreement with Millikan and White (1963) *J. Chem. Phys.* 39 3209-3213 (1963).



K. Owen, D.F. Davidson and R.K. Hanson, "Measurements of Oxygen Dissociation Using Laser Absorption," *AIAA J. of Thermophysics and Heat Transfer* 30 (2) 274-278 (2016) DOI 10.2514/1.T4506

Dissociation rates for 4400 to 7900 K and pressures of 0.2 to 1 atm. Values are in good agreement with Camac and Vaughn (1961) *J. Chem. Phys.* 34 460-470 (1961).

$$k_d = 2.87 \times 10^{18} T^{-1} \exp(-58580/T) (+/-7\%) \text{ cm}^3/\text{mol/s}.$$



M. Nations, S. Wang, C.S. Goldenstein, D.F. Davidson and R.K. Hanson, "Kinetics of Excited Oxygen Formation in Shock-Heated O₂-Ar Mixtures," *J. Phys. Chem. A* 120, 8234-8243, (2016). DOI 10.1021/acs.jpca.6b07274

Rate constants (k_M) for O atom electronic excitation from the ground state (3P) due to collisions with argon atoms were directly inferred using the reduced collisional-radiative model.

$$k_M(3P \rightarrow 5S^{\circ}) = 7.8 \times 10^{-17} T^{0.5} \exp(-1.061 \times 10^5 \text{ K}/T) \pm 25\% \text{ cm}^3 \text{ s}^{-1}$$

$$k_M(3P \rightarrow 3S^{\circ}) = 2.5 \times 10^{-17} T^{0.5} \exp(-1.105 \times 10^5 \text{ K}/T) \pm 25\% \text{ cm}^3 \text{ s}^{-1}$$

2019

$O_2(v) + M$

$O_2 + M = 2O + M$

J.W. Streicher, A. J. Krish, S. Wang, D.F. Davidson and R.K. Hanson, "Measurements of Oxygen Vibrational Relaxation and Dissociation using Ultraviolet Laser Absorption in Shock Tube Experiments," AIAA Paper 2019-0795, Aerospace Sciences Meeting, Jan. 2019.

<https://arc.aiaa.org/doi/pdf/10.2514/6.2019-0795>

This paper covers $M = O_2$ and Ar for temperatures of 2000-9000 K. An extended archival version of this paper is in preparation.

